CHAPTER

7

Group Reporting VI

Complex Consolidation Issues



LEARNING OBJECTIVES

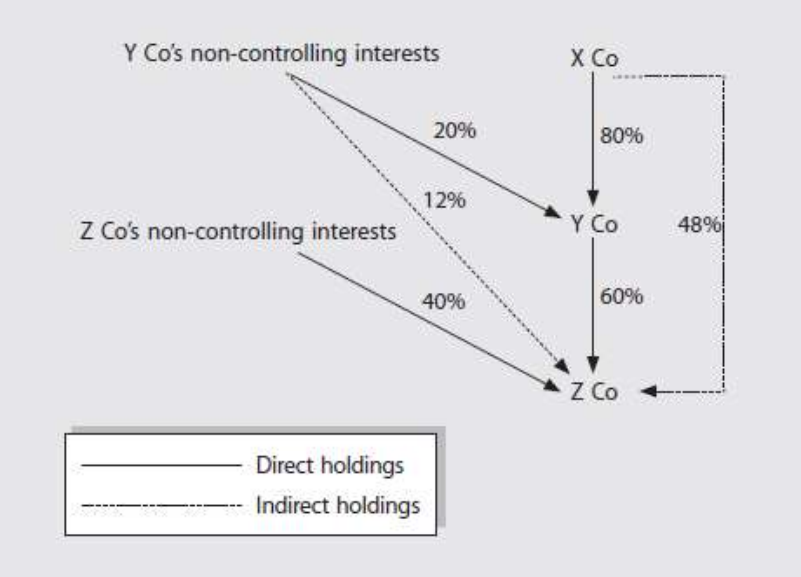
After reading this chapter, you should be able to:

- **LO1** Appreciate the implications of indirect ownership interests on consolidation and equity accounting;
- **LO2** Prepare consolidation adjustments and equity accounting entries for multi-tier group structures;
- **LO3** Appreciate the implications of business combinations achieved in stages and changes in ownership interests in a subsidiary;
- **LO4** Prepare consolidation adjustments for changes in ownership interests in a subsidiary, with and without change in control;
- **LO5** Analytically determine amounts in consolidated financial statements in more complex settings;
- **LO6** Prepare consolidation adjustments for asset transfers in more complex settings;
- **LO7** Understand the temporary differences in profit recognition arising from consolidation and the cost and equity methods; and
- **LO8** Appreciate the primary difference between a single entity's cash flow statement and consolidated cash flow statement.

INDIRECT OWNERSHIP INTERESTS

Chapter 2 discussed the concept of direct and indirect holdings. IFRS 10 *Consolidated Financial Statements* and IAS 28 *Investment in Associates and Joint Ventures* require an investor to consider indirect holdings to determine whether "control" or "significant influence" exists. A parent has an *indirect ownership interest* in a subsidiary when equity in that subsidiary is held through one or more of the parent's subsidiaries. In the same manner, non-controlling interests (NCI) or outside shareholders can also have an *indirect interest* in a subsidiary. Consider the ownership structure shown in Figure 7.1.

FIGURE 7.1 Group structure of X Co and subsidiaries



In the structure, X Co, which owns 80% of Y Co (which in turn owns 60% of Z Co), is known as the *ultimate parent company* while Y Co is the *intermediate parent company* of Z Co. Although X Co's indirect interest (or effective interest) is only 48% (less than 50%) of Z Co, X Co is the parent of Z Co because it is able to control Z Co through Y Co. Majority voting rights are present in all the linkages from X Co to Z Co. In the absence of evidence to the contrary, majority voting power (more than 50%) *in each linkage* is a basis for *de facto* control.

The affiliation structure includes non-controlling interests in Y Co and Z Co. Non-controlling interests may comprise several individual or corporate investors in Y Co and Z Co. Their unknown identity is not important as their interests are collectively represented in the income statement and statement of financial position as a single item. There are two types of non-controlling interests:

1. Direct non-controlling interests
2. Indirect non-controlling interests

In Figure 7.1, percentage ownership of *direct* non-controlling interests in Y Co is 20% while that in Z Co is 40%. Non-controlling shareholders of Y Co *indirectly* own 12% of the shares of Z Co through Y Co. Note that the sum of the majority shareholders' indirect interest in the lower-tiered subsidiary (i.e., 48% in Z Co) and the non-controlling

shareholders' indirect interest in the subsidiary (i.e., 12% in Z Co) is equal to the intermediate parent's share in the subsidiary (i.e., 60% that Y Co has in Z Co). For ease of reference, we describe the lower-tiered subsidiary that is controlled through an intermediate parent as an "indirect subsidiary."

Direct and Indirect Non-controlling Interests

Direct non-controlling interests will be subject to the usual consolidation procedures that Chapter 5 discusses. These procedures include allocating to direct non-controlling interests their share of the subsidiary's:

1. Fair value as at the date of acquisition;
2. Dividend payment;
3. Profit/(loss) after tax for the current period after adjustments for amortization of fair value adjustments and unrealized profit or loss from upstream sales; and
4. Change in post-acquisition retained earnings, other comprehensive income and other changes in equity.

However, *indirect non-controlling interests*, through their investment in the intermediate parent, will be entitled to their share of the indirect subsidiary's:

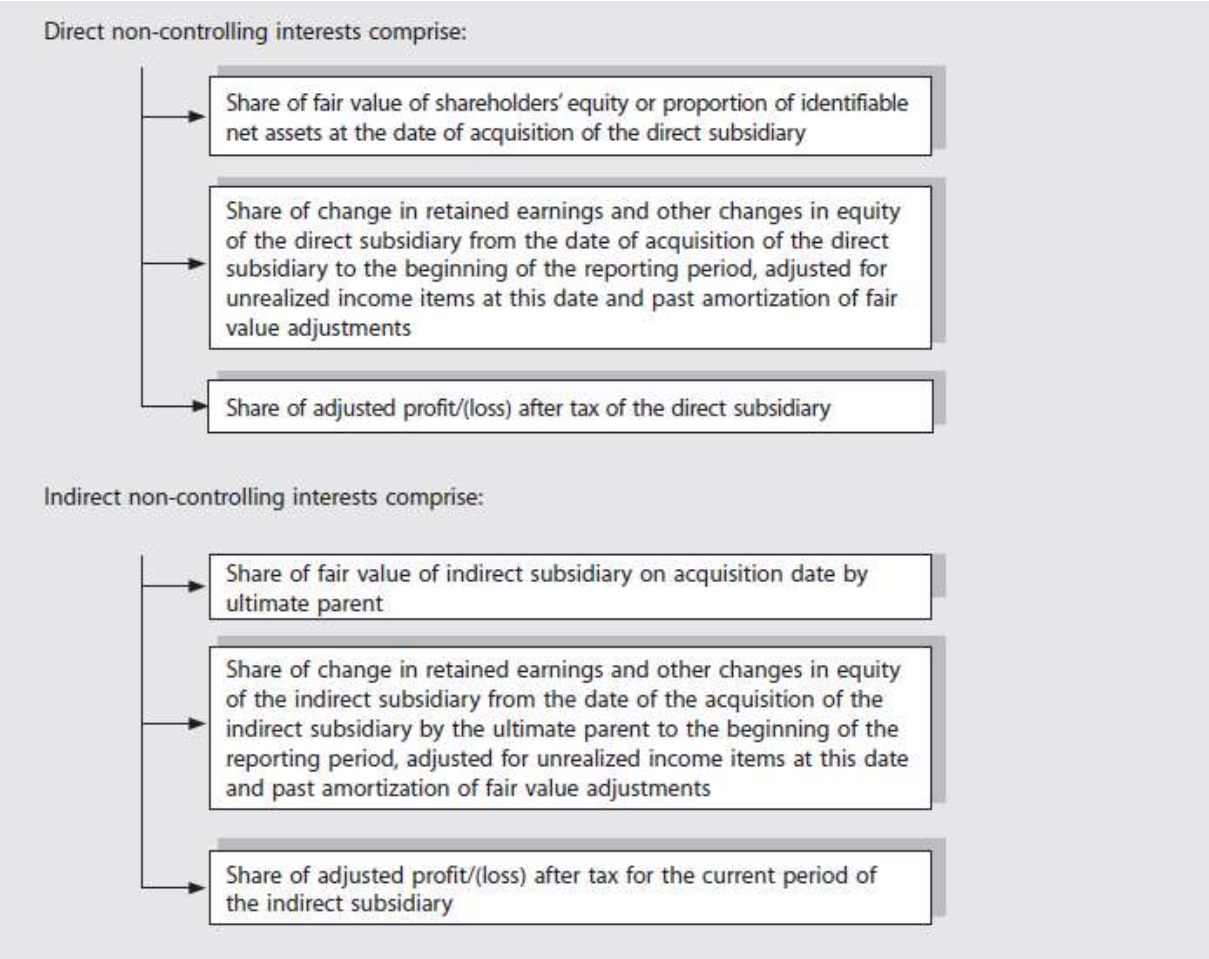
1. Fair value as at date of acquisition by the ultimate parent;
2. Profit/(loss) after tax for the current period after adjustments for amortization of fair value adjustments and unrealized profit or loss from upstream sales; and
3. Change in post-acquisition retained earnings, other comprehensive income and changes in equity only from the date when the ultimate parent acquires the indirect subsidiary.

Indirect non-controlling interests do *not* have a share of the indirect subsidiary's dividend payments as the payments are made to direct owners of the subsidiary only. Hence, the dividend elimination entry does not apply to indirect non-controlling interests.

Timing of Allocation of Share of Profit or Losses to Non-controlling Interests

Direct non-controlling interests have an interest in the subsidiary's profit or loss from the date when the subsidiary was incorporated. The changes in ownership between different non-controlling shareholders are not important as they are collectively seen as one body for purposes of consolidation. There is no distinction between pre-acquisition and post-acquisition retained earnings for direct non-controlling interests. Through consolidation procedures, non-controlling interests are built up chronologically, as shown in Figure 7.2.

FIGURE 7.2 Re-enactment of direct and indirect non-controlling interests



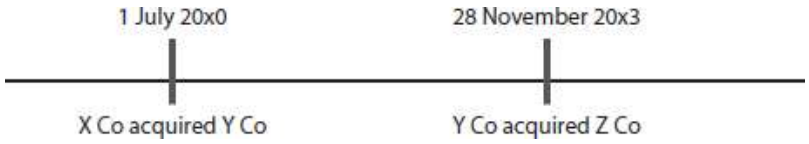
We can see from Figure 7.2 that the process of re-enactment of direct and indirect non-controlling interests involve recognizing share of equity of direct and indirect subsidiaries chronologically. Accounting for direct non-controlling interests should be familiar to us by now. Chapter 4 dealt with accounting for direct non-controlling interests at great length and in that chapter, we saw that non-controlling interests at the end of a reporting period comprises three components:

Non-controlling interests in a direct subsidiary = Share of book value of equity of the direct subsidiary at the end of a reporting period + Share of remaining excess of fair value (FV) over book value (BV) of identifiable net assets of the direct subsidiary + Goodwill (assuming Alternative 1 method of recognizing non-controlling interests) in the direct subsidiary.

Interestingly, the same relationship applies to indirect non-controlling interests as well. Indirect non-controlling interests also have a share in the three components of the indirect subsidiary. We consider two situations below and analyze how the three components can be worked out.

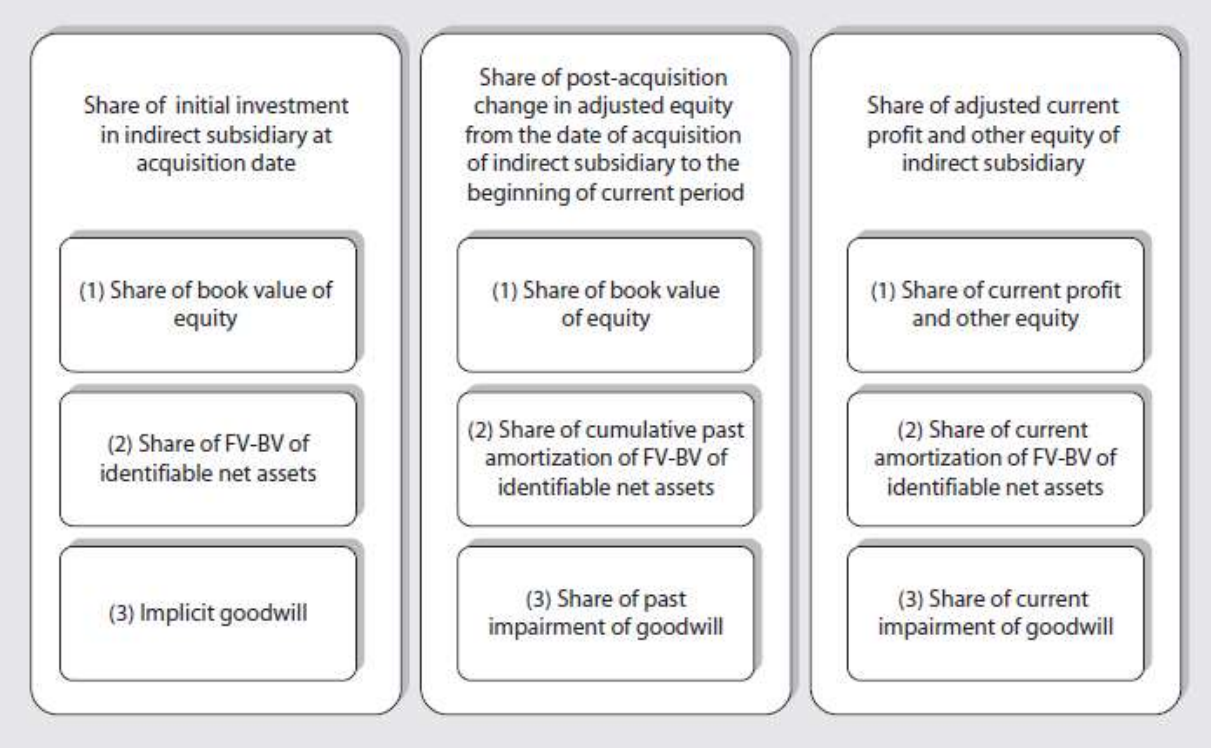
Acquisition of an Indirect Subsidiary After the Intermediate Parent is Acquired

Let us assume that X Co, the ultimate parent acquired Y Co first on 1 July 20x0. Subsequently, Y Co acquired Z Co as a stand-alone entity on 28 November 20x3. The non-controlling interests of Y Co would have a share of the initial investment that Y Co has in Z Co as at 28 November 20x3. In this sequence of events, the acquisition date of Z Co is 28 November 20x3. The timeline is shown below:



From Figure 7.3, we see that the indirect non-controlling interests in Z Co have a share of the initial investment in Z Co at acquisition date, 28 November 20x3. After 28 November 20x3, indirect interests will participate in the share of profits and changes in equity of Z Co.

FIGURE 7.3 Components of indirect non-controlling interests



Hence, at any point in time after acquisition of the indirect subsidiary, the following relationship holds.

Non-controlling interests in an indirect subsidiary = Share of book value of equity of the indirect subsidiary + Share of remaining excess of fair value (FV) over book value (BV) of identifiable net assets of the indirect subsidiary + Goodwill (assuming Alternative 1 method of recognizing non-controlling interests) in the indirect subsidiary.

However, in our analysis of non-controlling interests in an indirect subsidiary, we must be careful to avoid double counting of net assets. The investment in the indirect subsidiary that resides on the intermediate parent's statement of financial position must be removed in our analytical checks.

For example, in Figure 7.1, Y Co's non-controlling interests have stakes in both Y Co and Z Co.

Y Co's non-controlling interests have a share of 20% *direct interests* in the book value of Y Co and 20% *direct interests* of the remaining excess of value over book value of identifiable net assets of Y Co and goodwill in Y Co.

However, the book value of Y Co's net assets include investment in Z Co. To avoid double counting the interests in Z Co twice, we must remove the investment in Z Co from book value of Y Co.

Non-controlling interests in an intermediate parent (e.g., Y Co in Figure 7.1) = Share of book value of equity of the intermediate parent at the end of a reporting period ***less investment in the indirect subsidiary*** *+ Share of remaining excess*

of fair value (FV) over book value (BV) of identifiable net assets of the intermediate parent + Goodwill (assuming Alternative 1 method of recognizing non-controlling interests) in the intermediate parent.

Y Co's non-controlling interests also have a share of 12% *indirect interests* in the book value of Z Co and 12% *indirect interests* of the remaining excess of fair value over book of identifiable net assets of Z Co through Y Co.

However, Y Co's non-controlling interests will have a share of 20% of the goodwill that Y Co has in Z Co because goodwill in Z Co resides in Y Co's books, explained below in the arithmetic analysis.

Goodwill in Z Co attributable to Y Co = Investment in Z Co – 60% × Fair value of identifiable net assets of Z Co

Hence,

Investment in Z Co

= Goodwill in Z Co + 60% × Fair value of identifiable net assets of Z Co

Since, non-controlling interests in Y Co has a 20% interest in Y Co, it also has an interest of 20% in the investment in Z Co as shown below.

Indirect non-controlling interests in Z Co at acquisition date (28 November 20x3)

= 20% × Investment in Z Co

= 20% × ((Goodwill in Z Co attributable to Y Co) + (60% × Fair value of identifiable net assets of Z Co))

= 20% × Goodwill in Z Co + 12% × Fair value of identifiable net assets of Z Co

= 20% × Goodwill in Z Co + 12% × (Fair value – Book value) of identifiable net assets of Z Co + 12% × Book value of identifiable net assets of Z Co

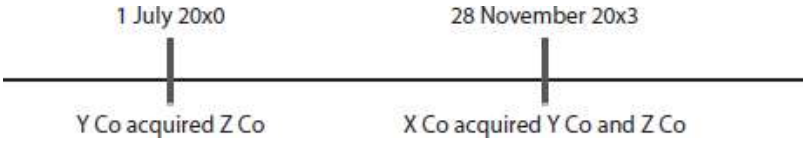
Hence,

Non-controlling interests in an indirect subsidiary = Indirect interests in the book value of equity of the indirect subsidiary + Indirect interests in the remaining excess of fair value (FV) over book value (BV) of identifiable net assets of the indirect subsidiary + Direct interests in the goodwill of the indirect subsidiary that is attributable to the intermediate parent

An analysis of the non-controlling interests in a situation where a subsidiary is acquired subsequent to the acquisition of an intermediate parent is shown in Illustration 7.1.

Acquisition of a Sub-group that Includes an Indirect Subsidiary

In many acquisitions, the acquirer obtains control of an intermediate parent that has subsidiaries of its own. In this instance, the indirect non-controlling interests would have had interests in the sub-group even before the acquirer obtains control of the intermediate parent and its subsidiaries. Consider the following timeline:



IFRS 3 *Business Combinations* requires the acquirer (X Co) to measure the non-controlling interests at either full fair value at acquisition date on 28 November 20x3 (including goodwill under Alternative 1) or as a proportion of the fair values of identifiable net assets (Alternative 2). In other words, it is a fresh start approach for non-controlling interests in the acquirer's consolidated financial statements.

Non-controlling interests in the direct and indirect subsidiary still comprise the three components (assuming that Alternative 1 is adopted). We will still need to deduct the investment balance in the indirect subsidiary from the net assets of the intermediate parent when analyzing the non-controlling interests' balance.

Non-controlling interests in an intermediate parent (e.g., Y Co in Figure 7.1) = Share of book value of equity of the intermediate parent at the end of a reporting period ***less investment in the indirect subsidiary*** *+ Share of remaining excess*

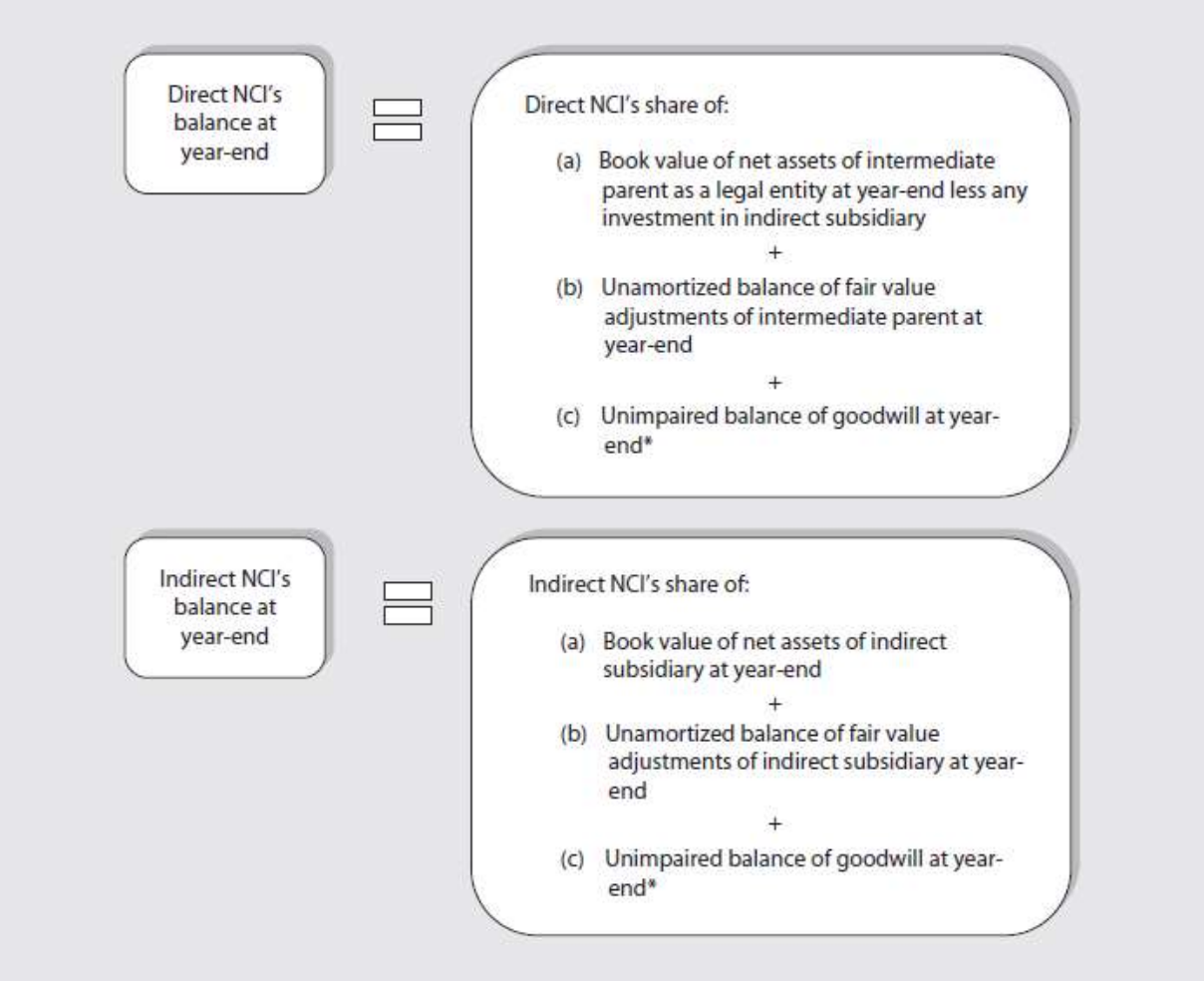
of fair value (FV) over book value (BV) of identifiable net assets of the intermediate parent + Goodwill (assuming Alternative 1 method of recognizing non-controlling interests) in the intermediate parent

However, using the fresh start approach, non-controlling interests in the acquired sub-group will be measured at fair value at acquisition date, 28 November 20x3. The old balances arising from 1 July 20x0 to 28 November 20x3 will be discarded and a new value imputed. Since non-controlling interests of the intermediate parent are also non-controlling interests of the indirect subsidiary, the fair value of non-controlling interests of the intermediate parent will include interests in the indirect subsidiary as well. Direct non-controlling interests will also be measured at its own fair value. Goodwill attributable to the non-controlling interests of the intermediate parent and subsidiaries will have to be determined on the basis of the imputed fair values at date of acquisition by the ultimate parent.

Non-controlling interests in an indirect subsidiary = Indirect interests in the book value of equity of the indirect subsidiary + Indirect interests in the remaining excess of fair value (FV) over book value (BV) of identifiable net assets of the indirect subsidiary + Goodwill in the indirect subsidiary

In Figure 7.4, we show diagrammatically the analytical checks on direct non-controlling interests in an intermediate parent and the indirect non-controlling interests in the intermediate parent's subsidiaries.

FIGURE 7.4 Analytical checks on non-controlling interests in an intermediate parent and its subsidiaries



* Goodwill may be combined as the fair value of non-controlling interests of the intermediate parent at acquisition date is often determined for the sub-group as a unit.

Practically, the goodwill is determined for the sub-group as a unit rather than the individual entities within the sub-group, as follows, using the structure in Figure 7.1:

Goodwill in Y Co and Z Co = Fair value of consideration to acquire Y + Fair value of non-controlling interests of Y and indirect non-controlling interests of Z + Fair value of direct non-controlling interests of Z – (Fair value of identifiable net assets of Y – Investment in Z) – Fair value of identifiable net assets of Z

Goodwill attributable to non-controlling interests in Y Co and indirect non-controlling interests in Z Co = Fair value of non-controlling interests of Y and indirect non-controlling interests of Z – (20% × (Fair value of identifiable net assets of Y – Investment in Z)) – (12% × Fair value of identifiable net assets of Z)

Goodwill attributable to direct non-controlling interests in Z Co = Fair value of direct non-controlling interests of Z – 40% × Fair value of identifiable net assets of Z}

An example of the analysis of non-controlling interests in a sub-group is shown in Illustration 7.2.

DUAL APPROACH TO CONSOLIDATION OF INDIRECT NON-CONTROLLING INTERESTS IN SUBSIDIARIES

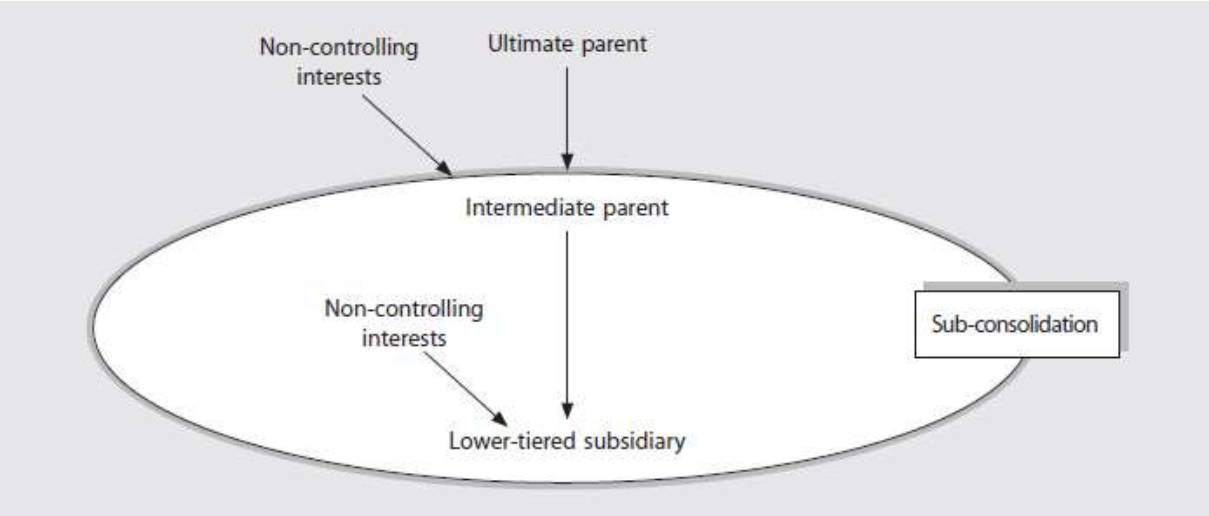
We can account for indirect non-controlling interests in subsidiaries in one of two ways:

1. Sequential or hierarchical consolidation or
2. Simultaneous or multiple consolidation

Sequential or Hierarchical Consolidation

This method involves a *series of sub-consolidations*, in a bottom-up approach beginning at the lowest level in the group and progressing upwards. For example, in Figure 7.5, the intermediate parent will consolidate the lower-tiered subsidiary, and the ultimate parent will consolidate the figures that arise from the first sub-consolidation.

FIGURE 7.5 Sequential or hierarchical consolidation



In Figure 7.1:

1. Y Co, the intermediate parent company with 60% interest in Z Co, will consolidate Z Co first.
2. Y Co will deduct non-controlling interests of 40% in Z Co's net profit after tax in the consolidated sub-group financial statements.
3. In the second consolidation, X Co, the ultimate parent company, will consolidate Y Co's consolidated sub-group figures.
4. Since X Co owns 80% of Y Co, non-controlling interests of 20% in Y Co's consolidated net profit after tax and non-controlling interests will be allocated to non-controlling interests of Y Co. Hence, effectively 12% of Z Co's net profit after tax is allocated to indirect non-controlling interests (i.e., 20% interest in Y Co multiplied by 60% that Y Co has in Z Co).
5. Through the series of consolidations, a total of 52% of Z Co's net profit after tax and 20% of Y Co's net profit after tax will be allocated to non-controlling interests in the higher-level consolidated income statement.

Simultaneous or Multiple Consolidation

The effects of sequential consolidation may be replicated through simultaneous or multiple consolidation. *Simultaneous consolidation* of all directly and indirectly held subsidiaries is carried out by the ultimate parent on the same consolidation worksheet. The consolidation worksheet will incorporate the income statements and statements of financial position of the ultimate parent, intermediate parent(s), and subsidiaries. Income of the lower-tiered subsidiary is allocated immediately to indirect non-controlling interests. Multiple consolidation is particularly useful and more efficient when there is no compelling reason to prepare consolidated financial statements for the intermediate parent company. IFRS 10 paragraph 4 exempts an intermediate parent company from preparing consolidated financial statements when the intermediate parent:

1. Is itself a wholly owned subsidiary;
2. Is a partially owned subsidiary but its non-controlling interests including those not otherwise entitled to vote have been informed about, and do not object to, the intermediate parent not presenting consolidated financial statements;
3. Has no debt or equity instruments that are publicly traded or traded on an over-the-market counter;
4. Has not filed or is not in the process of filing its financial statements with a securities commission or other regulatory organization for the purpose of issuing any class of instruments in a public market; and
5. The ultimate or any intermediate parent of the parent produces consolidated IFRS-compliant financial statements that are available for public use.

How Does Simultaneous Consolidation Differ from Sequential Consolidation?

Normal consolidation principles and procedures apply to simultaneous consolidation with a few critical exceptions as detailed below. The main thrust is to know when to apply the indirect ownership percentage in the elimination and allocation entries. In consolidating the financial statements of an ultimate parent and its direct and indirect subsidiaries simultaneously, we must remember that the consolidation adjustments are made from the point of view of the ultimate parent. For example, in Figure 7.1, consolidated retained earnings of the group must include 100% of retained earnings of X (ultimate parent), 80% share of post-acquisition retained earnings of Y Co (direct subsidiary of X), and 48% share of post-acquisition retained earnings of Z Co (indirect subsidiary of X). The acquisition date is also determined from the perspective of the ultimate parent. To determine the share of post-acquisition retained earnings of the indirect subsidiary, we have to determine the date when the ultimate parent acquired the indirect subsidiary through the direct subsidiary.

page 450

1. The fair value of consideration transferred is eliminated against the consolidated shareholders' equity of the direct subsidiary as at the date of acquisition. If the direct subsidiary owns another subsidiary or subsidiaries at the date of acquisition by the ultimate parent, elimination of investment is made against pre-acquisition retained earnings and other comprehensive income and other reserves as of date of acquisition by the ultimate parent. Thus, in the group structure in Figure 7.1:

(a) The investment in Y Co carried in X Co's books will be eliminated against the share capital, pre-acquisition retained earnings, other comprehensive income and other reserves of Y Co. However, if Y Co acquired Z Co prior to the acquisition of Y Co by X Co, the investment in Y Co will be eliminated against the share capital of Y Co, and the consolidated retained earnings and other reserves of Y Co and Z Co. A fair valuation of the sub-group is carried out at this point.

(b) To arrive at the consolidated equity of the sub-group, it is necessary to eliminate the investment in Z Co and the share capital and pre-acquisition retained earnings, other comprehensive income and other equity of Z Co. In simultaneous consolidation, investment in subsidiaries and indirect subsidiaries are eliminated at the same time. The investment in Z Co carried in Y Co's books will be eliminated against the share capital, pre-acquisition retained earnings, other comprehensive income and other reserves of Z Co as at date of acquisition by the ultimate parent.

2. In simultaneous consolidation, the sum of both direct and indirect non-controlling interests' share of current profit after tax and post-acquisition changes are allocated to non-controlling interests. In the group structure in Figure 7.1, the income allocated to non-controlling interests constitutes the sum of the:

   (a) *Direct non-controlling interests* in the profit after tax of the immediate subsidiary (i.e., 20% of Y Co's net profit after tax and 40% of Z Co's net profit after tax); and

   (b) *Indirect non-controlling interests* in the profit after tax of the lower-tiered subsidiaries (i.e., 12% of Z Co's net profit after tax is allocated to Z Co's indirect non-controlling interests).

3. Elimination of dividend income against dividends declared entails only direct ownership interests since dividend is paid to the legal owners of a subsidiary.

4. In determining indirect non-controlling interests' share of the profit or loss of an indirect subsidiary, dividend income from lower-tiered subsidiaries that is recognized in the income statement of the intermediate parent should be removed to avoid the double-counting of income allocated to indirect non-controlling interests. Under the simultaneous consolidation process, income from the lower-tiered subsidiaries is allocated to indirect non-controlling interests in step 2(b) above. Hence, dividend income from a direct subsidiary that is recorded in the separate financial statements of the intermediate parent should be removed to avoid the recognition of income in two different forms. The elimination of dividend income included in an intermediate parent's profit or loss is also consistent with the principle of eliminating intragroup income.

Illustration 7.1 explains the application of simultaneous consolidation to a multi-level group structure.

ILLUSTRATION 7.1 Simultaneous consolidation

Section B explains the step-by-step execution of simultaneous consolidation.

Section A: Question

The acquisition details are as follows:

	A Ltd	B Ltd
Acquired by	P Ltd	A Ltd
Date of acquisition	1 January 20x0	1 July 20x0
Equity at acquisition		
Share capital	$30,000	$30,000
Retained earnings	10,000	5,000
	$40,000	$35,000
Fair value of consideration transferred	$32,000	$35,000

Percentage acquired	75%	80%
Fair value of non-controlling interests	$10,000	$ 8,000

Income Statement and Partial Statement of Changes in Equity
For the Year Ended 31 December 20x2

	P Ltd	A Ltd	B Ltd
Operating profit	$20,000	$12,000	$19,000
Dividend income	6,000	4,000	
Profit before tax	$26,000	$16,000	$19,000
Tax	(4,000)	(2,400)	(3,800)
Profit after tax	$22,000	$13,600	$15,200
Retained earnings 1 January 20x2	21,000	17,000	6,000
Dividends declared	(12,000)	(8,000)	(5,000)
Retained earnings 31 December 20x2	$31,000	$22,600	$16,200

Statement of Financial Position
As at 31 December 20x2

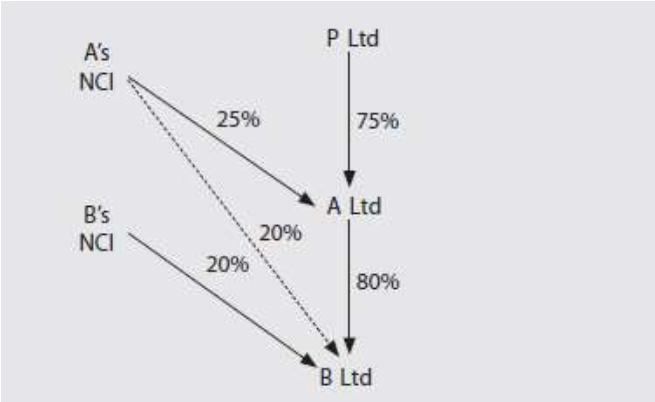
	P Ltd	A Ltd	B Ltd
Share capital	$60,000	$30,000	$30,000
Retained earnings	31,000	22,600	16,200
	$91,000	$52,600	$46,200
Investment in A Ltd	$32,000		
Investment in B Ltd		$35,000	
Other non-current assets	31,000	20,000	$23,000
Current assets	90,000	35,000	40,000
Current liabilities	(62,000)	(37,400)	(16,800)
	$91,000	$52,600	$46,200

Investments in subsidiaries are carried at cost. Book values of net identifiable assets are close to their fair values.

Section B: Solution

Step 1: Identify the direct and indirect non-controlling interests in the group structure.

Consider the following multi-level group structure:



	A Ltd	B Ltd
Direct non-controlling interests	25%	20%
Indirect non-controlling interests (Indirect non-controlling interests in B = 25% × 80%)		20%
Total non-controlling interests	25%	40%

Another way to arrive at B's total non-controlling interests is to subtract from 100%, P's effective interest in B of 60% (i.e., 75% × 80%). The remaining effective interest of 40% represents both the direct and indirect non-controlling interests in B.

What is the significance of the indirect non-controlling interests? Indirect non-controlling interests will have a share of post-acquisition retained earnings and current profits. However, only direct non-controlling interests feature in the elimination of dividends as the holding of investment and the payment of dividends relate to direct ownership interests only.

In this situation, A acquired B on 1 July 20x0 after P acquired A. Hence effectively, P acquired B on 1 July 20x0.

Step 2: Now work out the consolidation entry to eliminate the investment.

CJE1: Eliminate investment in A and A's shareholders' equity[1] as at the date of acquisition (1 January 20x0)

Dr Share capital	30,000	
Dr Retained earnings	10,000	
Dr Goodwill	2,000	
Cr Investment in A		32,000
Cr Non-controlling interests		10,000

$$\text{Goodwill} = \left(\begin{matrix}\text{Fair value of}\\ \text{acquisition cost}\end{matrix} + \begin{matrix}\text{Fair value of}\\ \text{non-controlling interests}\end{matrix}\right) - \text{Fair value of net identifiable assets}$$
$$= (\$32{,}000 + \$10{,}000) - \$40{,}000$$
$$= \$42{,}000 - \$40{,}000$$
$$= \$2{,}000$$

CJE2: Eliminate investment in B and B's shareholders' equity as at the date of acquisition (1 July 20x0)

Dr Share capital	30,000	

Dr Retained earnings	5,000	
Dr Goodwill	8,000	
Cr Investment in B		35,000
Cr Non-controlling interests		8,000

$$\text{Goodwill} = \left(\begin{matrix}\text{Fair value of}\\ \text{acquisition cost}\end{matrix} + \begin{matrix}\text{Fair value of}\\ \text{non-controlling interests}\end{matrix}\right) - \text{Fair value of net identifiable assets}$$
$$= (\$35{,}000 + \$8{,}000) - \$35{,}000$$
$$= \$43{,}000 - \$35{,}000$$
$$= \$8{,}000$$

Step 3: Work out non-controlling interests' share of post-acquisition retained earnings from the date of acquisition to the beginning of the year. Now be careful to apply both the direct and indirect interests in allocating post-acquisition retained earnings to non-controlling interests.

CJE3: Assign direct non-controlling interests' share of post-acquisition retained earnings of subsidiary A

Dr Retained earnings	1,750	
Cr Non-controlling interests		1,750 (25% × ($17,000 – $10,000))

CJE4: Assign direct and indirect non-controlling interests' share of post-acquisition retained earnings of subsidiary B

Dr Retained earnings	400	
Cr Non-controlling interests		400 (40% × ($6,000 – $5,000))

Step 4: Now work out non-controlling interests' share of current profit after tax. As in Step 3, be careful to apply the application of indirect non-controlling interests' percentage for the subsidiary that has more than one parent

CJE5: Assign direct non-controlling interests' share of current profit after tax for subsidiary A

Dr Income to non-controlling interests	2,400	
Cr Non-controlling interests		2,400
A's profit after tax for 20x2		$13,600
Less dividend income from B		(4,000)
		$ 9,600
Non-controlling interests' share at 25%		$ 2,400

Exclude dividend income because NCI will recognize share of profit of B in CJE6. Dividend is a repayment of profit rather than a basis for profit recognition. Assume dividend is tax-exempt.

CJE6: Assign direct and indirect non-controlling interests' share of current profit after tax of subsidiary B

Dr Income to non-controlling interests	6,080	
Cr Non-controlling interests		6,080
B's profit after tax for 20x2		$15,200
Direct and indirect non-controlling interests' share		40%

Total non-controlling interests' share			$ 6,080
CJE7: Elimination of dividends declared by B			
Dr Dividend income (A)	4,000		
Dr Non-controlling interests	1,000	(NCI % × Dividends declared = 20% × $5,000)	
Cr Dividends declared (B)			5,000
CJE8: Elimination of dividends declared by A			
Dr Dividend income (P)	6,000		
Dr Non-controlling interests	2,000	(NCI % × Dividends declared = 25% × $8,000)	
Cr Dividends declared (A)			8,000

Step 5: We compile the legal entities financial statements in one consolidation worksheet and enter the consolidation adjustments above. We also perform the analytical check of non-controlling interests.

Income Statement						
				Consolidation adjustments		
	P Ltd	A Ltd	B Ltd	Dr	Cr	Consolidated totals
Operating profit	$20,000	$12,000	$19,000			$51,000
Dividend income	6,000	4,000		$ 4,000		
				6,000		
Profit before tax	$26,000	$16,000	$19,000			$51,000
Tax	(4,000)	(2,400)	(3,800)			(10,200)
Profit after tax	$22,000	$13,600	$15,200			$40,800
Income to non-controlling interests				2,400		(8,480)
				6,080		
Retained profit						$32,320
Retained earnings, 1 Jan	21,000	17,000	6,000	10,000		26,850
				5,000		
				1,750		
				400		
Dividends declared	(12,000)	(8,000)	(5,000)		5,000	(12,000)
					8,000	
Retained earnings, 31 Dec	$31,000	$22,600	$16,200	$35,630	$13,000	$47,170

Statement of Financial Position						
				Consolidation adjustments		
	P Ltd	A Ltd	B Ltd	Dr	Cr	Consolidated totals
Share capital	$60,000	$30,000	$30,000	$ 30,000		$ 60,000
				30,000		
Retained earnings	31,000	22,600	16,200	35,630	$ 13,000	47,170
Non-controlling interests				1,000	10,000	25,630
				2,000	8,000	
					1,750	
					400	
					2,400	
					6,080	
	$91,000	$52,600	$46,200			$132,800
Investment in A Ltd	$32,000				32,000	
Investment in B Ltd		$35,000			35,000	
Goodwill				2,000		$ 10,000
				8,000		
Other non-current assets	31,000	20,000	$23,000			74,000
Current assets	90,000	35,000	40,000			165,000
Current liabilities	(62,000)	(37,400)	(16,800)			(116,200)
	$91,000	$52,600	$46,200			$132,800
				$108,630	$108,630	

Analytical check of non-controlling interests in B Ltd

Direct non-controlling interests	20%	
Book value of equity of B Ltd	46,200	
Share of book value of equity of B Ltd		9,240
Fair value of non-controlling interests	8,000	
Less share of fair value of identifiable net assets	7,000	
Goodwill attributable to non-controlling interests		1,000
		10,240

Analytical check of non-controlling interests in A Ltd				
Direct non-controlling interests in A Ltd		25%		
Book value of equity of A Ltd		$ 52,600		
Less Investment in B Ltd	Note 1	35,000		
		$ 17,600		
Share of book value of equity of A Ltd			$ 4,400	
Fair value of non-controlling interests		10,000		
Less share of fair value of identifiable net assets		10,000		
Goodwill attributable to non-controlling interests			0	
			4,400	
Indirect non-controlling interests in B Ltd	25%*80%	20%		
Book value of equity of B Ltd		46,200		
Share of book value of equity of B Ltd			9,240	20%*46,200
Investment in B Ltd		35,000		
Less share of fair value of B's identifiable net assets		28,000		80%*35,000
A's goodwill in B Ltd	Note 2	7,000		
Share of A's goodwill in B Ltd			1,750	25%*7,000
Total balance of non-controlling interests			$25,630	

Note 1: Investment in B Ltd is removed to avoid net assets of B being attributed to non-controlling interests in A Ltd twice. In the segment below, the same shareholders have been attributed their indirect interests in B Ltd's net assets.

Note 2: When A Ltd acquired B Ltd on 1 July 20x0, the NCI of A also effectively acquired a share of the goodwill arising from the business combination between A and B. A's NCI effectively has a 25% interest in the goodwill. Had there been under- or over-valued identifiable net assets, A's NCI would also have a share of 25% in these under- or over-valued identifiable net assets. We apply 25% interest because A's NCI has a 25% interest in the investment that A has in B.

ILLUSTRATION 7.2 Extension of sequential consolidation

We now show the sequential consolidation approach which will consolidate the lower tiered sub-group first. Two consolidation worksheets are required. Although this seems onerous, practically most groups perform this method of consolidation as it provides more information on a particular sub-group to non-controlling interests.

CJE1: Eliminate investment in B and B's shareholders' equity as at the date of acquisition

Dr Share capital	30,000		
Dr Retained earnings	5,000		
Dr Goodwill	8,000		
Cr Investment in B			35,000
Cr Non-controlling interests			8,000

CJE2: Assign direct non-controlling interests' share of post-acquisition retained earnings of subsidiary B

Dr Retained earnings	200		
Cr Non-controlling interests		200	(20% × ($6,000 – $5,000))

CJE3: Assign direct non-controlling interests' share of current profit after tax of subsidiary B

Dr Income to non-controlling interests	3,040	
Cr Non-controlling interests		3,040
B's profit after tax for 20x2		$15,200
Direct non-controlling interests' share		20%
Direct non-controlling interests' share		$ 3,040

CJE4: Elimination of dividends declared by B

Dr Dividend income (A)	4,000		
Dr Non-controlling interests	1,000	(NCI % × Dividends declared = 20% × $5,000)	
Cr Dividends declared (B)			5,000

Sub-consolidation of A Ltd and its subsidiary B Ltd

Income Statement					
			Consolidation adjustments		
	A Ltd	B Ltd	Dr	Cr	Consolidated totals
Operating profit	$12,000	$19,000			$31,000
Dividend income	4,000		$ 4,000		0
Profit before tax	$16,000	$19,000			$31,000
Tax	(2,400)	(3,800)			(6,200)
Profit after tax	$13,600	$15,200			$24,800
Income to non-controlling interests			3,040		(3,040)
Retained profit					$21,760
Retained earnings, 1 Jan	17,000	6,000	5,000		17,800
			200		
Dividends declared	(8,000)	(5,000)		$5,000	(8,000)
Retained earnings, 31 Dec	$22,600	$16,200	$12,240	$5,000	$31,560

Statement of Financial Position					
			Consolidation adjustments		Consolidated
	A Ltd	B Ltd	Dr	Cr	totals
Share capital	$30,000	$30,000	$30,000		$30,000
Retained earnings	22,600	16,200	12,240	$ 5,000	31,560
Non-controlling interests			1,000	8,000	10,240
				200	
				3,040	
	$52,600	$46,200			$71,800
Investment in B Ltd	$35,000			35,000	
Goodwill			8,000		$ 8,000
Other non-current assets	20,000	$23,000			43,000
Current assets	35,000	40,000			75,000
Current liabilities	(37,400)	(16,800)			(54,200)
	$52,600	$46,200	$51,240	$51,240	$71,800

Analytical check of non-controlling interests in B Ltd

Direct non-controlling interests	20%	
Book value of equity of B Ltd	46,200	
Share of book value of equity of B Ltd		9,240
Fair value of non-controlling interests	8,000	
Less share of fair value of identifiable net assets	7,000	
Goodwill attributable to non-controlling interests		1,000
		10,240

CJE5: Eliminate investment in A and A's shareholders' equity[2] as at the date of acquisition

Dr Share capital	30,000		
Dr Retained earnings	10,000		
Dr Goodwill	2,000		
Cr Investment in A		32,000	
Cr Non-controlling interests		10,000	

CJE6: Assign direct non-controlling interests' share of post-acquisition retained earnings of sub-group of subsidiary A

Dr Retained earnings	1,950		
Cr Non-controlling interests		1,950	(25% × ($17,800 – $10,000))

CJE7: Assign direct non-controlling interests' share of current profit after tax for subsidiary A

Dr Income to non-controlling interests	5,440	

Cr Non-controlling interests			5,440
A's consolidated profit after tax for 20x2			$24,800
Less income to direct non-controlling interests of B			(3,040)
			$21,760
Non-controlling interests' share at 25%			$ 5,440

CJE8: Elimination of dividends declared by A

Dr Dividend income (P)	6,000		
Dr Non-controlling interests	2,000	(NCI % × Dividends declared = 25% × $8,000)	
Cr Dividends declared (A)			8,000

Consolidation of P Ltd and A Ltd's Sub-Group

Income Statement

	P Ltd	A Ltd Sub-group	Consolidation adjustments Dr	Consolidation adjustments Cr	Consolidated totals
Operating profit	$20,000	$31,000			$51,000
Dividend income	6,000		$ 6,000		0
Profit before tax	$26,000	$31,000			$51,000
Tax	(4,000)	(6,200)			(10,200)
Profit after tax	$22,000	$24,800			$40,800
Income to non-controlling interests		(3,040)	5,440		(8,480)
Retained profit		$21,760			$32,320
Retained earnings, 1 Jan	21,000	17,800	10,000		26,850
			1,950		
Dividends declared	(12,000)	(8,000)		$8,000	(12,000)
Retained earnings, 31 Dec	$31,000	$31,560	$23,390	$8,000	$47,170

Statement of Financial Position					
			Consolidation adjustments		
	P Ltd	A Ltd Sub-group	Dr	Cr	Consolidated totals
Share capital	$60,000	$30,000	$30,000		$ 60,000
Retained earnings	31,000	31,560	23,390	$ 8,000	47,170
Non-controlling interests		10,240	2,000	10,000	25,630
				1,950	
				5,440	
	$91,000	$71,800			$132,800
Investment in A Ltd	$32,000			32,000	
Goodwill		$ 8,000	2,000		$ 10,000
Other non-current assets	31,000	43,000			74,000
Current assets	90,000	75,000			165,000
Current liabilities	(62,000)	(54,200)			(116,200)
	$91,000	$71,800	$57,390	$57,390	$132,800

Analytical check of non-controlling interests in B Ltd

Direct non-controlling interests	20%	
Book value of equity of B Ltd	46,200	
Share of book value of equity of B Ltd		9,240
Fair value of non-controlling interests	8,000	
Less share of fair value of identifiable net assets	7,000	
Goodwill attributable to non-controlling interests		1,000
		10,240

Analytical check of non-controlling interests in A Ltd

Direct non-controlling interests	25%	
Book value of consolidated equity of A Ltd (excluding non-controlling interests)	61,560	
Share of book value of equity of A Ltd		15,390
Fair value of non-controlling interests	10,000	
Less share of fair value of identifiable net assets	10,000	
Goodwill attributable to non-controlling interests		0
		15,390
Total balance of non-controlling interests		25,630

Sequence of Acquisition of the Intermediate Parent and the Indirect Subsidiary

Illustration 7.1 is an example of simultaneous and sequential consolidation of a three-tier group structure. In Illustration 7.1, the sequence is such that the ultimate parent acquires the intermediate parent, as a stand-alone entity, before the latter

acquires the indirect subsidiary.

In most situations, the ultimate parent acquires an existing group of companies. In such a scenario, the following procedures must be applied as at the date of acquisition of the intermediate parent by the ultimate parent.

1. The pre-acquisition retained earnings and other comprehensive income of each entity in the sub-group at the date of acquisition by the ultimate parent must be eliminated.
2. Goodwill on acquisition of the intermediate parent is the excess of the sum of the acquisition-date fair value of the consideration transferred and fair value of the non-controlling interests in the acquiree over the fair value of the consolidated net identifiable assets of the intermediate parent.

   We had earlier shown how goodwill is determined for a sub-group, using the structure in Figure 7.1 where Y Co is the intermediate parent and Z Co is the indirect subsidiary.

   Goodwill in Y and Z = Fair value of consideration transferred to acquire Y + Fair value of non-controlling interests of Y and indirect non-controlling interests of Z + Fair value of direct non-controlling interests of Z – (Fair value of identifiable net assets of Y – Investment in Z) – Fair value of identifiable net assets of Z.

   Fair values are determined as at the date of acquisition by the ultimate parent (X Co in Figure 7.1).

   As we have seen in earlier chapters, fair value of identifiable net assets of each entity comprises its book value of equity and differences between fair value and book values of identifiable net assets. Any fair valuation carried out in earlier periods (e.g., when intermediate parent acquired its subsidiaries) are discarded. This is elaborated in point 3 below.

   We need to deduct the investment in the indirect subsidiary to avoid double counting the net assets of the indirect subsidiary.

   Note also that the fair value of non-controlling interests in the intermediate parent would include their interests in the subsidiaries controlled by the intermediate parent.
3. Goodwill is determined for the sub-group at the date of the acquisition of the sub-group by the ultimate parent. Any goodwill that is recognized earlier in the sub-group as a result of the acquisition of the indirect subsidiary is ignored. The acquirer will also need to determine the fair value of identifiable net assets as at the acquisition date of the sub-group. Any earlier fair valuation will be adjusted to arrive at the updated fair values as at the acquisition date of the sub-group. page 461
4. Non-controlling interests in the intermediate parent have a share of the fair value of consolidated net assets of the intermediate parent. The fair value of the intermediate parent's non-controlling interests comprises the fair value of their direct interests in the intermediate parent and their indirect interests in the subsidiaries held by the intermediate parent at the date when the sub-group is acquired by the ultimate parent.

   The direct non-controlling interests in each subsidiary will also have to be measured at fair value, in the usual manner that we have seen in the earlier chapters. Under Alternative 1, the fair value of non-controlling interests includes goodwill and under Alternative 2, fair value of non-controlling interests is determined only with reference to their share of the fair values of identifiable net assets.
5. Subsequent to the acquisition date of the intermediate parent by the ultimate parent, non-controlling interests in the intermediate parent continue to have a share of the change in post-acquisition retained earnings and other comprehensive income of the indirect subsidiary. Other consolidation adjustments apply in the usual manner as seen in the earlier chapters.

Illustration 7.3 illustrates the scenario where an ultimate parent acquires an existing sub-group.

ILLUSTRATION 7.3 Simultaneous consolidation of an existing sub-group of companies

B acquired 70% of C on 1 January 20x0 when the shareholders' equity of C was $3,500,000. A acquired 90% of B on 1 January 20x3. The fair value (FV) of identifiable net assets (INA) of B was close to book value (BV) at date of acquisition. Land of C was undervalued at both dates. Land was unsold and proceeds, if any, are tax-exempt and deferred liability need not be recognized. Details are shown below. The timeline is as follows:



B acquired C

Percentage acquired	70%
Date of acquisition	1 January 20x0
Fair value of consideration transferred	$4,000,000
Fair value of non-controlling interests in C	$1,600,000
Fair value of land	$2,000,000
Carrying amount (book value) of land	$1,400,000

	1 January 20x0	1 January 20x3	1 January 20x5
Share capital of C	$1,500,000	$1,500,000	$1,500,000
Retained earnings of C	2,000,000	5,000,000	7,000,000
Shareholders' equity of C	$3,500,000	$6,500,000	$8,500,000

Net profit of C for the year ended 31 December 20x5	$1,000,000
Dividends declared by C during 20x5	(60,000)
Profit retained	$ 940,000
Retained earnings of C as at 31 December 20x5	$7,940,000

A acquired B

Percentage acquired	90%
Date of acquisition	1 January 20x3
Fair value of consideration transferred	$20,000,000
Fair value of non-controlling interests in B	$ 1,700,000
Fair value of direct non-controlling interests in C	$2,400,000
Fair value of land of C	$2,300,000
Carrying amount of land of C	$1,400,000

	1 January 20x3	1 January 20x5
Share capital of B	$ 6,000,000	$ 6,000,000
Retained earnings of B	5,900,000	7,200,000
Shareholders' equity of B	$11,900,000	$13,200,000

Net profit of B for the year ended 31 December 20x5	$2,000,000

Dividends declared by B during 20x5	(200,000)
Profit retained	$1,800,000
Retained earnings of B as at 31 December 20x5	$9,000,000

Consolidation adjustments as at 31 December 20x5 (Simultaneous method)

CJE1: Elimination of investment in B and investment in C as at 1 January 20x3

Dr Share capital (B)*	6,000,000	
Dr Share capital (C)	1,500,000	
Dr Retained earnings (B)	5,900,000	
Dr Retained earnings (C)	5,000,000	
Dr Land	900,000	
Dr Goodwill	8,800,000 (Note 1)	
Cr Investment in B (A)		20,000,000
Cr Investment in C (B)		4,000,000
Cr Non-controlling interests in B		1,700,000 (Note 3)
Cr Non-controlling interests in C		2,400,000

* Parenthesis indicates the legal entity in whose books the item appears

Note 1: Goodwill

= Fair value of consideration transferred + Fair value of non-controlling interests in B + Fair value of non-controlling interests in C – Fair value of identifiable net assets
= $20,000,000 + $1,700,000 + $2,400,000 – $15,300,000 (Note 2)
= $8,800,000

Note 2: Fair value of identifiable net assets

= Book value of net assets of B as at 1 January 20x3 (after deducting B's investment in C to avoid double counting of net assets) + Book value of net assets of C as at 1 January 20x3 + Excess of fair value of land of C as at 1 January 20x3
= ($11,900,000 – $4,000,000) + $6,500,000 + $900,000
= $15,300,000

We use the book value of equity comprising the components of equity (share capital and retained earnings in this example) to measure the book value of net assets as at acquisition date. We should note that the book value of net assets of B, the intermediate parent, includes investment in C, the indirect subsidiary. Since we have included the net assets of C separately in the equation, we should deduct the investment in C held by B to avoid double counting of net assets in the goodwill calculation.

Note 3: Non-controlling interests (NCI) in B has a fair value of $1,700,000 as at 1 January 20x3. The fair value comprises the non-controlling interests' share of net identifiable assets and goodwill.

	Total	NCI's share at 10%
Non-controlling interests in B and indirect non-controlling interests in C		
B's book value of equity or net assets as at 1 Jan 20x3	$11,900,000	

Less investment in C	(4,000,000)		
Adjusted Net assets of B	$ 7,900,000	$ 790,000	
C's book value of equity or net assets as at 1 Jan 20x3	6,500,000		
B's share	4,550,000	455,000	(or 7% x $6.5 million)
FV-BV of land of C	900,000		
B's share	630,000	63,000	(or 7% x $900,000)
Fair value of INA		1,308,000	
Goodwill attributable to B's NCI (residual)		392,000	
Fair value of NCI of B on 1 Jan 20x3		$1,700,000	

Note 4: Fair value of C's direct non-controlling interests as at 1 January 20x3 is $2,400,000. The fair value can be analyzed as follows:

	Total	NCI's share at 30%
C's shareholders' equity as at 1 January 20x3	$6,500,000	$1,950,000
Share of fair value excess of land		270,000
Share of goodwill		180,000*
NCI in C		$2,400,000

*NCI's goodwill = Fair value of NCI – Share of fair value of net identifiable assets of C
= $2,400,000 – (30% × $6,500,000) – (30% × $900,000)
= $2,400,000 – $1,950,000 – $270,000
= $180,000

CJE2: Allocation of change in B's retained earnings to direct non-controlling interests of B

Dr Opening retained earnings	130,000	
Cr Non-controlling interests		130,000

Retained earnings of B as at 1 January 20x5	$7,200,000
Retained earnings of B as at 1 January 20x3	5,900,000
Change in retained earnings of B	$1,300,000
Direct non-controlling interests in B	10%

CJE3: Allocation of change in C's retained earnings to direct non-controlling interests of C

Dr Opening retained earnings	600,000	
Cr Non-controlling interests		600,000

Retained earnings of C as at 1 January 20x5	$7,000,000
Retained earnings of C as at 1 January 20x3	5,000,000
Change in retained earnings of C	$2,000,000
Direct non-controlling interests in C	30%

(Direct non-controlling interests' share of retained earnings of C on 1 January 20x3 is accounted for in CJE1.)

CJE4: Allocation of change in C's retained earnings to indirect non-controlling interests of C

Dr Opening retained earnings	140,000	
Cr Non-controlling interests		140,000

Retained earnings of C as at 1 January 20x5	$7,000,000
Retained earnings of C as at 1 January 20x3	5,000,000
Change in retained earnings of C	$2,000,000
Indirect non-controlling interests in C	7% (10% × 70%)

(This entry can be combined with CJE3. Indirect non-controlling interests' share of change in retained earnings of C on 1 January 20x3 is accounted for in the recognition of fair value of non-controlling interest in B Co (and indirect non-controlling interest in C Co) in CJE1.)

CJE5: Allocation of current profit to total non-controlling interests of C

Dr Income to non-controlling interests	370,000	
Cr Non-controlling interests		370,000

Net income of C for 20x5	$1,000,000
Total non-controlling interests' share	37%

CJE6: Allocation of current profit to non-controlling interests of B

Dr Income to non-controlling interests	195,800	
Cr Non-controlling interests		195,800

Net profit of B for 20x5	$2,000,000
Less dividend income from C included in B's net profit	(42,000)
Adjusted net income of B for 20x5	$1,958,000
Direct non-controlling interests' share	10%

Dividend income from C is eliminated to avoid the recognition of intragroup income and double-counting of income in two forms (non-controlling interests have been allocated share of profit of C under CJE5).

CJE7: Eliminate dividends from C

Dr Dividend income (B)	42,000	
Dr Non-controlling interests	18,000	
Cr Dividends declared by C		60,000

CJE8: Eliminate dividends from B

Dr Dividend income (A)	180,000	
Dr Non-controlling interests	20,000	

Cr Dividends declared by B		200,000

Total non-controlling interests as at 31 December 20x5

	Total non-controlling interests	Non-controlling interests in B*	Direct non-controlling interests in C
CJE1: B's non-controlling interests at date of acquisition of B	$1,700,000	$1,700,000	
CJE1: C's non-controlling interests at date of acquisition of B	2,400,000		$2,400,000
CJE2: Allocation of change in B's retained earnings to direct non-controlling interests of B	130,000	130,000	
CJE3: Allocation of change in C's retained earnings to direct non-controlling interests of C	600,000		600,000
CJE4: Allocation of change in C's retained earnings to indirect non-controlling interests of C	140,000	140,000	
CJE5: Allocation of current profit of C	370,000	70,000	300,000
CJE6: Allocation of current profit of B	195,800	195,800	
CJE7: Elimination of dividends from C	(18,000)		(18,000)
CJE8: Elimination of dividends from B	(20,000)	(20,000)	
	$5,497,800	$2,215,800	$3,282,000

* Includes indirect non-controlling interests in B

Analytical check on non-controlling interests in B

	Total	NCI's share of B at 10%	
B's Net assets at 31 Dec 20x5	$15,000,000		
Less investment in C	(4,000,000)		
Adjusted Net assets of B	$11,000,000	$1,100,000	
C's book value of equity or net assets as at 31 Dec 20x5	9,440,000		
B's share	6,608,000	660,800	(or 7% x $9.440 million)
FV-BV of land of C	900,000		
B's share	630,000	63,000	(or 7% x $900,000)
Goodwill attributable to B's NCI (CJE1)		392,000	
NCI of B and indirect NCI of C on 31 Dec 20x5		$2,215,800	

Analytical check on direct non-controlling interests in C

	Total	Direct NCI's share of C at 30%
C's book value of equity or net assets as at 31 Dec 20x5	$9,440,000	$2,832,000
FV-BV of land of C	900,000	270,000
Goodwill attributable to C's NCI (CJE1)		180,000
Direct NCI of C on 31 Dec 20x5		$3,282,000

Analytical Checks on Direct and Indirect Non-controlling Interests in a Multi-tier Group Hierarchy

Essentially, the check on direct non-controlling interests is performed in the usual way as illustrated in previous chapters, and serves as an independent assessment[3] of the balance of the non-controlling interests at the end of a reporting period. However, a check on indirect non-controlling interests in a multi-tier hierarchy is more complex. By virtue of their interest through an intermediate parent, we need to remove the investment included on the intermediate parent's statement of financial position to avoid double counting of the net assets of the indirect subsidiary.

Schematic explanations on the allocation and elimination of retained earnings of direct and indirect subsidiary in Illustration 7.3 are shown in Figures 7.6 and 7.7.

FIGURE 7.6 Allocation and elimination of B's retained earnings in Illustration 7.3

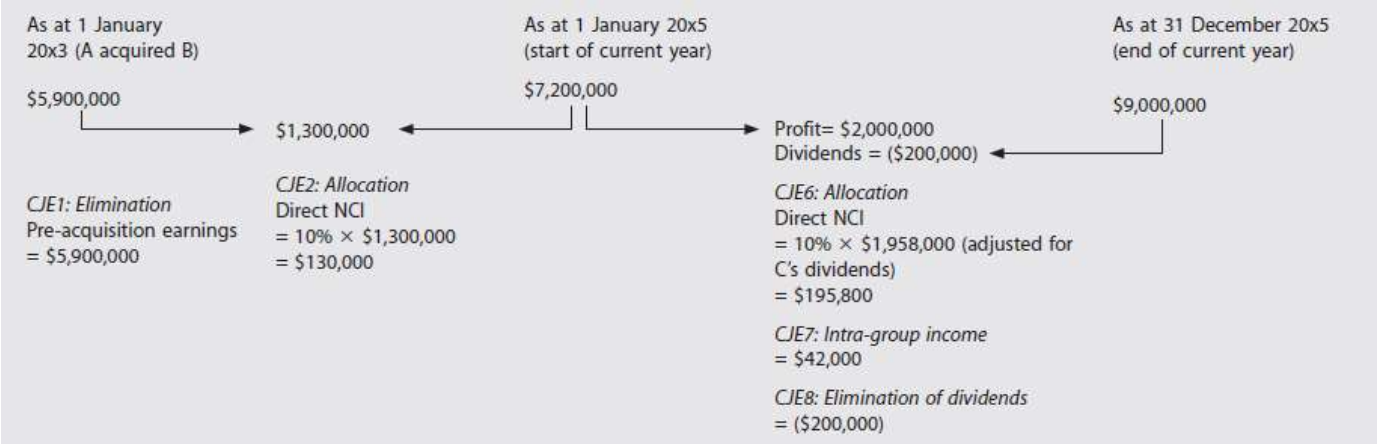
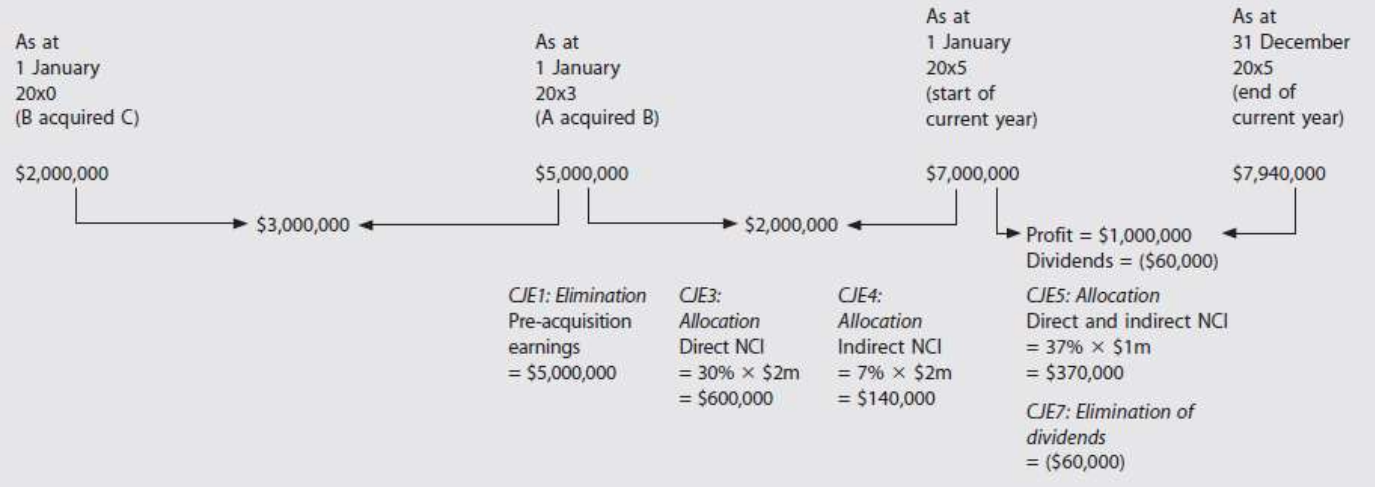


FIGURE 7.7 Allocation and elimination of C's retained earnings in Illustration 7.3



Impact of Fair Value Adjustments on Indirect Non-controlling Interests

Indirect non-controlling interests have a share of the net assets and the fair value adjustments of an indirect subsidiary at the date of acquisition. During the post-acquisition period, both the direct non-controlling interests and the intermediate parent have to bear a share of the amortization of fair value adjustments under simultaneous consolidation. Since indirect non-controlling interests have a share of the intermediate parent's profit or losses, it will also have to bear a share of the amortization of fair value adjustments that are borne by the intermediate parent. Hence, the effects of past cumulative and present amortization of fair value adjustments will be allocated to direct and indirect non-controlling interests. Illustration 7.3 shows and explains the allocation of the amortization of fair value adjustments, and provides another example of simultaneous consolidation.

Impact of Adjustments of Unrealized Profit on Indirect Non-controlling Interests

There is no fundamental difference in the treatment of unrealized profit on intercompany transfers. If the selling company within a group recognizes profit that is unrealized from the group perspective, the income allocated to non-controlling interests, both direct and indirect, of the selling company will be adjusted for any unrealized profit included in the net profit after tax. Furthermore, any unrealized profit included in opening retained earnings of the selling company will have to be borne by the controlling shareholders (through adjustment to opening retained earnings) and the non-controlling shareholders (both direct and indirect shareholders of the selling company). In a multi-tier group structure, the terms "upstream" and "downstream" should be interpreted in the broader sense. By "upstream" is meant a sale to the ultimate parent or its subsidiaries while "downstream" refers to a sale by the ultimate parent to its subsidiaries.

Analytical Checks on Consolidated Retained Earnings

In addition to analytically checking the direct and indirect non-controlling interests of subsidiaries, we can also check the consolidated retained earnings of direct and indirect subsidiaries from the ultimate parent's perspective. page 468
Essentially, the ultimate parent will have a share in the change of retained earnings from the date of acquisition to the end of the current reporting period, after eliminating current period dividend income and unrealized profit included in retained earnings.

page 469

The analytical check on consolidated retained earnings is a useful check to ascertain that only post-acquisition retained earnings of direct and indirect subsidiaries are included in consolidated retained earnings. This check may be used to indirectly affirm consolidated retained earnings and is a useful analytical check that may be applied, in addition to the analytical check on non-controlling interests.

The analytical check on consolidated retained earnings is the same as done in previous chapters. However, in a multi-level hierarchy, we include the ultimate parent's share of the indirect subsidiaries' post-acquisition retained earnings in the analysis. This is done by applying the ultimate parent's effective interest to the change in retained earnings of both direct and indirect subsidiaries from the date when the ultimate parent obtains control to the end of the year or reporting date.

The same principle applies to the other elements in the analytical check for consolidated retained earnings. We apply the ultimate parent's effective interest on the adjustments for unrealized profit from upstream transfers (for downstream transfers, the adjustment is 100%). We also apply the ultimate parent's effective percentage to work out the share of cumulative expensing or amortization of fair value adjustments for both direct and indirect subsidiaries.

Let us analyze the consolidated retained earnings in Illustration 7.3. In this example, there are no unrealized profit adjustments or cumulative expensing or amortization of fair value adjustments. In subsequent examples, we will show how these will be incorporated in the analytical process.

We determine the balance of the consolidated retained earnings by listing the legal entities' retained earnings balances and then showing the compilation of the consolidation adjustments that affect retained earnings.

Listing approach:

A's retained earnings as at 31 December 20x5		$12,000,000
B's retained earnings as at 31 December 20x5		9,000,000
C's retained earnings as at 31 December 20x5		7,940,000
CJE1	Elimination of pre-acquisition retained earnings of B	(5,900,000)
CJE1	Elimination of pre-acquisition retained earnings of C	(5,000,000)
CJE2	Allocation of post-acquisition retained earnings of B to B's NCI	(130,000)
CJE3	Allocation of post-acquisition retained earnings of C to direct NCI	(600,000)
CJE4	Allocation of post-acquisition retained earnings of C to indirect NCI	(140,000)
CJE5	Allocation of current profit to total NCI of C	(370,000)
CJE6	Allocation of current profit to B's NCI	(195,800)
CJE7	Elimination of dividends from C	18,000
CJE8	Elimination of dividends from B	20,000
Consolidated retained earnings as at 31 December 20x5		$16,642,200

Analytical check of consolidated retained earnings as at 31 December 20x5

A's retained earnings	$12,000,000
A's share of B's post-acquisition retained earnings (Note 1)	2,790,000
A's share of C's post-acquisition retained earnings (Note 2)	1,852,200
	$16,642,200

Note 1: 90% × ($9,000,000 – $5,900,000)

Note 2: 90% × 70% × ($7,940,000 – $5,000,000)

ILLUSTRATION 7.4 Simultaneous consolidation with fair value adjustments

Acquired company

	S Co	B Co
Acquirer	P Co	S Co
Date of acquisition	1 Jan 20x1	1 July 20x1
Percentage acquired	90%	60%
Direct non-controlling interests	10%	40%

The following information relates to the statements of financial position at the date of acquisition. (Net assets do not include goodwill and deferred tax on fair value adjustments).

	S Co Book value	S Co Fair value	B Co Book value	B Co Fair value
Intangible assets	$ 0	$ 250,000		
Inventory	400,000	450,000	$ 60,000	$ 55,000
Other net assets	900,000	900,000	100,000	100,000
Net assets	$1,300,000	$1,600,000	$160,000	$155,000
Share capital	1,000,000		50,000	
Retained earnings	300,000		110,000	
Equity	$1,300,000	$1,600,000	$160,000	$155,000
Fair value of NCI		$ 250,000		$ 80,000

The following financial statements relate to the year ended 31 December 20x3.

Income Statement and Partial Statement of Changes in Equity
For the Year Ended 31 December 20x3

	P Co	S Co	B Co
Sales	$4,000,000	$2,000,000	$400,000
Cost of sales	(1,500,000)	(1,000,000)	(240,000)
Gross profit	$2,500,000	$1,000,000	$160,000
Operating expenses net of other income	(930,000)	(700,000)	(100,000)
Net profit before tax	$1,570,000	$ 300,000	$ 60,000
Tax	(290,000)	(60,000)	(12,000)
Net profit after tax	$1,280,000	$ 240,000	$ 48,000
Dividends declared	(200,000)	(120,000)	(20,000)
Net profit attributable to shareholders	$1,080,000	$ 120,000	$ 28,000
Retained earnings, 1 January 20x3	2,500,000	500,000	250,000
Retained earnings, 31 December 20x3	$3,580,000	$ 620,000	$278,000

Statement of Financial Position
As at 31 December 20x3

	P Co	S Co	B Co
Investment in S Co	$2,500,000		
Investment in B Co		$ 300,000	

Other net assets	4,080,000	1,320,000	$328,000
	$6,580,000	$1,620,000	$328,000
Share capital	$3,000,000	$1,000,000	$ 50,000
Retained earnings	3,580,000	620,000	278,000
	$6,580,000	$1,620,000	$328,000

Additional information:

(a) Intangible assets of S Co were found to have an estimated useful life of five years from the date of acquisition by P Co. Amortization on a straight line basis was deemed to be the appropriate method of recognizing the decline in value.

(b) Inventory at the date of acquisition was sold off within the following financial period.

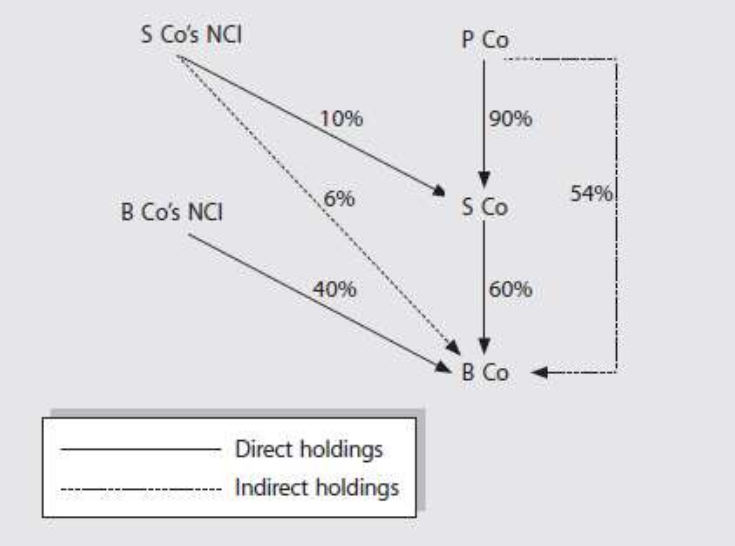
(c) There was no change in the share capital of the acquired companies from acquisition date.

(d) Assume a tax rate of 20% and recognize tax effects where appropriate.

Required:

1. Prepare consolidation adjustments for the year ended 31 December 20x3 using the simultaneous method of consolidation.
2. Perform an analytical check on the non-controlling interests of S Co and B Co as at 31 December 20x3.
3. Perform an analytical check on consolidated retained earnings as at 31 December 20x3.

Ownership structure of S Co and B Co



1. Consolidation adjustments

CJE1: Eliminate investment in S

Dr Share capital (S Co)	1,000,000	
Dr Retained earnings (S Co)	300,000	
Dr Goodwill	1,210,000	(Note 1)
Dr Intangible assets	250,000	

Dr Inventory	50,000		
Cr Deferred tax liability		60,000	
Cr Investment in S Co		2,500,000	
Cr Non-controlling interests		250,000	(Note 2)

Note 1:

$$\text{Goodwill} = \left(\begin{matrix}\text{Fair value of}\\ \text{acquisition cost}\end{matrix} + \begin{matrix}\text{Fair value of}\\ \text{non-controlling interests}\end{matrix}\right) - \text{Fair value of net identifiable assets}$$

$$= (\$2{,}500{,}000 + \$250{,}000) - (\$1{,}600{,}000 - \text{Deferred tax liability on fair value adjustments})$$
$$= \$2{,}750{,}000 - (\$1{,}600{,}000 - \$60{,}000)$$
$$= \$2{,}750{,}000 - \$1{,}540{,}000$$
$$= \$1{,}210{,}000$$

Deferred tax liability on fair value adjustments
= 20% * ($250,000 + $50,000)
= $60,000

Note 2:

Fair value of non-controlling interests as at acquisition date	$250,000
This comprises:	
Share of fair value of net identifiable assets (10% × $1,540,000)	$154,000
Share of goodwill ($250,000 – $154,000)	$ 96,000

CJE2: Recognize past amortization of intangible assets

Dr Opening retained earnings	90,000	
Dr Non-controlling interests	10,000	
Cr Accumulated amortization		100,000

CJE3: Recognize tax effects on past amortization of intangible assets

Dr Deferred tax liability	20,000	
Cr Opening retained earnings		18,000
Cr Non-controlling interests		2,000

CJE4: Recognize past cost of sales of undervalued inventory

Dr Opening retained earnings	45,000	
Dr Non-controlling interests	5,000	
Cr Inventory		50,000

CJE5: Recognize tax effects of past cost of sales of undervalued inventory

Dr Deferred tax liability	10,000	
Cr Opening retained earnings		9,000

Cr Non-controlling interests		1,000

CJE6: Recognize current amortization of recognized intangible assets

Dr Amortization of intangible assets	50,000	
Cr Accumulated amortization		50,000

CJE7: Recognize tax on current amortization of recognized intangible assets

Dr Deferred tax liability	10,000	
Cr Tax expense		10,000

CJE8: Allocate current profit after tax of S Co to non-controlling interests

Dr Income to non-controlling interests	18,800	
Cr Non-controlling interests		18,800

Net profit after tax (S)	$240,000
Less amortization of intangible assets (after-tax)	(40,000)
Less dividend income received from B Co	(12,000)
Adjusted profit after tax	$188,000
Non-controlling interests' share at 10%	$ 18,800

Dividend income received from B Co is included in the net profit after tax of S Co. Non-controlling interests of S Co recognize income from B based on the share of profit and not dividend income. The share of profit is recognized in CJE14; hence, dividend income included in net profit after tax of S should be removed to avoid double-counting of income.

CJE9: Eliminate dividend income against dividends declared

Dr Dividend income	108,000	
Dr Non-controlling interests	12,000	
Cr Dividends declared by S Co		120,000

CJE10: Recognize non-controlling interests' share of S Co's post-acquisition retained earnings up to 1 January 20x3

Dr Opening retained earnings	20,000	
Cr Non-controlling interests		20,000
Retained earnings of S Co at 1 January 20x3		$500,000
Retained earnings of S Co at acquisition date		300,000
Change in retained earnings		$200,000
Non-controlling interests' share at 10%		$ 20,000

CJE11: Eliminate investment in B

Dr Share capital (B Co)	50,000
Dr Retained earnings (B Co)	110,000

Dr Goodwill	224,000	(Note 1)		
Dr Deferred tax asset	1,000			
Cr Inventory			5,000	
Cr Investment in B Co			300,000	
Cr Non-controlling interests . . .			80,000	(Note 2)

Note 1:

$$\text{Goodwill} = \left(\begin{array}{c}\text{Fair value of}\\ \text{consideration transferred}\end{array} + \begin{array}{c}\text{Fair value of}\\ \text{non-controlling interests}\end{array}\right) - \text{Fair value of net identifiable assets}$$

$$= (\$300{,}000 + \$80{,}000) - (\$155{,}000 + \text{Deferred tax asset on fair value adjustments})$$
$$= \$380{,}000 - (\$155{,}000 + \$1{,}000)$$
$$= \$380{,}000 - \$156{,}000$$
$$= \$224{,}000$$

Deferred tax asset on overvaluation of inventory
$= 20\% \times \$5{,}000$
$= \$1{,}000$

Note 2:

Fair value of non-controlling interests as at acquisition date	$80,000
This comprises:	
Share of fair value of net identifiable assets (40% × $156,000)	$62,400
Share of goodwill ($80,000 – $62,400)	$17,600

CJE12: Adjust past decrease in cost of sales of overvalued inventory

Dr Inventory	5,000	
Cr Opening retained earnings		2,700
Cr Non-controlling interests		2,300

Opening retained earnings relate to the group's share of past profits, excluding non-controlling interests' share. As the group's interest in B is 54%, the adjustment to opening retained earnings is 54% of $5,000 or $2,700. Of the $5,000 amortization charge made in a previous period, 40% is allocated to direct non-controlling interests. Of the 60% that is borne by the intermediate parent, indirect non-controlling interests effectively shares 6% of the amortization of fair value adjustment. Hence, 46% of the past adjustment or $2,300 is allocated to non-controlling interests under simultaneous consolidation.

CJE13: Adjust tax on past cost of sales of overvalued inventory

Dr Opening retained earnings	540	
Dr Non-controlling interests	460	
Cr Deferred tax asset		1,000

CJE14: Allocate current profit after tax of B Co to non-controlling interests

Dr Income to non-controlling interests	22,080

Cr Non-controlling interests	22,080	
Net profit after tax (B Co)	$48,000	
Non-controlling interests' share of B Co	$22,080	(46% × $48,000)
Direct non-controlling interests' share	40%	
Indirect non-controlling interests' share	6%	
Total non-controlling interests	46%	

Total non-controlling interests of 46% is applied to B's net profit after tax. As overvalued inventory was sold in a previous year, current adjustment to net profit after tax is not necessary.

CJE15: Eliminate dividend income against dividends declared

Dr Dividend income	12,000	
Dr Non-controlling interests	8,000	
Cr Dividends declared by B		20,000

CJE16: Allocate change in retained earnings of B Co to non-controlling interests

Dr Opening retained earnings	64,400	
Cr Non-controlling interests		64,400

In this example, S acquired B on 1 July 20x1, after S was acquired by P. Non-controlling interests of S will have an interest in the change of retained earnings of B from 1 July 20x1. Both direct and indirect non-controlling interests may be combined.

Retained earnings of B at 1 January 20x3	$250,000
Retained earnings of B at 1 July 20x1	110,000
Change in retained earnings	$140,000
Direct and indirect non-controlling interests' share at 46%	$ 64,400

2. Analytical check on non-controlling interests of S Co and B Co

We show below the analytical check on the balances of non-controlling interests of S Co and B Co. S Co has an indirect interests in B Co. The analytical check on the balance of non-controlling interests in S Co can be done as a whole. There is no need to separate consolidation adjustments into direct and indirect non-controlling interests, since the investors are the same (i.e., investors in S Co are also indirect investors in B Co). The combined total of the consolidation adjustments can be analyzed as a unit. However, for purposes of a deeper explanation, we show the separation below with respect to the consolidation adjustments.

	Total non-controlling interests	Direct non-controlling interests	Indirect non-controlling interest
Relating to S Co			
CJE1: Non-controlling interests at acquisition date	$250,000	$250,000	
CJE2: Share of past amortization	(10,000)	(10,000)	
CJE3: Share of tax on past amortization	2,000	2,000	

CJE4: Share of adjustment to past cost of sales	(5,000)	(5,000)	
CJE5: Share of tax on adjustment to past cost of sales	1,000	1,000	
CJE8: Share of adjusted current profit after tax (Note 1)	18,800	20,000	(1,200)
CJE9: Dividends paid to non-controlling interests	(12,000)	(12,000)	
CJE10: Share of post-acquisition retained earnings	20,000	20,000	
Less Share of investment in B Co (Note 2)		(30,000)	30,000
	$264,800	$236,000	$28,800
Relating to B Co			
CJE11: Non-controlling interests at acquisition date	$ 80,000	$ 80,000	
CJE12: Share of past cost of sales adjustment of overvalued inventory	2,300	2,000	$ 300
CJE13: Share of tax on past cost of sales adjustment of overvalued inventory	(460)	(400)	(60)
CJE14: Share of adjusted current profit after tax	22,080	19,200	2,880
CJE15: Dividends paid to non-controlling interests	(8,000)	(8,000)	
CJE16: Non-controlling interests' share of post-acquisition retained earnings	64,400	56,000	8,400
Non-controlling interests as at 31 December 20x3	$160,320	$148,800	$11,520
Total non-controlling interests as at 31 December 20x3	$425,120	$384,800	$40,320

Note 1: Dividend income from B Co of $12,000 is eliminated from share of current profit of S Co. The reduction of $1,200 (10% of $12,000) is effectively a reduction of the indirect interests in B Co.

Note 2: When S Co acquired B Co, the non-controlling interests of S Co had a share of 10% of this initial investment in B Co. The share of initial investment in B Co relates to the indirect interests in B Co and is therefore presented under that column. The share of initial investment of $30,000 is analogous to the fair value of non-controlling interests in the indirect subsidiary at acquisition date. However, in this scenario, S Co acquired B Co after P Co acquired S Co.

Direct non-controlling interests' share of B Co as at 31 December 20x3

Book value of net identifiable assets of B Co	$328,000
Unamortized fair value adjustments	0
Adjusted book value of net identifiable assets	$328,000
Direct non-controlling interests' share at 40%	$131,200
Direct non-controlling interests' share of goodwill (refer to CJE11 Note 2)	17,600
Direct non-controlling interests' balance	$148,800

Direct non-controlling interests in S Co	10%	
Book value of equity of S Co	$1,620,000	
Less Investment in B Co	300,000	
	$1,320,000	
Share of book value of equity of S Co		$132,000
Unamortized balance of intangible assets, after-tax	80,000	

NCI's share of balance of intangible assets, after-tax		8,000	
Goodwill attributable to non-controlling interests		96,000	
		$236,000	
Indirect non-controlling interests in B Co 10%*60%	6%		
Book value of equity of B Co	$ 328,000		
Share of book value of equity of B Co		$ 19,680	6%*328,000
S's Investment in B Co	300,000		
Less S's share of fair value of B's identifiable net assets	93,600		60%*156,000
S's goodwill in B Co	206,400		
NCI's Share of S's goodwill in B Co		20,640	10%*206,400
		$ 40,320	
Total balance of non-controlling interests		$425,120	

Explanatory notes:

- Direct non-controlling interests of B Co can be checked in the normal way, that is, with reference to the book value of net assets, adjustments for unrealized profit at the end of the year and unamortized balance of fair value adjustments of B Co and non-controlling interests' share of goodwill. In this example, there are no adjustments for unrealized profit at the end of the year or unamortized balance of fair value adjustments. Hence, the year-end balance of direct non-controlling interests is solely explained by their share of the book value of net assets and goodwill.

3. Analytical check on consolidated retained earnings as at 31 December 20x3

The consolidated retained earnings balance is shown in the consolidation worksheets in the following pages. However, we can also compile the balance through the listings approach. In this illustration, we will use the balance as shown in the worksheet.

Consolidated retained earnings as		
per consolidation worksheets	$3,816,880	
Analytical check of consolidated retained earnings:		
P's retained earnings as at 31 December 20x3	$3,580,000	
P's share of S's post-acquisition retained earnings	288,000	90%*(620,000-300,000)
P's share of B's post-acquisition retained earnings	90,720	54%*(278,000-110,000)
Add/(less) P's share of:		
S's amortization of intangible assets, after tax	(108,000)	90%*80%*250,000/5*3
S's cost of sales of undervalued inventory, after tax	(36,000)	90%*80%*50,000
B's cost of sales of overvalued inventory, after tax	2,160	54%*80%*5,000
Consolidated retained earnings as at 31 December 20x3	$3,816,880	

Consolidation Worksheets with Simultaneous Consolidation

Worksheets are prepared in the normal way, except that in simultaneous consolidation, all the subsidiaries, including the indirectly owned subsidiaries, will be included in the same worksheet. The following consolidation worksheet example is based on Illustration 7.4.

page 479

Income Statement
For the Year Ended 31 December 20x3

				Consolidation adjustments				Consolidated
	P Co	S Co	B Co	Dr	CJE#	Cr	CJE#	I/S
Sales	$4,000,000	$2,000,000	$400,000					$6,400,000
Cost of sales	(1,500,000)	(1,000,000)	(240,000)					(2,740,000)
Gross profit	$2,500,000	$1,000,000	$160,000					$3,660,000
Operating expenses net of other income	(930,000)	(700,000)	(100,000)	$ 50,000	CJE6			(1,900,000)
				108,000	CJE9			
				12,000	CJE15			
Net profit before tax	$1,570,000	$ 300,000	$ 60,000					$1,760,000
Tax	(290,000)	(60,000)	(12,000)			$ 10,000	CJE7	(352,000)
Net profit after tax	$1,280,000	$ 240,000	$ 48,000					$1,408,000
Non-contolling interests				18,800	CJE8			(40,880)
				22,080	CJE14			
Net profit after non-controlling interests								$1,367,120
Dividends declared	(200,000)	(120,000)	(20,000)			120,000	CJE9	(200,000)
						20,000	CJE15	
Net profit attributable to shareholders	$1,080,000	$ 120,000	$ 28,000					$1,167,120
Retained earnings, 1 January 20x3	2,500,000	500,000	250,000	300,000	CJE1			2,649,760
				90,000	CJE2			
				45,000	CJE4			
				20,000	CJE10	18,000	CJE3	
				110,000	CJE11	9,000	CJE5	
				540	CJE13			
				64,400	CJE16	2,700	CJE12	
Retained earnings, 31 December 20x3	$3,580,000	$ 620,000	$278,000	$840,820	To SFP	$179,700	To SFP	$3,816,880

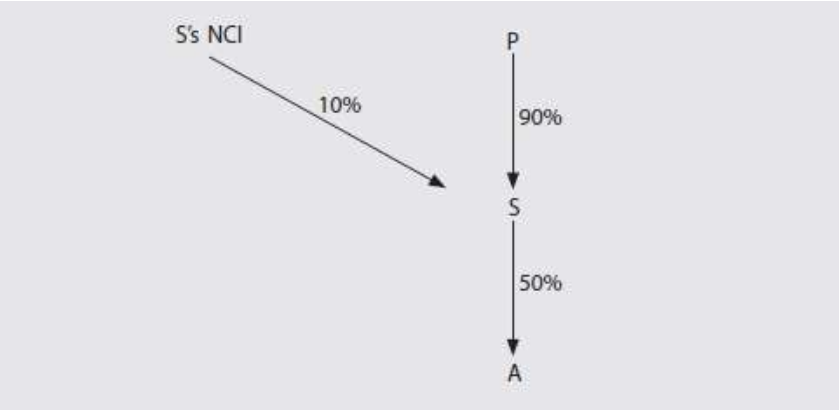
page 480

Statement of Financial Position As at 31 December 20x3									
				Consolidation adjustments					Consolidated
	P Co	S Co	B Co	Dr		Cr			SFP
Investment in S Co	$2,500,000					$2,500,000	CJE1		
Investment in B Co		$300,000				300,000	CJE11		
Other net assets	4,080,000	1,320,000	$328,000	$ 250,000	CJE1	60,000	CJE1		$5,808,000
				50,000	CJE1	100,000	CJE2		
				20,000	CJE3	50,000	CJE4		
				10,000	CJE5	50,000	CJE6		
				10,000	CJE7	5,000	CJE11		
				1,000	CJE11	1,000	CJE13		
				5,000	CJE12				
Goodwill				1,210,000	CJE1				1,434,000
				224,000	CJE11				
	$6,580,000	$1,620,000	$328,000	$1,780,000		$3,066,000			$7,242,000
Share capital	$3,000,000	$1,000,000	$50,000	$1,000,000	CJE1				$3,000,000
				50,000	CJE11				
Retained earnings	3,580,000	620,000	278,000	840,820	From I/S	$ 179,700	From I/S		3,816,880
Non-controlling interests				10,000	CJE2	250,000	CJE1		425,120
				5,000	CJE4	2,000	CJE3		
				12,000	CJE9	1,000	CJE5		
				460	CJE13	18,800	CJE8		
				8,000	CJE15	20,000	CJE10		
						80,000	CJE11		
						2,300	CJE12		
						22,080	CJE14		
						64,400	CJE16		
	$6,580,000	$1,620,000	$328,000	$1,926,280		$ 640,280			$7,242,000
				$3,706,280		$3,706,280			

INDIRECT HOLDING OF ASSOCIATES

An "indirect" associate may arise from the affiliation structures shown in Figures 7.8 and 7.9.

FIGURE 7.8 Indirect holding of an associate through a subsidiary

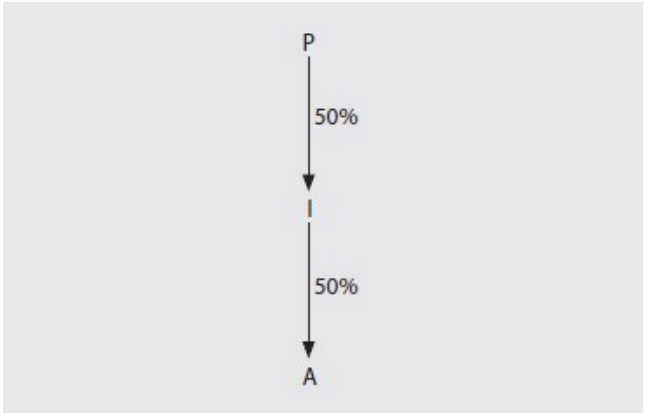


In Figure 7.8, A is an indirect associate of P, if we assume that control and significant influence are evidenced by the percentage of voting rights held by P and S, respectively.

1. As a result of the existence of non-controlling interests in S, P should not equity-account its indirect interest (i.e., 45%) in the net profit before tax and the tax expense of A. In recognizing only the net effects of P's indirect interest in A, the portion that relates to non-controlling interests is not included in the income allocated to non-controlling interests in the consolidated income statement of P. This presentation is not in accordance with the consolidation requirement that the consolidated profit after tax must show the total gross amount before allocating to non-controlling interests, and that non-controlling interests in the profit or loss of the consolidated subsidiaries are shown separately.
2. Thus, the consolidated profit before tax and the consolidated tax expense should include S's share (i.e., 50%) of A's profit before tax and tax expense, respectively. Income allocated to non-controlling interests in the consolidated income statement should include non-controlling interests' share of the equity-accounted profit after tax of A (i.e., 50% of the net profit after tax of A multiplied by 10% or effectively 5% of A's net profit after tax).
3. Since non-controlling interests' share in a subsidiary's profits is presented on an after-tax basis, the allocation of non-controlling interests' share of equity-accounted profit of a subsidiary's associate should also be on an after-tax basis.

Illustration 7.5 explains the application of the above principles. Figure 7.9 shows a different arrangement whereby P has an interest in A through an intermediate associate company, I.

FIGURE 7.9 Indirect holding through an associate



IAS 28 paragraph 5 presumes that an investor has significant influence if the investor holds, directly or indirectly (e.g., through subsidiaries), 20% or more of the voting power of the investee, unless it can be clearly demonstrated that no such influence exists. In Figure 7.9, P has significant influence but not control over I. Although I is not a subsidiary of P and the ownership structure is obviously not as strong as in Figure 7.8 to support P having indirect significant influence over A, all other available evidence as enumerated in IAS 28 paragraph 6 must be considered to determine whether A is an indirect associate of P.

Assuming that P has significant influence over A, the following procedures will apply if P applies "simultaneous" equity accounting:

1. Note that P needs to prepare two sets of financial statements, even though it does not have subsidiaries. One set is the separate financial statements of P as a legal entity, wherein investments in associates are carried at either cost or in accordance with IFRS 9 *Financial Instruments* or using the equity method. The other set is the financial statements of P and its associates as an economic entity in which investments in I and A are accounted for using the equity method.
2. Since I is an associate of P, non-controlling interests do not feature in the financial statements of P as an economic entity.
3. In the absence of non-controlling interests, P equity accounts its indirect interest in A's profit by applying 25% (50% × 50%) to A's profit after tax.
4. P also equity accounts its direct interest in I's profit in the usual way. However, any dividend income from A included in I's profit should be removed to avoid recognizing income in two forms since P is indirectly equity accounting the profit of A.
5. P shows only the investment in I on its statement of financial position. In its equity-accounted financial statements, the investment in I will include the share of post-acquisition retained earnings of both I and A.
6. The difference in effects between the structures in Figures 7.8 and 7.9 is essentially one of presentation whereby an indirect interest through a subsidiary requires recognition of the subsidiary's share of the associate's profit and a separate allocation to non-controlling interests. When indirect interest in an associate is held through another associate, the effective interest may be applied directly to the profit of the lower-tiered associate by the ultimate investor. The final effect to the economic entity's retained earnings is the same under either structure, although the presentation in the income statement differs.

ILLUSTRATION 7.5 Indirect holding of an associate held through a subsidiary

P acquired a 60% interest in S. S subsequently acquired a 30% interest in A. Shareholders' equity at the date of acquisition is shown below. Fair value adjustments at the date of acquisition were negligible. The fair value of S Co as an entity as at date of acquisition was $10,900,000 and the fair value of non-controlling interests at that date was $4,400,000.

	S	A
Share capital at acquisition date	$5,000,000	$1,000,000
Retained earnings at acquisition date	3,000,000	200,000
Shareholders' equity at acquisition date	$8,000,000	$1,200,000
Date of acquisition by investing company	30 July 20x2	4 May 20x3

Extracts of the financial statements of P, S, and A for the year ended 31 December 20x5 are shown below.

Income Statement
For the Year Ended 31 December 20x5

	P	S	A

Net profit before tax	$11,982,000	$ 997,000	$250,000
Tax	(2,382,000)	(197,000)	(50,000)
Net profit after tax	$ 9,600,000	$ 800,000	$200,000
Dividends declared	(1,500,000)	(120,000)	(40,000)
Profit retained	$ 8,100,000	$ 680,000	$160,000
Retained earnings, 1 January 20x5	30,000,000	4,500,000	300,000
Retained earnings, 31 December 20x5	$38,100,000	$5,180,000	$460,000

Statement of Financial Position
As at 31 December 20x5 (extract)

	P	S	A
Investment in S, cost	$ 6,500,000		
Investment in A, cost		$ 1,000,000	
Other net assets	81,600,000	9,180,000	$1,460,000
	$88,100,000	$10,180,000	$1,460,000
Shareholders' equity:			
Share capital	$50,000,000	$ 5,000,000	$1,000,000
Retained earnings	38,100,000	5,180,000	460,000
	$88,100,000	$10,180,000	$1,460,000

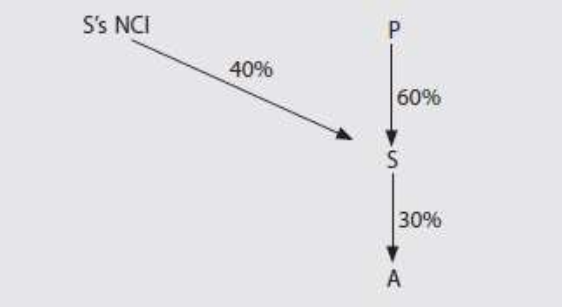
The tax rate was 20%.

page 484

Required:

1. Prepare the consolidation and equity accounting entries for 20x5.
2. Prepare the consolidation worksheets for 20x5.
3. Perform an analytical check on the balances as at 31 December 20x5 in the consolidated balance sheet with respect to non-controlling interests and the investment in A.

The affiliation structure is shown below. Non-controlling interest in S is 40%.



1. Consolidation and equity entries for 20x5

CJE1: Elimination of investment in S

Dr Share capital (S)	5,000,000			
Dr Retained earnings (S)	3,000,000			
Dr Goodwill	2,900,000	(Note 1)		
Cr Investment in S			6,500,000	
Cr Non-controlling interests			4,400,000	(Note 2)

Note 1:

$$\text{Goodwill} = \left(\begin{array}{c}\text{Fair value of}\\ \text{consideration transferred}\end{array} + \begin{array}{c}\text{Fair value of}\\ \text{non-controlling interests}\end{array}\right) - \text{Fair value of net identifiable assets}$$

$$= (\$6{,}500{,}000 + \$4{,}400{,}000) - \$8{,}000{,}000$$
$$= \$10{,}900{,}000 - \$8{,}000{,}000$$
$$= \$2{,}900{,}000$$

Note 2:

Fair value of non-controlling interests as at acquisition date	$4,400,000
This comprises:	
Share of fair value of net identifiable assets (40% × $8,000,000)	3,200,000
Share of goodwill ($4,400,000 – $3,200,000)	1,200,000

CJE2: Allocation of post-acquisition retained earnings of S to non-controlling interests

Dr Retained earnings (S)	600,000	
Cr Non-controlling interests		600,000

Retained earnings at 1 January 20x5	$4,500,000
Retained earnings at acquisition	(3,000,000)
Change	$1,500,000
Non-controlling interests' share at 40%	$ 600,000

CJE3: Elimination of dividend income received from S

Dr Dividend income (P)	72,000	($120,000 × 0.6)	
Dr Non-controlling interests	48,000	($120,000 × 0.4)	
Cr Dividends declared (S)			120,000

CJE4: Equity accounting of profits by S

Dr Investment in A	60,000	($200,000 × 0.3)	
Cr Share of profit of A			60,000

As explained in Chapter 6, the illustrative guidance of IAS 1 presents share of profit of associate as an after-tax basis. Although the guidance is not an integral part of the standard, this presentation has become general practice although conceptually confounding.

CJE5: Allocation of current profit after tax to non-controlling interests

Dr Income to non-controlling interests	339,200		
Cr Non-controlling interests		339,200	
Net profit after tax of S		$800,000	
Less dividend income from A		(12,000)	($40,000 × 0.3)
Add share of profit after tax of A		60,000	
Net profit after tax of S excluding dividend from A		$848,000	
Non-controlling interests' share at 40%		$339,200	

Dividend income from A is removed from profit for determination of non-controlling interests' share of current profit as profit is recognized in the form of equity-accounted profit.

CJE6: Reclassification of dividend from A

Dr Dividend income (S)	12,000	($40,000 × 0.3)	
Cr Investment in A			12,000

CJE7: Share of post-acquisition retained earnings of A

Dr Investment in A	30,000		
Cr Retained earnings		18,000	($30,000 × 0.6)
Cr Non-controlling interests		12,000	($30,000 × 0.4)
Retained earnings of A at 1 January 20x5		$300,000	
Retained earnings of A at acquisition date		(200,000)	
Change in retained earnings		$100,000	
S's share at 30%		$ 30,000	

S's share of the change in the post-acquisition retained earnings of A is allocated to the controlling interest in S and to its non-controlling interests. The allocation to the opening retained earnings represents the parent's share. The change in retained earnings in CJE7 includes the net effect of the cumulative profit of A and the cumulative dividends declared by A since the acquisition date. Hence, CJE7 re-enacts the cumulative effects of yearly equity accounting entries passed through the consolidation worksheet each year.

2. Consolidation worksheets for 20x5

Income Statement
For the Year Ended 31 December 20x5

	P	S	Combined P and S	Consolidation adjustments Dr		Consolidation adjustments Cr		Consolidated totals
Net profit before tax	$11,982,000	$ 997,000	$12,979,000	$ 72,000	CJE3			$12,955,000
				12,000	CJE6	$ 60,000	CJE4	
Tax	(2,382,000)	(197,000)	(2,579,000)					(2,579,000)
Net profit after tax	$ 9,600,000	$ 800,000	$10,400,000					$10,376,000
Non-controlling interests				339,200	CJE5			(339,200)
Net profit after non-controlling interests								$10,036,800
Dividends declared	(1,500,000)	(120,000)	(1,620,000)			120,000	CJE3	(1,500,000)
Profit retained	$ 8,100,000	$ 680,000	$ 8,780,000					$ 8,536,800
Retained profit, 1 January 20x5	30,000,000	4,500,000	34,500,000	3,000,000	CJE1	18,000	CJE7	30,918,000
				600,000	CJE2			
Retained profit, 31 December 20x5	$38,100,000	$5,180,000	$43,280,000	$4,038,200	To SFP	$213,000	To SFP	$39,454,800

Statement of Financial Position (extract)
As at 31 December 20x5

	P	S	Combined P and S	Consolidation adjustments Dr		Consolidation adjustments Cr		Consolidated totals
Investment in S, cost	$ 6,500,000		$ 6,500,000			$ 6,500,000	CJE1	$ 0
Investment in A, cost		$ 1,000,000	1,000,000	$ 60,000	CJE4	12,000	CJE6	1,078,000
				30,000	CJE7			
Goodwill on consolidation				2,900,000	CJE1			2,900,000
Other net assets	81,600,000	9,180,000	90,780,000					90,780,000
Net assets	$88,100,000	$10,180,000	$98,280,000	$2,990,000		$ 6,512,000		$94,758,000
Shareholders' equity:								
Share capital	$50,000,000	$ 5,000,000	$55,000,000	$ 5,000,000	CJE1			$50,000,000
Retained earnings	38,100,000	5,180,000	43,280,000	4,038,200	I/S	$ 213,000	I/S	39,454,800
	$88,100,000	$10,180,000	$98,280,000					
				48,000	CJE3	4,400,000	CJE1	
Non-controlling interests						600,000	CJE2	5,303,200
						339,200	CJE5	
						12,000	CJE7	
Total equity	$88,100,000	$10,180,000	$98,280,000	$ 9,086,200		$ 5,564,200		$94,758,000
				$12,076,200		$12,076,200		

3. Analytical checks

Analytical check on non-controlling interests of S as at 31 December 20x5

Share capital (S) . . . . . $ 5,000,000

Retained earnings (S)	5,180,000	
Unadjusted shareholders' equity of S	$10,180,000	
S's share of A's post-acquisition retained earnings	78,000	(Note 1)
Adjusted shareholders' equity of S	$10,258,000	
Non-controlling interests' share at 40%	$ 4,103,200	
Non-controlling interests' share of goodwill (refer to CJE 1 Note 2)	1,200,000	
Non-controlling interests as at 31 December 20x5	**$ 5,303,200**	

Note 1: S's share of change in A's post-acquisition retained earnings = 30% * ($460,000 — $200,000) = $78,000.

Analytical check on investment in A as at 31 December 20x5

Shareholders' equity of A as at 31 December 20x5	$1,460,000
S's interest in the net book value as at 31 December 20x5 (at 30%)	$ 438,000
Goodwill implicit in the investment account ($1,000,000 – (30% × $1,200,000))	640,000
Investment in A	**$1,078,000**

Note: If there is a fair value differential, S's share of the unamortized balance should be included in the check.

CHANGES IN OWNERSHIP INTERESTS

Intercorporate relationships change with facts and circumstances relating to the rights that the investor has over the investee. We can think of the strength and intensity of the relationships along a continuum. We saw in Chapter 2 that existing rights that give the investor the current ability to direct relevant activities and expose or entitle the investor to variable returns leads to control over the investee. Control is the basis for the strongest intercorporate relationship between investor and investee. IFRS 10 *Consolidated Financial Statements* provides the principles to determine the existence of control. When significant influence ends, control begins.

We saw in Chapter 6 that significant influence gives the investor the power to participate in financial and operating policy-making processes but does not give the investor the right to control or jointly control the outcome of those processes. IAS 28 *Investments in Associates and Joint Ventures* provides the basis for determining significant influence. In the absence of control, significant influence or joint control, the investor holds the investments as financial assets and applies the requirements of IFRS 9 *Financial Instruments* to those investments. Figure 7.10 shows the intercorporate relationships along a continuum. For simplicity, we do not show joint control as a separate component.

FIGURE 7.10 Intercorporate relationships



Intercorporate relationships change over time when facts and circumstances change the rights of the investor. We focus on four broad changes:

1. Business combination achieved in stages
2. Loss of control
3. Changes in ownership interests without change in control
4. No change in significant influence

BUSINESS COMBINATION ACHIEVED IN STAGES

A parent may achieve control in a subsidiary only after acquiring additional block(s) of shares to complement its existing holdings. Achieving control through incremental purchases is sometimes referred to as a "business combination achieved in stages" or a "step acquisition." Significant changes have been made in the revised IFRS 3 (2008) to determine fair value of goodwill at acquisition date when control is obtained as opposed to IFRS 3 (2004) that recognizes goodwill in a piecemeal manner at the date of each incremental purchase.

Remeasurement of Previously Held Equity Interest

The revised IFRS 3 (2008) requires the following remeasurement procedures to be applied to a business combination achieved in stages:

1. Previously held equity interest in the acquiree must be remeasured to fair value at acquisition date when control is obtained. Hence, even though there is no exchange transaction relating to previously held equity interest, the revised IFRS 3 requires the remeasurement of existing holdings to fair value. The remeasurement is necessary to enable the determination of a goodwill figure that is up-to-date and consistent with the fair value information existing at acquisition date when control is obtained. Under the previous IFRS 3 (2004), goodwill on the statement of financial position is a combination of historical goodwill arising from past holdings and current goodwill arising from the latest incremental purchase. Thus, under IFRS 3 (2004), goodwill was fragmented and did not reflect fair value at acquisition date.
2. Gains or losses arising from the remeasurement to fair value of previously held equity interest above are recognized in profit or loss in accordance with IFRS 3 paragraph 42.

   Remeasurement gains or losses = Remeasured fair value of previously held equity interest
   – Carrying amount of previously held equity interest

   In the consolidated financial statements of the acquirer, investment held of existing equity is remeasured to fair value at subsequent acquisition date(s) and the unrealized gains or losses are taken to income statement. page 490 Thus, an investment in a subsidiary is carried at the fair value at the date of acquisition of control. A noteworthy feature is that gains and losses are taken to income statement and not directly to equity. The rationale for the remeasurement is that a significant economic event has occurred, that is, the acquisition of control. Hence, in substance, there are effectively two transactions – effectively a sale of a previously held interests and a purchase of a new interest in a subsidiary. There is a fundamental change in relationship between the investor and the investee. The investee is no longer a passive investment or an associate. It has become a subsidiary through the incremental purchases by the investor. The moment it crosses the control boundary, it becomes a subsidiary. Hence, we can see it as two transactions. A previously held investment (e.g., an associate) is "sold" and a new investment (a subsidiary) is "bought." Let us take an example. Parent Co bought 20% interest in S Co on 1 July 20x1 and another 25% on 16 October 20x3. On 5 March 20x4, it acquired an additional 40% which gave Parent Co control over S Co. On 5 March 20x4, the previously held interests of 45% have to be remeasured to fair value and a remeasurement gain or loss has to be recognized. There are two effects of this remeasurement. First, the effect is as if Parent Co had sold 45% of the investment, recognizing a gain or loss in income statement. The second effect is that the cumulative investment of 85% is now carried at fair value on acquisition date.
3. If the acquirer has, in prior periods, recognized changes in value of its equity interest in the acquiree in other comprehensive income, the cumulative amount that was recognized directly in other comprehensive income will be accounted for in a manner as if the acquirer had disposed of the previously held equity interests (IFRS 3:42). Under IFRS 9, the fair value change of an equity investment that is classified as fair value through other comprehensive income would not be reclassified to income statement on disposal. If the entity elects to measure the cost of

investment using fair value as deemed cost in the separate financial statements, the entity may transfer the cumulative fair value gains or losses to retained earnings on "deemed disposal" of the investment if these changes were previously recognized in a separate component of equity. Under this accounting policy choice, the cost of investment in the subsidiary is derived as the sum of the fair value of the initial interest at the date of obtaining control of the subsidiary, plus any consideration paid for the additional interest. Conversely, if the entity elects to use the accumulated cost approach in measuring its cost of investment, the entity will have to 'unwind' the fair value changes previously recognized for the original interest in the subsidiary. Under this approach, the cost of investment is the sum of the consideration paid for the initial interest and the consideration paid for the additional interest. The difference between the fair value of the initial interest at the date of obtaining control of the subsidiary and its original consideration paid is recognized in profit or loss.

Determination of Goodwill in a Business Combination Achieved in Stages

If business combination is achieved in stages, the goodwill calculation will include a new element, the fair value of previously held interests as follows:

$$\text{Goodwill} = \left(\begin{array}{c} \text{Fair value of} \\ \text{consideration} \\ \text{transferred} \end{array} + \begin{array}{c} \text{Amount of} \\ \text{non-controlling} \\ \text{interests} \end{array} + \begin{array}{c} \text{Acquisition-date fair value of the} \\ \text{acquirer's previously held equity} \\ \text{interest in the acquiree} \end{array} \right) - \text{Fair value of net identifiable assets of the acquiree at the acquisition date}$$

Illustration 7.6 shows the application of IFRS 3 in the consolidated financial statements of an acquirer in a business combination achieved in stages.

ILLUSTRATION 7.6 Business combination achieved in stages

P acquired S in two successive purchases as follows:

Date	Acquisition cost paid for incremental investment	Fair value of S as an entity	Percentage acquired in incremental investment	Cumulative percentage held by P
1 Jan 20x0	$ 3,000,000	$15,000,000	20%	20%
31 Dec 20x0	$12,000,000	$20,000,000	60%	80%

Non-controlling interests are measured at fair value and are deemed to have a proportionate interest in the fair value of S as an entity. Information relating to S is as follows:

	1 Jan 20x0 Book value	1 Jan 20x0 Fair value	31 Dec 20x0 Book value	31 Dec 20x0 Fair value
Intangible assets	$ 0	$ 1,800,000	$ 0	$ 1,600,000
Inventory	1,000,000	1,200,000	2,000,000	2,400,000
Other net assets	9,000,000	9,000,000	12,000,000	12,000,000
Net identifiable assets	$10,000,000	$12,000,000	$14,000,000	$16,000,000
Share capital	$ 4,000,000		$ 4,000,000	
Retained earnings	6,000,000		10,000,000	
	$10,000,000		$14,000,000	

Net profit before tax for the year ended 31 December 20x0 . . . . . $6,250,000

Tax expense for the year ended 31 December 20x0 . . . . . $1,250,000

Dividends declared during the year ended 31 December 20x0 . . . . . $1,000,000

Intangible assets had a remaining useful life of six years from 1 January 20x0
Inventories were sold within three months from the acquisition date

In the separate financial statements of P, the journal entries are as follows:

1 January 20x0

Dr Investment in S	3,000,000	
Cr Cash		3,000,000

Acquisition of 20% of S at fair value

31 December 20x0

Dr Investment in S	12,000,000	
Cr Cash		12,000,000

Acquisition of an additional 60% of S at fair value

Since P had 20% equity interest in S during 20x0 and assuming that P had significant influence over S, P had to equity account S's profit for the year ended 31 December 20x0. The equity accounting (EA) entries and consolidation journal entries (CJE) are as follows:

EA1: Share of profit for the year ended 31 December 20x0

Dr Investment in S	920,000	
Cr Share of profit		920,000
Profit after tax of S	$5,000,000	
Less amortization of intangible assets, after tax	(240,000)	
Less cost of sales from undervalued inventory, after tax	(160,000)	
Adjusted profit after tax	$4,600,000	
Share of S's profit after tax at 20%	$ 920,000	

EA2: Reclassification of dividend income received during 20x0

Dr Dividend income	200,000	
Cr Investment in S		200,000

The reconstructed investment in S account as at 31 December 20x0 is as follows:

Investment in S (Consolidated financial statements)					
1 Jan 20x0	Cash	3,000,000	31 Dec 20x0	Dividends	200,000
31 Dec 20x0	Cash	12,000,000		Balance	16,000,000
	Share of profit after tax	920,000			
	Remeasurement gain	280,000			

Carrying amount of the previously held equity interests at consolidated level
= $3,000,000 + $920,000 – $200,000
= $3,720,000

Fair value of previously-held equity interests = $4,000,000 (20% × $20,000,000)

Remeasurement gain on previously held equity interests at consolidated level
= $4,000,000 – $3,720,000
= $280,000

For simplicity, we assume that the remeasurement gain is tax-exempt. The next entry recognizes the remeasurement gain at the consolidated level.

EA3: Recognition of remeasurement gain in the consolidated financial statements

Dr Investment in S	280,000	
Cr Gain on remeasurement		280,000

The balance of investment in S, just prior to elimination in CJE1, is shown below. Note that the balance is $16,000,000 which is proportionate to fair value of S as an entity. Because of the control premium that a parent may pay to acquire a controlling interest in a subsidiary, the parent's share (and hence the non-controlling interests' share) need not be proportionate to the fair value of the subsidiary. At acquisition date,

Balance of an investment in a subsidiary acquired in stages = Fair value of consideration transferred on acquisition date + Fair value of previously held equity interests

Investment in S (Consolidated financial statements)					
1 Jan 20x0	Cash	3,000,000	31 Dec 20x0	Dividends	200,000
31 Dec 20x0	Cash	12,000,000		Balance	16,000,000
	Share of profit after tax	920,000			
	Remeasurement gain	280,000			

With the obtaining of control on 31 December 20x0, the investment balance has to be eliminated and goodwill recognized on acquisition date. The elimination entry is as follows:

CJE1: Elimination of investment and

recognition of goodwill

Dr Share capital	4,000,000			
Dr Retained earnings	10,000,000			
Dr Goodwill	4,400,000	(Note 3)		
Dr Intangible assets	1,600,000			
Dr Inventory	400,000			
Cr Deferred tax liability			400,000	(Note 1)
Cr Investment in S			16,000,000	
Cr Non-controlling interests			4,000,000	(Note 2)

Note 1: Deferred tax liability = 20% × Fair value adjustments of $2,000,000
= $400,000

Note 2: Non-controlling interests have to be recognized on 31 December 20x0. The non-controlling interests in this question have a proportionate interest in the fair value of S as an entity.

Non-controlling interests = 20% × Fair value of S as an entity
= 20% × $20,000,000
= $4,000,000

Note 3: Fair value of net identifiable assets = $16,000,000 – Deferred tax liability of $400,000
= $15,600,000

$$\text{Goodwill} = \left(\begin{array}{l} \text{Fair value of consideration transferred at acquisition date} \quad \text{Fair value of non-controlling interests} \quad \text{Fair value of the acquirer's previously held equity interest in the acquiree at acquisition date} \\ - \text{ Fair value of net identifiable assets} \end{array} \right)$$

= ($12,000,000 + $4,000,000 +$4,000,000) – $15,600,000
= $20,000,000 – $15,600,000
= $4,400,000

Fair value of P's previously held interests is 20% of the fair value of S as an entity of $20,000,000.

In the subsequent year ended 31 December 20x1, equity accounting entries are re-enacted to retained earnings to reinstate opening consolidated retained earnings.

EA1: Equity accounting of post-acquisition retained earnings

Dr Investment in S	720,000	
Cr Opening retained earnings		720,000

Retained earnings as at 31 December 20x0	$10,000,000
Retained earnings as at 1 January 20x0	6,000,000
Change in retained earnings	$ 4,000,000

Less amortization and cost of sales adjustments (after-tax)	(400,000)
Adjusted change in retained earnings	$ 3,600,000
P's share at 20%	$ 720,000

The above entry re-enacts the combined effects of EA1 and EA2 from the previous year.

EA2: Recognition of remeasurement gain on previously held equity interests

Dr Investment in S	280,000	
Cr Opening retained earnings		280,000

This entry re-enacts the reinstatement of the remeasurement gain at the consolidated level ($280,000). Hence, the investment in S balance after putting through the re-enacted equity accounting entries in EA1 and EA2 stands at $16,000,000. Since S became a subsidiary on 31 December 20x0, this investment amount has to be eliminated against the share capital and retained earnings of S. The following consolidation adjustment re-enacts the elimination entry.

CJE1: Elimination of investment and recognition of goodwill

Dr Share capital	4,000,000	
Dr Retained earnings	10,000,000	
Dr Goodwill	4,400,000	
Dr Intangible assets	1,600,000	
Dr Inventory	400,000	
Cr Deferred tax liability		400,000
Cr Investment in S		16,000,000
Cr Non-controlling interests		4,000,000

In subsequent years after 20x1, the re-enactment entries will need to be passed each year to the consolidated financial statements until such time when S is disposed of or when P loses control of S.

LOSS OF CONTROL

A parent loses control of a subsidiary through total or partial divestment of ownership or through loss of rights to direct the most significant activity of the subsidiary. In some cases, the divestment is partial and the investor retains some interests in the former subsidiary. The retained interests result in a new relationship for the investor. The former parent may now be an investor in an associate or a passive investor in a financial asset.

The International Accounting Standards Board (the Board) expressed its view that the loss of a control of a subsidiary is a significant economic event[4] and requires the investor to measure the retained investment at fair value. The new investor-investee relationship is significantly different from the former parent-subsidiary relationship.

Since a loss of control is a significant economic event, we can see the transaction as being a sale of a subsidiary and purchase of an associate or financial asset. For example, if the investor decreases its ownership interests from 70% to 20% by selling 50% of its ownership interests, in substance, the investor is selling 70% and buying 20%. The income statement effect would be on 70% of the investment comprising the gain or loss from the actual sale of 50% and a "remeasurement" gain or loss from the retained 20% interests. Because of the significance of this event, IFRS 10 requires the investor to remeasure the retained interests to fair value and to recognize a remeasurement gain or loss in the income statement.

When a parent loses control of a subsidiary, it will determine the following amounts at the date when control is lost and do the following:[5]

1. Derecognize the assets and liabilities of the subsidiary. The assets and liabilities will include goodwill and any remaining balance of fair value adjustments.
2. Derecognize the carrying amount of any non-controlling interests.
3. Recognize the fair value of consideration received for the sale of the ownership interests and the gain or loss on sale.
4. Recognize any distribution of shares.
5. Remeasure any retained interests at fair value and recognize the gain or loss on remeasurement.

We compare this scenario with a gain in control when an associate or financial asset becomes a subsidiary. This situation was discussed in the previous section but we reinforce the accounting requirements here again. In a step acquisition or a business combination achieved in stages, the investor has previously held interests before obtaining control. In economic terms, one can see the change in relationship as divesting an associate/passive investment and acquiring a subsidiary. In substance, the effective (not actual) divestment of the previously held interests results in a gain or loss for the investor.

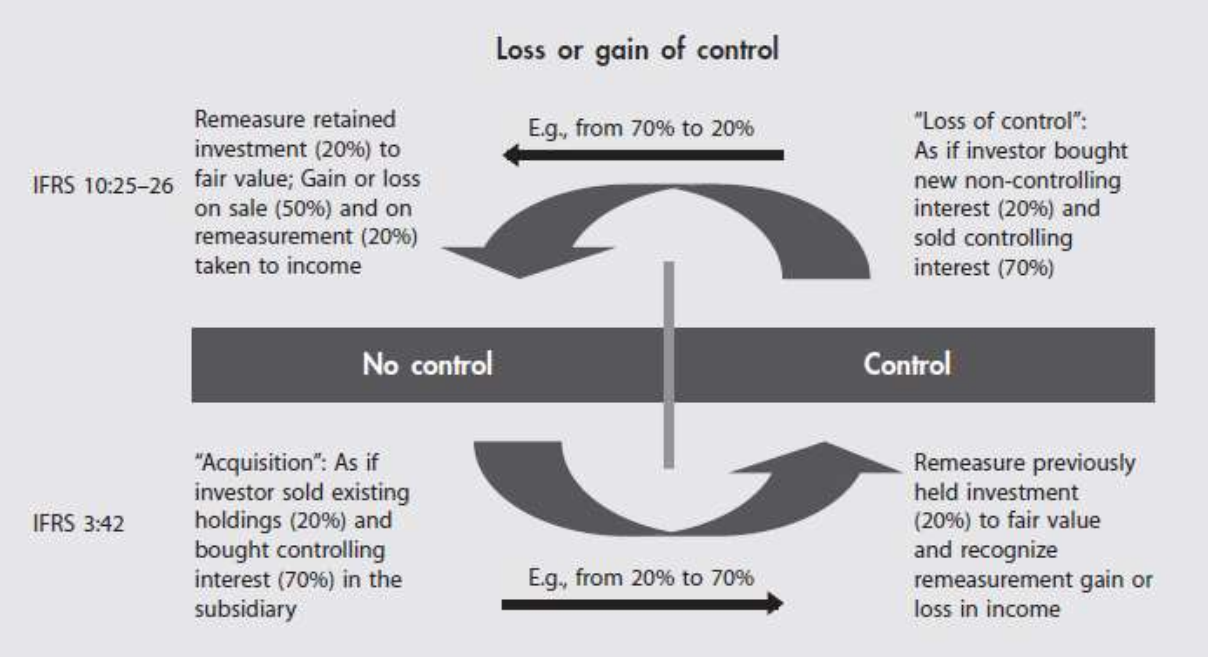
For example, if the investor increases its ownership interests from 20% to 70% by buying an additional 50% interests, it is as if the investor is selling its existing 20% interests and buying a new interest in 70% of the investee. IFRS 3 requires the investor to remeasure the previously held interests of 20% to fair value and recognize a remeasurement gain or loss on the 20%.

page 496

The same underlying principle applies in both situations. When control is obtained, a major significant event has arisen that changes the relationship between the investor and the investee. The previously held interests is remeasured resulting in a gain or loss to income. When control is lost, any retained interests is remeasured resulting in a gain or loss to income.

We show the pictorial diagram below in Figure 7.11 to represent the changes in ownership interests resulting in loss or gain of control:

FIGURE 7.11 Loss or gain of control

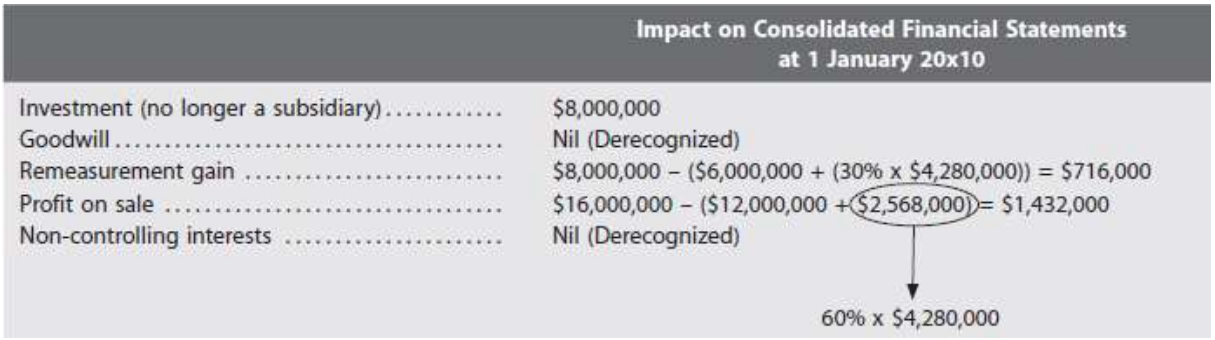


An example of an accounting policy note on accounting for loss of control is shown below. The extract is from the 2018 Annual Report of Keppel Corporation Limited.

When control of a subsidiary is lost as a result of a transaction, event or other circumstance, the Group derecognizes all assets (including any goodwill), liabilities and non-controlling interests at their carrying amounts. Amounts previously recognized in other comprehensive income in respect of that former subsidiary are reclassified to the profit and loss account or transferred directly to revenue reserves if required by a specific standard. Any retained interest in the former subsidiary is recognized at its fair value at the date control is lost, with the gain or loss arising recognized in the profit and loss account.

Scenario 1 and Scenario 2 relate to situations when control is lost or gained. For simplicity, we assume in each scenario that ownership interests of more than 50% results in control for the investor.

Scenario 1: Loss of Control P Co decreases ownership on 1 January 20x10 from 90% to 30% by reducing investment from $18 million to $6 million. The proceeds are $16,000,000 and the fair value of retained investment is $8,000,000. S Co's post-acquisition change in retained earnings is $4,280,000. There are no other post-acquisition changes in equity.



The consolidation adjustment in the year when control is lost is shown below:

CJE1: Adjustment to profit on sale and recognition of remeasurement gain

Dr Investment in S	2,000,000	(from $6 million to $8 million)
Dr Profit on sale (PL)	2,568,000	Note 1
Cr Remeasurement gain (PL)		716,000
Cr Opening retained earnings		3,852,000

Alternatively, we can analyze in two parts

Dr Investment in S	2,000,000	
Cr Remeasurement gain (PL)		2,000,000

Remeasurement from cost to fair value

Dr Profit on sale (PL)	2,568,000	
Dr Remeasurement gain (PL)	1,284,000	
Cr Opening retained earnings		3,852,000

Share of post-acquisition equity "foregone" on loss of control

Note 1:	Group	Legal entity	Difference
Sales proceeds	16,000,000	16,000,000	
Initial investment disposed	(12,000,000)	(12,000,000)	

Share of post-acquisition equity in S forgone	(2,568,000)		
Profit on sale (PL)	1,432,000	4,000,000	2,568,000

The profit on sale calculated at the separate and consolidated financial statements differ. At the consolidated level, the cost of investment sold is not simply the initial investment given up but also the parent's share of post-acquisition change in equity that had been consolidated up to the date of sale. By selling the investment, the investor gives up the right to future dividends or distributions that would have flowed out of the post-acquisition change in equity (for example, retained earnings, other comprehensive income and other equity).

The difference between the profit on sale at the legal entity and group level is recognized as a consolidation adjustment. The remeasurement gain is not recognized in the separate financial statements. Hence, the consolidation adjustment will recognize the remeasurement gain in the income statement. The calculation for the remeasurement gain is consistent with that for the gain on sale. The parent's share of post-acquisition profits has to be incorporated in the carrying amount of the investment when determining the remeasurement gain or loss. The post-acquisition retained profits and other post-acquisition changes have to be re-enacted before this consolidation adjustment is passed. In Appendix 7A, we show full length questions that show the complete process involving changes in ownership interests. Suffice now to understand the main principles.

page 498

We increase investment by $2,000,000 because in the legal entity's books the investment is carried at initial cost of $6 million. However, in the consolidated financial statements, the investment has to be carried at fair value of $8 million on the date when control is lost. Finally, the opening retained earnings is increased by $3,852,000 to re-enact the past consolidation of P's share of the subsidiary's retained earnings up to the date when control is lost.

In the subsequent year (20x11), the impact on retained earnings is $2 million. This amount represents the net change in opening retained earnings in 20x11 arising from the loss of control (i.e. $3,852,000 retained earnings at 1 January 20x10 that is "washed out" by the lower profit on sale of $2,568,000 and increased by the remeasurement gain of $716,000).

Re-enactment during 20x11:

Dr Investment	2,000,000	
Cr Opening retained earnings		2,000,000

We should note a few special features. If the subsidiary had other comprehensive income (e.g., revaluation reserves), a similar entry must also be reenacted to recognize P's share of the subsidiary's revaluation reserves at the date when control is lost.

The remeasurement entry has to be re-enacted at each subsequent reporting period for as long as P Co is required to prepare consolidated financial statements (if P has other subsidiaries) or combined financial statements (if P has no other subsidiary but has associates or joint ventures). Without the entry, subsequent consolidated retained earnings will not be the same as the amount reported in the year when control was lost.

If control is lost during the year (as is often the case), we need to consolidate the subsidiary up to the date when control is lost.

For example, in Scenario 1, let us assume that the change in ownership interests was 1 July 20x10 and the post-acquisition profit is analyzed as follows:

Share of post-acquisition retained earnings as at 1 Jan 20x10	$3,852,000	
Share of post-acquisition revaluation reserves as at 1 Jan 20x10	$ 350,000	
Share of net profit for first half ended 30 June 20x10	$ 450,000	(Note 1)

Note 1: Share of net profit retained comprises the following:

Sales	$2,000,000
Less Cost of sales	$1,200,000
Gross profit	$ 800,000

Less Expenses (including tax)	$ 300,000
Net profit after tax	$ 500,000
Income attributable to non-controlling interests	$ 50,000
Profit retained	$ 450,000

The consolidation adjustment is as follows:

Dr Investment	2,000,000		
Dr Profit on sale (PL)	3,101,333	(Note 2)	
Cr Remeasurement gain (PL)		449,333	(Note 2)
Cr Opening retained earnings		3,852,000	
Cr Opening revaluation reserves		350,000	
Cr Retained earnings		450,000	(Note 3)

Note 2: Share of post-acquisition change in equity includes share of post-acquisition revaluation reserves of $350,000 and share of current net profit after tax of $450,000. Profit on sale and remeasurement gain in Scenario 1 is reduced by this additional consolidated share of equity forgone.

	Reduction in profit on sale	Remeasurement gain
In Scenario 1	($2,568,000)	$716,000
Share of post-acquisition revaluation reserves given up	($233,333) (60/90 × $350,000)	($116,667) (30/90 × $350,000)
Share of current net profit after tax given up	($300,000) (60/90 × $450,000)	($150,000) (30/90 × $450,000)
	($3,101,333)	$449,333

Note 3: It is not sufficient to show a single line entry in retained earnings. The investor has to consolidate the income statement on a line-by-line basis, showing the items in Note 1, for the period when it has control. However, the net effect of the line-by-line consolidation is to increase retained earnings by $450,000 up to the date when control is lost.

Scenario 2: Gain on Control P Co increases ownership from 30% to 80% on 1 Jan 20x10 by increasing investment from $2 million to $17 million. The fair value of previously held investment is $6,000,000. Investment in associate (equity-accounted) as at 31 Dec 20x9 is $3,500,000. The fair value of identifiable net assets on 1 Jan 20x10 after deferred tax adjustment is $20,000,000, share capital is $10,000,000, pre-acquisition retained earnings are $6,000,000, unrecognized intangible asset is $5,000,000, and tax rate is 20%. The fair value of non-controlling interests on 1 Jan 20x10 is $4,000,000.

	Impact on Consolidated Financial Statements at 1 January 20x10
Investment in subsidiary	Nil
Goodwill	($15,000,000 + $6,000,000 + $4,000,000) –$20,000,000 = $5,000,000
Income (Remeasurement gain)	$6,000,000 – $3,500,000 = $2,500,000
Non-controlling interests	$4,000,000

The consolidation adjustments are shown below:

Re-enactment of equity accounting of post-acquisition retained earnings as at 1 Jan 20x10:

Dr Investment in Associate	1,500,000	
Cr Opening retained earnings		1,500,000

We can infer the share of post-acquisition retained earnings as follows (we can assume from the question that there are no revaluation reserves or other comprehensive income at or prior to acquisition date):

Investment in Associate, equity method	$3,500,000
Investment in Associate, initial cost	2,000,000
Share of post-acquisition retained earnings	$1,500,000

Remeasure previously-held interests of 30%

Dr Investment in subsidiary	2,500,000	
Cr Remeasurement gain (PL)		2,500,000

Recognize goodwill as of acquisition date

Dr Goodwill	5,000,000	
Dr Share capital	10,000,000	
Dr Retained earnings	6,000,000	
Dr Intangible asset	5,000,000	
Cr Investment		21,000,000
Cr Non-controlling interests		4,000,000
Cr Deferred tax liability		1,000,000

The investment in subsidiary of $21 million comprises fair value of consideration transferred to obtain control ($15 million) and the fair value of previously-held interests ($6 million). We are able to arrive at fair value of previously-held interests of $6 million through the re-enactment and remeasurement process.

Goodwill of $5 million is determined with reference to fair values at acquisition date and is not a layering of the results of different cost measures.

In Appendix 7A, we show a full–length illustration of such a scenario.

CHANGES IN OWNERSHIP INTERESTS WITHOUT CHANGE IN CONTROL

A parent may divest or increase ownership stakes in a subsidiary without losing or gaining control. Changes within the control continuum do not change the relationship with the subsidiary. In this scenario, there is no change in control that warrants a remeasurement gain or loss on retained interests or previously held interests. However, there is a transaction between the controlling and non-controlling interests and a re-balancing of their ownership interests. Transactions among owners in their capacity as owners are purely equity transactions. IFRS 10 requires the investor to recognize the gain or loss on sale or purchase directly in equity and not in the income statement. Any new economic goodwill or fair value adjustments that are included in successive purchases after the acquisition date should not be recognized as goodwill or

fair value adjustments. After the acquisition date, new goodwill or fair value adjustments are recognized in equity (these transactions will reduce equity).

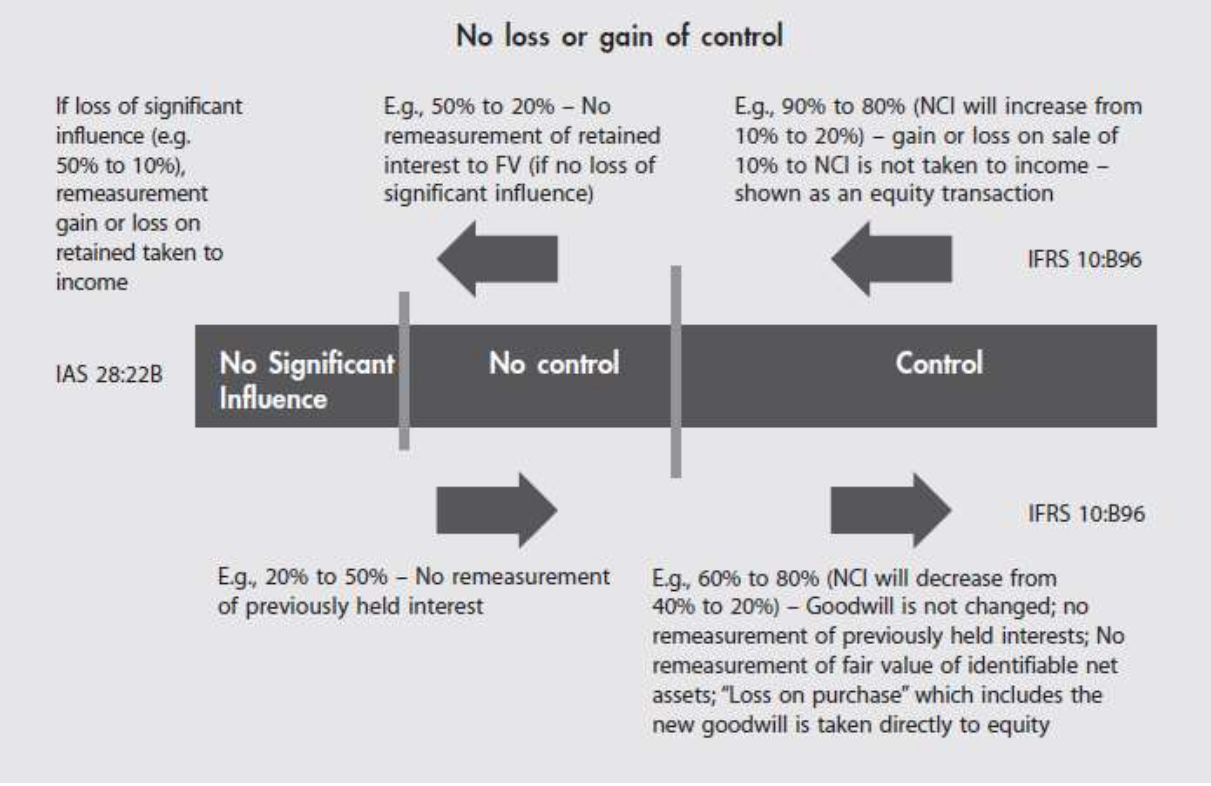
IFRS 10 paragraph B96 requires changes in ownership interests in a subsidiary that do not result in a loss of control to be accounted for as equity transactions (i.e., transactions between the controlling equity holders and non-controlling interests in their capacity as owners). When such transactions occur:

1. The carrying amounts of the controlling and non-controlling interests are adjusted to reflect the changes in their relative interests in the subsidiary. For example, if non-controlling interests increase to 20% from 5%, page 501 the equity balance of non-controlling interests will be adjusted upwards to be 20% of the subsidiary's equity. The carrying amounts refer to the amounts of the subsidiary's equity or net assets that have been recognized in the consolidated (economic entity) financial statements.
2. However, the amounts paid or received for the additional interests bought or sold will not be at the carrying amounts. The difference between the amount by which the non-controlling interests are adjusted and the fair value of consideration paid or received is taken directly to equity (not income). The difference is attributed to the owners of the parent (i.e., it is not shared with non-controlling interests).

In the 2018 Annual Report of Keppel Corporation Limited, the Group explained its accounting treatment on changes in Group's interest in a subsidiary that do not result in a loss of control.

> Changes in the Group's interest in a subsidiary that do not result in a loss of control are accounted for as equity transactions. The carrying amounts of the Group's interests and the non-controlling interests are adjusted and the difference between the change of the carrying amounts of the non-controlling interests and the fair value of the consideration paid or received is recognized directly in equity and attributed to owners of the Company.

We show the change in ownership interests without change in control in the following diagram.



Scenario 3 and Scenario 4 show how the changes in ownership interests without change in control are accounted for under IFRS 10. Full-length self-study illustrations are included in Appendix 7A. The diagram also shows change in significant influence. Appendix 7B deals with this in greater depth.

Scenario 3: Acquisition of non-controlling interests – No change in control P Co acquired 90% of S Co on 1 Jan 20x8 for $18,000,000. On 1 Jan 20x8, S Co's share capital is $8,000,000 and retained earnings is $1,520,000. On that date, S Co has an unrecognized intangible asset of $600,000. The intangible asset has an indefinite useful life and its benefits are taxable at 20%. Fair value of non-controlling interest on 1 Jan 20x8 is $1,600,000. P Co increases ownership from 90% to 95% on 1 January 20x10 by increasing investment from $18,000,000 to $20,000,000. Fair value of identifiable net assets (after tax) on 1 Jan 20x10 is $15,000,000. Fair value of non-controlling interest on 1 Jan 20x10 is $4,000,000. Retained earnings of S Co as at 1 January 20x10 is $5,800,000. Net profit after tax of S Co for 20x10 is $1,700,000. S Co has no other changes in equity in 20x10. The intangible asset acquired on 1 January 20x8 remains unimpaired as at 31 December 20x10.

page 502

	Impact on Consolidated Financial Statements at 1 January 20x10
Investment in subsidiary	Nil (Eliminated)
Goodwill	$18,000,000 + $1,600,000 – $10,000,000 = $9,600,000
Non-controlling interests (NCI)	$1,014,000 (refer CJE3 workings)
Equity (Loss on purchase)	$ 986,000 (refer CJE3 workings)

The consolidation adjustments passed in 20x10 are as follows:

CJE1: Elimination of investment

Dr Share capital	8,000,000	
Dr Retained earnings	1,520,000	
Dr Intangible asset	600,000	
Dr Goodwill	9,600,000	
Cr Deferred tax liability		120,000
Cr Investment in S Co		18,000,000
Cr NCI		1,600,000

Explanatory note:

Goodwill does not change after acquisition date. Hence, CJE1 is static and reflects the fair value information as at acquisition date. Subsequent increases in goodwill and fair value of identifiable net assets are not recognized. The investment amount eliminated reflects the amount as at acquisition date. Note that the investment amount is "under-eliminated" as there is an additional $2 million not eliminated in CJE1. The additional investment is eliminated in CJE3.

CJE2: Allocation of post-acquisition change in retained earnings to NCI

Dr Retained earnings	428,000	
Cr NCI		428,000
Retained earnings at date of second investment	5,800,000	
Retained earnings at acquisition date	(1,520,000)	
Change in retained earnings	4,280,000	

Explanatory note:

This entry re-enacts the allocation of past profits to NCI; hence the historical ownership interests have to be applied. If S Co has other comprehensive income (e.g. revaluation reserves), the same principle applies. The use of historical ownership interests also applies to any past amortization or expensing of fair value adjustments and adjustment of intra-group transactions that affect historical retained earnings (Refer Appendix 7A for more complex examples). The re-enactment

builds up the NCI balance to arrive at the historical balance just before the change in ownership interests. CJE3 then "re-balances" NCI downwards so that NCI balance is exactly proportional to the latest ownership interest.

CJE3: Loss on purchase of additional 5% with no change in control

Dr NCI	1,014,000	
Dr Equity (Loss on purchase)	986,000	
Cr Investment in S Co		2,000,000

Loss on purchase = Consideration paid for 5% – Carrying amount of 5% of NCI
= 2,000,000 – 1,014,000
= 986,000

Explanatory note:

In this entry, we eliminate the additional $2 million (not eliminated in CJE1) and also reduce NCI's carrying amount proportionately for the percentage disposed by NCI. The loss on purchase is the difference between the additional investment paid by P Co and the carrying amount of non-controlling interests reduced. The premium represents additional goodwill and fair value changes of identifiable net assets since the date when P Co obtained control of S Co. However, P Co is not permitted to recognize additional goodwill or new fair value adjustments on identifiable net assets after 1 January 20x8, the date when control was obtained. The difference is written off to equity and effectively represents a capital loss, although it may have an economic benefit to P Co. The loss on purchase may be taken to capital reserve. It should not be taken to the income statement as the loss arises from a transaction between owners.

The carrying amount of non-controlling interests can be determined in one of two ways as follows:

In the first method, we can reconstruct the NCI balance just before divestment through the chronological listing of consolidation entries.

CJE1: Initial fair value of NCI	1,600,000
CJE2: Change in retained earnings attributable to NCI	428,000
NCI balance as at 1 Jan 20x10 (before divestment)	2,028,000
5% is acquired by P Co, i.e. 50% of NCI	1,014,000

Alternatively, NCI may be derived analytically on 1 Jan 20x10 with reference to the three components:

Book value of equity of S Co	8,000,000 + 5,800,000	13,800,000
Unimpaired balance of intangible asset, after tax	600,000 × 80%	480,000
		14,280,000
NCI's share of identifiable net assets		1,428,000
NCI's goodwill	1,600,000 – 10%*10,000,000	600,000
NCI balance as at 1 Jan 20x10 (before divestment)		2,028,000

CJE4: Allocation of current income to NCI

Dr Income to NCI	85,000	
Cr NCI		85,000

Current income allocated to NCI (5% x $1,700,000)

Explanatory note:

The new ownership interests applies to allocation to NCI after the change in ownership interests. The same principle applies to allocation of other comprehensive income to NCI or adjustments for intra-group transactions and amortization or expensing of fair value adjustments.

Analytical check of NCI as at 31 December 20x10	
CJE1: Initial fair value of NCI	1,600,000
CJE2: Allocation of post-acquisition change in retained earnings to NCI	428,000
CJE3: Re-attribution from NCI	(1,014,000)
CJE4: Allocation of current income to NCI	85,000
NCI balance as at 31 Dec 20x10	1,099,000

Analyzed as follows:		
Book value of equity as at 31 Dec 20x10		15,500,000
Unimpaired balance of intangible asset, net of tax		480,000
Adjusted book value of equity		15,980,000
NCI's share of adjusted book value of equity		799,000
NCI's goodwill	800,000 – (5% x 10,000,000)	300,000
NCI as at 31 Dec 20x10		1,099,000

Explanatory note:

The analytical check is carried out in the usual manner, except that the proportion applied is the latest ownership interests. NCI has been proportionally reduced. Hence, the analytical check works neatly in the usual way but with a different percentage and goodwill that is attributable to NCI is a proportion of the original goodwill attributable to NCI.

Scenario 4: Partial disposal of controlling interest – No loss of control P Co acquired 90% of S Co on 1 Jan 20x8 for $18,000,000. On 1 Jan 20x8, S Co's share capital is $8,000,000 and retained earnings is $1,520,000. On that date, S Co has an unrecognized intangible asset of $600,000. The intangible asset has an indefinite useful life and its benefits are taxable at 20%.

Fair value of non-controlling interests on 1 Jan 20x8 is $1,600,000. P Co decreased ownership to 60% on 1 Jan 20x10. The proceeds were $15,000,000. Retained earnings of S Co as at 1 January 20x10 is $5,800,000. Net profit after tax of S Co for 20x10 is $1,700,000. S Co has no other changes in equity in 20x10. The intangible asset remains unimpaired as at 31 December 20x10.

	Impact on Consolidated Financial Statements at 1 January 20x10
Investment in subsidiary	Nil (Eliminated)
Goodwill	$18,000,000 + $1,600,000 – $10,000,000 = $9,600,000*
Income (Gain on sale)	Nil
Non-controlling interests	Increased by $7,284,000** (30% ÷ 90% × $21,852,000)***
Equity (Gain on sale)	$15,000,000 – $7,284,000 = $7,716,000

* Total goodwill does not change after the divestment of 30% by P Co.

** Adjustment for the change in relative interests.

*** CJE3

The following journal entry is recorded in the separate financial statements of P Co:

Dr Cash	15,000,000	
Cr Investment ($18,000,000 × 30/90)		6,000,000
Cr Gain on sale		9,000,000

The consolidation adjustments passed in 20x10 are as follows:

CJE1: Elimination of investment

Dr Share capital	8,000,000	
Dr Retained earnings	1,520,000	
Dr Intangible asset	600,000	
Dr Goodwill	9,600,000	
Cr Deferred tax liability		120,000
Cr Investment in S Co		18,000,000
Cr NCI		1,600,000

Explanatory note:

As explained in the previous scenario, goodwill does not change after acquisition date. Hence, CJE1 remains the same for as long as control is retained by P Co and reflects the fair value information as at acquisition date. Subsequent disposal of P Co's interest to NCI simply results in the transfer of a proportion of P Co's goodwill and fair value adjustments to NCI. Total goodwill remains the same. The investment amount eliminated reflects the amount as at acquisition date. Note that the investment amount is "over-eliminated" as investment in S Co by P Co at latest balance sheet date is only $12 million (60/90 x $18 million). The over-elimination is corrected in CJE3.

CJE2: Allocation of post-acquisition change in retained earnings to NCI

Dr Retained earnings	428,000	
Cr NCI		428,000
Retained earnings at date of partial divestment by P Co		5,800,000
Retained earnings at acquisition date		(1,520,000)
Change in retained earnings		4,280,000

Explanatory note:

As explained in the previous scenario, this entry re-enacts the allocation of past profits to NCI; hence the historical ownership interests have to be applied. If S Co has other comprehensive income (e.g. revaluation reserves), the same principle applies. The use of historical ownership interests also applies to any past amortization or expensing of fair value adjustments and adjustment of intra-group transactions that affect historical retained earnings (Refer Appendix 7A for more complex examples). The re-enactment builds up the NCI balance to arrive at the historical balance just before the change in ownership interests. Since there is an increase in the percentage held by NCI, CJE3 "re-balances" NCI upwards so that the NCI balance is exactly proportional to the latest ownership interest.

CJE3: Profit on sale of 30% (no change in control) taken to equity

Dr Investment in S Co	Note 1	6,000,000	
Dr Profit on sale (P Co's books)	Note 2	9,000,000	
Cr NCI	Note 3		7,284,000
Cr Equity (e.g. Capital Reserve)	Note 4		7,716,000
Workings:		Legal entity perspective	Group perspective
Sales proceeds		15,000,000	15,000,000
Investment in S Co disposed (Note 5)		(6,000,000)	(7,284,000)
Profit on sale		9,000,000	7,716,000

Explanatory notes:

The workings show the profit on sale calculations at the legal entity and the group level. The difference lies in the carrying amount of the investment disposed. While the investment in S Co is the cost of the initial investment for the legal entity, the effective investment in S Co at the group level is the cost of the initial investment and the consolidated profits given up. Hence, the group's profit on sale is lower. Note that the profit on sale at the group level is not taken to the income statement because the profit arises from transactions with owners in their capacity as owners. The following notes explain the individual legs of the journal entry.

Note 1: This entry corrects the "over-elimination" of the investment in S Co In CJE1.

Note 2: The profit on sale in P's legal entity financial statements is eliminated from income statement as it relates to a transaction between owners and should not be shown in the income statement.

Note 3: Since NCI's interest has risen from 10% to 40%, the NCI balance has to be adjusted upwards. Effectively, P Co's share of 30% of the equity of S Co is transferred to NCI. The workings for the adjustment are shown in Note 5.

Note 4: The profit on sale at the Group level has to be taken directly to equity (e.g. to Capital Reserve) because it arises from transactions between owners. It should not be taken to income statement. Hence, both the amount and the account is different at the Group level.

Note 5: The carrying amount of the investment in S Co at the Group level is the initial investment plus P's share of the post-acquisition changes in equity of S Co. Through the disposal, P Co gives up not only a proportion of the initial investment in S Co but also a proportion of the share of post-acquisition changes in equity of S Co. A proportion of the carrying amount relating to the ownership interests disposed is effectively transferred to NCI through a re-attribution or re-balancing process. The workings are shown below.

Initial investment by P Co		18,000,000
P Co's share of post-acquisition change in retained earnings	90% × 4,280,000	3,852,000
P Co's effective investment in S Co before divestment		21,852,000
P Co's effective investment in S Co transferred to NCI	30/90 × 21,852,000	7,284,000

CJE4: Allocation of current income to NCI

Dr Income to NCI	680,000	
Cr NCI		680,000
Current income allocated to NCI (40% x $1,700,000)		

Explanatory note:

The new ownership interests applies to allocation to NCI after the change in ownership interests. The same principle applies to allocation of other comprehensive income to NCI or adjustments for intra-group transactions and amortization or expensing of fair value adjustments.

Analytical check of NCI

CJE1: Initial fair value of NCI	1,600,000
CJE2: Allocation of post-acquisition change in retained earnings to NCI	428,000
CJE3: Re-attribution to NCI	7,284,000
CJE4: Allocation of current income to NCI	680,000
NCI balance as at 31 Dec 20x10	9,992,000

Analyzed as follows:

Book value of equity as at 31 Dec 20x10		15,500,000
Unimpaired balance of intangible asset, net of tax		480,000
Adjusted book value of equity		15,980,000
NCI's share of adjusted book value of equity	40% × 15,980,000	6,392,000
NCI's goodwill	Note 5	3,600,000
NCI as at 31 Dec 20x10		9,992,000

Note 5: NCI's goodwill is made up of two tranches. One tranche comprises the original goodwill attributable to NCI. The second tranche comprises of the goodwill that is effectively bought over from P Co when NCI increased its ownership interests by an additional 30%. The calculation is as follows:

	As at acquisition date	**Re-balancing**	**As at date of change**
Investment by P Co	18,000,000		
Fair value of NCI	1,600,000		
	19,600,000		
Percentage held by P Co	90%	(30%)	60%
NCI percentage	10%	30%	40%
Share capital	8,000,000		
Pre-acquisition retained earnings	1,520,000		
Intangible asset	600,000		
Deferred tax liability	(120,000)		
Fair value of identifiable net assets	10,000,000		
P Co's share of identifiable net assets	9,000,000		
NCI's share of identifiable net assets	1,000,000		
	10,000,000		
Goodwill	9,600,000		
P Co's goodwill 18,000,000-9,000,000	9,000,000	(3,000,000)	6,000,000
NCI's goodwill 1,600,000-1,000,000	600,000	3,000,000	3,600,000
	9,600,000		9,600,000

Explanatory note:

The analytical check relates to the three components. The latest ownership interest is applied to book value of equity and the remaining balance of any fair value adjustment. However, the goodwill calculation requires special consideration. A proportion of P Co's goodwill is effectively sold to NCI at the date of change of ownership interests. NCI's share of goodwill increases by 30/90 x $9,000,000 or $3,000,000. However, total goodwill does not change but the individual components of equity changes. The changes are reflected in equity through the re-balancing of NCI upwards and the reduction for P Co's equity through a smaller profit that is taken to equity.

NO CHANGE IN SIGNIFICANT INFLUENCE

Similar to the situation of no change in control, an investor may divest or increase ownership stakes in an associate without changing the relationship with its associate. No special accounting requirements apply to such transactions. Divestment of interests without loss of significant influence will not require remeasurement of the remaining interests. page 508 Similarly, increase in ownership interests without changing the existing relationship with the associate will not require the remeasurement of previously held interests. The accounting for such changes is explained in Appendix 7B.

ASSET TRANSFERS IN MORE COMPLEX SETTINGS

In earlier chapters, the following principles relating to intragroup transfers have been established:

1. Intragroup balances, transactions, income, and expenses must be eliminated in full.
2. Unrealized profit or loss included in inventory, fixed assets or other transferred assets must be eliminated in full. If the unrealized loss indicates an asset impairment, the loss should be recognized in the consolidated financial statements.
3. Where the seller or transferor is a partially owned subsidiary, non-controlling interests in the subsidiary must bear a portion of the adjustment of unrealized profit or loss. The approach taken in this text is that profit recognition by non-controlling interests must be aligned with that of the controlling interests. This alignment of profit recognition is underpinned by the entity theory that non-controlling interests are equity stakeholders of the subsidiary and consistent accounting bases should be applied to both controlling and non-controlling stakeholders. This view is consistent with the underlying trends in IFRS 3 and IFRS 10 to adopt the same accounting treatments for both groups of stakeholders. Hence, if a transfer originates from a partially owned subsidiary, the recognition of profit by non-controlling interests is deferred until the transferred asset is re-sold to third parties or is depreciated.
4. A sale from a parent to its subsidiaries is commonly described as a "downstream" sale and a sale from a subsidiary to its parent is described as an "upstream" sale. However, there are certain intragroup transfers that are not clearly "downstream" or "upstream." For example, if a subsidiary sells an asset to another subsidiary, it appears to be a lateral or horizontal transfer rather than an "upstream" sale. Nonetheless, the same principle applies. If the selling subsidiary is partially owned, the profit adjustments must be borne by both the group and the non-controlling interests of the selling subsidiary.

We consider three types of transfers within complex structures:

Asset Transfers between Parent and Indirect Subsidiaries

The "upstream" and "downstream" analysis applies to situations where assets are transferred between parent and indirect subsidiaries. In a "downstream" scenario where a parent transfers an asset to an indirect subsidiary, the adjustment is made for the unrealized profit and tax effects included in the parent's profit. No adjustment to non-controlling interests is necessary. However, in an "upstream" scenario, non-controlling interests have to be adjusted for the unrealized profit included in current profit of the transferring subsidiary and any realized profit that originated from previous periods. Since indirect subsidiaries have both direct and indirect non-controlling interests, adjustment will affect total non-controlling interests. Illustration 7.7 shows the accounting entries for an "upstream" transfer from an indirect subsidiary to a parent.

Asset Transfers between Fellow Subsidiaries

Although these transfers are more lateral or horizontal than "upstream," the non-controlling interests in the transferor bear a portion of adjustments of unrealized profit or loss. The non-controlling interests of the buying subsidiary are not affected by such adjustments.

Asset Transfers between a Subsidiary and an Associate

If a group company (parent or any of its subsidiaries) sells to an associate or if an associate sells to a group company, the group recognizes the unrelated investors' share of the unrealized profit under IAS 28. Only the group's share of the unrealized profit is not recognized. For example, if an investor has 30% interest in an associate, the investor should recognize 70% of unrealized profit arising from asset transfers, whether upstream or downstream.

The approach taken in this text as explained in Chapter 6 is that adjustments for unrealized profit or loss, whether arising from upstream or downstream sales, are made against equity-accounted profits. Since equity accounting is a compressed form of consolidation (commonly known as "one-line" consolidation), the adjustments are made against the one-line share of profit and share of tax entries rather than the separate line items such as sales or cost of sales. Proportionate adjustments against separate line items are, in our view, more in line with proportionate consolidation and would be inconsistent with the concept of equity accounting.[6]

Similarly, the adjustment for unrealized profit included in inventory or fixed assets of the associate is made against the investment in the associate account and not the individual asset. The investment in associate account is a proxy for the individual assets and liabilities held by the associate. In an "upstream sale," the same approach is taken whereby adjustments are made against the investment in associate and share of profit or loss of the associate.

If a partially owned subsidiary sells to an associate, the non-controlling interests in the subsidiary will also bear their share of the adjustment. Overall, the adjustment is borne proportionately by the group and the non-controlling interests of the subsidiary. The view taken in our text is that profit recognition by non-controlling interests must be consistent with that of the majority interests.

If the transfer is made from an associate to a subsidiary, no adjustment to non-controlling interests is necessary since the selling company is the associate. Illustration 7.7 shows and explains a scenario where a subsidiary transfers inventory to an associate of its parent.

ILLUSTRATION 7.7 Transfer of asset between a subsidiary and an associate

P Company has direct and indirect interests in S, A, and B. Details of the group structure and the profit before tax and tax expense of the companies for the years ended 31 December 20x0 and 31 December 20x1 are shown below. Profit and tax for the year ended 31 December 20x0 are as follows:

Company	Profit before tax	Tax expense	Direct % held	Direct investor
S	$1,000,000	$200,000	80%	P
A	400,000	80,000	30%	P
B	500,000	100,000	70%	S

An intragroup transfer was made during 20x0 as follows:

Transfer price	$100,000
Original cost	(60,000)

Profit on transfer	$ 40,000
Percentage resold to third parties during 20x0	60%
Percentage unsold as at 31 December 20x0	40%
Percentage unsold as at 31 December 20x1	0%

Profit and tax for the year ended 31 December 20x1 are as follows:

Company	Profit before tax	Tax expense
S	$1,200,000	$240,000
A	600,000	120,000
B	400,000	80,000

Consider two scenarios below.

Scenario 1: S sells to A

P's share of A's profit is adjusted for the group's share of unrealized profit. However, S's non-controlling interests must also bear a portion of the adjustment of the unrealized profit. In the consolidated financial statements, the equity accounting entry as at 31 December 20x0 are as follows:

EA1: Equity accounting of profit and tax for 20x0

Dr Investment in A	92,160	
Cr Share of profit of A		92,160

Unrealized profit is adjusted from the profit before tax of A and tax expense of A. The group is permitted by IAS 28 to recognize the unrelated investor's share of the profit.

Profit before tax of A	$400,000	
Unrealized profit at year-end	(16,000)	(40% × $40,000)
Adjusted profit before tax of A	384,000	
P's share of adjusted profit	$115,200	(30% × $384,000)
Tax expense of A	$ 80,000	
Tax on unrealized profit	(3,200)	(40% × $8,000)
Adjusted tax expense of A	$ 76,800	
P's share of adjusted tax	$ 23,040	(30% × $76,800)

Thus, the group's share of profit after tax (before allocation to non-controlling interests) is reduced by $3,840. In the consolidation adjustment, non-controlling interests will have a share of this reduction (as the seller is S) and the net reduction suffered by the group is $3,072 (i.e., 80% × $3,840).

Consolidation adjustments for 20x0

Dr Income to non-controlling interests	159,232	
Cr Non-controlling interests		159,232

Since the sale originated from S, the non-controlling interests in S bears a share of the adjustment for the unrealized profit.

Profit after tax of S	$ 800,000	
Less unrealized profit arising from transfer from S to A	(3,840)	
Adjusted profit after tax of S	$796,160	
Non-controlling interests at 20%	$159,232	
Original profit	$ 32,000	(Note 1)
Adjustment to share of A's profit after tax	(3,840)	(Note 2)
Adjustment to non-controlling interests	768	(Note 3)
Non-controlling interests' share	(6,400)	(Note 4)
Profit recognized from inventory transfer	$ 22,528	

If the sale is made to a third party, the profit recognized by the group would be $25,600 (i.e., interest in S of 80% multiplied with the after-tax profit on transfer of $32,000). However, because the transfer is made to an associate, the group can only recognize up to the unrelated investor's portion (i.e., not its own 30% share) of the unrealized profit. The realized profit can be recognized in full. Hence, the final profit recognized is $22,528 which excludes $3,072 which is P's share of the unrealized profit (refer to the explanatory notes under equity accounting below).

Note 1: Total profit on transfer is included in S's profit. Adjustment for unrealized profit is included in items below.

Note 2: Adjustment for proportion of unrealized profit relating to P's interest in A. P's interest of 30% is applied to after-tax unrealized profit (i.e. 30% × 80% × $16,000). Only A's unrelated investor's share (70%) of the unrealized profit is recognized.

Note 3: Adjustment for unrealized profit borne by non-controlling interests of S (i.e., 20% × $3,840).

Note 4: Share of profit on transfer allocated to non-controlling interests (i.e., 20% × $32,000). The combined effects of Note 3 and Note 4 is that non-controlling interests of S (as with the group) recognizes profit on the transfer to the extent of the unrelated investor's share in A.

We now show the equity accounting entries as at 31 December 20x1.

EA1: Reinstating the unrealized loss adjustment in opening retained earnings		
Dr Opening retained earnings	3,072	
Dr Non-controlling interests	768	
Cr Investment in A		3,840

The inventory in A on 1 January 20x1 is over-stated by the unrealized profit. However, in view of the fact that A is only an associate, it is not possible to correct the overvalued inventory which is not consolidated with P's financial statements. The proxy for inventory is the investment in A's account. Likewise, it is not possible to adjust directly the deferred tax asset account of A. The adjustment to the deferred tax asset account is taken to the investment in A account instead. Since the seller is S, the non-controlling interests in S bear a portion of the profit adjustment.

EA2: Equity accounting of profit and tax for the year ended 31 December 20x1		
Dr Investment in A	147,840	

Cr Share of profit of A		147,840

Since inventory held by A is resold to third parties during 20x1, the realized profit from the previous year's transfer of inventory can be added back to profit before tax. The same applies to the tax expense.

Profit before tax of A	$600,000	
Add back realized profit	16,000	(40% × $40,000)
Adjusted profit before tax of A	$616,000	
P's share of adjusted profit	$184,800	(30% × $616,000)
Tax expense of A	$120,000	
Tax on unrealized profit	3,200	(20% × $16,000)
Adjusted tax expense of A	$123,200	
P's share of adjusted tax	$ 36,960	(30% × $123,200)

Consolidation adjustments for 20x1

Dr Income to non-controlling interests	192,768	
Cr Non-controlling interests		192,768

Applying the concept that profit recognition for non-controlling interests is aligned with majority interests where the seller is the partially owned subsidiary, the realized profit is recognized by non-controlling interests in 20x1 as follows:

page 513

Profit after tax of S	$960,000	
Add realized profit arising from transfer from S to A	3,840	
Adjusted profit after tax of S	$963,840	
Non-controlling interests at 20%	$192,768	
Adjustment to share of A's profit after tax		$3,840
Adjustment to non-controlling interests		(768)
Profit recognized from inventory transfer		$3,072

Scenario 2: B sells to P

When B sells to P, the unrealized profit included in B's profit after tax has to be eliminated. The adjustment of the unrealized profit has to be also borne by the non-controlling interests of B. Since B is owned directly and indirectly by non-controlling interests, total non-controlling interests will bear their portion of the adjustment.

Direct non-controlling interests of B	30%
Indirect non-controlling interests of B	14%
Total non-controlling interests of B	44%

Total non-controlling interests of B is also the complement of P's indirect interests in B = 80% × 70% = 56%.

The consolidation adjustments for 20x0 are shown below:

CJE 1: Elimination of sales and unrealized profit on transfer from B to P

Dr Sales	100,000	
Cr Cost of sales		84,000
Cr Inventory		16,000

Sales is eliminated in full. Cost of sales comprises the cost of sales of realized sales and reversal of the cost of sales of the unrealized portion. Inventory is reduced by the amount of unrealized profit at the year end.

CJE2: Tax effects of CJE 1

Dr Deferred tax asset	3,200	
Cr Tax expense		3,200

CJE 3: Allocation of profit to non-controlling interests

Dr Income to non-controlling interests	170,368	
Cr Non-controlling interests		170,368

Since the sale originated from B, the lower-tiered subsidiary, any adjustment of unrealized profit will be borne by both direct and indirect non-controlling interests. The allocation of current profit after tax to non-controlling interests is shown below:

Profit after tax of B	$400,000	
Unrealized profit (after-tax)	(12,800)	(40% × 80% × $40,000)
Adjusted profit after tax of B	$387,200	
Total non-controlling interests	44%	

The consolidation adjustments for 20x1 are shown below:

CJE1: Recognition of realized profit in 20x1

Dr Opening retained earnings	8,960	
Dr Non-controlling interests	7,040	
Cr Cost of sales		16,000

CJE2: Tax effects of CJE1

Dr Tax expense	3,200	
Cr Opening retained earnings		1,792
Cr Non-controlling interests		1,408

CJE3: Allocation of profit to non-controlling interests

Dr Income to non-controlling interests	146,432	
Cr Non-controlling interests		146,432

Subsequently, the allocation of current profit after tax to non-controlling interests is:

Profit after tax of B	$320,000
Realized profit (after-tax)	12,800
Adjusted profit after tax of B	$332,800
Total non-controlling interests	44%
Income to non-controlling interests	$146,432

IMPACT OF CONSOLIDATION AND THE COST AND EQUITY METHODS ON PROFIT UPON THE DISPOSAL OF SUBSIDIARIES

When a parent sells a subsidiary, the "cost of sale" includes the sacrifice made by the parent, in giving up its rights to the distribution of post-acquisition profits. The profit (or loss) on sale is determined at two levels as follows.

At the group level, if the investee is consolidated:

$$\text{Profit/(loss) on sale of investment} = \text{Sales proceeds} - \begin{pmatrix}\text{Original cost}\\ \text{of investment}\end{pmatrix} + \begin{pmatrix}\text{Post-acquisition}\\ \text{profits}\end{pmatrix}$$

If equity accounting is used,

$$\text{Profit/(loss) on sale of investment} = \text{Sales proceeds} - \begin{pmatrix}\text{Original cost}\\ \text{of investment}\end{pmatrix} + \begin{pmatrix}\text{Post-acquisition}\\ \text{profits}\end{pmatrix}$$

Compare this with the effect in the separate financial statements if the cost method is used:

$$\text{Profit/(loss) on sale of investment} = \text{Sales proceeds} - \text{Original cost of investment}$$

If fair value (under IFRS 9) is used,

$$\text{Profit/(loss) on sale of investment} = \text{Sales proceeds} - \begin{matrix}\text{Carrying amount of investment}\\ \text{(at fair value)}\end{matrix}$$

Equity accounting has the effect of aligning the company's carrying amount of the investment with the group's carrying amount. As such, the profit on sale under equity accounting is the same as that under consolidation.

If fair value is used to determine the carrying amount of the investment, realized capital gain at the date of sale is smaller than under the cost method. Instead of a large terminal profit, unrealized capital profit has been systematically recognized over the life of the investment and credited directly to equity (if the investment is classified as a FVOCI equity security).

Illustration 7.8 shows profit recognition under the cost and equity methods and consolidation.

ILLUSTRATION 7.8 Comparison of the cost and equity methods and consolidation

To illustrate that the differences between consolidation, cost and equity methods are only temporary in nature, consider the following scenario when a sale is made of an investment prepared under each of these three methods. Total profit earned during the investment-holding period (share of post-acquisition profits and profit on sale) is the same under each method.

	Cost method	Equity method	Consolidation
Sales proceeds	$10,000,000	$10,000,000	$10,000,000
Original cost	$ 6,000,000	$ 6,000,000	$ 6,000,000
Share of post-acquisition profits	0	3,500,000	3,500,000
Cost of sales	$ 6,000,000	$ 9,500,000	$ 9,500,000
Profit on sale	$ 4,000,000	$ 500,000	$ 500,000
Total profit	$ 4,000,000	$ 4,000,000	$ 4,000,000

Under the cost method,

Terminal entry (cost method):		
Dr Cash	10,000,000	
Cr Investment		6,000,000
Cr Profit on sale		4,000,000

The equity accounting adjustment to adjust the terminal entry from the cost to the equity method is shown below:

Dr Profit on sale	3,500,000	
Cr Investment		3,500,000

A terminal consolidation journal entry is required to reflect the fact that the group had already recognized, in previous years, the profit now recognized by the parent on disposal of the investment under the cost method. The following shows the consolidation journal entry:

Dr Profit on sale	3,500,000	
Cr Retained earnings		3,500,000

OVERVIEW OF CONSOLIDATED CASH FLOW STATEMENTS

The preparation of a consolidated cash flow statement generally follows the same procedures as applied to the cash flow statement of a stand-alone entity. The information for the preparation of a consolidated cash flow statement comes from the following sources:

1. Consolidated income statement;
2. Consolidated balance sheets (for two consecutive years); and
3. Consolidated notes to the accounts.

Under IAS 7 *Cash Flow Statements*, a cash flow statement can be prepared using either one of the two methods: the direct method or the indirect method. For our purpose, we shall use the indirect method to illustrate the preparation of the consolidated cash flow statement.

As with the cash flow statement of a stand-alone company, the starting point is the net profit (in this case, the consolidated net profit), and adjustments are made for all non-cash items (in this case, items that have a non-cash impact on the consolidated net income).

The following points should be noted with respect to the preparation of consolidated cash flow statements:

1. Depreciation and amortization of a purchase differential (difference between fair value and book value) together with the depreciation and amortization expenses of the individual companies in the group are added back to consolidated net profit as they do not involve any cash inflow or outflow.
2. All transfers between group companies and any unrealized profits resulting from these transfers would have been eliminated in the preparation of the consolidated income statement and consolidated balance sheets and thus no further adjustments are needed.

page 517

3. Non-controlling interests' share of net profit is a non-cash adjustment that is added back to consolidated net income to arrive at net cash flows from operating activities.
4. Payments to non-controlling shareholders (for example, dividends) and receipts from non-controlling shareholders (for example, subscription of additional shares) are disclosed as uses or sources of cash under financing activities.

Illustration 7.9 provides an example of how the consolidated cash flow statement is prepared.

ILLUSTRATION 7.9 Consolidated cash flow statements

On 1 January 20x5, Parco acquired 80% of the share capital of Beta in exchange for a fresh issue of 2,000,000 of its own ordinary shares at a fair value of $2.00 per share and $500,000 in cash. The fair value of non-controlling interests of Beta was $1,125,000 on acquisition date. The fair values of the following identifiable assets and liabilities of Beta at the date of acquisition are as follows:

	$'000
Property, plant, and equipment	3,800
Inventories	1,350
Receivables	1,600
Cash	150
Payables	(1,450)
Long-term loan	(250)
	5,200

Additional information:

(a) Prior to the acquisition of Beta, Parco acquired 75% of the issued share capital of another subsidiary, Alpha.

(b) In 20x5, Parco disposed of a non-current property asset for $2,380,000. The carrying amount of the property at the date of disposal was $1,200,000. There were no other disposals of non-current assets.

(c) Goodwill in both subsidiaries was unimpaired.

(d) During the year, Parco paid a dividend of $7,000,000, and Alpha paid a dividend of $800,000. No dividends were declared by Beta.

The consolidated income statement for the year ended 31 December 20x5 and the consolidated statements of financial position as at 31 December 20x4 and 31 December 20x5 of Parco Group are shown on the following pages.

page 518

PARCO GROUP
Consolidated Income Statement and Partial Statement of Changes in Equity
For the Year Ended 31 December 20x5

	$'000
Revenue	$122,000
Cost of sales	(86,350)
Gross profit	$ 35,650
Operating expenses	(6,780)
Depreciation expense	(5,350)
Interest expense	(1,540)
Operating profit	$ 21,980
Gain on disposal of fixed asset	1,180
Profit before tax	$ 23,160
Taxation	(4,830)
Profit after tax	$ 18,330
Profit allocated to non-controlling interests	(800)
Profit attributable to the group	$ 17,530
Dividends declared	(7,000)
Retained earnings, 1 January 20x5	13,350
Retained earnings, 31 December 20x5	$ 23,880

PARCO GROUP
Statements of Financial Position
As at 31 December 20x4 and 31 December 20x5

	20x5 $'000	**20x4 $'000**
Property, plant and equipment	$ 48,150	$ 38,280
Goodwill	4,585	4,160
Current assets:		
Inventories	38,500	32,300
Accounts receivable	29,840	28,280
Cash	1,070	4,050
Total assets	$122,145	$107,070
Current liabilities:		
Accounts payable	$ 40,000	$ 38,790
Interest payable	1,500	1,350
Tax payable	5,500	4,840
Long-term loan	16,000	19,200
Non-controlling interests	3,265	1,540
Share capital	32,000	28,000
Retained earnings	23,880	13,350

Total liabilities and equity	$122,145	$107,070

The consolidated cash flow statement for 20x5 is shown below.

PARCO GROUP
Statement of Cash Flow
For the Year Ended 31 December 20x5

	$'000
Cash flows from operating activities	
Net profit after tax and non-controlling interests	$ 17,530
Add (deduct):	
Depreciation	5,350
Interest expense	1,540
Taxation	4,830
Gain on disposal of fixed asset	(1,180)
Increase in inventories (Note 1)	(4,850)
Decrease in receivables (Note 2)	40
Decrease in payables (Note 3)	(240)
Non-controlling interests' share of net profit	800
Interest paid (Note 4)	(1,390)
Tax paid (Note 5)	(4,170)
Net cash flows from operations	$ 18,260
Cash flows from investing activities	
Sale of fixed assets (Note 6)	2,380
Acquisition of subsidiary (Note 7)	(350)
Purchase of fixed assets (Note 8)	(12,620)
Net cash flows from investing activities	$(10,590)
Cash flows from financing activities	
Repayment of long-term loan (Note 9)	(3,450)
Dividends paid	(7,000)
Dividends paid to non-controlling shareholders (Note 10)	(200)
Net cash flows from financing activities	$(10,650)
Cash surplus (deficit)	(2,980)
Cash and cash equivalents at the beginning of the year	4,050
Cash and cash equivalents at the end of the year	$ 1,070

All figures in notes 1 to 10 are in thousands:

Note 1:

$$\begin{aligned}\text{Increase in inventories} &= \text{Opening inventories} + \text{Inventories from acquired subsidiary} - \text{Closing inventories}\\ &= \$32{,}300 + \$1{,}350 - \$38{,}500 = \$(4{,}850)\end{aligned}$$

Note 2:

$$\begin{aligned}\text{Decrease in receivables} &= \text{Opening receivables} + \text{Receivables from acquired subsidiary} - \text{Closing receivables}\\ &= \$28{,}280 + \$1{,}600 - \$29{,}840 = \$40\end{aligned}$$

Note 3:

$$\begin{aligned}\text{Decrease in payables} &= \text{Opening payables} + \text{Payables from acquired subsidiary} - \text{Closing payables}\\ &= \$38{,}790 + \$1{,}450 - \$40{,}000 = \$(240)\end{aligned}$$

Note 4:

$$\begin{aligned}\text{Interest paid} &= \text{Opening interest payable} + \text{Interest expense} - \text{Closing interest payable}\\ &= \$1{,}350 + \$1{,}540 - \$1{,}500 = \$1{,}390\end{aligned}$$

Note 5:

$$\begin{aligned}\text{Tax paid} &= \text{Opening tax payable} + \text{Tax expense} - \text{Closing tax payable}\\ &= \$4{,}840 + \$4{,}830 - \$5{,}500 = \$4{,}170\end{aligned}$$

Note 6: Proceeds from sale of fixed asset: information as provided in item (b) under *Additional information*.

Note 7:

$$\begin{aligned}\text{Cash outflows from acquisition of subsidiary} &= \text{Cash payment on acquisition} - \text{Cash balances of acquired subsidiary}\\ &= \$500 - \$150\\ &= \$350\end{aligned}$$

Note 8:

$$\begin{aligned}\text{Purchase of fixed assets} &= \text{Opening fixed assets} - \text{Disposal of fixed assets} + \text{Fixed assets from acquired subsidiary} - \text{Current depreciation} - \text{Closing fixed assets}\\ &= \$38{,}280 - \$1{,}200 + \$3{,}800 - \$5{,}350 - \$48{,}150\\ &= \$(12{,}620)\end{aligned}$$

Note 9:

$$\begin{aligned}\text{Repayment of long-term loan} &= \text{Opening loan payable} + \text{Loan from acquired subsidiary} - \text{Closing loan payable}\\ &= \$19{,}200 + \$250 - \$16{,}000\\ &= \$3{,}450\end{aligned}$$

Note 10:

Dividends paid to non-controlling shareholders of Alpha = 25% × $800 = $200

Note that the non-controlling interests' share of net profit does not affect the group's cash flow position, but was subtracted to arrive at the group's consolidated net profit. Dividends paid to non-controlling shareholders by a partially owned subsidiary result in an outflow of cash under financing activities.

Non-controlling interests comprises:	
Balance as at 1 January 20x5	$1,540,000
Fair value of non-controlling interests of Beta as at 1 January 20x5	1,125,000
Share of current profit after tax	800,000
Dividends paid to non-controlling interests	(200,000)
Balance as at 31 December 20x5	$3,265,000

Note also that where there is an acquisition or sale made during the year, IAS 7 paragraph 40 requires the following disclosures to be made:

1. The total purchase or sale consideration;
2. The portion of the purchase/sale consideration paid for in cash;
3. The amount of cash and cash equivalents in the subsidiary acquired/sold; and
4. The amount of the assets and liabilities other than cash or cash equivalent in the subsidiary acquired/sold summarized by each major category.

The disclosures under IAS 7 paragraph 40 as applied to the above illustration are provided below for the acquisition of a subsidiary:

	$'000	
Property, plant and equipment	$3,800	
Inventories	1,350	
Receivables	1,600	
Cash	150	
Payables	(1,450)	
Long-term loan	(250)	
Fair value of net identifiable assets	$5,200	
Goodwill on acquisition	425	(Note 1)
Fair value of subsidiary	$5,625	
Less fair value of non-controlling interests	(1,125)	
Purchase consideration	$4,500	
Payment by issue of shares	(4,000)	
Cash paid	$ 500	
Cash and cash equivalents acquired	(150)	
Net cash paid on acquisition	$ 350	

Note 1: Goodwill on acquisition

	$'000
Purchase consideration	$4,500
Fair value of non-controlling interests	1,125
	$5,625
Less fair value of net identifiable assets	(5,200)
Goodwill on acquisition	$ 425

Consolidated Cash Flow Statements Involving Foreign Operations

Consolidated cash flow statements involving a foreign subsidiary have its unique features. However, translating cash flows of a foreign subsidiary involves elements from this chapter and Chapter 8. Chapter 8 deals with the translation of foreign operations. To appreciate the process, you will need to be familiar with the principles in this chapter and Chapter 8. We apply the principles that we have discussed above with respect to consolidating the cash flows of a local subsidiary. However, special considerations are required with respect to consolidating the cash flow statement of a foreign operation.

Cash flows from operations, investing and financing activities of a foreign subsidiary should be translated at the actual rates on the dates of the cash flows. While cash flows from investing and financing activities are generally linked to specific transactions, cash flows from operations occur throughout a period. Using the indirect method, the starting point is the net profit figure. Adjustments involving working capital changes and non-cash adjustments are necessary to arrive at operating cash flows. These adjustments should therefore be translated at the actual rates on the dates of the operating cash flows that they are meant to adjust. It would be incorrect to translate the working capital balances at the closing rates at the respective year end dates.

A full-length example of the preparation of a consolidated cash flow statement for a local parent and a foreign subsidiary is shown in Appendix 7E.

A real-life example of a consolidated cash flow statement follows. The extracts are from the 2018 Annual Report of Keppel Corporation Limited.

Consolidated Statement of Cash Flows

For the financial year ended 31 December 2018

	Note	2018 $'000	2017 $'000
Operating activities		1,042,566	801,046
Operating profit			
Adjustments:			
Depreciation and amortisation		182,386	212,380
Share-based payment expenses		34,885	32,583
Profit on sale of fixed assets and an investment property		(2,795)	(20,142)
Gain on disposal of subsidiaries		(604,638)	(146,542)
Gain on disposal of associated companies		(48,783)	(62,673)
Impairment/write-off of fixed assets		6,911	15,530
Impairment/(write-back of impairment) of associated companies		60,782	(39,192)
Impairment of investments		—	14,330
Fair value gain on investment properties		(84,886)	(177,939)
Profit on sale of investments		(2,232)	(35,294)
(Gain)/loss from change in interest in associated companies		(63,622)	13,075
Unrealised foreign exchange differences		27,622	(87,745)
Operational cash flow before changes in working capital		548,196	519,417
Working capital changes:			
Stocks		(394,258)	438,670
Contract assets		357,046	478,634
Debtors		543,245	122,556
Creditors		(696,015)	(217,728)
Contract liabilities		12,430	357,652
Investments		(5,448)	(17,549)
Intangibles		(561)	(731)
Amount due to/from associated companies		177	(60,578)
		364,812	1,620,343
Interest received		154,482	130,832
Interest paid		(198,637)	(184,841)
Net income taxes paid		(195,904)	(363,377)
Net cash from operating activities		124,753	1,202,957
Investing activities			
Acquisition of subsidiaries	A	(38,052)	—
Acquisition and further investment in associated companies		(365,818)	(291,356)
Acquisition of fixed assets and investment properties		(254,511)	(392,991)
Disposal of subsidiaries		1,085,671	878,873
Proceeds from disposal of associated companies and return of capital	B	179,342	96,954
Proceeds from disposal of fixed assets		5,524	37,385
Advances to/from associated companies		(216,636)	(42,555)
Dividends received from investments and associated companies		281,375	270,199
Net cash from investing activities		676,895	556,509
Financing activities			
Acquisition of additional interest in subsidiaries		(3,337)	(66,380)
Proceeds from share issues		412	2,916
Proceeds from reissuance of treasury shares pursuant to share option scheme		5,324	1,374
Proceeds from non-controlling shareholders of subsidiaries		–	77
Proceeds from term loans		1,549,445	1,700,023
Repayment of term loans		(1,939,475)	(2,707,102)
Purchase of treasury shares		(90,758)	(19,428)
Dividend paid to shareholders of the Company		(526,152)	(363,531)
Dividend paid to non-controlling shareholders of subsidiaries		(20,321)	(26,574)
Net cash used in financing activities		(1,024,862)	(1,478,625)
Net (decrease)/increase in cash and cash equivalents	C	(223,214)	280,841
Cash and cash equivalents as at beginning of year		2,241,448	2,018,772
Effects of exchange rate changes on the balance of cash held in foreign currencies		(46,390)	(58,165)
Cash and cash equivalents as at end of year		1,971,844	2,241,448

Consolidated Statement of Cash Flows

Reconciliation of liabilities arising from financing activities

2018

	1 January 2018 $'000	Principal and interest payments (net of proceeds) $'000	Non-cash changes: Acquisition of subsidiaries $'000	Non-cash changes: Disposal of subsidiaries $'000	Non-cash changes: Interest expense (Note 26) $'000	Non-cash changes: Foreign exchange movement $'000	31 December 2018 $'000
Term loans	7,793,003	(588,667)	297,923	(171,380)	199,464	18,166	7,548,509

2017

	1 January 2017 $'000	Principal and interest payments (net of proceeds) $'000	Non-cash changes: Disposal of subsidiaries $'000	Non-cash changes: Interest expense (Note 26) $'000	Non-cash changes: Foreign exchange movement $'000	31 December 2017 $'000
Term loans	9,053,042	(1,191,920)	(138,288)	189,223	(119,054)	7,793,003

Notes to Consolidated Statement of Cash Flows

A. Acquisition of Subsidiaries

During the financial year, net assets of subsidiaries acquired at their fair values were as follows:

	2018 $'000	2017 $'000
Fixed assets	47	–
Investment Properties	360,000	–
Debtors and other assets	530	–
Bank balances and cash	18,521	–
Creditors	(6,778)	–
Borrowings	(297,923)	–
Current and deferred taxation	(3,827)	–
Total identifiable net assets at fair value	70,570	–
Amount previously accounted for as associated companies	(32,484)	–
Loss on remeasurement of previously held equity interest at fair value at acquisition date	18,487	–
Net assets acquired	56,573	–
Total purchase consideration	56,573	–
Less: Bank balances and cash acquired	(18,521)	–
Cash outflow on acquisition	38,052	–

Acquisition of subsidiaries during the year relates mainly to the acquisition of 77.6% interest in PRE 1 Investments Pte Ltd on 20 December 2018.

B. Disposal of Subsidiaries

During the financial year, the book values of net assets of subsidiaries disposed were as follows:

	2018 $'000	2017 $'000
Fixed assets	(4,272)	(129,536)
Investment properties	(948,613)	(405,604)
Long term investments	—	(2,102)
Stocks	(692,651)	(282,344)
Debtors and other assets	(7,939)	(159,030)
Bank balances and cash	(39,194)	(36,374)
Creditors and other liabilities	446,973	77,431
Borrowings	171,380	138,288
Current and deferred taxation	139,863	13,280
Non-controlling interests	210,166	69,451
	(724,287)	(716,540)
Amount accounted for as associated company	—	73,593
Net assets disposed of	(724,287)	(642,947)
Net profit on disposal	(604,638)	(146,542)
Realisation of foreign currency translation reserve	(7,575)	9,698
Sale proceeds	(1,336,500)	(779,791)
Add: Payments received in advance	—	(174,538)
Less: Advance payments received in prior year	174,538	—
Less: Bank balances and cash disposed	39,194	36,374
Less: Deferred proceeds	37,097	39,082
Cash outflow on acquisition	(1,085,671)	(878,873)

Significant disposal of subsidiaries during the year relates to the sale of Keppel China Marina Holdings Pte Ltd, Keppel Township Development (Shenyang) Co. Ltd, Keppel Bay Property Development (Shenyang) Co. Ltd and Aether Limited.

Significant disposal in the prior year relates to the sale of Keppel Lakefront (Nantong) Property Development Co Ltd, Wiseland Investment (Myanmar) Limited, 80% interest in PT Sentral Tunjungan Perkasa, Keppel DC Singapore 4, 90% interest in Keppel DC Singapore 3, Keppel Verolme and Kepwealth Property Phils., Inc. In addition, the Group lost control of some entities in the prior year but continued to retain significant influence. These entities were deconsolidated from the Group's financial statements for the financial year ended 31 December 2017 and were accounted for as associated companies using the equity method from their respective dates of ceasing control.

C. Cash and Cash Equivalents

Cash and cash equivalents consist of cash on hand and balances with banks. Cash and cash equivalents in the consolidated statement of cash flows comprise the following balance sheet amounts:

	2018 $'000	2017 $'000
Bank balances, deposits and cash	1,981,406	2,273,788
Amounts held under escrow accounts for overseas acquisition of land, payment of construction cost and liabilities	(9,562)	(32,340)
	1,971,844	2,241,448

Reproduced with the kind permission of Keppel Corporation Limited.

Self-Study Illustrations on Complex Consolidation

Question 1

The financial statements of P Co and its subsidiaries and associate are shown below:

Income Statement for year ended 31 December 20x6	P Co	B Co	D Co	C Co	Consolidated
Profit before tax	4,300,000	1,400,000	620,000	480,000	
Tax	(860,000)	(280,000)	(124,000)	(96,000)	
Profit after tax	3,440,000	1,120,000	496,000	384,000	
Income to non-controlling interests					
Dividends declared	(200,000)	(130,000)	(50,000)	(80,000)	
Profit retained	3,240,000	990,000	446,000	304,000	
Retained earnings, 1 Jan 20x6	2,356,000	800,200	800,000	450,000	
Retained earnings, 31 Dec 20x6	5,596,000	1,790,200	1,246,000	754,000	

Statement of Financial Position as at 31 December 20x6	P Co	B Co	D Co	C Co	Consolidated
Goodwill					
Intangible assets, net book value					
Fixed assets, net book value	2,789,000	1,954,000	1,340,000	890,000	
Investment in B Co	2,200,000				
Investment in D Co	1,200,000				
Investment in C Co		900,000			
Inventory	790,000	600,000	450,000	247,000	
Intercompany receivable	1,500,000				
Other net assets	1,970,000	1,265,000	803,450	520,000	
Cash	300,000	256,000	120,000	120,000	
	10,749,000	4,975,000	2,713,450	1,777,000	
Accounts payable	2,453,000	484,800	667,450	403,000	
Bank loans	1,400,000				
Intercompany payable		1,500,000			
Share capital	1,300,000	1,200,000	800,000	520,000	
Retained earnings	5,596,000	1,790,200	1,246,000	754,000	
Revaluation reserves				100,000	
Non-controlling interests					
	10,749,000	4,975,000	2,713,450	1,777,000	

	B Co	D Co	C Co
Date of acquisition by direct investor	1 Aug 20x4	1 Jan 20x5	1 Jan 20x3
Percentage acquired by P Co	90%	60%	
Percentage of retained interests by P Co		40%	
Percentage acquired by B Co			70%

Shareholders' equity at date of acquisition by direct investor			
Share capital	1,200,000	800,000	520,000
Retained earnings	650,000	390,000	311,000
Revaluation reserves			85,000
	1,850,000	1,190,000	916,000
	B Co		**C Co**
Fair value of non-controlling interest at 1 Aug 20x4	226,900		305,500
Fair value of non-controlling interest at 1 Jan 20x3			294,800
Retained earnings of C Co at 1 Aug 20x4 (when B Co was acquired by P Co)			320,000
Revaluation reserves of C Co at 1 Aug 20x4			95,000
Additional revaluation reserves of C Co as at 31 Dec 20x6			5,000

Fair and book values of net assets of each company at date of acquisition by P Co

	B Co →		D Co →		C Co →	
	Book value	**Fair value**	**Book value**	**Fair value**	**Book value**	**Fair value**
Intangible assets	—	150,000				
Inventory					120,000	200,000
Other net assets	1,850,000	1,850,000	1,190,000	1,190,000	815,000	815,000
Total net assets	1,850,000	2,000,000	1,190,000	1,190,000	935,000	1,015,000

Date of acquisition of D as a subsidiary	1 July 20x2
Fair value of consideration transferred	1,800,000
Fair value of NCI of D Co on 1 July 20x2	800,000
P Co's share of post-acquisition change in equity of D Co from 1 July 20x2 to 1 Jan 20x5	280,000
Date of loss of control	1 Jan 20x5
Sales proceeds for 20% interest	700,000
Fair value of retained interests	1,400,000

Additional information:

1. Intangible assets of B Co had an indefinite useful life. On 31 December 20x6, the annual impairment test revealed a loss of $100,000.
2. Ninety percent of the undervalued inventory of C Co was sold to third parties during 20x5 and the remaining was written down to zero value during 20x6.
3. P provided architectural services to B on an extension of a building. The extension was completed on 1 July 20x6 and had an estimated useful life of ten years. All progress billings have been paid and the project was closed on 1 July 20x6. The following transactions were recorded in P's books:

Project fee revenue	
1 Sept 20x5 to 31 Dec 20x5	190,000
1 Jan 20x6 to 30 June 20x6	60,000

Project costs	
1 Sept 20x5 to 31 Dec 20x5	160,000
1 Jan 20x6 to 30 June 20x6	50,000

4. Apply a tax rate of 20% on all appropriate adjustments. Recognize tax effects on fair value adjustments. Companies recognize impairment losses, if any, at the financial year-end.

Required:

1. Prepare the consolidation and equity accounting entries for the year ended 31 December 20x6.
2. Analytically determine the following balances in the consolidated financial statements:
   a. Investment in associate
   b. Non-controlling interests
   c. Retained earnings as at 31 December 20x6.
3. Complete the consolidated totals for the income statement for the year ended 31 December 20x6 and the statement of financial position as at 31 December 20x6.

Question 2

The financial statements of P Co and its subsidiaries and associate are as follows:

Income Statement for year ended 31 December 20x3					
	P Co	X Co	Z Co	Y Co	Consolidated
Profit before tax	6,000,000	2,500,000	1,000,000	500,000	
Tax	(1,200,000)	(500,000)	(180,000)	(100,000)	
Profit after tax	4,800,000	2,000,000	820,000	400,000	
Income to Non-controlling interest					
Dividends declared	(320,000)	(240,000)	(60,000)	(65,000)	
Profit retained	4,480,000	1,760,000	760,000	335,000	
Retained earnings, 1 Jan 20x3	4,500,000	1,250,000	468,000	800,000	
Retained earnings, 31 Dec 20x3	8,980,000	3,010,000	1,228,000	1,135,000	

Statement of Financial Position as at 31 December 20x3					
	P Co	X Co	Z Co	Y Co	Consolidated
Goodwill					
Fixed assets, net book value	3,500,000	1,934,790	1,450,000	620,000	
Investment in X Co	3,200,000				
Investment in Z Co	980,000				
Investment in Y Co		790,000			
Inventory	450,000	620,000	300,000	900,000	
Intercompany receivable		100,000			
Other net assets	2,790,000	1,290,000	1,620,000	800,000	
Cash	360,000	200,000	145,000	90,730	
	11,280,000	4,934,790	3,515,000	2,410,730	
Accounts payable	740,000	724,790	1,487,000	825,730	
Intercompany payable	100,000				
Share capital	1,460,000	1,200,000	800,000	350,000	
Retained earnings	8,980,000	3,010,000	1,228,000	1,135,000	
Revaluation reserves				100,000	
Capital reserve					
Non-controlling interests					
	11,280,000	4,934,790	3,515,000	2,410,730	

	X Co	Z Co	Y Co
Date of acquisition by direct investor	1 Jan 20x1	1 Jan 20x1	1 July 20x0
Percentage acquired by P Co	90%	30%	
Percentage of retained interests by P Co	70%		
Percentage acquired by X Co			60%
Shareholders' equity at date of acquisition by direct investor			
Share capital	1,200,000	800,000	350,000
Retained earnings	700,000	320,000	520,000
Revaluation reserves			90,000
	1,900,000	1,120,000	960,000

	X Co	Y Co
Fair value of non-controlling interests at 1 January 20x1	400,000	444,000
Fair value of non-controlling interests at 1 July 20x0		320,000

Retained earnings of Y Co at 1 January 20x1		
(when X Co was acquired by P Co)		590,000
Revaluation reserves of Y Co at 1 January 20x1		120,000
Revaluation reserves of Y Co decreased on 31 Dec 20x3		(20,000)
Proceeds on sale of 20% of X Co at 31 Dec 20x2	1,100,000	

Fair and book values of net assets of each company at date of acquisition by P

	X Co		Z Co		Y Co	
	Book value	Fair value	Book value	Fair value	Book value	Fair value
Provision for claims		(100,000)				
Intangible asset				400,000		
Other net assets	1,900,000	1,900,000	1,120,000	1,120,000	1,060,000	1,060,000
Total net assets	1,900,000	1,800,000	1,120,000	1,520,000	1,060,000	1,060,000

Additional information:

1. The fair values of identifiable net assets of Y Co were close to their book values on 1 July 20x0 and 1 January 20x1. The estimated useful life of intangible assets of Z Co was five years.
2. The provision for claims of X Co was settled in full during 20x1 for $130,000.
3. On 1 July 20x3, Z provided new electrical equipment to P Co at the invoiced price of $120,000. The original cost was $200,000 and the net book value was $60,000. The estimated useful life of the equipment was three years. Z Co recognized a profit of $$60,000 on the sale. Due to operational changes, the recoverable amount of the equipment as at 31 December 20x3 was estimated as $60,000.
4. On 1 October 20x2, Y sold inventory to P Co at the transfer price of $100,000. The original cost and carrying amount of the inventory was $70,000. Percentage resold to third parties were as follows:

20x2	10%
20x3	80%

5. Apply a tax rate of 20% on all appropriate adjustments. Recognize tax effects on fair value adjustments. Companies recognize impairment losses, if any, at the financial year-end.

Required:

1. Prepare the consolidation and equity accounting entries for the year ended 31 December 20x3.
2. Analytically determine the following balances in the consolidated financial statements:
   a. Investment in associate
   b. Non-controlling interests
   c. Retained earnings as at 31 December 20x3
3. Complete the consolidated totals for the income statement for the year ended 31 December 20x3 and the statement of financial position as at 31 December 20x3.

Question 3

The financial statements of P Co and its subsidiaries and associate are shown below:

Income Statement for year ended 31 December 20x3

	P Co	X Co	Z Co	Y Co	Consolidated
Profit before tax	5,790,000	2,234,900	1,000,000	1,200,000	
Tax	(1,158,000)	(446,980)	(200,000)	(240,000)	
Profit after tax	4,632,000	1,787,920	800,000	960,000	
Income to non-controlling interests					
Dividends declared	(230,000)	(180,000)	(46,000)	(120,000)	
Profit retained	4,402,000	1,607,920	754,000	840,000	
Retained earnings, 1 Jan 20x3	4,321,000	1,387,000	327,000	960,000	
Retained earnings, 31 Dec 20x3	8,723,000	2,994,920	1,081,000	1,800,000	

Statement of Financial Position as at 31 December 20x3

	P Co	X Co	Z Co	Y Co
Goodwill				
Fixed assets, net book value	4,890,000	2,371,000	1,760,000	1,325,000
Investment in X Co	4,500,000			
Investment in Z Co	1,200,000			
Investment in Y Co		850,000		
Inventory	589,000	563,000	256,000	328,000
Intercompany receivable	200,000			
Other net assets	3,200,000	980,000	1,769,000	732,000
Cash	120,000	216,700	27,000	123,760
	14,699,000	4,980,700	3,812,000	2,508,760
Accounts payable	3,176,000	285,780	2,011,000	208,760
Intercompany payable		200,000		
Share capital	2,800,000	1,500,000	720,000	300,000
Retained earnings	8,723,000	2,994,920	1,081,000	1,800,000
Revaluation reserves				200,000
Capital reserve				
Non-controlling interests				
	14,699,000	4,980,700	3,812,000	2,508,760

page 530

	X Co	Z Co	Y Co
Date of acquisition by direct investor	1 Jan 20x1	1 Jan 20x1	1 July 20x0
Percentage acquired by P Co	70%	30%	
Percentage of revised interests by P Co	90%		
Percentage acquired by X Co			60%
Shareholders' equity at date of acquisition by direct investor			
Share capital	1,500,000	720,000	300,000
Retained earnings	600,000	240,000	400,000
Revaluation reserves			90,000

	2,100,000	960,000	790,000
	X Co		**Y Co**
Fair value of non-controlling interests at 1 January 20x1	1,311,429		400,000
Fair value of non-controlling interests at 1 July 20x0			280,000
Retained earnings of Y Co at 1 January 20x1 (when X Co was acquired by P Co)			430,000
Revaluation reserves of Y Co at 1 January 20x1			120,000
Revaluation reserves of Y Co arising on 31 Dec 20x3			80,000

Consideration paid for additional interests of 20% in X Co at 31 Dec 20x2 is $1,300,000

Fair and book values of net assets of each company at date of acquisition by P

	X Co		Z Co		Y Co	
	Book value	**Fair value**	**Book value**	**Fair value**	**Book value**	**Fair value**
Inventory	20,000	35,000				
Intangible asset				50,000		100,000
Other net assets	2,080,000	2,080,000	960,000	960,000	850,000	850,000
Total net assets	2,100,000	2,115,000	960,000	1,010,000	850,000	950,000

Additional information:

1. The undervalued inventory of X Co was sold to third parties during 20x2.
2. The unrecognized intangible asset of Z Co and Y Co had an estimated useful life of five years from acquisition date by P Co.
3. On 1 October 20x2, Y sold to X inventory at transfer price of $50,000. The original cost was $30,000. Ten percent was resold to third parties during 20x2 and an additional 80% resold during 20x3.
4. On 1 July 20x3, Z transferred to P fixed assets at transfer price of $130,000. The original cost was $180,000 and the accumulated depreciation was $126,000. The remaining useful life as at the date of transfer was three years.
5. Apply a tax rate of 20% on all appropriate adjustments. Recognize tax effects on fair value adjustments. Companies recognize impairment losses, if any, at the financial year-end.

Required:

1. Prepare the consolidation and equity accounting entries for the year ended 31 December 20x3.
2. Analytically determine the following balances in the consolidated financial statements as at 31 December 20x3:
   a. Investment in associate
   b. Non-controlling interests
   c. Retained earnings
3. Complete the consolidated totals for the income statement for the year ended 31 December 20x3 and the statement of financial position as at 31 December 20x3.

Question 4

The financial statements of P Co and its subsidiaries are shown below (adapted from Question 1):

Income Statement for year ended 31 December 20x6					
	P Co	B Co	D Co	C Co	Consolidated
Profit before tax	4,300,000	1,400,000	620,000	480,000	
Tax	(860,000)	(280,000)	(124,000)	(96,000)	
Profit after tax	3,440,000	1,120,000	496,000	384,000	
Income to non-controlling interests					
Dividends declared	(200,000)	(130,000)	(50,000)	(80,000)	
Profit retained	3,240,000	990,000	446,000	304,000	
Retained earnings, 1 Jan 20x6	2,356,000	800,200	800,000	450,000	
Retained earnings, 31 Dec 20x6	5,596,000	1,790,200	1,246,000	754,000	

Statement of Financial Position as at 31 December 20x6					
	P Co	B Co	D Co	C Co	Consolidated
Goodwill					
Intangible assets, net book value					
Fixed assets, net book value	2,789,000	1,954,000	1,340,000	890,000	
Investment in B Co	2,200,000				
Investment in D Co	720,000				
Investment in C Co		900,000			
Inventory	790,000	600,000	450,000	247,000	
Intercompany receivable	1,500,000				
Other net assets	1,970,000	1,265,000	803,450	520,000	
Cash	300,000	256,000	120,000	120,000	
	10,269,000	4,975,000	2,713,450	1,777,000	
Accounts payable	1,973,000	484,800	667,450	403,000	
Bank loans	1,400,000				
Intercompany payable		1,500,000			
Share capital	1,300,000	1,200,000	800,000	520,000	
Retained earnings	5,596,000	1,790,200	1,246,000	754,000	
Revaluation reserves				100,000	
Non-controlling interests					
	10,269,000	4,975,000	2,713,450	1,777,000	

	B Co	D Co	C Co
Date of acquisition by direct investor	1 Aug 20x4	1 Jan 20x5	1 July 20x3
Percentage acquired by P Co	90%	60%	(Note A)
Percentage acquired by B Co			70%
Shareholders' equity at date of acquisition by direct investor			
Share capital	1,200,000	800,000	520,000
Retained earnings	650,000	390,000	311,000
Revaluation reserves			85,000
	1,850,000	1,190,000	916,000

	B Co	C Co

Fair value of non-controlling interests at 1 Aug 20x4	226,900	305,500
Fair value of non-controlling interests at 1 July 20x3		294,800
Retained earnings of C Co at 1 Aug 20x4 (when B Co was acquired by P Co)		320,000
Revaluation reserves of C Co at 1 Aug 20x4		95,000
Additional revaluation reserves of C Co as at 31 Dec 20x6		5,000

Fair and book values of net assets of each company at date of acquisition by P Co

	B Co		D Co		C Co	
	Book value	Fair value	Book value	Fair value	Book value	Fair value
Intangible assets	—	150,000				
Inventory					120,000	200,000
Other net assets	1,850,000	1,850,000	1,190,000	1,190,000	815,000	815,000
Total net assets	1,850,000	2,000,000	1,190,000	1,190,000	935,000	1,015,000

Note A:

Date of acquisition of 40% interests in D (significant influence)	1 July 20x2
Date of additional investment of 20% in D (gain of control)	1 January 20x5
Fair value of consideration transferred on 1 Jan 20x5	400,000
Fair value of non-controlling interests on 1 Jan 20x5	700,000
P Co's share of post-acquisition change in equity of D Co from 1 July 20x2 to 1 Jan 20x5	280,000
Fair value of previously-held interests on 1 Jan 20x5	700,000
Initial investment of previously-held interests	320,000

Additional information:

1. Intangible assets of B Co had an infinite useful life. On 1 January 20x6, new licensing requirements resulted in a restricted useful life of 10 years from this date.
2. Ninety percent of the undervalued inventory of C Co was sold to third parties during 20x5; ten percent was sold during 20x6.
3. P provided financing to B on the self-construction of a building. The building was completed on 1 July 20x6 and had an estimated useful life of 10 years. The following transactions were recorded in P's books:

Interest income	
1 Sept 20x5 to 31 Dec 20x5	120,000
1 Jan 20x6 to 30 June 20x6	100,000
Interest expense	
1 Sept 20x5 to 31 Dec 20x5	80,000
1 Jan 20x6 to 30 June 20x6	60,000

4. Apply a tax rate of 20% on all appropriate adjustments. Recognize tax effects on fair value adjustments. Companies recognize impairment losses, if any, at the financial year-end.

Required:

1. Prepare the consolidation and equity accounting entries for the year ended 31 December 20x6.
2. Analytically determine the following balances in the consolidated financial statements:
   a. Non-controlling interests
   b. Retained earnings as at 31 December 20x6.
3. Complete the consolidated totals for the income statement for the year ended 31 December 20x6 and the balance sheet as at 31 December 20x6.

SOLUTIONS

The following abbreviations are used throughout the solutions:

AC: Analytical check

Acq: Acquisition

DTA: Deferred tax assets

DTL: Deferred tax liabilities

FV: Fair value

INA: Identifiable Net Assets

NPAT: Net profit after tax

NPBT: Net profit before tax

NCI: Non-controlling interests

RE: Retained earnings

ORE: Opening retained earnings

Solution to Question 1

Income Statement for year ended 31 December 20x6						
	P Co	B Co	D Co	C Co	Consolidated	
Profit before tax	4,300,000	1,400,000	620,000	480,000	6,069,400	Note 1
Tax	(860,000)	(280,000)	(124,000)	(96,000)	(1,212,800)	Note 2
Profit after tax	3,440,000	1,120,000	496,000	384,000	4,856,600	
Income to NCI					(238,112)	Note 3
Dividends declared	(200,000)	(130,000)	(50,000)	(80,000)	(200,000)	
Profit retained	3,240,000	990,000	446,000	304,000	4,418,488	
Retained earnings, 1 Jan20x6	2,356,000	800,200	800,000	450,000	2,876,792	Note 4
Retained earnings, 31 Dec20x6	5,596,000	1,790,200	1,246,000	754,000	7,295,280	AC

Statement of Financial Position as at 31 December 20x6						
	P Co	B Co	D Co	C Co	Consolidated	
Goodwill					663,400	CJE1
Intangible assets, net book value					50,000	
Fixed assets, net book value	2,789,000	1,954,000	1,340,000	890,000	5,595,000	Note 5
Investment in B Co	2,200,000					
Investment in D Co	1,200,000				1,742,400	AC
Investment in C Co		900,000				
Inventory	790,000	600,000	450,000	247,000	1,637,000	
Intercompany receivable	1,500,000					
Other net assets	1,970,000	1,265,000	803,450	520,000	3,752,600	Note 6
Cash	300,000	256,000	120,000	120,000	676,000	
	10,749,000	4,975,000	2,713,450	1,777,000	14,116,400	
Accounts payable	2,453,000	484,800	667,450	403,000	3,340,800	
Bank loans	1,400,000				1,400,000	
Intercompany payable		1,500,000				
Share capital	1,300,000	1,200,000	800,000	520,000	1,300,000	
Retained earnings	5,596,000	1,790,200	1,246,000	754,000	7,295,280	
Revaluation reserves				100,000	3,150	
Non-controlling interests					777,170	AC
	10,749,000	4,975,000	2,713,450	1,777,000	14,116,400	

Note 1:			
Combined P and subsidiaries	6,180,000	4,300,000 + 1,400,000 + 480,000
Impairment loss of intangible asset	(100,000)	
Current cost of sales of undervalued inventory of C	(8,000)	10%*80,000
Adjustment for unrealized profit (P to B)	(8,000)	10,000 – (30,000 + 10,000)/10*6/12
Less dividend income from B Co	(117,000)	90%*130,000
from C Co	(56,000)	70%*80,000
from D Co	(20,000)	40%*50,000
Add share of profit of associate	198,400	
Consolidated profit before tax	6,069,400	

Note 2:	Combined P and subsidiaries	(1,236,000)	860,000 + 280,000 + 96,000
	Tax on:		
	Impairment loss of intangible asset	20,000	
	Current cost of sales of undervalued inventory of C	1,600	
	Adjustment for unrealized profit (P to B)	1,600	
		(1,212,800)	

Note 3:	Income to NCI of B	98,400
	Income to NCI of C	139,712
		238,112

Note 4:	P's RE at 1 Jan 20x6	2,356,000	
	P's share of B's post-acquisition RE	135,180	90%*(800,200−650,000)
	P's share of C's post-acquisition RE	81,900	63%*(450,000−320,000)
	P's share of D's post-acquisition RE	164,000	40%*(800,000−390,000)
	P's share of past cost of sales of undervalued inventory of C	(36,288)	63%*80%*90%*80,000
	P's share of unrealized construction, 1 Jan (P to B)	(24,000)	80%*(190,000−160,000)
	Remeasurement gain on loss of control of D Co	200,000	
		2,876,792	

Note 5: 2,789,000 + 1,954,000 + 8,90,000 – (40,000*9.5/10)

Note 6: 1,970,000 + 1,265,000 + 520,000 + (20%*40,000*9.5/10) – 20%*50,000

CJE1: Elimination of investment in B Co and C Co

Dr Share capital (B Co)	1,200,000	
Dr Share capital (C Co)	520,000	
Dr Retained earnings (B Co)	650,000	
Dr Retained earnings (C Co)	320,000	
Dr Revaluation reserve (C Co)	95,000	
Dr Inventory	80,000	
Dr Intangible assets	150,000	
Dr Goodwill	663,400	
Cr Deferred tax liability		46,000
Cr Investment in B Co		2,200,000
Cr Investment in C Co		900,000
Cr Non-controlling interests in B Co		226,900
Cr Non-controlling interests in C Co		305,500
	3,678,400	3,678,400

Goodwill in sub-group				
Fair value of INA of sub-group at 1 Aug 20x4				
Share capital (B Co)		1,200,000		
Retained earnings (B Co)		650,000		
Book value of B Co			1,850,000	
Less B Co's investment in C Co			(900,000)	
Add intangible assets			150,000	
less deferred tax liability			(30,000)	
Fair value of INA of B Co			1,070,000	
Share capital (C Co)		520,000		
Retained earnings (C Co)		320,000		
Revaluation reserve (C Co)		95,000		
Add undervalued inventory		80,000		
Deferred tax liability		(16,000)		
Fair value of INA of C Co			999,000	
B Co's consolidated equity or net assets			2,069,000	
Fair value of consideration transferred		2,200,000		
Fair value of non-controlling interests (B Co)		226,900		
Fair value of non-controlling interests (C Co)		305,500		
Less fair value of INA of sub-group		(2,069,000)	(see workings above)	
Goodwill			663,400	
P Co's goodwill			90%	
Fair value of consideration transferred			2,200,000	
Less share of fair value of INA of sub-group				
Share of B Co's FV of INA	963,000	90%*1,070,000		
Share of C Co's FV of INA	629,370	90%*70%*999,000	(1,592,370)	
Goodwill attributable to P Co			607,630	
		Total		
B Co's NCI			226,900	
Less fair value of INA of sub-group				
Share of B Co's FV of INA	107,000	10%*1,070,000		
Share of C Co's FV of INA	69,930	10%*70%*999,000		
B Co's consolidated equity as at 1 January 20x4			176,930	
B Co's NCI's goodwill			49,970	30%*999,000
C Co's NCI			305,500	
Less Share of C Co's FV of INA			299,700	
C Co's NCI's goodwill			5,800	
Total Goodwill			663,400	

CJE2: Loss of control of D Co

Dr Investment	200,000	
Cr ORE		200,000

Workings:

	Group	Legal

Sales proceeds on sale of 20%	700,000	700,000	
Fair value of retained interests (40%)	1,400,000		
Fair value of consideration transferred on 1 July 20x2	(1,800,000)	(600,000)	1,800,000/60%*20%
P Co's share of post-acquisition change in equity of D Co (1 Jul 20x2 to 1 Jan 20x5)	(280,000)		
Profit recognized on 1 Jan 20x5	20,000	100,000	

	Group profit on sale	Group remeasurement gain
Sales proceeds on sale of 20%	700,000	
Fair value of retained interests (40%)		1,400,000
Fair value of consideration transferred on 1 July 20x2	(600,000)	(1,200,000)
P Co's share of post-acquisition change in equity of D Co (1 Jul 20x2 to 1 Jan 20x5)	(93,333)	(186,667)
P&L effect	6,667	13,333

During 20x5, the following entry was passed

Dr Investment	200,000	1,400,000−1,200,000	
Dr Profit on sale	93,333	100,000−6,667	
Cr Remeasurement gain			13,333
Cr ORE			280,000
	293,333		293,333

CJE3: Impairment of intangible asset

Dr Impairment expense	100,000	
Cr Accumulated impairment		100,000

CJE4: Tax effects of CJE3

Dr Deferred tax liability	20,000	
Cr Tax expense		20,000

CJE5: Allocate share of post-acq RE of B Co to NCI of B Co

Dr Opening RE	15,020		
Cr NCI (BS)		15,020	
RE at 1 Jan 20x6		800,200	
RE at date of acquisition		650,000	
Change in RE		150,200	
NCI's share		15,020	10%*150,200

CJE6: Eliminate dividends declared by B Co

Dr Dividend income	117,000		
Dr Non-controlling interests	13,000		10%*130,000
Cr Dividend declared			130,000

CJE7: Allocate share of current income of B Co to NCI of B Co

Dr Income to NCI	98,400		
Cr NCI (BS)		98,400	
NPAT of B Co		1,120,000	
Impairment expense, after tax		(80,000)	
Less dividend received from C Co		(56,000)	
Adjusted NPAT		984,000	

CJE8: Adjustment of construction profit (transfer from P to B)

Dr ORE	30,000		
Dr Construction revenue	60,000		
Cr Construction expense		50,000	
Cr Fixed asset		40,000	

CJE9: Tax effects of CJE8

Dr DTA	8,000		
Cr Tax expense		2,000	
Cr ORE		6,000	

CJE10: Adjustment of excess depreciation on fixed assets

Dr Accumulated depreciation	2,000		6/12*40,000*1/10
Cr Depreciation expense		2,000	

CJE11: Tax effects of CJE10

Dr Tax expense	400		
Cr DTA		400	

CJE12: Adjustment of cost of sales of undervalued inventory

Dr ORE	45,360		
Dr NCI	26,640		37%*90%*80,000
Cr Inventory		72,000	

CJE13: Tax effects of CJE12

Dr DTL	14,400		
Cr ORE		9,072	
Cr NCI		5,328	

CJE14: Cost of sales of undervalued inventory

Dr Cost of sales (PL)	8,000		
Cr Inventory		8,000	

CJE15: Tax effects of CJE14

Dr DTL	1,600	
Cr Tax expense		1,600

CJE16: Allocation of post-acquisition RE to total NCI of C Co

Dr Opening RE	48,100			
Cr Non-controlling interests		48,100		
Retained earnings at 1 Jan 20x6				450,000
Retained earnings at 1 Aug 20x4				320,000
Change in retained earnings				130,000
Direct NCI			30%	39,000
Indirect NCI			7.00%	9,100
Total NCI			37%	48,100

CJE17: Allocation of current profit after tax to total NCI of C Co

Dr Income to NCI	139,712				
Cr Non-controlling Interests		139,712			
Net profit after tax of C Co			384,000		
Less cost of sale of undervalued inventory, after tax			(6,400)	10%*80%*80,000	
Adjusted NPAT			377,600		
Direct NCI			30%	113,280	
Indirect NCI			7%	26,432	
Total NCI			37%	139,712	

CJE18: Elimination of dividends declared by C Co

Dr Dividend income	56,000		70%*80,000
Dr Non-controlling interests	24,000		30%*80,000
Cr Dividend declared		80,000	

CJE19: Allocation of post-acq revaluation reserves to total NCI of C Co

Dr Share of Revaluation reserves to NCI	1,850			
Cr Non-controlling interests		1,850		
Revaluation reserves at 31 Dec 20x6				100,000
Revaluation reserves at 1 Jan 20x6				95,000
Change in revaluation reserves				5,000
Direct NCI			30%	1,500
Indirect NCI			7%	350
Total NCI			37%	1,850

CJE20: Eliminate intercompany payable and receivable

Dr Intercompany payable	1,500,000	
Cr Intercompany receivable		1,500,000

Analytical Check of Non-controlling Interests

Total Non-controlling interests as at 31 December 20x6

	Total NCI	NCI of B Co and indirect NCI of C Co	Direct NCI of C Co
CJE1: Elimination of investment in B Co and C Co	532,400	226,900	305,500
CJE5: Allocate share of post-acq RE of B Co to NCI of B Co	15,020	15,020	
CJE6: Eliminate dividends declared by B Co	(13,000)	(13,000)	
CJE7: Allocate share of current income of B Co to NCI of B Co	98,400	98,400	
CJE12: Adjustment of cost of sales of undervalued inventory	(26,640)	(5,040)	(21,600)
CJE13: Tax effects of CJE12	5,328	1,008	4,320
CJE16: Allocation of post-acquisition RE to total NCI of C Co	48,100	9,100	39,000
CJE17: Allocation of current profit after tax to total NCI of C Co	139,712	26,432	113,280
CJE18: Elimination of dividends declared by C Co	(24,000)		(24,000)
CJE19: Allocation of post-acq revaluation reserves to total NCI of C Co	1,850	350	1,500
	777,170	359,170	418,000

Analytical check on non-controlling interests of B Co.		
B Co's shareholders' equity as at 31 December 20x6	2,990,200	1,200,000+1,790,200
Less investment in C Co	(900,000)	
	2,090,200	
NCI's share of book value of equity of B as at 31 Dec 20x6	209,020	10%*2,090,200
NCI's share of intangible asset, after tax as at 31 Dec 20x6	4,000	10%*80%*50,000
NCI of B Co's goodwill as at 31 Dec 20x6	49,970	
NCI of B Co and indirect NCI of C Co	262,990	

Analytical check of NCI of C Co			**Indirect NCI**	**Direct NCI**
C Co's shareholders' equity as at 31 December 20x6	1,374,000	520,000+754,000+100,000		
C Co's total NCI share of equity as at 31 December 20x6	508,380	37%*1,374,000	96,180	412,200
Goodwill attributable to C Co's NCI	5,800			5,800
Total NCI of C Co as at 31 December 20x6	514,180		96,180	418,000
Total NCI	**777,170**		359,170	418,000

EA1: Recognize share of post-acq RE of D Co

Dr Investment in D Co 164,000

Cr ORE 164,000

RE of D as at 1 Jan 20x6 800,000

RE of D as at date of acquisition	390,000
Change in RE	410,000
Share of D's change in RE	164,000

EA2: Reclassify dividend income as a reduction of investment

Dr Dividend income	20,000	
Cr Investment in D Co		20,000

EA3: Recognize share of current profit after tax of D

Dr Investment in D Co	198,400	
Cr Share of profit of D		198,400

Workings:

NPAT	496,000
Share of NPAT of D	198,400

Analytical check of Investment in D Co:

Book value of shareholders' equity of D		2,046,000	800,000 + 1,246,000
P's share of D's identifiable net assets		818,400	
Implicit goodwill in Investment in D Co:			
Investment in D Co		1,400,000	
BV of net assets of D at acq	1,190,000		
FV-BV	0		
FV of net assets of D at acq	1,190,000		
Less Share of FV of net assets of D at acq		476,000	
Goodwill implicit in the Investment in D Co		924,000	
		1,742,400	

Investment in D Co, at cost	1,200,000
CJE2: Loss of control of D Co	200,000
EA1: Recognize share of post-acq RE of D Co	164,000
EA2: Reclassify dividend income as a reduction of investment	(20,000)
EA3: Recognize share of current profit after tax of D	198,400
Investment in D Co as at 31 Dec 20x6	**1,742,400**

Analytical check of closing retained earnings

P's RE	5,596,000
B's RE	1,790,200
C's RE	754,000
CJE1: Elimination of investment in B Co and C Co	(970,000)

CJE2: Loss of control of D Co	200,000
CJE3: Impairment of intangible asset	(100,000)
CJE4: Tax effects of CJE3	20,000
CJE5: Allocate share of post-acq RE of B Co to NCI of B Co	(15,020)
CJE6: Eliminate dividends declared by B Co	13,000
CJE7: Allocate share of current income of B Co to NCI of B Co	(98,400)
CJE8: Adjustment of construction profit (transfer from P to B)	(40,000)
CJE9: Tax effects of CJE8	8,000
CJE10: Adjustment of excess depreciation on fixed assets	2,000
CJE11: Tax effects of CJE10	(400)
CJE12: Adjustment of cost of sales of undervalued inventory	(45,360)
CJE13: Tax effects of CJE12	9,072
CJE14: Cost of sales of undervalued inventory	(8,000)
CJE15: Tax effects of CJE14	1,600
CJE16: Allocation of post-acquisition RE to total NCI of C Co	(48,100)
CJE17: Allocation of current profit after tax to total NCI of C Co	(139,712)
CJE18: Elimination of dividends declared by C Co	24,000
EA1: Recognize share of post-acq RE of D Co	164,000
EA2: Reclassify dividend income as a reduction of investment	(20,000)
EA3: Recognize share of current profit after tax of D	198,400
Consolidated RE at 31 Dec 20x6	**7,295,280**

Analytical check

P's RE	5,596,000	
P's share of B Co's post-acquisition RE	1,026,180	90%*(1,790,200 − 650,000)
P's share of C Co's post-acquisition RE	273,420	63%*(754,000 − 320,000)
P's share of D Co's post-acquisition RE	342,400	40%*(1,246,000 − 390,000)
P's share of impairment loss of intangible asset of B Co	(72,000)	90%*80%*100,000
P's share of cost of sales of undervalued inventory of C Co	(40,320)	63%*80%*80,000
P's share of unrealized construction profit (P to B Co)	(30,400)	80%*(190,000 − 160,000 +60,000 − 50,000)*9.5/10
Remeasurement gain	200,000	
Consolidated RE at 31 Dec 20x6	**7,295,280**	

Solution to Question 2

Income Statement for year ended 31 December 20x3						
	P Co	X Co	Z Co	Y Co	Consolidated	
Profit before tax	6,000,000	2,500,000	1,000,000	500,000	9,023,400	Note 1
Tax	(1,200,000)	(500,000)	(180,000)	(100,000)	(1,804,800)	Note 2
Profit after tax	4,800,000	2,000,000	820,000	400,000	7,218,600	
Income to NCI					(831,436)	Note 3
Dividends declared	(320,000)	(240,000)	(60,000)	(65,000)	(320,000)	
Profit retained	4,480,000	1,760,000	760,000	335,000	6,067,164	
Retained earnings, 1 Jan 20x3	4,500,000	1,250,000	468,000	800,000	4,989,022	Note 4
Retained earnings, 31 Dec 20x3	8,980,000	3,010,000	1,228,000	1,135,000	11,056,186	AC

Statement of Financial Position as at 31 December 20x3						
	P Co	X Co	Z Co	Y Co	Consolidated	
Goodwill					2,868,286	CJE1
Fixed assets, net book value	3,500,000	1,934,790	1,450,000	620,000	6,054,790	
Investment in X Co	3,200,000					
Investment in Z Co	980,000				1,192,400	AC
Investment in Y Co		790,000				
Inventory	450,000	620,000	300,000	900,000	1,967,000	Note 5
Intercompany receivable		100,000				
Other net assets	2,790,000	1,290,000	1,620,000	800,000	4,880,600	Note 6
Cash	360,000	200,000	145,000	90,730	650,730	
	11,280,000	4,934,790	3,515,000	2,410,730	17,613,806	
Accounts payable	740,000	724,790	1,487,000	825,730	2,290,520	
Intercompany payable	100,000					
Share capital	1,460,000	1,200,000	800,000	350,000	1,460,000	
Retained earnings	8,980,000	3,010,000	1,228,000	1,135,000	11,056,186	
Revaluation reserves				100,000	(8,400)	
Capital reserve					37,106	
Non-controlling interests					2,778,394	AC
	11,280,000	4,934,790	3,515,000	2,410,730	17,613,806	

Note 1:	Combined P and subsidiaries	9,000,000	
	Add realized profit from transfer	24,000	
	Less dividend income from X Co	(168,000)	70%*240,000
	from Y Co	(39,000)	60%*65,000
	from Z Co	(18,000)	30%*60,000
	Add share of profit of associate	224,400	
	Consolidated profit before tax	9,023,400	

Note 2:	Combined P and subsidiaries	(1,800,000)
	Add tax on realized profit from transfer	(4,800)
		(1,804,800)

Note 3:	Income to NCI of X	588,300
	Income to NCI of Y	243,136
		831,436

Note 4: Opening RE need not be reconstructed. It can be derived as a balancing figure if the consolidated closing RE is available.
For completeness, the analytical check for opening RE is shown here.
However, for explanations of the plug below, refer to the analytical check in the consolidated closing RE.

P's RE at 1 Jan 20x3	4,500,000	
P's share of X's post-acquisition RE	385,000	70%*(1,250,000 – 700,000)
P's share of Y's post-acquisition RE	88,200	70%*60%*(800,000 – 590,000)
P's share of Z's post-acquisition RE	44,400	30%*(468,000 – 320,000)
P"s share of past expensing of claims	56,000	70%*80%*100,000
P's share of past amortization of intangible assets	(38,400)	30%*80%*400,000*2/5
P's share of unrealized profit in inventory, 1 Jan	(9,072)	70%*60%*80%*27,000
Reversal of profit on sale in P's RE	(185,714)	} (37,106)
Share of RE transferred to NCI (included in capital reserve)	148,608	
	4,989,022	

Note 5: Reduced by unrealized profit in ending inventory 3,000

Note 6: Increased by deferred tax asset on unrealized profit 600

CJE1: Elimination of investment in X Co and Y Co

Dr Share capital (X Co)	1,200,000		
Dr Share capital (Y Co)	350,000		
Dr Retained earnings (X Co)	700,000		
Dr Retained earnings (Y Co)	590,000		
Dr Revaluation reserve (Y Co)	120,000		
Dr Deferred tax asset	20,000		
Dr Goodwill	2,868,286		
Cr Provision for claim		100,000	
Cr Investment in X Co		4,114,286	(3,200,000/70%*90%)
Cr Investment in Y Co		790,000	
Cr Non-controlling interests in X Co		400,000	
Cr Non-controlling interests in Y Co		444,000	
	5,848,286	5,848,286	

Goodwill in sub-group

Fair value of INA of sub-group at 1 Jan 20x1		
Share capital (X Co)	1,200,000	
Retained earnings (X Co)	700,000	
Book value of X Co		1,900,000
Less X Co's investment in Y Co		(790,000)
Undervalued provision for claims		(100,000)
Deferred tax asset		20,000
Fair value of INA of X Co		1,030,000
Share capital (Y Co)	350,000	
Retained earnings (Y Co)	590,000	
Revaluation reserve (Y Co)	120,000	
Fair value of INA of Y Co		1,060,000
X Co's consolidated equity or net assets		2,090,000
Fair value of consideration transferred		4,114,286
Fair value of non-controlling interests (X Co)		400,000
Fair value of non-controlling interests (Y Co)		444,000
Less fair value of INA of sub-group (see workings above)		(2,090,000)
Goodwill		2,868,286

	1 Jan 20x1	Divested	Remaining
P Co's share of goodwill	90%	20%	70%
Fair value of consideration transferred	4,114,286	914,286	3,200,000
Less share of fair value of INA of sub-group			
Share of X Co's FV of INA	(927,000)	(206,000)	
Share of Y Co's FV of INA	(572,400)	(127,200)	
Goodwill	2,614,886	581,086	

X Co's NCI as at 1 January 20x1 (before divestment)	Total	10%
X Co's shareholders' equity as at 1 January 20x1	1,900,000	
Less investment in X Co	(790,000)	
Share of undervalued provision for claim, after-tax	(80,000)	
X Co's share of Y Co's shareholders'equity as at 1 January 20x1	636,000	
X Co's consolidated equity as at 1 January 20x1	1,666,000	166,600
X Co's NCI's goodwill (residual)		233,400
X Co's NCI		400,000

		Total	40%
Y Co's shareholder's equity as at 1 January 20x1		1,060,000	424,000
Share capital	350,000		
Retained earnings	590,000		
Revaluation reserves	120,000		
	1,060,000		
Y Co's undervaluation of identifiable net assets			0
Goodwill (residual)			20,000
NCI of Y Co			444,000
Total Goodwill			2,868,286

CJE2: Effects of the decrease in investment in X Co and Y Co

Dr Investment	914,286		(a)
Dr ORE	185,714		(b)
Cr Capital reserve (Equity)		37,106	(c)
Cr Non-controlling interests		1,062,894	(d)
	1,100,000	1,100,000	

Work out P's share of equity that is transferred to NCI through divestment of 20%

P's share of equity of the sub-group as at 31 Dec 20x2

P's share of post-acquisition RE of X Co	495,000	90%*(1,250,000 − 700,000)	
P's share of post-acquisition RE of Y Co	113,400	90%*60%*(800,000 − 590,000)	
P's share of past expensing of claims of X Co	72,000	90%*80%*100,000	
P's share of unrealized profit of Y Co	(11,664)	90%*60%*80%*90%*30,000	
P's share of sub-group's post-acquisition equity	668,736		
20% of initial investment in X Co	914,286	20/70*3,200,000	(a)
20% of P's share of post-acq change in equity	148,608	20/90*668,736	
20% of equity in X Co transferred to NCI by P Co	1,062,894	(d)	

Workings:

	Parent's PL	**Group equity**
Sales proceeds	1,100,000	1,100,000
Investment in S Co	(914,286)	(1,062,894)
Profit	185,714	37,106
	(b)	(c)

This check of NCI at 31 Dec 20x2 is optional - it is only done to show that the re-allocation results in a correct NCI balance

Analytical check of NCI on 31 Dec 20x2 (after re-allocation of 20% to NCI)

CJE1: Elimination of investment in X Co and Y Co	844,000
CJE2: Effects of the decrease in investment in X Co and Y Co	1,062,894
CJE3: Settlement of undervalued provision for claims	10,000
CJE4: Tax effects of CJE3	(2,000)
CJE5: Allocate share of post-acq RE of X Co to NCI of X Co	55,000
CJE8: Adjustment for upstream sale of inventory from Y Co to P Co	(12,420)
CJE9: Tax effects on CJE8	2,484
CJE10: Allocation of post-acquisition RE to total NCI of Y Co	96,600
Per Listings approach	**2,056,558**

Analytical approach

Book value of net assets of X Co on 31 Dec 20x2	2,450,000	1,200,000 + 1,250,000
Less investment in Y Co	(790,000)	
	1,660,000	
NCI's share of X Co (30%)	498,000	30%*1,660,000
Book value of net assets of Y Co on 31 Dec 20x2	1,270,000	350,000 + 800,000 + 120,000
Less unrealized profit on 31 Dec 20x2	(21,600)	30,000*80%*90%
	1,248,400	
NCI's share of Y Co (40%+(30%*60%))	724,072	58%*1,248,400
NCI's goodwill at acquisition	253,400	233,400 + 20,000
P Co's goodwill transferred to NCI of X Co	581,086	2,614,886*20/90
	834,486	
NCI balance as at 31 Dec 20x2	**2,056,558**	498,000 + 724,072 + 834,486

CJE3: Settlement of undervalued provision for claims

Dr Provision for claims	100,000	
Cr Opening RE		90,000
Cr Non-controlling interests		10,000

CJE4: Tax effects of CJE3

Dr Opening RE	18,000	
Dr Non-controlling interests	2,000	
Cr Deferred tax asset		20,000

CJE5: Allocate share of post-acq RE of X Co to NCI of X Co

Dr Opening RE	55,000	

Cr NCI (BS)			55,000
RE at 1 Jan 20x3			1,250,000
RE at date of acquisition			700,000
Change in RE			550,000
NCI's share (before divestment by P)			55,000

CJE6: Eliminate dividends declared by X Co

Dr Dividend income	168,000		
Dr Non-controlling interests	72,000	30%*240,000	
Cr Dividend declared			240,000

CJE7: Allocate share of current income of X Co to NCI of X Co

Dr Income to NCI	588,300	30%*1,961,000	
Cr NCI (BS)			588,300
NPAT of X Co			2,000,000
Less dividend received from Y Co			(39,000)
Adjusted NPAT			1,961,000

CJE8: Adjustment for upstream sale of inventory from Y Co to P Co

Dr Opening RE	14,580	90%*30,000*90%*60%		
Dr Non-controlling interests	12,420	90%*30,000*46%		
Cr Cost of sales (P's PL)			24,000	80%*30,000
Cr Inventory			3,000	10%*30,000

Direct NCI	40%	10,800
Indirect NCI (before divestment by P)	6%	1,620
Total NCI	46%	12,420

CJE9: Tax effects on CJE8

Dr Tax expense	4,800	
Dr Deferred tax asset	600	
Cr Opening RE		2,916
Cr Non-controlling interests		2,484

Direct NCI	40%	2,160
Indirect NCI (before divestment by P)	6%	324
Total NCI	46%	2,484

CJE10: Allocation of post-acquisition RE to total NCI of Y Co				
Dr Opening RE	96,600			
Cr Non-controlling interests		96,600		
Retained earnings at 1 Jan 20x3				800,000
Retained earnings at 1 Jan 20x1				590,000
Change in retained earnings				210,000
Direct NCI			40%	84,000
Indirect NCI (before divestment by P)			6%	12,600
Total NCI			46%	96,600

CJE11: Allocation of current profit after tax to total NCI of Y Co				
Dr Income to NCI	243,136			
Cr Non-controlling Interests		243,136		
Net profit after tax of Y Co				400,000
Add realized profit from upstream sale of inventory, after tax				19,200
Adjusted NPAT				419,200
Direct NCI			40%	167,680
Indirect NCI (after divestment by P)			18%	75,456
Total NCI			58%	243,136

CJE12: Elimination of dividends declared by Y Co

Dr Dividend income	39,000		60%*65,000
Dr Non-controlling interests	26,000		40%*65,000
Cr Dividend declared		65,000	

CJE13: Allocation of post-acq revaluation reserves to total NCI of Y Co				
Dr NCI	11,600			
Cr Share of revaluation reserves		11,600		
Revaluation reserves at 31 Dec 20x3				100,000
Revaluation reserves at 1 Jan 20x3				120,000
Change in revaluation reserves				20,000
Direct NCI			40%	8,000
Indirect NCI (after divestment by P)			18%	3,600
Total NCI			58%	11,600

CJE14: Eliminate intercompany payable and receivable

Dr Intercompany payable	100,000	
Cr Intercompany receivable		100,000

Analytical check of Non-controlling interests

Total Non-controlling interests as at 31 December 20x3

	Total NCI	NCI of X Co and indirect NCI of Y Co	Direct NCI of Y Co
CJE1: Elimination of investment in X Co and Y Co	844,000	400,000	444,000

CJE2: Effects of the decrease in investment in X Co and Y Co	1,062,894	1,062,894	
CJE3: Settlement of undervalued provision for claims	10,000	10,000	
CJE4: Tax effects of CJE3	(2,000)	(2,000)	
CJE5: Allocate share of post-acq RE of X Co to NCI of X Co	55,000	55,000	
CJE6: Eliminate dividends declared by X Co	(72,000)	(72,000)	
CJE7: Allocate share of current income of X Co to NCI of X Co	588,300	588,300	
CJE8: Adjustment for upstream sale of inventory from Y Co to P Co	(12,420)	(1,620)	(10,800)
CJE9: Tax effects on CJE8	2,484	324	2,160
CJE10: Allocation of post-acquisition RE to total NCI of Y Co	96,600	12,600	84,000
CJE11: Allocation of current profit after tax to total NCI of Y Co	243,136	75,456	167,680
CJE12: Elimination of dividends declared by Y Co	(26,000)		(26,000)
CJE13: Allocation of post-acq revaluation reserves to total NCI of Y Co	(11,600)	(3,600)	(8,000)
	2,778,394	2,125,354	653,040

Analytical check on non-controlling interests of X Co

X Co's shareholders' equity as at 31 December 20x3	4,210,000	1,200,000 + 3,010,000
Less investment in Y Co	(790,000)	
	3,420,000	
NCI's share of book value of equity as at 31 Dec 20x3	1,026,000	30%*3,420,000
NCI of X Co's goodwill as at 31 Dec 20x3	814,486	233,400 + 581,086
NCI of X Co	1,840,486	

Analytical check of NCI of Y Co

			Indirect NCI	Direct NCI
Y Co's shareholders' equity as at 31 December 20x3	1,585,000	350,000+1,135,000+100,000		
Less unrealized profit on upstream sale of inventory	(2,400)			
Adjusted shareholders' equity	1,582,600			
Y Co's total NCI share of equity as at 31 December 20x3	917,908	58%*1,582,600	284,868	633,040
Goodwill attributable to Y Co's NCI	20,000			20,000
Total NCI of Y Co as at 31 December 20x3	937,908		284,868	653,040
Total NCI	**2,778,394**		2,125,354	653,040

EA1: Recognize share of post-acquisition RE of Z

Dr Investment in Z	44,400		
Cr Opening RE		44,400	
RE of Z as at 1 Jan 20x3			468,000
RE of Z as at date of acquisition			320,000

Change in RE		148,000
Share of Z's change in RE		44,400

EA2: Reclassify dividend income as a reduction of investment

Dr Dividend income	18,000		
Cr Investment in Z		18,000	

EA3: Amortization of undervalued intangible asset

Dr Opening RE	38,400	400,000/5*2*80%*30%	
Cr Investment in Z		38,400	

EA4: Recognize share of current profit after tax of Z

Dr Investment in Z	224,400		
Cr Share of profit of Z		224,400	

Workings:

NPBT	1,000,000	
Less amortization of FV-BV of intangible asset	(80,000)	400,000/5
Less unrealized profit on sale of fixed asset	(60,000)	Note 1
Add excess depreciation	10,000	60,000/3*1/2
Add back excess impairment loss	40,000	Note 2
Adjusted NPBT of Z	910,000	
Share of adjusted NPBT of Z	273,000	
Tax expense of Z	180,000	
Less tax on amortization of FV-BV of intangible	(16,000)	
Less on unrealized profit on sale of fixed asset	(12,000)	
Add back tax on excess depreciation and excess impairment loss	10,000	
Adjusted tax expense of Z	162,000	
Share of adjusted tax expense of Z	48,600	

Note 1: Transfer of fixed assets (from Z to P)

Profit on sale of fixed assets	60,000
Excess depreciation for half-year 20x3	10,000
Remaining unrealized profit, 31 Dec 20x3	50,000
Recoverable amount	60,000

Note 2: Excess impairment loss

	Legal entity	**Group**	**Difference**
Transfer price	120,000		

Original cost			
NBV, 31 Dec 20x3	100,000	50,000	
Recoverable amount	60,000	60,000	
Impairment loss	40,000	0	40,000
Revised balance	60,000	50,000	(10,000)

Analytical check of Investment in Z

Book value of shareholders' equity of Z		2,028,000	
Add unamortized intangible asset		128,000	400,000*2/5*80%
Less unrealized profit of fixed asset as at end of 20x3		(8,000)	10,000*80%
		2,148,000	
P's share of Z's identifiable net assets		644,400	
Implicit goodwill in investment in Z:			
Investment in Z		980,000	
BV of net assets of Z at acq	1,120,000		
Unrecognized intangible asset, after-tax	320,000		
FV of net assets of Z at acq	1,440,000		
Less Share of FV of net assets of Z at acq		432,000	
Goodwill in Z implicit in the investment in Z		548,000	
		1,192,400	
Investment in Z, at cost		980,000	
EA1: Recognize share of post-acq RE of Z		44,400	
EA2: Reclassify dividend income as a reduction of investment		(18,000)	
EA3: Amortization of undervalued intangible asset		(38,400)	
EA4: Recognize share of current profit after tax of Z		224,400	
Investment in Z as at 31 Dec 20x3		**1,192,400**	

Analytical check of closing retained earnings

P's RE	8,980,000	
X's RE	3,010,000	
Y's RE	1,135,000	
CJE1: Elimination of investment in X Co and Y Co	(1,290,000)	700,000 + 590,000
CJE2: Effects of the decrease in investment in X Co and Y Co	(185,714)	
CJE3: Settlement of undervalued provision for claims	90,000	
CJE4: Tax effects of CJE3	(18,000)	
CJE5: Allocate share of post-acq RE of X Co to NCI of X Co	(55,000)	
CJE6: Eliminate dividends declared by X Co	72,000	
CJE7: Allocate share of current income of X Co to NCI of X Co	(588,300)	
CJE8: Adjustment for upstream sale of inventory from Y Co to P Co	9,420	24,000 − 14,580

CJE9: Tax effects on CJE8	(1,884)	2,916 − 4,800
CJE10: Allocation of post-acquisition RE to total NCI of Y Co	(96,600)	
CJE11: Allocation of current profit after tax to total NCI of Y Co	(243,136)	
CJE12: Elimination of dividends declared by Y Co	26,000	
EA1: Recognize share of post-acq RE of Z	44,400	
EA2: Reclassify dividend income as a reduction of investment	(18,000)	
EA3: Amortization of undervalued intangible	(38,400)	
EA4: Recognize share of current profit after tax of Z	224,400	
Consolidated RE at 31 Dec 20x3	**11,056,186**	

Analytical check (Method 1 - show latest ownership interests and a plug that is the capital reserve adjustment)

P's RE	8,980,000	
P's share of post-acquisition RE of X Co	1,617,000	70%*(3,010,000 − 700,000)
P's share of post-acquisition RE of Y Co	228,900	70%*60%*(1,135,000 − 590,000)
P's share of post-acquisition RE of Z Co	272,400	30%*(1,228,000 − 320,000)
P's share of past expensing of claims of X Co	56,000	70%*80%*100,000
P's share of amortization of intangible asset of Z Co	(57,600)	30%*400,000*80%*3/5
P's share of unrealized profit of Y Co (inventory)	(1,008)	42%*10%*80%*30,000
P's share of unrealized profit of fixed asset transfer to Z Co	(2,400)	30%*80%*10,000
Plug—balancing figure adjusted to capital reserve	(37,106)	Note 1
Consolidated RE at 31 Dec 20x3	**11,056,186**	

Note 1: RE captures P's share at the historical ownership interests (i.e., 90% from acquisition of X to 31 Dec 20x2, 70% from 1 Jan 20x3 to 31 Dec 20x3).

To enable a more expedient check, we use the latest ownership interests in the analytical check.

The analytical check using the latest ownership interests would not agree with the compilation of CJE.

The difference is the profit on sale taken to equity (capital reserve) which is shown in the analytical check as a plug.

The plug represents the following:

Reversal of profit on divestment of 20% in P's legal entity income statement	(185,714)	
20% of P's share of post-acquisition change in equity up to divestment date	148,608	Note 2
Amount taken to capital reserve	(37,106)	

Note 2: Why is this positive? Arithmetically, the RE analytical check for simplicity, uses the latest ownership interests (70%) when in reality, the RE is made of a layering of historical profits (90% up to 31 Dec 20x2 and 70% thereafter). Refer to workings under CJE2.

Had the plug been taken to RE (and not Capital Reserve, a separate equity account), the RE reconciliation as shown with latest ownership interests would balance nicely with the listings approach. However, we are of the view that retained earnings is a historical compilation of profits and should not be adjusted retrospectively. The same approach applies to other comprehensive income.

But since the plug is taken to Capital Reserve, an imbalance arises – we use the plug here to show the difference between the listings approach (based on historical percentages) and the analytical approach using the latest ownership interests. Method 1 is a short cut method. But if that approach is confusing, use Method 2 which is a more detailed compilation to show the historical ownership interests at different points in time.

Analytical check (Method 2 - show the layering of RE using the ownership interests as applicable at that point in time)

P's RE	8,980,000	
P's share of post-acquisition RE of X Co (before divestment of 20%)	495,000	90%*(1,250,000 − 700,000)
P's share of post-acquisition RE of X Co (after divestment of 20%)	1,232,000	70%*(3,010,000 − 1,250,000)
P's share of post-acquisition RE of Y Co (before divestment of 20%)	113,400	90%*60%*(800,000 − 590,000)
P's share of post-acquisition RE of X Co (after divestment of 20%)	140,700	70%*60%*(1,135,000 − 800,000)
P's share of post-acquisition RE of Z Co	272,400	30%*(1,228,000 − 320,000)
P's share of past expensing of claims of X Co (before divestment)	72,000	90%*80%*100,000
P's share of amortization of intangible asset of Z Co	(57,600)	30%*400,000*80%*3/5
P's share of after tax unrealized profit from sale to Y Co (before divestment)	(11,664)	90%*60%*80%*27,000
P's share of after tax realized profit from sale to Y Co (after divestment) . .	8,064	70%*60%*80%*24,000
P's share of unrealized profit of fixed asset transfer to Z Co	(2,400)	30%*80%*10,000
Reversal of profit on divestment recognized in P's separate PL	(185,714)	
	11,056,186	

Solution to Question 3

Income Statement for year ended 31 December 20x3						
	P Co	X Co	Z Co	Y Co	Consolidated	
Profit before tax	5,790,000	2,234,900	1,000,000	1,200,000	9,195,500	Note 1
Tax	(1,158,000)	(446,980)	(200,000)	(240,000)	(1,844,180)	Note 2
Profit after tax	4,632,000	1,787,920	800,000	960,000	7,351,320	
Income to NCI					(611,720)	Note 3
Dividends declared	(230,000)	(180,000)	(46,000)	(120,000)	(230,000)	
Profit retained	4,402,000	1,607,920	754,000	840,000	6,509,600	
Retained earnings, 1 Jan 20x3	4,321,000	1,387,000	327,000	960,000	5,087,912	Note 4
Retained earnings, 31 Dec 20x3	8,723,000	2,994,920	1,081,000	1,800,000	11,597,512	AC

Statement of Financial Position as at 31 December 20x3						
	P Co	X Co	Z Co	Y Co	Consolidated	
Goodwill					2,719,429	CJE1
Fixed assets, net book value	4,890,000	2,371,000	1,760,000	1,325,000	8,586,000	
Investment in X Co	4,500,000					
Investment in Z Co	1,200,000				1,429,900	AC
Investment in Y Co		850,000				
Inventory	589,000	563,000	256,000	328,000	1,478,000	Note 5
Intercompany receivable	200,000					
Other net assets	3,200,000	980,000	1,769,000	732,000	4,944,400	Note 6
Cash	120,000	216,700	27,000	123,760	460,460	
	14,699,000	4,980,700	3,812,000	2,508,760	19,618,189	
Accounts payable	3,176,000	285,780	2,011,000	208,760	3,670,540	
Intercompany payable		200,000				
Share capital	2,800,000	1,500,000	720,000	300,000	2,800,000	
Retained earnings	8,723,000	2,994,920	1,081,000	1,800,000	11,597,512	AC
Revaluation reserves				200,000	43,200	Note 7
Capital reserve					(212,682)	CJE1
Non-controlling interests					1,719,619	AC
	14,699,000	4,980,700	3,812,000	2,508,760	19,618,189	

Note 1:	Combined P and subsidiaries	9,224,900	
	Less amortization of intangible assets of Y Co	(20,000)	
	Add realized profit from transfer	16,000	
	Less dividend income from X Co	(162,000)	90%*180,000
	from Y Co	(72,000)	60%*120,000
	from Z Co	(13,800)	30%*46,000
	Add share of profit of associate	222,400	
	Consolidated profit before tax	9,195,500	

Note 2:	Combined P and subsidiaries	(1,844,980)
	Less tax on amortization of intangible assets	4,000
	Add tax on realized profit from transfer	(3,200)
		(1,844,180)

Note 3:	Income to NCI of X	171,592
	Income to NCI of Y	440,128
		611,720

Note 4: Opening RE need not be reconstructed. It can be derived as a balancing figure if the consolidated closing RE is available.

For completeness, the analytical check on opening RE is shown here.

However, for explanations of the plug below, refer to the analytical check in the consolidated closing RE.

P's RE at 1 Jan 20x3	4,321,000	
P's share of X's post-acquisition RE	708,300	90%*(1,387,000 − 600,000)
P's share of Y's post-acquisition RE	286,200	90%*60%*(960,000 − 430,000)
P's share of Z's post-acquisition RE	26,100	30%*(327,000 − 240,000)
P's share of past cost of sales of undervalued inventory	(10,800)	90%*80%*15,000
P's share of past amortization of Y Co's intangible assets	(17,280)	90%*60%*80%*100,000/5*2
P's share of past amortization of Z Co's intangible assets	(4,800)	30%*80%*50,000*2/5
P's share of unrealized profit in inventory, 1 Jan	(7,776)	90%*60%*80%*18,000
Difference–NCI's share of RE taken to Capital Reserve	(213,032)	Refer explanatory note in AC on closing RE
	5,087,912	

Note 5:	Reduced by unrealized profit in ending inventory		2,000
Note 6:	Combined P and subsidiaries	4,912,000	
	Remaining balance of intangible asset of Y	40,000	100,000/5*2
	Deferred tax liability on intangible asset	(8,000)	
	Deferred tax asset on unrealized profit in inventory	400	
		4,944,400	
Note 7:	P's share of post-acquisition revaluation reserves of Y		90%*60%*80,000

CJE1: Elimination of investment in X Co and Y Co

Dr Share capital (X Co)	1,500,000	
Dr Share capital (Y Co)	300,000	
Dr Retained earnings (X Co)	600,000	
Dr Retained earnings (Y Co)	430,000	
Dr Revaluation reserve (Y Co)	120,000	
Dr Inventory (X Co)	15,000	
Dr Intangible asset (Y Co)	100,000	
Dr Goodwill	2,719,429	
Cr Deferred tax liability		23,000

Cr Investment in X Co		3,200,000
Cr Investment in Y Co		850,000
Cr Non-controlling interests in X Co		1,311,429
Cr Non-controlling interests in Y Co		400,000
	5,784,429	5,784,429

Goodwill in sub-group (this calculation is optional - suffice to show the goodwill as a residual in CJE1)

Fair value of INA of sub-group at 1 Jan 20x1		
Share capital (X Co)	1,500,000	
Retained earnings (X Co)	600,000	
Book value of X Co		2,100,000
Less X Co's investment in Y Co		(850,000)
Undervalued inventory		15,000
Deferred tax liability		(3,000)
Fair value of INA of X Co		1,262,000
Share capital (Y Co)	300,000	
Retained earnings (Y Co)	430,000	
Revaluation reserve (Y Co)	120,000	
Intangible asset	100,000	
Deferred tax liability	(20,000)	
Fair value of INA of Y Co		930,000
X Co's consolidated equity or net assets		2,192,000
Fair value of consideration transferred (4,500,000 − 1,300,000)		3,200,000
Fair value of non-controlling interests (X Co)		1,311,429
Fair value of non-controlling interests (Y Co)		400,000
Less fair value of INA of sub-group (see workings above)		(2,192,000)
Goodwill		2,719,429

Goodwill attributable to P Co	**1 Jan 20x1**
P Co's share of goodwill	70%
Fair value of consideration transferred	3,200,000
Less share of fair value of INA of sub-group	
Share of X Co's FV of INA	(883,400)
Share of Y Co's FV of INA	(390,600)
Goodwill attributable to P Co	1,926,000

X Co's NCI as at 1 January 20x1 (before additional investment)	**Total**	**30%**
X Co's shareholders' equity as at 1 January 20x1	2,100,000	
Less investment in Y Co	(850,000)	

Share of undervalued inventory, after-tax	12,000		
X Co's share of Y Co's shareholders' equity as at 1 January 20x1	558,000		60%*930,000 below
X Co's consolidated equity as at 1 January 20x1	1,820,000	546,000	30%*1,820,000
X Co's NCI's goodwill		765,429	Residual
X Co's NCI		1,311,429	

Y Co's NCI as at 1 January 20x1 (before additional divestment)		Total	40%	
Y Co's shareholder's equity as at 1 January 20x1		850,000	340,000	
Share capital	300,000			
Retained earnings	430,000			
Revaluation reserves	120,000			
	850,000			
Y Co's unrecognized intangible asset, after tax		80,000	32,000	
Goodwill			28,000	Residual
NCI of Y Co		930,000	400,000	
Total Goodwill attributable to P and NCI (1,926,000+765,429+28,000)			2,719,429	

CJE2: Effects of the increase in investment in X Co by P Co

Dr Non-controlling interests	1,087,318	
Dr Capital reserve (Equity)	212,682	
Cr Investment		1,300,000

Note 1: X's NCI's share of equity of the sub-group as at 31 Dec 20x2

NCI's share of post-acquisition RE of X Co	236,100	30%*(1,387,000 − 600,000)
NCI's share of post-acquisition RE of Y Co	95,400	18%*(960,000 − 430,000)
NCI 's share of past cost of sales of X Co	(3,600)	30%*80%*15,000
NCI 's share of past amortization of intangible asset of Y Co	(5,760)	18%*80%*100,000/5*2
NCI 's share of unrealized profit of Y Co (sale to X)	(2,592)	18%*80%*90%*20,000
NCI 's share of sub-group's post-acquisition equity	319,548	Note 2
20% of initial FV of NCI in X Co	874,286	20/30*1,311,429
20% of NCI's share of post-acq change in equity	213,032	20/30*319,548
20% of equity in X Co transferred by NCI to P Co	1,087,318	

Note 2: In this problem, the change in equity is attributable solely to retained earnings.
There was no post-acquisition change in revaluation reserves.

Workings:

	Group equity
Additional investment in X Co by P Co	1,300,000
Acquisition of equity from NCI by P Co	(1,087,318)

Loss on purchase taken to capital reserve	212,682

This check of NCI at 31 Dec 20x2 is optional–it is only done to show that the re-allocation results in a correct NCI balance

Analytical check of NCI on 31 Dec 20x2 (after re-allocation of 20% to P)

CJE1: Elimination of investment in X Co and Y Co	1,311,429	
CJE2: Effects of the increase in investment in X Co by P Co	(1,087,318)	
CJE3: Past cost of sales of undervalued inventory	(4,500)	
CJE4: Tax effects of CJE3	900	
CJE5: Allocate share of post-acq RE of X Co to NCI of X Co	236,100	
CJE8: Adjustment for upstream sale of inventory from Y Co to X Co	(3,240)	18%*90%*20,000
CJE9: Tax effects on CJE8	648	20%*3,240
CJE10: Amortization of intangible asset	(7,200)	18%*100,000/5*2
CJE11: Tax effects of CJE10	1,440	20%*7,200
CJE12: Allocation of post-acquisition RE to total NCI of Y Co	95,400	18%*(960,000 − 430,000)
	543,659	
Book value of net assets of X Co on 31 Dec 20x2	2,887,000	1,500,000 + 1,387,000
Less investment in Y Co	(850,000)	
	2,037,000	
NCI's share of X Co (after divestment)	203,700	10%*2,037,000
Book value of net assets of Y Co on 31 Dec 20x2	1,380,000	300,000 + 960,000 + 120,000
Add remaining balance of intangible asset, after tax	48,000	100,000/5*3*80%
Less unrealized profit on 31 Dec 20x2	(14,400)	90%*80%*20,000
	1,413,600	
NCI's share of Y Co (after divestment)	84,816	6%*1,413,600
NCI's goodwill at acquisition	765,429	
NCI's goodwill transferred to P on divestment	(510,286)	20/30*765,429
	255,143	
NCI balance as at 31 Dec 20x2 (203,700+84,816+255,143)	**543,659**	

CJE3: Past cost of sales of undervalued inventory

Dr Opening RE	10,500	
Dr NCI	4,500	
Cr Inventory		15,000

CJE4: Tax effects of CJE3

Dr Deferred tax liability	3,000	
Cr Opening RE		2,100
Cr NCI		900

CJE5: Allocate share of post-acq RE of X Co to NCI of X Co

Dr Opening RE	236,100		
Cr NCI (BS)		236,100	
RE at 1 Jan 20x3		1,387,000	
RE at date of acquisition		600,000	
Change in RE		787,000	
NCI's share (before divestment to P)		236,100	30%*787,000

CJE6: Eliminate dividends declared by X Co

Dr Dividend income		162,000	
Dr Non-controlling interests		18,000	10%*180,000
Cr Dividend declared			180,000

CJE7: Allocate share of current income of X Co to NCI of X Co

Dr Income to NCI	171,592		10%*1,717,920
Cr NCI (BS)			171,592
NPAT of X Co			1,787,920
Less dividend received from Y Co			(72,000)
Adjusted NPAT (after divestment to P)			1,715,920

CJE8: Adjustment for upstream sale of inventory from Y Co to X Co

Dr Opening RE	7,560	70%*60%*90%*20,000			
Dr Non-controlling interests	10,440	58%*90%*20,000			
Cr Cost of sales			16,000	80%*20,000	
Cr Inventory			2,000	10%*20,000	
Direct NCI				40%	7,200
Indirect NCI (before divestment to P)				18%	3,240
Total NCI				58%	10,440

CJE9: Tax effects on CJE8

Dr Tax expense	3,200				
Dr Deferred tax asset	400				
Cr Opening RE			1,512		
Cr Non-controlling interests			2,088		
Direct NCI				40%	1,440
Indirect NCI (before divestment to P)				18%	648
Total NCI				58%	2,088

CJE10: Amortization of intangible asset

Dr ORE	16,800	70%*60%*100,000/5*2	
Dr NCI	23,200	58%*100,000/5*2	
Dr Amortization	20,000	100,000/5	
Cr Accumulated amortization			60,000

CJE11: Tax effects of CJE10

Dr Deferred tax liability	12,000	
Cr ORE		3,360
Cr NCI		4,640
Cr Tax expense		4,000

CJE12: Allocation of post-acquisition RE to total NCI of Y Co

Dr Opening RE	307,400			
Cr Non-controlling interests		307,400		
Retained earnings at 1 Jan 20x3				960,000
Retained earnings at 1 Jan 20x1				430,000
Change in retained earnings				530,000
Direct NCI			40%	212,000
Indirect NCI (before divestment to P)			18%	95,400
Total NCI			58%	307,400

CJE13: Allocation of current profit after tax to total NCI of Y Co

Dr Income to NCI	440,128			
Cr Non-controlling Interests		440,128		
Net profit after tax of Y Co				960,000
Less amortization of intangible asset, after tax				(16,000)
Add realized profit from upstream sale of inventory, after tax				12,800
Adjusted NPAT				956,800
Direct NCI			40%	382,720
Indirect NCI (after divestment to P)			6%	57,408
Total NCI			46%	440,128

CJE14: Elimination of dividends declared by Y Co

Dr Dividend income	72,000		
Dr Non-controlling interests	48,000	40%*120,000	
Cr Dividend declared		120,000	

CJE15: Allocation of post-acquisition revaluation reserves to total NCI of Y Co

Dr Share of revaluation reserves	36,800			
Cr NCI		36,800		
Revaluation reserves at 31 Dec 20x3				200,000
Revaluation reserves at 1 Jan 20x3				120,000
Change in revaluation reserves				80,000
Direct NCI			40%	32,000
Indirect NCI (after divestment to P)			6%	4,800
Total NCI			46%	36,800

CJE16: Eliminate intercompany payable and receivable

Dr Intercompany payable	200,000	
Cr Intercompany receivable		200,000

Analytical check of Non-controlling interests

Total Non-controlling interests as at 31 December 20x3

	Total NCI	NCI of X Co and indirect NCI of Y Co	Direct NCI of Y Co
CJE1: Elimination of investment in X Co and Y Co	1,711,429	1,311,429	400,000
CJE2: Effects of the increase in investment in X Co by P Co	(1,087,318)	(1,087,318)	
CJE3: Past cost of sales of undervalued inventory	(4,500)	(4,500)	
CJE4: Tax effects of CJE3	900	900	
CJE5: Allocate share of post-acq RE of X Co to NCI of X Co	236,100	236,100	
CJE6: Eliminate dividends declared by X Co	(18,000)	(18,000)	
CJE7: Allocate share of current income of X Co to NCI of X Co	171,592	171,592	
CJE8: Adjustment for upstream sale of inventory from Y Co to X Co	(10,440)	(3,240)	(7,200)
CJE9: Tax effects on CJE8	2,088	648	1,440
CJE10: Amortization of intangible asset	(23,200)	(7,200)	(16,000)
CJE11: Tax effects of CJE10	4,640	1,440	3,200
CJE12: Allocation of post-acquisition RE to total NCI of Y Co	307,400	95,400	212,000
CJE13: Allocation of current profit after tax to total NCI of Y Co	440,128	57,408	382,720
CJE14: Elimination of dividends declared by Y Co	(48,000)		(48,000)
CJE15: Allocation of post-acq revaluation reserves to total NCI of Y Co	36,800	4,800	32,000
	1,719,619	759,459	960,160

Analytical check on non-controlling interests of X Co

X Co's shareholders' equity as at 31 December 20x3	4,494,920	1,500,000 + 2,994,920
Less investment in Y Co	(850,000)	
	3,644,920	
NCI's share of book value of equity as at 31 Dec 20x3	364,492	10%*3,644,920
X Co's NCI's goodwill as at 31 Dec 20x3	255,143	765,429*10/30
	619,635	

Analytical Check of NCI of Y Co

			Indirect NCI of Y	Direct NCI of Y
Y Co's shareholders' equity as at 31 December 20x3	2,300,000	300,000+1,800,000+200,000		
Add remaining balance of intangible asset, after tax	32,000	100,000*2/5*80%		
Less unrealized profit on upstream sale of inventory	(1,600)	10%*80%*20,000		
Adjusted shareholders' equity	2,330,400			
Y Co's total NCI share of equity as at 31 December 20x3	1,071,984	46%*2,330,400	139,824	932,160
Goodwill attributable to Y Co's NCI	28,000			28,000
NCI of Y Co as at 31 December 20x3	1,099,984		139,824	960,160
Total NCI	**1,719,619**		759,459	960,160

EA1: Recognize share of post-acq RE of Z

Dr Investment in Z	26,100		
Cr Opening RE		26,100	
RE of Z as at 1 Jan 20x3			327,000
RE of Z as at date of acquisition			240,000
Change in RE			87,000
Share of Z's change in RE			26,100

EA2: Reclassify dividend income as a reduction of investment

Dr Dividend income	13,800	
Cr Investment in Z		13,800

EA3: Amortization of undervalued intangible

Dr Opening RE	4,800	50,000/5*2*80%*30%
Cr Investment in Z		4,800

EA4: Recognize share of current profit after tax of Z

Dr Investment in Z	222,400	
Cr Share of profit of Z		222,400

Workings:

NPBT	1,000,000	
Less amortization of FV-BV of intangible	(10,000)	50,000/5
Less unrealized profit on sale of fixed asset	(76,000)	Note 1
Add excess depreciation	12,667	(130,000-54,000)/3*1/2
Adjusted NPBT of Z	926,667	

Share of adjusted NPBT of Z	278,000
Tax expense of Z	200,000
Less tax on amortization of FV-BV of intangible	(2,000)
Less on unrealized profit on sale of fixed asset	(15,200)
Add tax on excess depreciation	2,533
Adjusted tax expense of Z	185,333
Share of adjusted tax expense of Z	55,600

Note 1: Transfer of fixed assets (from Z to P)

Profit on sale of fixed assets	76,000
Excess depreciation for half-year 20x3	12,667
Remaining unrealized profit, 31 Dec 20x3 (Note 2)	63,333

Note 2: Difference in carrying amounts

	Legal entity	Group	Difference
Transfer price	130,000		
Original cost		180,000	
NBV, 31 Dec 20x3	108,333	45,000	(63,333)

Analytical check of Investment in Z

Book value of shareholders' equity of Z		1,801,000	
Add unamortized intangible		16,000	50,000*2/5*80%
Less unrealized profit of fixed asset as at end of 20x3		(50,667)	63,333*80%
		1,766,333	
P's share of Z's identifiable net assets		529,900	
Implicit goodwill in investment in Z:			
Investment in Z		1,200,000	
BV of net assets of Z at acq	960,000		
Unrecognized intangible, after-tax	40,000		
FV of net assets of Z at acq	1,000,000		
Less Share of FV of net assets of Z at acq		300,000	
Goodwill in Z implicit in the investment in Z		900,000	
		1,429,900	
Investment in Z, at cost		1,200,000	
EA1: Recognize share of post-acq RE of Z		26,100	
EA2: Reclassify dividend income as a reduction of investment		(13,800)	
EA3: Amortization of undervalued intangible		(4,800)	

EA4: Recognize share of current profit after tax of Z	222,400	
Investment in Z as at 31 Dec 20x6	**1,429,900**	

Analytical check of closing retained earnings

P's RE	8,723,000	
X's RE	2,994,920	
Y's RE	1,800,000	
CJE1: Elimination of investment in X Co and Y Co	(1,030,000)	600,000 + 430,000
CJE3: Past cost of sales of undervalued inventory	(10,500)	
CJE4: Tax effects of CJE3	2,100	
CJE5: Allocate share of post-acq RE of X Co to NCI of X Co	(236,100)	
CJE6: Eliminate dividends declared by X Co	18,000	
CJE7: Allocate share of current income of X Co to NCI of X Co	(171,592)	
CJE8: Adjustment for upstream sale of inventory from Y Co to X Co	8,440	16,000 − 7,560
CJE9: Tax effects on CJE8	(1,688)	1,512 − 3,200
CJE10: Amortization of intangible asset	(36,800)	16,800 + 20,000
CJE11: Tax effects of CJE10	7,360	3,360 + 4,000
CJE12: Allocation of post-acquisition RE to total NCI of Y Co	(307,400)	
CJE13: Allocation of current profit after tax to total NCI of Y Co	(440,128)	
CJE14: Elimination of dividends declared by Y Co	48,000	
EA1: Recognize share of post-acq RE of Z	26,100	
EA2: Reclassify dividend income as a reduction of investment	(13,800)	
EA3: Amortization of undervalued intangible	(4,800)	
EA4: Recognize share of current profit after tax of Z	222,400	
Consolidated RE at 31 Dec 20x3	**11,597,512**	

Analytical check (Method 1 - show latest ownership interests and a plug that is transfer from NCI's equity that is taken to capital reserve)

P's RE	8,723,000	
P's share of post-acquisition RE of X Co	2,155,428	90%*(2,994,920 − 600,000)
P's share of post-acquisition RE of Y Co	739,800	90%*60%*(1,800,000 − 430,000)
P's share of post-acquisition RE of Z Co	252,300	30%*(1,081,000 − 240,000)
P's share of past cost of sales of X Co	(10,800)	90%*80%*15,000
P's share of amortization of intangible asset of Y Co	(25,920)	90%*60%*80%*100,000/5*3
P's share of amortization of intangible asset of Z Co	(7,200)	30%*50,000*80%*3/5
P's share of unrealized profit of Y Co (inventory)	(864)	90%*60%*80%*2,000
P's share of unrealized profit of fixed asset transfer from Z Co	(15,200)	30%*80%*63,333
Plug - balancing figure adjusted to capital reserve	(213,032)	Note 1

Consolidated RE at 31 Dec 20x3 **11,597,512**

Note 1:

Consolidated retained earnings captures P's share of subsidiaries profits at the historical ownership interests (i.e., 70% from acquisition of X to 31 Dec 20x2, 90% from 1 Jan 20x3 to 31 Dec 20x3).

To enable a more expedient check, we use the latest ownership interests in the analytical check rather than a layering of historical profit figures.

The analytical check using the latest ownership interests would not agree with the listings approach.

The difference is the share of post-acquisition change in RE taken over from NCI (the 20% share from acquisition date to 31 Dec 20x2).

In this example, the difference is:

20% of P's share of post-acq change in equity up to divestment date 213,032 Note 2

As there is no change in post-acq RR in this question, the entire change in post-acquisition equity transferred from NCI is due to post-acq change in RE.

Note 2: Why is this negative? Arithmetically, the RE analytical check, for simplicity, uses the latest ownership interests (90%) when in reality, the RE is made of a layering of historical profits (70% up to 31 Dec 20x2 and 90% thereafter). So the negative figure indicates that the RE should be lower than the short-cut analytical check. Refer to workings under CJE2.

Had the plug been taken to RE (and not Capital Reserve, a separate equity account), the RE reconciliation as shown with the latest ownership interests would balance nicely with the listings approach. But since the plug is taken to Capital Reserve, an imbalance arises - we use the plug here to show the difference between the listings approach (based on historical percentages) and the analytical approach using the latest ownership interests. However, we are of the view that retained earnings is a historical compilation of profits and should not be adjusted retrospectively. The same approach applies to other comprehensive income. Method 1 is a short-cut method. But if that approach is confusing, use Method 2 which is a more detailed compilation to show the historical ownership interests at different points in time.

The plug is not the same amount as the Capital Reserve entry of $212,682 (CJE2). CJE2 is the loss on purchase or the excess payment over the carrying amount of the NCI.

This scenario is different from Question 2 where the plug is taken directly from Capital Reserve. The Capital Reserve in that situation comprises purely the profit on sale at group level (a retained earnings item).

Analytical check (Method 2 - show the layering of RE using the ownership interests as applicable at that point in time)

P's RE	8,723,000	
P's share of post-acquisition RE of X Co (before investment of 20%)	550,900	70%*(1,387,000 – 600,000)
P's share of post-acquisition RE of X Co (after investment of 20%)	1,447,128	90%*(2,994,920 – 1,387,000)
P's share of post-acquisition RE of Y Co (before investment of 20%)	222,600	70%*60%*(960,000 – 430,000)
P's share of post-acquisition RE of Y Co (after investment of 20%)	453,600	90%*60%*(1,800,000 – 960,000)
P's share of post-acquisition RE of Z Co	252,300	30%*(1,081,000 – 240,000)
P's share of past cost of sales of X Co (before investment of 20%)	(8,400)	70%*80%*15,000
P's share of amortization of intangible asset of Y Co (before investment)	(13,440)	70%*60%*80%*100,000/5*2
P's share of amortization of intangible asset of Y Co (after investment)	(8,640)	90%*60%*80%*100,000/5
P's share of amortization of intangible asset of Z Co	(7,200)	30%*50,000*80%*3/5
P's share of after tax unrealized profit from sale of Y Co (before investment)	(6,048)	70%*60%*80%*90%*20,000
P's share of after tax realized profit from sale of Y Co (after investment)	6,912	90%*60%*80%*16,000
P's share of unrealized profit of fixed asset transfer from Z Co	(15,200)	30%*80%*63,333
	11,597,512	

Solution to Question 4

Income Statement for year ended 31 December 20x6	P Co	B Co	D Co	C Co	Consolidated	
Profit before tax	4,300,000	1,400,000	620,000	480,000	6,538,000	Note 1
Tax	(860,000)	(280,000)	(124,000)	(96,000)	(1,348,200)	Note 2
Profit after tax	3,440,000	1,120,000	496,000	384,000	5,189,800	
Income to NCI					(443,312)	Note 3
Dividends declared	(200,000)	(130,000)	(50,000)	(80,000)	(200,000)	
Profit retained	3,240,000	990,000	446,000	304,000	4,546,488	
Retained earnings, 1 Jan 20x6	2,356,000	800,200	800,000	450,000	3,130,792	Note 4
Retained earnings, 31 Dec 20x6	5,596,000	1,790,200	1,246,000	754,000	7,677,280	AC

Statement of Financial Position as at 31 December 20x6	P Co	B Co	D Co	C Co	Consolidated	
Goodwill					1,273,400	CJE1
Intangible assets, net book value					135,000	
Fixed assets, net book value	2,789,000	1,954,000	1,340,000	890,000	6,897,000	Note 5
Investment in B Co	2,200,000					
Investment in D Co	720,000					
Investment in C Co		900,000				
Inventory	790,000	600,000	450,000	247,000	2,087,000	
Intercompany receivable	1,500,000					
Other net assets	1,970,000	1,265,000	803,450	520,000	4,546,650	Note 6
Cash	300,000	256,000	120,000	120,000	796,000	
	10,269,000	4,975,000	2,713,450	1,777,000	15,735,050	
Accounts payable	1,973,000	484,800	667,450	403,000	3,528,250	
Bank loans	1,400,000				1,400,000	
Intercompany payable		1,500,000				
Share capital	1,300,000	1,200,000	800,000	520,000	1,300,000	
Retained earnings	5,596,000	1,790,200	1,246,000	754,000	7,677,280	AC
Revaluation reserves				100,000	3,150	
Non-controlling interests					1,826,370	Note 7
	10,269,000	4,975,000	2,713,450	1,777,000	15,735,050	

Note 1:	Combined P and subsidiaries		6,800,000	4,300,000 + 1,400,000 + 620,000 + 480,000
	Amortization of intangible asset		(15,000)	
	Current cost of sales of undervalued inventory of C		(8,000)	10%*80,000
	Adjustment for unrealized profit (P to B)		(36,000)	40,000-(80,000/10*6/12)
	Less dividend income	from B Co	(117,000)	90%*130,000
		from C Co	(56,000)	70%*80,000
		from D Co	(30,000)	60%*50,000
	Consolidated profit before tax		6,538,000	

Note 2:	Combined P and subsidiaries	(1,360,000)	860,000 + 280,000 + 124,000 + 96,000
Tax on:	Amortization of intangible asset	3,000	
	Current cost of sales of undervalued inventory of C	1,600	
	Adjustment for unrealized profit (P to B)	7,200	

	(1,348,200)	

Note 3:

Income to NCI of B	105,200
Income to NCI of C	139,712
Income to NCI of D	198,400
	443,312

Note 4:

P's RE at 1 Jan 20x6	2,356,000	
P's share of B's post-acquisition RE	135,180	90%*(800,200 − 650,000)
P's share of C's post-acquisition RE	81,900	63%*(450,000 − 320,000)
P's share of D's post-acquisition RE	246,000	60%*(800,000 − 390,000)
P's share of past cost of sales of undervalued inventory of C	(36,288)	63%*80%*90%*80,000
P's share of unrealized interest income, 1 Jan (P to B)	(32,000)	80%*(120,000-80,000)
Effects from gain of control of D Co	380,000	
	3,130,792	

Note 5: 2,789,000 + 1,954,000 + 1,340,000 + 890,000−(80,000*9.5/10)

Note 6: 1,970,000 + 1,265,000+803,450 + 520,000 + (20%*80,000*9.5/10)−(20%*150,000*9/10)

Note 7:

NCI of B	365,970	AC
NCI of C	418,000	AC
NCI of D	1,042,400	AC
	1,826,370	

CJE1: Elimination of investment in B Co, C Co and D Co

Dr Share capital (B Co)	1,200,000		
Dr Share capital (C Co)	520,000		
Dr Share capital (D Co)	800,000		
Dr Retained earnings (B Co)	650,000		
Dr Retained earnings (C Co)	320,000		
Dr Retained earnings (D Co)	390,000		
Dr Revaluation reserve (C Co)	95,000		
Dr Inventory (C Co)	80,000		
Dr Intangible assets	150,000		
Dr Goodwill	1,273,400		
Cr Deferred tax liability		46,000	
Cr Investment in B Co		2,200,000	
Cr Investment in C Co		900,000	
Cr Investment in D Co		1,100,000	400,000 + 700,000
Cr Non-controlling interests in B Co		226,900	

Cr Non-controlling interests in C Co		305,500
Cr Non-controlling interests in D Co		700,000
	5,478,400	5,478,400

Goodwill in sub-group

Fair value of INA of sub-group at 1 Aug 20x4			
Share capital (B Co)	1,200,000		
Retained earnings (B Co)	650,000		
Book value of B Co		1,850,000	
Less B Co's investment in C Co		(900,000)	
Add intangible assets		150,000	
less deferred tax liability		(30,000)	
Fair value of INA of B Co		1,070,000	
Share capital (C Co)	520,000		
Retained earnings (C Co)	320,000		
Revaluation reserve (C Co)	95,000		
Add undervalued inventory	80,000		
Deferred tax liability	(16,000)		
Fair value of INA of C Co		999,000	
B Co's consolidated equity or net assets		2,069,000	
Fair value of consideration transferred	2,200,000		
Fair value of non-controlling interests (B Co)	226,900		
Fair value of non-controlling interests (C Co)	305,500		
Less fair value of INA of sub-group	(2,069,000)	(see workings above)	
Goodwill in B Co and C Co		663,400	

P Co's share of B			90%	
Fair value of consideration transferred			2,200,000	
Less share of fair value of INA of sub-group				
Share of B Co's FV of INA	963,000	90%*1,070,000		
Share of C Co's FV of INA	629,370	90%*70%*999,000	(1,592,370)	
Goodwill of B Co attributable to P Co			607,630	
B Co's NCI			226,900	
Less fair value of INA of sub-group				
Share of B Co's FV of INA	107,000	10%*1,070,000		
Share of C Co's FV of INA	69,930	10%*70%*999,000		
B Co's consolidated equity as at 1 January 20x4			176,930	
B Co's NCI's goodwill			49,970	
C Co's NCI			305,500	
Less Share of C Co's FV of INA			299,700	30%*999,000
C Co's NCI's goodwill			5,800	
Goodwill of B Co and C Co			663,400	

Goodwill of D Co

Fair value of consideration transferred	400,000
Fair value of previously-held interests	700,000
Fair value of NCI	700,000
Less Share of D Co's FV of INA	(1,190,000)
Goodwill of D Co	610,000

Goodwill of D Co attributable to NCI	
Fair value of NCI	700,000
Less Share of D Co's FV of INA	(476,000)
Goodwill of D Co attributable to NCI	224,000
Total goodwill (663,400 + 610,000)	**1,273,400**

CJE2: Remeasurement gain in previously held interests of D Co

Dr Investment	380,000	700,000 − 320,000
Cr ORE		380,000

Workings:

	Group
Fair value of previously held interests (40%)	700,000
Initial investment in 40% of D Co	(320,000)
P Co's share of post-acquisition change in equity of D Co (1 Jul 20x2 to 1 Jan 20x5)	(280,000)
Remeasurement gain on 1 Jan 20x5	100,000

During 20x5, the following entry was passed

Dr Investment	380,000		
Cr Remeasurement gain		100,000	
Cr ORE		280,000	Re-enact the post-acquisition change in RE

CJE3: Amortization of intangible asset

Dr Amortization		15,000	
Cr Accumulated amortization			15,000

CJE4: Tax effects of CJE3

Dr Deferred tax liability		3,000	
Cr Tax expense			3,000

CJE5: Allocate share of post-acq RE of B Co to NCI of B Co

Dr Opening RE	15,020		
Cr NCI (BS)		15,020	
RE at 1 Jan 20x6		800,200	
RE at date of acquisition		650,000	
Change in RE		150,200	
NCI's share		15,020	10%*150,200

CJE6: Eliminate dividends declared by B Co

Dr Dividend income		117,000	
Dr Non-controlling interests		13,000	10%*130,000
Cr Dividend declared			130,000

CJE7: Allocate share of current income of B Co to NCI of B Co

Dr Income to NCI		105,200	
Cr NCI (BS)			105,200
NPAT of B Co			1,120,000
Amortization of intangible asset, after tax			(12,000)
Less dividend received from C Co			(56,000)
Adjusted NPAT			1,052,000

CJE8: Adjustment of interest profit (transfer from P to B)

Dr ORE		40,000	
Dr Interest income		100,000	
Cr Interest expense			60,000

Cr Fixed asset			80,000

CJE9: Tax effects of CJE8

Dr DTA	16,000		
Cr Tax expense			8,000
Cr ORE			8,000

CJE10: Adjustment of excess depreciation on fixed assets

Dr Accumulated depreciation	4,000	6/12*80,000*1/10	
Cr Depreciation expense			4,000

CJE11: Tax effects of CJE10

Dr Tax expense	800		
Cr DTA			800

CJE12: Adjustment of cost of sales of undervalued inventory

Dr ORE	45,360		
Dr NCI	26,640	37%*90%*80,000	
Cr Inventory			72,000

CJE13: Tax effects of CJE12

Dr DTL	14,400		
Cr ORE			9,072
Cr NCI			5,328

CJE14: Cost of sales of undervalued inventory

Dr Cost of sales	8,000		
Cr Inventory			8,000

CJE15: Tax effects of CJE14

Dr DTL	1,600		
Cr Tax expense			1,600

CJE16: Allocation of post-acquisition RE to total NCI of C Co

Dr Opening RE	48,100			
Cr Non-controlling interests		48,100		
Retained earnings at 1 Jan 20x6				450,000
Retained earnings at 1 Aug 20x4				320,000
Change in retained earnings				130,000
Direct NCI			30%	39,000
Indirect NCI			7%	9,100
Total NCI			37%	48,100

CJE17: Allocation of current profit after tax to total NCI of C Co					
Dr Income to NCI	139,712				
Cr Non-controlling Interests		139,712			
Net profit after tax of C Co				384,000	
Less cost of sale of undervalued inventory, after tax				(6,400)	10%*80%*80,000
Adjusted NPAT				377,600	
Direct NCI			30%	113,280	
Indirect NCI			7%	26,432	
Total NCI			37%	139,712	

CJE18: Elimination of dividends declared by C Co

Dr Dividend income	56,000		70%*80,000
Dr Non-controlling interests	24,000		30%*80,000
Cr Dividend declared		80,000	

CJE19: Allocation of post-acq revaluation reserves to total NCI of C Co				
Dr Share of Revaluation reserves to NCI	1,850			
Cr Non-controlling interests		1,850		
Revaluation reserves at 31 Dec 20x6				100,000
Revaluation reserves at 1 Aug 20x4				95,000
Change in revaluation reserves				5,000
Direct NCI			30%	1,500
Indirect NCI			7%	350
Total NCI			37%	1,850

CJE20: Eliminate intercompany payable and receivable

Dr Intercompany payable	1,500,000		
Cr Intercompany receivable		1,500,000	

Analytical check of Non-controlling interests

Total Non-controlling interests as at 31 December 20x6	**Total NCI**	**NCI of B Co and indirect NCI of C Co**	**Direct NCI of C Co**
CJE1: Elimination of investment in B Co, C Co and D Co	532,400	226,900	305,500
CJE5: Allocate share of post-acq RE of B Co to NCI of B Co	15,020	15,020	
CJE6: Eliminate dividends declared by B Co	(13,000)	(13,000)	
CJE7: Allocate share of current income of B Co to NCI of B Co	105,200	105,200	
CJE12: Adjustment of cost of sales of undervalued inventory	(26,640)	(5,040)	(21,600)
CJE13: Tax effects of CJE12	5,328	1,008	4,320
CJE16: Allocation of post-acquisition RE to total NCI of C Co	48,100	9,100	39,000
CJE17: Allocation of current profit after tax to total NCI of C Co	139,712	26,432	113,280
CJE18: Elimination of dividends declared by C Co	(24,000)		(24,000)
CJE19: Allocation of post-acq revaluation reserves to total NCI of	1,850	350	1,500

C Co			
	783,970	365,970	418,000

Analytical check on non-controlling interests of B Co

B Co's shareholders' equity as at 31 December 20x6	2,990,200	1,200,000 + 1,790,200
Less investment in C Co	(900,000)	
	2,090,200	
NCI's share of book value of equity of B as at 31 Dec 20x6	209,020	10%*2,090,200
NCI's share of intangible asset, after tax as at 31 Dec 20x6	10,800	10%*9/10*80%*150,000
NCI of B Co's goodwill as at 31 Dec 20x6	49,970	
NCI of B Co	269,790	

Analytical check of NCI of C Co

		Indirect NCI	**Direct NCI**
C Co's shareholders' equity as at 31 December 20x6	1,374,000	520,000 + 754,000	+ 100,000
C Co's total NCI share of equity as at 31 December 20x6	508,380	96,180	412,200
Goodwill attributable to C Co's NCI	5,800		5,800
Total NCI of C Co as at 31 December 20x6	514,180	96,180	418,000
NCI of B Co and C Co	**783,970**	365,970	418,000

CJE21: Allocate share of post-acquisition RE of D Co to NCI

Dr ORE	164,000		
Cr NCI		164,000	
RE of D as at 1 Jan 20x6			800,000
RE of D as at date of acquisition			390,000
Change in RE			410,000
Share of D's change in RE			164,000

CJE22: Elimination of dividends declared by D Co

Dr Dividend income		30,000	
Dr NCI		20,000	
Cr Dividend declared			50,000

CJE23: Share of current income to NCI

Dr Share of current income to NCI	198,400		
Cr NCI		198,400	

Workings:

NPAT			496,000
Share of adjusted NPAT of D			198,400

Analytical check of NCI in D Co:

Book value of shareholders' equity of D	2,046,000	800,000+1,246,000
NCI's share of D's identifiable net assets	818,400	
NCI's goodwill	224,000	
	1,042,400	
CJE1: Elimination of investment in B Co, C Co and D Co	700,000	
CJE21: Allocate share of post-acq RE of D Co to NCI	164,000	
CJE22: Elimination of dividends declared by D Co	(20,000)	
CJE23: Share of current income to NCI	198,400	
NCI in D Co as at 31 Dec 20x6	**1,042,400**	-

Analytical check of closing retained earnings

P's RE	5,596,000
B's RE	1,790,200
C's RE	754,000
D's RE	1,246,000
CJE1: Elimination of investment in B Co, C Co and D Co	(1,360,000)
CJE2: Remeasurement gain in previously held interests of D Co	380,000
CJE3: Amortization of intangible asset	(15,000)
CJE4: Tax effects of CJE3	3,000
CJE5: Allocate share of post-acq RE of B Co to NCI of B Co	(15,020)
CJE6: Eliminate dividends declared by B Co	13,000
CJE7: Allocate share of current income of B Co to NCI of B Co	(105,200)
CJE8: Adjustment of interest profit (transfer from P to B)	(80,000)
CJE9: Tax effects of CJE8	16,000
CJE10: Adjustment of excess depreciation on fixed assets	4,000
CJE11: Tax effects of CJE10	(800)
CJE12: Adjustment of cost of sales of undervalued inventory	(45,360)
CJE13: Tax effects of CJE12	9,072
CJE14: Cost of sales of undervalued inventory	(8,000)
CJE15: Tax effects of CJE14	1,600
CJE16: Allocation of post-acquisition RE to total NCI of C Co	(48,100)
CJE17: Allocation of current profit after tax to total NCI of C Co	(139,712)
CJE18: Elimination of dividends declared by C Co	24,000
CJE21: Allocate share of post-acq RE of D Co to NCI	(164,000)
CJE22: Elimination of dividends declared by D Co	20,000
CJE23: Share of current income to NCI	(198,400)
Consolidated RE at 31 Dec 20x6	**7,677,280**

Analytical check

P's RE	5,596,000	
P's share of B Co's post-acquisition RE	1,026,180	90%*(1,790,200 − 650,000)
P's share of C Co's post-acquisition RE	273,420	63%*(754,000 − 320,000)
P's share of D Co's post-acquisition RE	513,600	60%*(1,246,000 − 390,000)
P's share of cumulative amortization of intangible asset of B Co	(10,800)	90%*80%*150,000/10
P's share of cost of sales of undervalued inventory of C Co	(40,320)	63%*80%*80,000
P's share of unrealized interest income (P to B Co)	(60,800)	80%*(120,000 − 80,000+ 100,000-60,000)*9.5/10
Remeasurement gain in D Co	380,000	
Consolidated RE at 31 Dec 20x6	**7,677,280**	

APPENDIX 7B

Change in Significant Influence

ACQUISITION OF AN ASSOCIATE IN STAGES

Just as an acquirer may obtain control of an acquiree in stages as discussed in Chapter 7, an investor may also obtain significant influence over an associate in stages.[7] This can be achieved either by the investor acquiring additional equity interests or it could be due to circumstantial changes that result in the investor obtaining significant influence. Such changes in circumstances will be further discussed in the section below.

While IFRS 3 provides detailed guidance on how the acquirer accounts for business combination achieved in stages, IAS 28 is silent on how piecemeal acquisition of associates[8] should be accounted for by the investor. Paragraph 10 of IAS 28 requires the investment in associate to be recognized at cost on initial recognition when applying the equity method. However, neither the IFRS Standard nor other IFRS Standards define cost and accordingly how cost is to be measured for associates acquired in stages.

In our view, an entity has an accounting policy choice to account for a step acquisition of an associate using either the cost approach or the fair value approach. The entity will have to apply the accounting policy consistently for similar transactions, other events and conditions as required under paragraph 13 of IAS 8 *Accounting policies, Changes in Accounting Estimates and Errors*.

Cost approach

In applying the cost approach in accounting for acquisition of associates in stages, an entity has to determine the measurement for three components in the investment in associate. They are namely:

1. Cost of the investment in associate
2. Share of post-acquisition results and other comprehensive income for the previously held tranche of equity interests – specifically whether the investor should recognize the share of post-acquisition results and other comprehensive

income for the initial tranche held when it first applies equity accounting after obtaining significant influence in the investee

3. Goodwill in the investment (or conversely gain on bargain purchase)

Insofar as the cost of investment of the associate is concerned, it is generally accepted that the cost of investments in both the consolidated and separate financial statements[9] of the investor is the aggregate of the considerations given in exchange for the various tranches of equity interests plus any acquisition-related costs. To the extent that the initial tranche of investment is accounted for as a fair value through other comprehensive instrument (under IFRS 9) for which fair value changes are taken to other comprehensive income, the cumulative changes in fair value recognized in other comprehensive income is reversed through profit or loss to restate the carrying value of the initial tranche of investment back to cost when significant influence is obtained with the acquisition of the subsequent tranche.

page 575

Conversely, if the initial tranche of investment is measured at fair value through profit or loss, the fair value changes previously recognized in retained earnings are reversed through profit or loss[10] to restate the initial investment back to cost when significant influence is obtained with the acquisition of the subsequent tranche of equity interest.

In determining the second and third components above, there is diversity in practice. The diversity mainly stems from how the requirements of IAS 28 are interpreted in the context of step acquisition of associates. In particular,

1. Paragraph 32 of IAS 28 requires an investment to be accounted for using the equity method from the date significant influence is attained. The question is, in the case where an associate is acquired in stages, is a catch-up adjustment required in respect of the initial tranche of investment to reflect the application of the equity method when the investor obtains significant influence on acquiring the subsequent tranche of equity interests, that is, should the investor recognize the share of post-acquisition results and other comprehensive income for the initial tranche of equity interests at the point where significant influence is attained.
2. Paragraph 32 of IAS 28 further requires the difference between the cost of investment and the investor's share of the net fair value of the investee's identifiable assets and liabilities to be accounted for as (a) goodwill relating to an associate or a joint venture which is included in the carrying amount of the investment or (b) as income in the determination of the investor's share of the associate or joint venture's profit or loss in the period in which the investment is acquired, any excess of the investor's share of the net fair value of the investee's identifiable assets and liabilities over the cost of the investment. However, the standard is silent as to which date the fair value of the identifiable assets and liabilities should be determined in the case of a step acquisition. Should it be determined only at the date when the investment becomes an associate or should the fair value be determined at every date for which consideration was given that is, fair value of the identifiable assets should be determined for every tranche of investment until significant influence is attained. This question similarly extends to the calculation of the investor's share of the fair value adjustments in respect of its investment in the associate.

The interplay between the various interpretations of the responses to the questions above results in different variations of the cost approach. One of the acceptable approaches is shown in Illustration 7B.1. In this approach, the cost of the investment in associate acquired on a piecemeal basis is the aggregate of the considerations paid for each tranche and a catch-up adjustment of the share of investee's profit or loss as well as other comprehensive income for the first tranche is recorded. Changes in fair value of the net assets are also accounted for the first tranche.[11] Goodwill and the investor's share of any fair value adjustments are calculated at each stage of the acquisition.[12] Acquisition related costs are capitalized as part of the investment in associate.

page 576

ILLUSTRATION 7B.1 Applying the cost approach in step acquisition of associate

On 1 January 20x3, Entity X acquired a 10% interest in Entity Y for $100,000. The investment was designated and accounted for as a fair value through other comprehensive income financial instrument for which fair value changes are taken to other comprehensive income. On 1 January 20x5, Entity X purchased an additional 30% interest for $750,000 and obtained significant influence in Entity Y. Transaction costs associated with the purchase of the second tranche amounted to $15,000. The summarized financial information of Entity Y from 20x3 to 20x5 is set out below.

Changes in net assets of Entity Y		$
1 Jan 20x3	Net assets at book value	800,000
	Net assets at fair value	*820,000*
	Profit for the year	200,000
	Dividends paid	(50,000)
	Asset revaluation for the year — OCI	30,000
	Fair value changes of financial instruments for the year	8,000
31 Dec 20x3/1 Jan 20x4	Net assets at book value	988,000
31 Dec 20x4	Profit for the year	180,000
	Dividends paid	(10,000)
	Fair value changes of financial instruments for the year	5,000
31 Dec 20x4/1 Jan 20x5	Net assets at book value	1,163,000
	Net assets at fair value	*2,000,000*
31 Dec 20x5	Profit for the year	300,000
	Dividends paid	(20,000)
	Asset revaluation for the year — OCI	2,000
	Fair value changes of financial instruments for the year	3,000
31 Dec 20x5	Net assets at book value	1,448,000

The fair values of the 10% shareholdings in Entity Y as at 31 December 20x3 and 31 December 20x4 are $120,000 and $160,000, respectively. Assume that Entity X accounts for the investment in associate at cost in the separate financial statements. The effects of taxes are ignored in this illustration.

Calculate the financial impact of the step acquisition of associate and prepare the accounting entries for the transaction.

Analysis

Calculate the goodwill arising from the acquisition for each tranche

	$	
1st Tranche (Acquisition of 10% interest)		
Consideration paid	100,000	
Less: Fair value of net identifiable assets	(82,000)	(10%*820,000)
Goodwill	18,000	
2nd Tranche (Acquisition of 30% interest)		
Consideration paid	750,000	
Less: Fair value of net identifiable assets	(600,000)	(30%*2,000,000)

Goodwill	150,000
Total goodwill	168,000

In this variant of the cost approach, goodwill is calculated for each tranche of acquisition as the difference between the fair value of the consideration paid and the proportionate share of the fair value of the net identifiable assets of Entity Y at the date of each purchase. In other variations, goodwill is calculated only at the acquisition where significant influence is attained.

Calculate the fair value adjustments arising from acquisition for each tranche

	$	
1st Tranche (Acquisition of 10% interest)		
Proportionate share of fair value of net assets	82,000	(10%*820,000)
Proportionate share of book value of net assets	(80,000)	(10%*800,000)
Excess of fair value over net book value (Note 1)	2,000	
Amortization per year	1,000	
2nd Tranche (Acquisition of 30% interest)		
Proportionate share of fair value of net assets	600,000	(30%*2,000,000)
Proportionate share of book value of net assets	(348,900)	(30%*1,163,000)
Excess of fair value over net book value (Note 2)	251,100	
Amortization per year	25,110	

Note 1: Excess of fair value of net book value relates to sales order book. This intangible asset is amortized over two years.
Note 2: Excess of fair value of net book value relates to trademark. This intangible asset is amortized over ten years.

Calculation of the share of profits and other comprehensive income to be recognized for first tranche

	$	
FY2013		
Share of associate's profits for the year	20,000	(10%*200,000)
Less: Amortization of intangible asset – sales order book	(1,000)	
	19,000	
Share of other comprehensive income		
– asset revaluation for the year	3,000	(10%*30,000)
– fair value changes for the year	800	(10%*8,000)
	3,800	
FY2014		
Share of associate's profits for the year	18,000	(10%*180,000)

Less: Amortization of intangible asset – sales order book	(1,000)	
	17,000	
Share of other comprehensive income		
– fair value changes for the year	500	(10%*5,000)

In this variant of the cost approach, the investor will account for its share of associate's results and other comprehensive income based on its shareholdings interest for the first tranche. In other variants, the investor does not account for its share of the associate's results and other comprehensive income for the initial tranche of investment.

Calculation of the share of profits and other comprehensive income for FY2015

	$	
FY2015		
Share of associate's profits for the year	120,000	(40%*300,000)
Less: Amortization of intangible asset – trademark	(25,110)	
	94,890	
Share of other comprehensive income		
– asset revaluation for the year	800	(40%*2,000)
– fair value changes for the year	1,200	(40%*3,000)
	2,000	

Prepare accounting entries

In the separate financial statements

1 Jan 20x3	Dr Investment in FVOCI asset	100,000	
	Cr Cash		100,000
	Being cash paid for 10% investment		
31 Dec 20x3	Dr Investment in FVOCI asset	20,000	
	Cr Fair value reserves		20,000
	Being fair value changes for the year		
	Dr Cash	5,000	
	Cr Dividend income		5,000
	Being dividend income received		
31 Dec 20x4	Dr Investment in FVOCI asset	40,000	
	Cr Fair value reserves		40,000
	Being fair value changes for the year		

	Dr Cash	1,000	
	Cr Dividend income		1,000
	Being dividend income received		
1 Jan 20x5	Dr Profit or loss	60,000	
	Cr Investment in FVOCI asset		60,000
	Being reinstatement of FVOCI at fair value back to cost – Note 3		
	Dr Investment in associate	100,000	
	Cr Investment in FVOCI asset		100,000
	Being reclassification of investment in FVOCI asset to investment in associate		
	Dr Investment in associate	750,000	
	Cr Cash		750,000
	Being payment for 2nd tranche of investment in associate		
	Dr Investment in associate	15,000	
	Cr Cash		15,000
	Being capitalization of transaction costs		
31 Dec 20x5	Dr Cash	8,000	
	Cr Dividend income		8,000
	Being dividend income received		
	Dr Fair value reserves	60,000	
	Cr Retained earnings		60,000
	Being reclassification of fair value reserves to retained earnings		

Note 3: The cumulative changes in fair value recognized in other comprehensive income is reversed through profit or loss to restate the carrying value of the initial tranche of 10% investment back to cost when significant influence is obtained with the acquisition of the subsequent 30% interest.

In the consolidated financial statements

1 Jan 20x5	Dr Investment in associate	40,300		
	Cr Opening retained earnings		36,000	(19,000 + 17,000)
	Cr Share of associate's OCI		4,300	
	Being equity accounting for 1st tranche of investment			
	Dr Opening retained earnings	6,000		
	Cr Investment in associate		6,000	
	Being dividend income received from associate for FY2013 and FY2014			
31 Dec 20x5	Dr Investment in associate	96,890		
	Cr Share of associate's profits		94,890	
	Cr Share of associate's OCI		2,000	
	Being equity accounting for the year			
	Dr Dividend income	8,000		
	Cr Investment in associate		8,000	
	Being dividend income received from associate			

Compute the balance in the investment in associate account

Investment in associate account		$
1 Jan 20x3	Payment for 1st tranche of investment	100,000
31 Dec 20x3	Fair value changes for FY2013	20,000
31 Dec 20x4	Fair value changes for FY2014	40,000
1 Jan 20x5	Reversal of fair value changes on attainment of significant influence	(60,000)
1 Jan 20x5	Payment for 2nd tranche of investment	750,000
1 Jan 20x5	Transaction costs capitalized	15,000
1 Jan 20x5	Equity accounting for 1st tranche up to FY2014	40,300
1 Jan 20x5	Dividend income from associate for FY2013 and FY2014	(6,000)
31 Dec 20x5	Equity accounting for FY2015	96,890
31 Dec 20x5	Dividend income from associate	(8,000)
31 Dec 20x5	Balance as at end	988,190

page 580

Analytical check on the investment in associate account

Investment in associate account	$	$	
Book value of net assets of associate at 31 December 20x5			
– 1st tranche	144,840		(10%*1,448,000)
– 2nd tranche	434,400		(30%*1,448,000)
		579,200	
Goodwill			
– 1st tranche	18,000		
– 2nd tranche	150,000		

		168,000	
Unamortized fair value adjustments			
– 1st tranche	–		(2,000 – (2*1,000))
– 2nd tranche	225,990		(251,100 – 25,110)
		225,990	
Transaction costs capitalized		15,000	
		988,190	

Fair Value Approach

Under the fair value approach, the cost of the investment in associate is measured as the sum of the fair value of the previously held interest determined at the date when significant influence is obtained and the fair value of the consideration transferred in exchange for additional equity interests at the date when the investment becomes an associate.

The conceptual underpinning for the fair value approach is predicated on paragraph 26 in IAS 28 which explains that many of the consolidation procedures and the concepts used in accounting for the acquisition of a subsidiary (in IFRS 3) are also appropriate in the accounting for the acquisition of an investment in an associate or a joint venture and the application of the equity method. Since IFRS 3 requires previously held interests, in the case of a business combination achieved in stages, to be remeasured to fair value at the date where control is obtained with the resulting gain or loss recognized in profit or loss, this requirement can be applied by analogy to the previously held interests in cases where the associate is acquired in stages.

Hence, at the date where significant influence is attained, previously held interests are measured at fair value. As this initial investment should either be accounted for at fair value through profit or loss or fair value through other comprehensive income under IFRS 9, there would be no further changes to the carrying value. If the initial investment was accounted for at fair value through profit or loss, the gain or loss on fair value measurement would have been accounted for at profit or loss which is consistent with the requirement of IFRS 3. Hence, no other adjustment is required. Conversely, if the initial investment was accounted for as fair value through other comprehensive income under IFRS 9, the investor would reclassify the cumulative fair value changes that were recognized in other comprehensive income to profit or loss as if the investor had disposed of the initial investment directly.

In the case of goodwill determination, consistent with the approach in IFRS 3, goodwill (or gain on bargain purchase) is calculated only at the date when significant influence is achieved by comparing the cost of investment (which is the sum of the fair value of previously held interest and the fair value of the consideration transferred for the additional equity interests) with the investor's share of the fair value of the identifiable assets and liabilities at the date where significant influence is achieved.

ILLUSTRATION 7B.2 Applying the fair value approach in step acquisition of associate

Assume the same facts as Illustration 7B.1. Instead of using the cost approach, Entity X adopts the fair value approach to account for the step acquisition of associate – Entity Y.

Analysis

Calculate the goodwill on acquisition of associate

Goodwill	=	[Fair value of consideration transferred	+	Fair value of previously held interest at date where significant influence is attained]	−	Investor's share of fair value of net identifiable assets of associate at date where significant influence is attained
	=	(750,000	+	160,000)	−	(40%*2,000,000)
	=	910,000	−	800,000		
	=	110,000				

Calculate the fair value adjustments arising from acquisition of associate

	$	
Proportionate share of fair value of net assets	800,000	(40%*2,000,000)
Proportionate share of book value of net assets	(465,200)	(40%*1,163,000)
Excess of fair value over net book value (Note 1)	334,800	
Amortization per year	33,480	

Note 1: Excess of fair value of net book value relates to trademark. This intangible asset is amortized over 10 years.

Calculation of the share of profits and other comprehensive income for FY2015

	$	
FY2015		
Share of associate's profits for the year	120,000	(40%*300,000)
Less: Amortization of intangible asset – trademark	(33,480)	
	86,520	
Share of associate's other comprehensive income		
– asset revaluation for the year	800	(40%*2,000)
– fair value changes for the year	1,200	(40%*3,000)
	2,000	

Prepare accounting entries

In the separate financial statements

The accounting entries in the separate financial statements are the same as the entries passed in the cost approach.

In the consolidated financial statements

1 Jan 20x5	Dr Fair value reserves	60,000	
	Cr Other income–Profit or loss		60,000
	Being reclassification of amounts accumulated in fair value reserves to profit or loss		

	Dr Transaction costs–profit or loss	15,000	
	Cr Investment in associate		15,000
	Being transaction costs expensed in the consolidated financial statements		
31 Dec 20x5	Dr Investment in associate	88,520	
	Cr Share of associate's profits		86,520
	Cr Share of associate's OCI		2,000
	Being equity accounting for the year		
	Dr Opening retained earnings	8,000	
	Cr Investment in associate		8,000
	Being dividend income received from associate		

Under the fair value approach, equity accounting for the first tranche of investment is not performed and transaction costs are expensed instead of being capitalized in the investment in associate.

Compute the balance in the investment in associate account

Investment in associate account		$
1 Jan 20x3	Payment for 1st tranche of investment	100,000
31 Dec 20x3	Fair value changes for FY2013	20,000
31 Dec 20x4	Fair value changes for FY2014	40,000
1 Jan 20x5	Payment for 2nd tranche of investment	750,000
31 Dec 20x5	Equity accounting for FY2015	88,520
31 Dec 20x5	Dividend income from associate	(8,000)
31 Dec 20x5	Balance as at end	990,520

Analytical check on the investment in associate account

Investment in associate account	$	
Book value of net assets of associate at 31 December 20x5	579,200	(40%*1,448,000)
Goodwill	110,000	
Unamortized fair value adjustments	301,320	(334,800–33,480)
	990,520	

ACQUISITION OF ADDITIONAL INTERESTS IN AN ASSOCIATE

After an investor has obtained significant influence in an associate, that investor may acquire additional equity interests in that associate without gaining control, that is, the investor continues to have significant influence after the acquisition of the additional interests.

Similarly, IAS 28 does not provide guidance on such transactions. However, in this case, we believe that the transaction should be accounted for as follows:

1. Fair value of the consideration paid for the additional equity interests is added to the carrying value of the cost of investment in the associate.
2. Goodwill and the investor's share of any new fair value adjustments arising from the acquisition of this additional interest is calculated based on the fair value of the consideration given in exchange for the additional interests and the investor's incremental share of the fair value of identifiable assets and liabilities at the date of the additional purchase. The goodwill and unamortized portion of the fair value adjustments for this purchase of additional interests is aggregated with the goodwill and unamortized portion of the fair value adjustments for the existing tranche at the end of the financial period.
3. No remeasurement of the previously held interest in the associate to fair value is performed. Instead, the investor will continue to apply equity accounting in respect of the previously held interests by recognizing its share of profits and other comprehensive income to the extent of its previously held interests up to the date where the additional interests are acquired. Appropriate adjustments to the investor's share of the associate's profit or loss after acquisition should continue to be made in respect of the investor's share of the fair value adjustments recognized at the date where the associate was first acquired (for example additional depreciation based on the fair value of the assets at the date of acquisition of associate).
4. Subsequent to the acquisition to additional interests, the investor will recognize its share of results and other comprehensive income based on its new shareholding interests in the associate. Adjustments pertaining to the amortization of the fair value adjustments related to both the initial acquisition and subsequent purchase of additional interests should continue to be made.

This accounting treatment is appropriate regardless of which accounting policy the investor elects in accounting for the acquisition of associates in stages as discussed above.

The rationale for the accounting treatment described above is attributable to the fact that there is no change in the nature of the investee post-acquisition of the additional interests. The investee continues to be an associate for which the investor has significant influence. As control is not obtained with the purchase of the additional interests, the transaction does not constitute a significant economic event which warrants a change in the classification and measurement of the investment as set out in IFRS 3. Hence, no remeasurement of the previously held interests to fair value is carried out and the cost accumulation approach where the fair value of the consideration paid for the additional interest is "accumulated" in the cost of investment in associate is the most appropriate method of accounting.

ILLUSTRATION 7B.3 Acquisition of additional interests in associate while maintaining significant influence

Assume the base facts as set out in Illustration 7B.1. On 1 July 20x6, Entity X acquired an additional 5% interest in Entity Y for $120,000. This is in addition to the 40% it already holds. Entity X determines that it continues to have significant influence after the additional acquisition of 5% interest. Transaction costs amounted to $8,000. The summarized financial information of Entity Y from 20x5 to 20x6 is given below. The effects of taxes are ignored in this illustration.

page 584

Changes in net assets of Entity Y		$
31 Dec 20x5/1 Jan 20x6	Net assets at book value	1,448,000
	Profit for the period from 1 Jan to 30 Jun 20x6 .	200,000
	Fair value changes of financial instruments for the period from 1 Jan to 30 Jun 20x6	1,500

30 Jun 20x6/1 Jul 20x6	Net assets at book value	1,649,500
	Net assets at fair value	*1,800,000*
	Profit for period from 1 Jul to 31 Dec 20x6	250,000
	Dividends paid	(10,000)
	Fair value changes of financial instruments from 1 Jul to 31 Dec 20x6	2,000
31 Dec 20x6	Net assets at book value	1,891,500

Calculate the financial impact of the acquisition of additional interests and prepare the accounting entries for the transaction.

Analysis

Calculate the goodwill arising from the additional acquisition of 5% interest

	$	
Acquisition of additional 5% interest		
Consideration paid	120,000	
Less: Fair value of net identifiable assets	(90,000)	(5%*1,800,000)
Goodwill	30,000	
Goodwill from 1st and 2nd tranches	168,000	(From Illustration 7B.1)
Total goodwill	198,000	

Calculate the fair value adjustments arising from the acquisition of additional 5% interest

	$	
Acquisition of additional 5% interest		
Proportionate share of fair value of net assets on 1 Jul 20x6	90,000	(5%*1,800,000)
Proportionate share of book value of net assets on 1 Jul 20x6	(82,475)	(5%*1,649,500)
Excess of fair value over net book value (Note 1)	7,525	
Amortization per year	1,505	

Note 1: Excess of fair value of net book value relates to patented technology. This intangible asset is amortized over 5 years.

Calculation of the share of profits and other comprehensive income for the year

	$	
FY2016		
Share of associate's profits for the year (Note 2)	192,500	((40%*450,000) + (5%*250,000))
Less: Amortization of intangible asset – trademark	(25,110)	(From Illustration 7B.1)
Less: Amortization of intangible asset – patented	(753)	(1,505/2) – Amortization for half year

technology			
	166,637		
Share of associate's other comprehensive income – fair value changes for the year (Note 2)	1,500	((40%*(1,500 + 2,000)) + (5%*2,000))	

Note 2: This relates to Entity X's 40% share of profits and other comprehensive income in Entity Y from 1 January 20x6 to 31 December 20x6 + 5% of profits and other comprehensive income from 1 July 20x6 to 31 December 20x6.

Prepare accounting entries

In the separate financial statements

1 Jul 20x6	Dr Investment in associate	120,000	
	Cr Cash		120,000
	Being payment for additional 5% interest in associate		
	Dr Investment in associate	8,000	
	Cr Cash		8,000
	Being capitalization of transaction costs		
31 Dec 20x6	Dr Cash	4,500	
	Cr Dividend income		4,500
	Being dividend income received		

In the consolidated financial statements

31 Dec 20x6	Dr Investment in associate	168,137	
	Cr Share of associate's profits		166,637
	Cr Share of associate's OCI		1,500
	Being equity accounting for the year		
	Dr Dividend income	4,500	
	Cr Investment in associate		4,500
	Being dividend income received from associate		

Compute the balance in the investment in associate account

Investment in associate account		$
31 Dec 20x5/1 Jan 20x6	Balance as at beginning (From Illustration 7B.1)	988,190
1 Jul 20x6	Payment for additional 5% interests	120,000
1 Jul 20x6	Transaction costs capitalized	8,000
31 Dec 20x6	Equity accounting for FY2016	168,137
31 Dec 20x6	Dividend income from associate	(4,500)
31 Dec 20x6	Balance as at end	1,279,827

Analytical check on the investment in associate account

Investment in associate account	$	$	
Book value of net assets of associate at 31 December 20x6			
– Initial shareholdings (40%)	756,600		(40%*1,891,500)
– Additional shareholdings (5%)	94,575		(5%*1,891,500)
		851,175	
Goodwill			
– Initial shareholdings (40%)	168,000		(From Illustration 7B.1)
– Additional shareholdings (5%)	30,000		(From Illustration 7B.1)
		198,000	
Unamortized fair value adjustments			
– Initial shareholdings (40%)	200,880		(251,100 - (2*25,110))
– Additional shareholdings (5%)	6,772		(7,525 - (1,505/2))
		207,652	
Transaction costs capitalized		23,000	(15,000 + 8,000)
		1,279,827	

LOSS OF SIGNIFICANT INFLUENCE

Paragraph 22 of IAS 28 requires the equity method to be discontinued from the date when the investment ceases to be an associate, that is, when the investor loses significant influence. The situations in which this could happen would include the following scenarios:

1. When the investment becomes a subsidiary, i.e. control is obtained.
2. When the interests in the investment is disposed of and the investor retains a remaining interest which is a financial asset in the former associate.
3. When the interests in the investment is fully disposed.

INVESTMENT IN ASSOCIATE BECOMES SUBSIDIARY

As discussed above, an investor may obtain control over its existing associate through the acquisition of additional interests such that the associate becomes a subsidiary. In this case, the investor has effectively acquired a subsidiary in stages. Paragraph 22(a) of IAS 28 specifically requires the investor to discontinue the use of equity method and account for the investment in accordance with IFRS 3 and IFRS 10. Refer to Chapter 7 for a more detailed discussion of the accounting treatment for business combination achieved in stages.

PARTIAL DISPOSAL OF ASSOCIATE WHILE MAINTAINING SIGNIFICANT INFLUENCE

An investor may dispose its equity interests in an associate partially and continue to maintain significant influence based on its remaining shareholding interest, that is, the investee remains as an associate. Conversely, the decrease in interest in the associate may also be brought about by changes in circumstances. Such changes in circumstances will be discussed further in the section below.

IAS 28 is silent as to how the cost of the investment partially disposed should be measured and whether a gain or loss on disposal should be recognized in such transactions. While IFRS 10 requires a partial disposal of a subsidiary in which the parent does not lose control to be accounted for as an equity transaction, we believe that this principle cannot be extended to that of a partial disposal of an associate where the investor does not loss significant influence. This is because the associate is not part of the group as defined in IFRS 10. Therefore, the partial disposal of the associate is not a transaction with equity owners. Instead, it is a transaction (i.e., disposal) with third parties. Accordingly, the proportionate cost of investment of the associate disposed would be derecognized with the gain or loss on disposal recognized in the profit or loss. This argument is further augmented by the paragraph 25 of IAS 28 which requires the investor to "reclassify to profit or loss, the proportion of the gain or loss that had been previously recognized in other comprehensive income relating to the reduction in the ownership interests if that gain or loss would be required to be reclassified to profit or loss on the disposal of the related assets or liabilities" in transactions when the investor's ownership interests in an associate is reduced but the investor continues to apply the equity method.

Similarly, paragraph 48C of IAS 21 also requires the proportionate share of the cumulative amount of exchange differences recognized in other comprehensive income and accumulated in equity to be reclassified to profit or loss in the case of partial disposals of foreign operations. Partial disposal of foreign operations in this case would include partial disposal of interests in a foreign associate or joint arrangement. Accordingly, we believe that a gain or loss on disposal should be recognized in the profit or loss when an investor makes a partial disposal of its investment in the associate.

Specifically, the investor should:

1. Derecognize the cost of investment in the associate on a proportionate basis on the basis of the percentage of shareholdings disposed. The cost of investment derecognized would include the proportionate share of goodwill and the investor's share of unamortized portion of the fair value adjustments at the date of disposal.
2. Recognize the fair value of consideration received from the transaction, event, or circumstance that result in the partial disposal.
3. Reclassify on a proportionate basis to profit or loss[13] or transfer directly to retained earnings[14] if required by other IFRS Standards, the amounts recognized in other comprehensive income in relation to the disposed interests.
4. Recognize any resulting difference as a gain or loss in profit or loss.
5. Account for the remaining interests in the associate in accordance with IAS 28.

ILLUSTRATION 7B.4 Partial disposal of associate while maintaining significant influence

Entity I has 45% interest in Entity A. The shareholdings were previously acquired on 31 December 20x2 for consideration amounting to $65,000. Entity I's share of the fair value of the net identifiable assets and the book value of the net assets of Entity A at the date of acquisition were $55,000 and $49,500, respectively.

On 1 January 20x5, Entity I disposed 10% of its interest in Entity A at $65,000. Entity I has assessed that it continues to maintain significant influence over Entity A subsequent to the disposal. The carrying value of the net assets in Entity A at the date of acquisition and disposal is set out below.

Statement of Financial Position	$	$

	1 Jan 20x5	31 Dec 20x2
Other assets	700,000	310,000
Other liabilities	(150,000)	(200,000)
	550,000	110,000
Represented by:		
Share capital	10,000	10,000
Retained earnings	500,000	100,000
Asset revaluation reserves	25,000	–
Fair value reserves	15,000	–
	550,000	110,000

Calculate the financial impact of the partial disposal of associate and prepare the accounting entries for the transaction. The effects of taxes are ignored in this illustration.

Analysis

Calculate the goodwill and fair value adjustments on acquisition of the associate

	$	
Consideration paid	65,000	
Less: Fair value of net identifiable assets	(55,000)	
Goodwill	10,000	
Proportionate share of fair value of net assets	55,000	
Proportionate share of book value of net assets	(49,500)	(45%*110,000)
Excess of fair value over net book value (Note 1)	5,500	
Amortization per year	1,100	

Note 1: Excess of fair value of net book value relates to patented technology. This intangible asset is amortized over 5 years.

Calculate the gain on disposal in the separate and consolidated financial statements

In the separate financial statements

	$
Sales proceeds	65,000
Less: Cost of investment in separate financial statements (Note 2)	(14,444)
Gain on disposal	50,556

Note 2: Derived as (10%/45%)* initial cost of US$65,000.

In the consolidated financial statements

	$
Sales proceeds	65,000
Less: Carrying value of net assets (Note 3)	(57,956)
Gain on disposal (Note 4)	7,044

Note 3: The carrying value of net assets is derived as:

Investment in associate before disposal	$	
Cost	65,000	
Share of post-acquisition profits	177,800	(45%*(500,000 − 100,000) − (1,100*2))
Share of associate's OCI	18,000	(45%*(25,000 + 15,000))
	260,800	
Analytical check:		
Book value of net assets of associate	247,500	(45%*550,000)
Goodwill	10,000	
Unamortized intangible asset	3,300	(5,500 − (1,100*2))
	260,800	
Therefore:		
10% of carrying value disposed	57,956	(10%/45%*260,800)

Note 4: Gain on disposal at the consolidated financial statements level is effectively as follows:

Gain on disposal at consolidated financial statements level	=	Gain on disposal in separate financial statements	- Entity I's disposed share of post-acquisition profits and OCI of Entity A
	=	50,556	- (10%*(500,000 − 100,000)) − ((10%/45%)*(2*1,100))
	=	50,556	- 43,512
	=	7,044	

Prepare the accounting entries

In the separate financial statements

1 Jan 20x5	Dr Cash	65,000	
	Cr Cost of investment		14,444
	Cr Gain on disposal		50,556
	Being gain on disposal of investment in the separate financial statements		

In the consolidated financial statements

1 Jan 20x5	Dr Gain on disposal	43,512	

	Cr Beginning retained profits	43,512

Being gain on disposal of investment in the consolidated financial statements

Calculate the balance in the investment in the associate in the consolidated financial statements post-disposal and perform an analytical check

Investment in associate account		$
1 Jan 20x5	Balance as at beginning	260,800
1 Jan 20x5	Disposal of 10% interest	(57,955)
31 Dec 20x6	Balance as at end	202,845

Analytical check on the investment in associate account

Investment in associate account	$	
Book value of net assets of associate at 1 January 20x5	192,500	(35%*550,000)
Goodwill	7,778	((35%/45%)*10,000)
Unamortized fair value adjustments	2,567	((35%/45%)*3,300)
	202,845	

RETAINED INTEREST IN FORMER ASSOCIATE IS A FINANCIAL ASSET

An entity may dispose a significant portion of its interests in the investment in the associate such that it loses significant influence but continues to retain some interest in the former associate.

In this situation, paragraph 22 of IAS 28 requires the equity method to be discontinued from the date when the investment ceases to be an associate. If the retained interest in the former associate is a financial asset, that retained interest shall be measured at fair value at the date where significant influence is lost. This fair value of the retained interest is regarded as its fair value on initial recognition as a financial asset in accordance with IFRS 9. The entity will recognize in profit or loss any difference between (1) the fair value of any retained interest and any proceeds received from disposing part of the interest in the associate and (2) the carrying amount of the investment at the date the equity method was discontinued.

Paragraph 22(c) of IAS 28 further requires the entity to account for all amounts previously recognized in other comprehensive income in relation to that investment on the same basis as would have been required if the investee had directly disposed of the related assets or liabilities when an entity discontinues the use of the equity method.

ILLUSTRATION 7B.5 Disposal of interest while retaining financial asset

Assume the same fact pattern in Illustration 7B.4. Instead of disposing 10% interest, Entity I sold off 35% shareholding interests in the associate to a third party for $250,000 while retaining 10% interest in Entity A on 1 January 20x5. As a result of the disposal, Entity I assessed that it lost significant influence over Entity A. The fair value of the remaining 10% interest at the date of disposal is $25,000.

Analysis

Calculate the gain on disposal in the separate and consolidated financial statements

In the separate financial statements

	$
Sales proceeds	250,000
Less: Cost of investment in separate financial statements (Note 1)	(50,556)
Gain on disposal	199,444

Note 1: In the separate financial statements, the carrying value of the investment derecognized is based on the proportionate share of interests disposed. This is derived as (35%/45%)* initial cost of US$65,000.

In the consolidated financial statements

	$
Sales proceeds	250,000
Add: Fair value of 10% interest retained	25,000
Add: Share of associate's OCI reclassified to profit or loss – fair value changes (Note 2)	6,750
Less: Carrying value of net assets (Note 3)	(260,800)
Gain on disposal (Note 4)	20,950

Note 2: The cumulative balance in the fair value reserve is reclassified from equity to profit or loss upon disposal of the associate for which significant influence is lost in accordance with paragraph 22(c) of IAS 28.

Note 3: As Entity I has lost significant influence with the disposal of the 35% interest, the entire carrying amount of investment in associate is derecognized. The carrying value of net assets is derived as:

Investment in associate before disposal	$	
Cost	65,000	
Share of post-acquisition profits	177,800	(45%*(500,000 − 100,000) − (1,100*2))
Share of associate's OCI	18,000	(45%*(25,000 + 15,000))
	260,800	

Note 4: Gain on disposal at the consolidated financial statements level is effectively as follows:

Gain on disposal at consolidated financial statements level	=	Gain on disposal in separate financial statements	−	Entity I's disposed share of post-acquisition profits and OCI of Entity A	+	Fair value uplift of retained interest at date of disposal
	=	199,444	−	(45%*(500,000 − 100,000) (1,100*2)) + (45%*25,000)[+]	+	(25,000 − 14,444)
	=	199,444	−	(177,800 + 11,250)	+	10,556
	=	20,950				

+Only Entity I's share of asset revaluation reserve is included here as paragraph 41 of IAS 16 stipulates that the revaluation surplus may be transferred directly to retained earnings when the asset is derecognized and such transfers are not made through profit or loss. Entity I's share of fair value reserves were already included in the computation of gain or loss on disposal as explained in Note 2 above.

Prepare the accounting entries

In the separate financial statements

1 Jan 20x5	Dr Cash	250,000	
	Cr Cost of investment		50,556
	Cr Gain on disposal		199,444
	Being gain on disposal of investment in the separate financial statements		
	Dr Investment in FVOCI equity investment	10,556	
	Cr Profit or loss		10,556
	Being difference between cost and fair value of FVOCI instrument taken to profit or loss		

In the consolidated financial statements

1 Jan 20x5	Dr Investment in FVOCI equity investment	10,556	
	Dr Gain on disposal	178,494	
	Cr Opening retained earnings − post acquisition profits (Note 5)		177,800
	Cr Opening retained earnings − reclassification of asset revaluation reserves to beginning retained profit (Note 5)		11,250
	Being gain on disposal of investment in the consolidated financial statements		
	Dr Profit or loss	10,556	
	Cr Investment in FVOCI equity investment		10,556
	Being reversal of accounting entry passed in the separate financial statements		

Note 5: In practice, these two entries are aggregated together. We have separated the entries here for purposes of demonstrating the reclassification of the asset revaluation reserves to the beginning retained profits without recycling through profit or loss.

FULL DISPOSAL OF ASSOCIATE

When the shareholdings in an associate is fully disposed, the entity will discontinue the use of the equity method from the date when its investment ceases to be an associate in accordance with paragraph 22 of IAS 28. The carrying value of the associate at the date of disposal will be derecognized and the entity will recognize in profit or loss any difference between (1) the fair value of the consideration received from the disposal of the associate and (2) the carrying amount of the investment at the date of disposal. Furthermore, all amounts previously recognized in other comprehensive income in respect of that investment in associate will be accounted for on the same basis as it would have been required had the investee disposed of the related assets or liabilities directly in accordance with paragraph 22(c) of IAS 28.

DEEMED ACQUISITION OR DISPOSAL OF ASSOCIATES

As discussed above, apart from a direct acquisition or disposal of equity interests, an investor's interest in an associate can also be increased or reduced by way of changes in circumstances. Such changes in ownership interests due to circumstantial changes are commonly known as deemed acquisitions or deemed disposals in practice.

Deemed acquisitions and disposals may arise due to a number of reasons. Set out in the Table 7B.1 are some situations that could give rise to such transactions.

TABLE 7B.1 Scenarios on deemed acquisitions and deemed disposals

Deemed acquisitions	Deemed disposals
• Associate repurchases its ordinary shares held by other parties resulting in the investor increasing its relative shareholdings • Rights issue by associate is subscribed in full by investor but not the rest of the shareholders • Changes in composition of board of directors such that investor attains significant influence or control • Capitalization of amounts and loans due to the investor by the associate	• Investor does not subscribe or subscribes partially the rights issue undertaken by the associate • Associate issues ordinary shares upon the exercise of options held by parties other than the investor (e.g., employees on exercise of the vested employee share options) • Associate issues additional ordinary shares to Third parties other than the investor to raise funding • Changes in composition of board of directors such that investor loses significant influence • Conversion of convertible loans issued by associate into ordinary shares by debt holders

In order to identify and correctly account for these transactions, the entity must analyze the circumstances that lead to the deemed acquisitions or disposals carefully. As evident from the table above, consideration may or may not be transferred and the investor's absolute shareholding in the associate may or may not change after the transaction. We will discuss the accounting implications of deemed acquisitions and disposals below.

DEEMED ACQUISITIONS

An associate may undertake a share buy-back exercise in which ordinary shares held by other shareholders are repurchased. Subsequent to the exercise, there is an overall reduction in the shares in issue for the associate. In this case, there is no change to the investor's absolute shareholdings as the shares bought back are those of the other shareholders and not that of the investor. However, given the smaller share capital base after the share buy-back exercise, this results in an increase in the investor's shareholding interests relative to other shareholders. In substance, there has been an acquisition of "additional" interests in the associate, albeit with no consideration paid by the investor in this case. In this scenario, the consideration paid for the acquisition effectively came from the associate as the associate is the party that bought back the shares.

Conversely, consideration may also be transferred by the investor in another scenario. For instance, the associate may increase its paid up capital by undertaking a rights issue exercise to its existing shareholders. The investor subscribes for the rights in full in exchange for cash consideration paid. However, the rest of the shareholders may not take up their proportionate share of the rights. In this case, the investor's absolute shareholdings (in terms of the actual number of shares held in the associate) and its relative shareholdings to other shareholders increases, thereby resulting in an effective acquisition.

In all cases, the investor must analyze the nature of its shareholdings subsequent to the deemed acquisition of additional interests in the associate. The investor may continue to exert significant influence over the associate post-transaction. In this situation, the transaction will be an acquisition of additional interest in an associate while significant influence is still maintained. Conversely, the investor may gain control as a result of the deemed acquisition such that the associate becomes a subsidiary. In this case, the transaction will be accounted for as a business combination achieved in stages or business combination achieved without the transfer of consideration[15] under IFRS 3.

ILLUSTRATION 7B.6 Share-buyback by associate

On 1 January 20x3, Entity I acquired 30,000 shares which represents 30% interest in Entity A at consideration of $200,000. The book and fair value of the net assets of Entity A at the date of acquisition was $500,000 and $620,000, respectively.

On 1 July 20x4, Entity A embarked on a share buy-back exercise in which 25,000 of the shares held by the other investors are repurchased by Entity A at consideration of $250,000. The shares bought back are cancelled. Entity A assessed and concluded that it continues to have significant influence in Entity A after the share buy-back.

The following table shows the movement of net assets of Entity A from 20x3 to 20x4.

Changes in net assets of Entity A		$
1 Jan 20x3	Net assets at book value	500,000
	Net assets at fair value	*620,000*
	Profit for the year	200,000
	Asset revaluation for the year	50,000
	Fair value changes of financial instruments for the year	9,000
31 Dec 20x3/1 Jan 20x4	Net assets at book value	759,000
	Profit for period from 1 Jan 20x4 to 30 Jun 20x4	120,000
	Fair value changes of financial instruments from 1 Jan 20x4 to 30 Jun 20x4	5,000
30 Jun 20x4	Net assets at book value	884,000
	Less: Cash paid to repurchase shares	(250,000)
	Net assets at book value after share buyback	634,000
	Net assets at fair value	*700,000*
31 Dec 20x4	Profit for period from 1 Jul 20x4 to 31 Dec 20x4	300,000
	Fair value changes of financial instruments from 1 Jul 20x4 to 31 Dec 20x4	(2,000)
	Net assets at book value	932,000

Calculate the financial impact of the share-buyback and prepare the accounting entries in both the consolidated and separate financial statements.

Analysis

Calculate the changes in the effective shareholding interest subsequent to the share buy-back by Entity A and analyze the impact of the transaction

	Before transaction	After transaction
Shares held by Entity I	30,000	30,000
Shares in issue by Entity A	100,000	75,000
Percentage shareholdings	30%	40%

Subsequent to the share buyback, Entity I's shareholding interest in Entity A relative to other shareholders increases from 30% to 40%, that is, effectively there was a deemed acquisition. This is notwithstanding that there was no increase in terms of the absolute shareholdings held by Entity I, that is, it continues to hold 30,000 shares before and after the transaction.

In essence, it can be argued that Entity I's cost of the deemed acquisition of the 10% is essentially it share of the consideration paid by Entity A for the shares repurchased, that is, $75,000 which is derived as 30%*$250,000. Effectively, Entity I has exchanged it share of cash of $75,000 in Entity A in exchange for the additional 10% relative interest in the net assets of the associate.

Therefore, the deemed cost of the acquisition to be added to the investment in Entity A in Entity I's consolidated financial statements is the amount that Entity I has "paid" (via Entity A) to acquire the additional shareholding interest in Entity A of $75,000 (i.e., 30%*$250,000).

At the same time, Entity I will reduce the carrying value of the investment in Entity A in the consolidated financial statements by its share of the decrease in the net assets of Entity A following the cash consideration paid to the other shareholders of $75,000 (i.e., 30%*$250,000). Therefore, on a net basis, there is no change to the carrying value of the investment in Entity A at the point of the deemed acquisition.

In this case, as Entity I continues to have significant influence post transaction, Entity I will account for the deemed acquisition in accordance with the guidance discussed in the section on acquisition of additional interests in associate.

In Entity I's separate financial statements, there is no impact to the carrying value of investment in Entity A as there is no change to Entity I's absolute shareholding interest in Entity A. Accordingly, no accounting entries are effected in the separate financial statements.

Calculate the goodwill on acquisition of the separate tranches

	$	
1 st Tranche (Acquisition of 30% interest)		
Consideration paid	200,000	
Less: Fair value of net identifiable assets	(186,000)	(30%*620,000)
Goodwill	14,000	
2nd Tranche (Deemed acquisition of 10% interest)		
Deemed consideration paid	75,000	See explanation above.
Less: Fair value of net identifiable assets	(70,000)	(10%*700,000)
Goodwill	5,000	
Total goodwill	19,000	

Calculate the fair value adjustments arising from the transaction

$

1st Tranche (Acquisition of 30% interest)		
Proportionate share of fair value of net assets	186,000	(30%*620,000)
Proportionate share of book value of net assets	(150,000)	(30%*500,000)
Excess of fair value over net book value (Note 1)	36,000	
Amortization per year	7,200	
2nd Tranche (Deemed acquisition of 10% interest)		
Proportionate share of fair value of net assets	70,000	(10%*700,000)
Proportionate share of book value of net assets	(63,400)	(10%*634,000)
Excess of fair value over net book value (Note 2)	6,600	
Amortization per year	660	

Note 1: Excess of fair value of net book value relates to patented technology. This intangible asset is amortized over five years.

Note 2: Excess of fair value of net book value relates to trademark. This intangible asset is amortized over ten years.

Calculation of the share of results and other comprehensive income for FY2013 and FY2014

	$	
FY2013		
Share of associate's profits for the year	60,000	(30%*200,000)
Less: Amortization of intangible asset – patented technology	(7,200)	
	52,800	
Share of other comprehensive income		
– asset revaluation for the year	15,000	(30%*50,000)
– fair value changes for the year	2,700	(30%*9,000)
	17,700	
FY2014 – 1 Jan 20x4 up to 30 June 20x4		
Share of associate's profits for the period	36,000	(30%*120,000)
Less: Amortization of intangible asset – patented technology	(3,600)	(7,200/2)
	32,400	
Share of other comprehensive income		
– fair value changes for the period	1,500	(30%*5,000)
FY2014 – 1 Jul 20x4 to 31 Dec 20x4		
Share of associate's profits for the period	120,000	(40%*300,000)
Less: Amortization of intangible asset – patented technology	(3,600)	(7,200/2)
Less: Amortization of intangible asset – trademark	(330)	(660/2)
	116,070	

Share of other comprehensive income		
– fair value changes for the period	800	(40%*2,000)

Prepare accounting entries

In the separate financial statements

1 Jan 20x3	Dr Investment in associate	200,000	
	Cr Cash		200,000
	Being payment for 30% interest in associate		

In the consolidated financial statements

31 Dec 20x3	Dr Investment in associate	70,500	
	Cr Share of associate's profits		52,800
	Cr Share of associate's OCI		17,700
	Being equity accounting for the year		
30 June 20x4	Dr Investment in associate	33,900	
	Cr Share of associate's profits		32,400
	Cr Share of associate's OCI		1,500
	Being equity accounting for the period from 1 Jan 20x4 to 30 June 20x4		
	Dr Investment in associate (Note 3)	75,000	
	Cr Investment in associate (Note 3)		75,000
	Being deemed acquisition of additional interest in associate		
31 Dec 20x4	Dr Investment in associate	115,270	
	Dr Share of associate's OCI	800	
	Cr Share of associate's profits		116,070
	Being equity accounting for the period from 1 Jul 20x4 to 31 Dec 20x4		

Note 3: This is a set of memorandum accounting entries passed to reflect the deemed acquisition. There is no net impact to the carrying value of the investment in associate account.

Compute the balance in the investment in associate account

Investment in associate account		$
1 Jan 20x3	Payment for 30% interest in associate	200,000
31 Dec 20x3	Equity accounting for FY2013	70,500
30 Jun 20x4	Equity accounting for FY2014 – 1 Jan to 30 Jun 20x4	33,900

31 Dec 20x4	Equity accounting for FY2014 – 1 Jul to 31 Dec 20x4	115,270
31 Dec 20x4	Balance as at end	419,670

Analytical check on the investment in associate account

Investment in associate account	$	$	
Book value of net assets of associate as at 31 December 20x4			
– 30% interest	279,600		(30%*932,000)
– 10% deemed interest	93,200		(10%*932,000)
		372,800	
Goodwill			
– 30% interest	14,000		
– 10% deemed interest	5,000		
		19,000	
Unamortized fair value adjustments			
– 30% interest	21,600		(36,000 − (2*7,200))
– 10% deemed interest	6,270		(6,600 − 330)
		27,870	
		419,670	

ILLUSTRATION 7B.7 Participation of rights issue by investor resulting in control being obtained

On 1 January 20x5, Entity I acquired 25,000 shares which represents 25% interest in Entity A at consideration of $150,000. The book and fair value of the net assets of Entity A at the date of acquisition were $350,000 and $450,000, respectively.

On 31 December 20x5, Entity A issued additional shares with the consent of other shareholders to Entity I to raise additional financing for an impending acquisition. Specifically, 55,000 new shares were issued to Entity I in exchange for consideration of $200,000. Entity I assessed and concluded that it has obtained control in Entity A subsequent to the transaction.

The following table shows the movement of net assets of Entity A from 1 January 20x5 to 31 December 20x5.

Changes in net assets of Entity A		$
1 Jan 20x5	Net assets at book value	350,000
	Net assets at fair value	*450,000*
31 Dec 20x5	Profit for the year	150,000
	Effective portion of hedging instrument taken to OCI for the year	60,000
	Net assets at book value	560,000

Net assets at fair value	*600,000*
Additional cash injection	200,000
Net assets at book value	760,000
Net assets at fair value	*800,000*

The fair value of Entity A as at 31 December 20x5 is $920,000.

The extract of the financial statements as at 31 December 20x5 subsequent to the transaction is as follows:

	$
Assets	900,000
Less: Liabilities	*(140,000)*
Net assets	760,000
This is represented by:	
Share capital	425,000
Retained earnings	275,000
Hedging reserve	60,000
Net assets at book value	760,000

Calculate the financial impact of the issuance of additional shares on Entity I's shareholding interests and prepare the accounting entries in both the consolidated and separate financial statements. Ignore tax effects. Assume that non-controlling interests is carried at fair value at the date of acquisition in the financial statements of Entity I.

Analysis

Calculate the changes in the effective shareholding interest subsequent to the issuance of additional shares by Entity A and analyze the impact of the transaction

	Before transaction	After transaction
Shares held by Entity I	25,000	80,000
Shares in issue by Entity A	100,000	155,000
Percentage shareholdings	25%	52%

Subsequent to the issuance of the additional shares, Entity I's shareholding interests in Entity A increases from 25% to 52%. Entity I assessed that it has obtained control with 52%, that is, effectively there was an acquisition. As Entity I had 25% interest previously, this transaction is therefore accounted for as a step acquisition in accordance with IFRS 3. Non-controlling interests at the date of acquisition is 48%.

Calculate the goodwill on acquisition

	$	
Consideration paid	200,000	
Fair value of previously held interests	230,000	(25%*920,000)

Non-controlling interests at fair value	445,161	(48%*920,000)
	875,161	
Less: Fair value of net identifiable assets	(800,000)	
Goodwill	75,161	

Calculate the fair value adjustment arising from the transaction

	$
Fair value of net identifiable assets at date of acquisition	800,000
Less: Book value of net assets at date of acquisition	(760,000)
Excess of fair value over net book value (Note 1)	40,000

Note 1: Excess of fair value of net book value relates to patented technology. This intangible asset is amortized over 5 years.

Calculate the remeasurement gain on previously held interests

	$	
Fair value of previously held interests at the date of acquisition	230,000	(25%*920,000)
Add: Share of associate's OCI reclassified to profit or loss	15,000	(Note 2)
Less: Carrying value of previously held interests of 25% at date of acquisition	(197,500)	(Note 3)
Remeasurement gain (charged to profit or loss)	47,500	

Note 2: The share of Entity A's other comprehensive income that was recognized in Entity I's other comprehensive income is reclassified to profit or loss in accordance with paragraph 42 of IFRS 3. This is calculated as 25%*carrying value of $60,000 in the hedging reserves as at 31 December 20x5.

Note 3: This is derived as:

Investment in Entity A		$	
1 Jan 20x5	Balance as at beginning	150,000	
31 Dec 20x5	Equity accounting for FY2015	47,500	(See computation below)
31 Dec 20x5	Balance as at end	197,500	

Calculation of the share of profit and other comprehensive income for FY2015

	$	
FY2015		
Share of associate's profits for the year	37,500	(25%*150,000)
Less: Amortization of intangible asset – patented technology	(5,000)	((25%*(450,000-350,000))/5^)
	32,500	
Share of associate's OCI	15,000	(25%*60,000)

^ The excess of fair value of net book value which was determined on acquisition of the Entity A on 1 January 20x5 relates to patented technology and the estimated useful life was previously determined to be five years.

Prepare accounting entries

In the separate financial statements

1 Jan 20x5	Dr Investment in associate	150,000	
	Cr Cash		150,000
	Being payment for 25% interest in associate – Entity A		
31 Dec 20x5	Dr Investment in associate	200,000	
	Cr Cash		200,000
	Being payment for additional shares issued in Entity A		
	Dr Investment in subsidiary	350,000	
	Cr Investment in associate		350,000
	Being reclassification of investment in associate to investment in subsidiary with control being obtained		

In the consolidated financial statements

31 Dec 20x5	Dr Investment in associate	47,500	
	Cr Share of profit in associate		32,500
	Cr Share of OCI in associate		15,000
	Being equity accounting for the year		
31 Dec 20x5	Dr Share capital	425,000	
	Dr Retained earnings	275,000	
	Dr Hedging reserve	60,000	
	Dr Goodwill	75,161	
	Dr Intangible asset – patented technology	40,000	
	Dr Share of Entity A's OCI	15,000	
	Cr Investment in subsidiary (Note 3)		397,500
	Cr Remeasurement gain (profit or loss)		47,500
	Cr Non-controlling interests (Note 4)		445,161
	Being elimination of share capital and pre-acquisition reserves with control being obtained over Entity A		

Note 3: The carrying value of investment in subsidiary which is eliminated on preparation of the consolidated financial is derived as the sum of the cost of investment of $350,000 in the separate financial statement and the equity accounted share of profits and OCI of Entity A of $47,500 prior to it becoming a subsidiary.

Note 4: Non-controlling interests are recognized at fair value. They are computed by applying the proportionate interest of non-controlling interest to the fair value of Entity A as an entity, that is, 48%*$920,000.

DEEMED DISPOSALS

Conversely, an associate may issue additional shares to other shareholders or even new investors apart from the investor itself. This results in a dilution of the investor's relative interests in the associate as compared to other shareholders notwithstanding that its absolute shareholdings (in the number of shares held in the associate) has not changed. In substance, this is effectively a disposal of the investor's interest in the associate. Subsequent to the issuance of the additional shares, the investor may assess that it continues to have significant influence and the associate remains as an associate of the investor. In this case, the transaction constitutes a partial disposal of associate while significant influence is maintained. Conversely, the additional shares issued by associate may severely dilute the investor's interest such that significant influence is lost. In this case, the investor will have to account for this transaction as a disposal with a retained interest in a financial asset as described in the section above.

Inevitably, deemed disposals which result in a reduction in stake in the associate will give rise to gains or losses which are referred to in practice as "dilution gains or losses" or "gains or losses on deemed disposals." Consistent with the arguments set out under the section on partial disposal of associate while maintaining significant influence, by virtue of the fact that the associate is not part of the group (as defined in IFRS 10), when an investor reduces its stake in an associate indirectly other than direct disposal of the shareholdings interests, it is not a transaction with owners. Rather, it is a transaction (i.e., disposal) with third parties. Accordingly, similar to gains or losses on disposal, we believe gains or losses on deemed disposal should also be accounted for in the profit or loss.

Additionally, the requirement in paragraph 25 of IAS 28 also applies in deemed disposal transactions. Specifically, the investor will have to "reclassify to profit or loss, the proportion of the gain or loss relating to the reduction in the ownership interests that had been previously recognized in other comprehensive income if that gain or loss would be required to be reclassified to profit or loss on the disposal of the related assets or liabilities". This was specifically discussed by the IFRIC in its May 2009 discussions. In the IFRIC Update, the IFRIC noted that there is no specific guidance on the recognition of a gain or loss resulting from a reduction in the investor's ownership interest resulting from the issue of shares by the associate. However, the IFRIC also noted that reclassification of amounts to profit or loss from other comprehensive income is generally required as part of determining the gain or loss on a disposal. Paragraph 19A of IAS 28 (now paragraph 25 of IAS 28 (2011)) applies to all reductions in the investor's ownership interest, regardless of the cause.

ILLUSTRATION 7B.8 Non-participation of associate's rights issue by investor

On 1 January 20x6, Entity I acquired 25,000 shares which represents 25% interest in Entity A at consideration of $75,000. The book value and fair value of the net assets of Entity A at the date of acquisition were $150,000 and $200,000, respectively.

On 31 December 20x6, Entity A embarked on a 2 for 5 rights issue exercise to raise additional funds for expansion. Entity I did not participate in the exercise but other shareholders did. A total of 20,000 new shares were issued in exchange for cash consideration of $100,000. Entity I's effective interest in Entity A was diluted after the transaction. Subsequent to assessment, Entity I concluded that it continues to have significant influence in Entity A after the rights issue.

The following table shows the movement of net asset position of Entity A from 1 January 20x6 to 31 December 20x6.

Changes in net assets of Entity A		$
1 Jan 20x6	Net assets at book value	150,000

	Net assets at fair value	*250,000*
31 Dec 20x6	Profit for the year	80,000
	Net assets at book value	230,000
	Net assets at fair value	*300,000*
	Additional cash injection	100,000
	Net assets at book value	330,000

Calculate the financial impact of the rights issue on the Entity I's shareholding interests and prepare the accounting entries in both the consolidated and separate financial statements. Ignore tax effects.

Analysis

Calculate the changes in the effective shareholding interest subsequent to the rights issue by Entity A and analyze the financial impact of the share buyback on Entity I

	Before transaction	After transaction
Shares held by Entity I	25,000	25,000
Shares in issue by Entity A	100,000	120,000
Percentage shareholdings	25%	21%

Subsequent to the rights issue, Entity I's relative shareholding interests in Entity A decreases from 25% to 21%, that is, effectively there was a disposal of 4% shareholding interests. This is notwithstanding that its absolute shareholding interests remains unchanged at 25,000 shares. As mentioned in the fact pattern, Entity I assessed that it continues to have significant influence after the transaction. Consequently, the transaction is accounted for as a disposal while significant influence is maintained.

In analyzing this transaction, there are, in essence two offsetting financial impacts to Entity I's shareholding interests in Entity A with the rights issue.

First, with the dilution of relative interests in Entity A from 25% to 21%, Entity I's effective share in the net assets of Entity A also reduces by 4%, that is, $14,800 which is derived as (25%-21%)/25%*the carrying value of the Entity A in the consolidated financial statements of Entity I at the date of deemed disposal of $92,500. Refer to the computation below for the carrying value of Entity A.

Second, with the increase in the net assets of Entity A from the consideration of $100,000 received from the rights issue, Entity I is entitled to 21% interest of the increase in the net assets, that is, $21,000 which is derived as 21%*$100,000.

The net impact of the above is a gain of $6,200 which is referred to as a gain on deemed disposal in practice.

Calculate the goodwill and fair value adjustments on acquisition of the associate

	$	
Consideration paid	75,000	
Less: Fair value of net identifiable assets	(50,000)	(25%*200,000)
Goodwill	25,000	
Proportionate share of fair value of net assets	50,000	

Proportionate share of book value of net assets	(37,500)	(25%*150,000)
Excess of fair value over net book value (Note 1)	12,500	
Amortization per year	2,500	

Note 1: Excess of fair value of net book value relates to patented technology. This intangible asset is amortized over 5 years.

Calculation of the Entity I's share of profit in Entity A for FY2016 prior to rights issue

	$	
FY2016		
Share of associate's profits for the year	20,000	(25%*80,000)
Less: Amortization of intangible asset – patented technology	(2,500)	
	17,500	

Calculation of investment in Entity A prior to the rights issue

Investment in Entity A		$
1 Jan 20x6	Cost of investment	75,000
31 Dec 20x6	Equity accounting for FY2016	17,500
31 Dec 20x6	Balance as at end	92,500

Calculate the gain on disposal in the separate and consolidated financial statements

In the separate financial statements

From the perspective of the separate financial statements, there is no disposal as Entity I's absolute shareholdings interest in Entity A has not changed.

In the consolidated financial statements

	$	
Sales proceeds	–	
Less: Share of associate derecognized	(14,800)	((25%-21%)/25%*92,500)
Add: Share of proceeds from rights issue	21,000	(21%*100,000)
Gain on deemed disposal	6,200	

Prepare the accounting entries

In the separate financial statements

No entry required.

In the consolidated financial statements

31 Dec 20x6	Dr Investment in associate	6,200

	Cr Gain on deemed disposal	6,200
	Being gain on deemed disposal of investment in the consolidated financial statements	

Calculate the balance in the investment in associate in the consolidated financial statements post-deemed disposal and perform an analytical check

Investment in associate account		$
31 Dec 20x6	Balance prior to deemed disposal	92,500
31 Dec 20x6	Gain on deemed disposal	6,200
31 Dec 20x6	Balance as at end	98,700

Analytical check on the investment in associate account

Investment in associate account	$	
Book value of net assets of associate at 31 Dec 20x6	69,300	(21%*330,000)
Goodwill	21,000	((21%/25%)*25,000)
Unamortized fair value adjustments	8,400	((21%/25%)*(12,500 − 2,500))
	98,700	

However, if the fact pattern were to be amended such that a total of 30,000 new shares were issued at $150,000 instead of 20,000 new shares being issued for cash consideration of $100,000, Entity I's relative shareholding interests will be diluted to 19% from 25% and Entity I assessed that it had lost significant influence in Entity A as a result of the rights issue.

In this case, the operation of paragraph 22 of IAS 28 would require Entity I to discontinue the use of the equity method and recognize the retained interest of 19% in Entity A at fair value on 31 December 20x6 when significant influence is lost. Entity I will recognize in profit or loss any difference between the fair value of any retained interest and the carrying amount of the Entity A on 31 December 20x6 where the equity method was discontinued as shown below.

In the consolidated financial statements

	$
Sales proceeds	–
Less: Share of associate derecognized	(92,500)
Add: Fair value of 19% interest in Entity A*	80,000
Loss on deemed disposal	(12,500)

* Assume that the fair value of the financial asset of 19% interest in Entity A is valued at $80,000.

The accounting entry to be passed in the consolidated financial statements of Entity I would be:

31 Dec 20x6	Dr Investment in FVOCI asset	80,000	
	Cr Loss on deemed disposal		12,500
	Cr Investment in associate		92,500
	Being deemed disposal of interest in Entity A in the consolidated financial statements		

ACCOUNTING FOR ACQUISITION OF ADDITIONAL OR PARTIAL DISPOSAL OF INTERESTS IN THE SEPARATE FINANCIAL STATEMENTS

The discussions in the preceding sections have centered around the financial reporting effects of changes in ownership interests in the associates in the consolidated financial statements or stand-alone financial statements[16] of the investor. In this section, we will discuss the impact of such transactions on the separate financial statements of the investor.

Paragraph 10 of IAS 27 *Separate Financial Statements* allows an entity to account for its investments in subsidiaries, joint ventures and associates either (a) at cost, (b) in accordance IFRS 9; or (c) using the equity method as described in IAS 28 when it prepares separate financial statements.

In accordance with IFRS 9

When an associate or joint venture is acquired in stages or additional interests are procured, if the accounting policy of the investor is to account for the investment in accordance with IFRS 9, the accounting is straight-forward as any amounts paid for as consideration for additional interests is capitalized in the investment account and the entire investment is measured at fair value at each reporting period.

Cost Method

However, the question arises as to how the carrying value should be measured if the accounting policy for the measurement of these investments in the separate financial statements is the cost method.

As discussed in Chapter 8, cost is neither defined in IAS 27 nor in other IFRS Standards. However, the glossary[17] to IFRS Standards explains cost as "the amount of cash or cash equivalent paid or the fair value of the other consideration given to acquire an asset at the time of its acquisition or construction." The IFRIC in its May and July 2009 discussion specifically discussed how the initial carrying value of an equity method investment should be determined. As set out in the July 2009 IFRIC Update, "the IFRIC noted that IFRSs consistently require assets not measured at fair value through profit or loss to be measured at initial recognition at cost. Generally stated, cost includes the purchase price and other costs directly attributable to the acquisition or issuance of the asset such as professional fees for legal services, transfer taxes, and other transaction costs. Therefore, the cost of an investment in an associate determined in accordance with paragraph 11 of IAS 28 (now paragraph 10 in IAS 28 (2011)) comprises its purchase price and any directly attributable expenditures necessary to obtain it."

Accordingly, we are of the view that when the cost method is adopted, in cases where an associate or joint venture is acquired in stages or additional interests are procured, the cost of investment in the associate or joint venture is accounted for as the aggregate amount of consideration given in exchange for the various tranches of equity interests plus any acquisition-related costs.

To the extent that the initial tranche of investment is accounted for as fair value through other comprehensive income financial asset for which fair value changes are taken to other comprehensive income, the cumulative changes in fair value recognized in other comprehensive income is reversed through profit or loss to restate the initial tranche of investment back to cost when significant influence is obtained with the acquisition of the subsequent tranche in the separate financial statements.

Conversely, if the initial tranche of investment is measured at fair value through profit and loss account, the fair value changes previously recognized in retained earnings are reversed through profit or loss to restate the initial investment back to cost when significant influence is obtained with the acquisition of the subsequent tranche.

The rationale for reversing the fair value adjustments through either other comprehensive income or retained earnings to reinstate the accounting for the investment back to original cost is because such adjustments are neither part of

consideration given in exchange for the investment nor acquisition costs as envisaged by the IFRIC discussion. Accordingly, these adjustments will have to be unwound to restate the carrying value of the underlying investment back to its "true" cost, that is, the fair value of the consideration given up before the amount paid for the subsequent tranche is added to it.

In the case of partial disposal of the investment in associate and joint venture, IAS 27 is also silent on the accounting treatment for the derecognition of the interests disposed. Consistent with the arguments set out in the sections above, we believe that symmetrical accounting treatment should be accorded in the separate financial statements that is, the proportionate cost of investment of the associate disposed recognized in the separate financial statements should be derecognized with the gain or loss on disposal recognized in the profit or loss. In case the investor loses significant influence, the entire carrying value of the investment in the associate or joint venture should be derecognized.

Equity Method

As discussed above, IAS 28 is silent on the accounting treatment for piecemeal acquisition of associates and acquisition of additional interests in associate while maintaining significant influence consequently; this has resulted in different acceptable variants of accounting treatment in piecemeal acquisition of associate.

However, if the investor elects to account for the investment in associate and joint venture using the equity method as described in IAS 28 in its separate financial statements, it is our view that the investor should apply the accounting treatment that it had elected in the consolidated financial statements to its separate financial statements.

The discussions above would apply to direct acquisition or disposal of additional interests. In the case of indirect or deemed acquisitions or disposals of interests, care must be taken to analyze the transactions to determine if there is any accounting impact in the separate financial statements. Generally, if there is no change to the absolute interests held by the investor, there should be no corresponding accounting impact in the separate financial statements.

Refer to the accounting entries in the illustrations above for the demonstration of the accounting impact in the separate financial statements.

APPENDIX 7C

Changes in Ownership Interests for Joint Arrangements

Paragraphs 13 and 19 of IFRS 11 require an entity to make a reassessment in 2 specific aspects when facts and circumstances change. They are namely:

1. Whether the entity continues to have joint control over the joint arrangement
2. Whether the type of joint arrangement in which the entity is involved in (either joint operation or joint venture) has changed

Therefore, when the ownership interests in the joint arrangement changes, the entity have to re-evaluate the nature of its involvement in the joint arrangement post-transaction. In the following section, the accounting treatment when ownership interest in joint arrangements changes is discussed.

CHANGES IN OWNERSHIP INTERESTS IN JOINT OPERATIONS

Acquisition of Interests in Joint Operations

Instead of establishing a joint operation with other joint operators from inception, an entity may elect to obtain joint control by acquiring an interest in an existing joint operation. For instance, an entity may acquire a stake in the joint operation from its existing joint operators. As part of this transaction, the contractual agreement is amended to require unanimous consent from all parties including the entity on decisions over the relevant activities pertaining to the joint operation.

When the activities of the joint operation constitutes a business as defined in IFRS 3, paragraph B33A of IFRS 11 requires an entity that acquires an interest in the joint operation to apply all the principles on accounting for business combinations set out in IFRS 3 and other IFRS[18] to the extent of its share in the joint operation. In other words, the entity is required to apply acquisition accounting in its financial statements based on its share of assets and liabilities in the joint operation[19].

Acquisition of an interest in a joint operation does not meet the definition of a business combination in IFRS 3 as the entity purchasing the interest does not obtain control of the joint operation. However, the Board is persuaded that acquisition accounting is the most appropriate method to account for such transactions when the underlying joint operation meets the definition of business under IFRS 3. This is because the Board is of the view that goodwill, when present, should be separately recognized and the accounting requirements set out in IFRS 3 and other IFRSs provide a comprehensive and consistent basis for accounting for acquisitions of businesses.

Paragraph B33A of IFRS 11 further sets out a list of non-exhaustive principles in IFRS 3 that applies for acquisition of interests in joint operations. They are:

(a) measuring identifiable assets and liabilities at fair value, other than items for which exceptions are given in IFRS 3 and other IFRSs;

(b) recognizing acquisition-related costs as expenses in the periods in which the costs are incurred and the services are received, with the exception that the costs to issue debt or equity securities are recognized in accordance with IAS 32 *Financial Instruments: Presentation* and IFRS 9;

(c) recognizing deferred tax assets and deferred tax liabilities that arise from the initial recognition of assets or liabilities, except for deferred tax liabilities that arise from the initial recognition of goodwill, as required by IFRS 3 and IAS 12 *Income Taxes* for business combinations;

(d) recognizing the excess of the consideration transferred over the net of the acquisition-date amounts of the identifiable assets acquired and the liabilities assumed, if any, as goodwill; and

(e) testing for impairment a cash-generating unit to which goodwill has been allocated at least annually, and whenever there is an indication that the unit may be impaired, as required by IAS 36 *Impairment of Assets* for goodwill acquired in a business combination.

Illustration 7C.1, which is adapted from IE 53 that accompanies IFRS 11, demonstrates the application of the principles above.

ILLUSTRATION 7C.1 Acquisition of interests in a joint operation

Three companies, Alpha, Bravo and Charlie exercise joint control over Joint Operation Delta. Their ownership interests in Joint Operation Delta are 40%, 30% and 20% respectively. The remaining interest of 10% is held by Company Golf. The activity of Joint Operation Delta constitutes a business, as defined in IFRS 3 *Business Combinations*.

On 1 January 20x7, Company Echo acquires Company Alpha's 40% ownership interest in Joint Operation Delta at cash consideration of $300,000. Acquisition-related costs amounted to $50,000. Subsequent to the transaction, Company Alpha ceased to be and Company Echo became a joint operator.

The contractual arrangement between the parties establishes that Company Echo's shares in several assets and liabilities differ from its 40% ownership interest in Joint Operation Delta.

The following table sets out Company Echo's share in the assets and liabilities related to Joint Operation Delta as stipulated in the contractual arrangement between the parties. Company Echo's share of the net carrying value of the assets and liabilities on 1 January 20x7 amounted to approximately $132,000.

A valuation was performed and the fair values of the respective assets and liabilities of the Joint Operation Delta as at the date of acquisition on 1 Jan 20x7 are included in the table below. Corporate tax rate is 20%.

Assets/liabilities	Company Echo's share in contractual agreement	Fair value (100%) at 1 Jan 20x7 ($'000)
Property, plant and equipment	48%	288
Intangible assets (excluding goodwill)	90%	80
Accounts receivable	40%	210
Inventory	40%	175
Retirement benefit obligations	15%	(80)
Accounts payable	40%	(120)
Contingent liabilities	56%	(93)
		460

Company Echo has subsidiaries and prepares both consolidated and separate financial statements.

Calculate the financial impact of the acquisition of interest in Joint Operation Delta on Company Echo's consolidated and separate financial statements and prepare the accounting entries for the transaction.

Analysis

Company Echo will recognize its share of the assets and liabilities in Joint Operation Delta resulting from the contractual arrangement in both its separate and consolidated financial statements. Specifically, on the acquisition of the interest in Joint Operation Delta, Company Echo applies the principles on business combination accounting in IFRS 3 and other IFRSs for identifying, recognizing, measuring and classifying the assets acquired, and the liabilities assumed. However, in doing so, Company Echo does not apply principles in those IFRS Standards that conflict with the guidance in IFRS 11. Therefore, Company Echo recognizes, and measures only its share in each of the assets that are jointly held and its share in each of the liabilities that are incurred jointly, as stated in the contractual arrangement. The share of assets and liabilities of the other parties in Joint Operation Delta are not included in both the consolidated and separate financial statements of Company Echo.

Calculate Company Echo's share of the fair value of assets and liabilities in Joint Operation Delta on 1 January 20x7

Assets/liabilities	Company Echo's contractual share	Fair value (100%) at 1 Jan 20x7 ($'000)	Company Echo's share ($'000)
Property, plant and equipment	48%	288	138
Intangible assets (excluding goodwill)	90%	80	72

Accounts receivable	40%	210	84
Inventory	40%	175	70
Retirement benefit obligations	15%	(80)	(12)
Accounts payable	40%	(120)	(48)
Contingent liabilities	56%	(93)	(52)
			252
Deferred tax liability (Note 1)			(24)
			228

Note 1: Deferred tax liability is derived as follows:

	$'000
Fair value of identifiable assets and liabilities	252
Less: Carrying value of assets and liabilities	(132)
Excess of fair value over carrying value	120
Tax rate	20%
Deferred tax liability	24

Calculate the goodwill on acquisition of interest in the joint operation

	$'000
Consideration paid	300
Less: Fair value of net identifiable assets	(228)
Goodwill	72

Prepare the accounting entries

In the separate and consolidated financial statements of Company Echo

		$'000	$'000
1 Jan 20x7	Dr Property, plant and equipment	138	
	Dr Intangible asset	72	
	Dr Goodwill	72	
	Dr Accounts receivable	84	
	Dr Inventory	70	
	Cr Retirement benefit obligation		12
	Cr Accounts payable		48

Cr Contingent liabilities		52
Cr Deferred tax liability		24
Cr Cash at bank		300

Being acquisition of interest in joint operation

Dr Transaction costs – profit or loss	50	
Cr Cash at bank		50

Payment for transaction costs

However, IFRS 11 sets out an exception to the principle above. If the joint operators including the entity that acquires the interest in the joint operation are under the common control of the same ultimate controlling parties or parties both before and after the acquisition and that control is not transitory, the accounting treatment set out above will not apply.

Acquisition of Additional Interests in Joint Operations

After joint control has been obtained, a joint operator may acquire additional interests in that joint operation without gaining control. In other words, the joint operator continues to exercise joint control with other joint operators after the acquisition of the additional interests.

If the joint operation meets the definition of business under IFRS 3, the joint operator is required to apply acquisition accounting in respect of the additional interests acquired in accordance with paragraph 21A of IFRS 11. Consistent with the approach in accounting for initial acquisition of interests in joint operations as discussed above, the joint operator applies the accounting guidance in IFRS 3 and other IFRSs pertaining to business combinations that does not conflict with the requirements of IFRS 11. Paragraph B33C of IFRS 11 further stipulates that previously held interests in such cases are not remeasured.

The rationale behind this accounting treatment for the previously held interests is because the Board is of the view that acquisition of additional interests in a joint operation (that is a business) that is already jointly controlled by its joint operators is analogous to an acquisition of additional interests in a business that is already controlled by its acquirer. Specifically, preexisting interests in equity transactions under paragraph 23 of IFRS 10 are not remeasured. Accordingly, the Board is of the view that paragraph 23 of IFRS 10 which sets out the accounting treatment for changes in ownership interests in which the parent maintains control should be applied by analogy to acquisitions of additional interests in joint operations for which joint control is maintained.

ILLUSTRATION 7C.2 Acquisition of additional interests in a joint operation after joint control is obtained

Assume the same base facts as set out in Illustration 7C.2. On 1 June 20x8, Company Echo acquired an additional 10% interest in Joint Operation Delta from Company Bravo for $50,000. This is in addition to the 40% it already holds. As a result of the acquisition of the additional interest, the contractual agreement amongst the other joint operators was revised and Company Echo's share in the assets and liabilities in Joint Operation Delta was increased as set out in the table below. Included in the table below are also the fair values of the respective assets and liabilities of the Joint Operation Delta as at 1 June 20x8. Company Echo continues to have joint control with Company Bravo and Charlie.

Assets/liabilities	Company Echo's original share	Company Echo's revised share	Fair value (100%) at 1 Jun 20x8 ($'000)
Property, plant and equipment	48%	50%	350
Intangible assets (excluding goodwill)	90%	90%	85

Accounts receivable	40%	55%	300
Inventory	40%	45%	275
Retirement benefit obligations	15%	30%	(120)
Accounts payable	40%	45%	(250)
Contingent liabilities	56%	56%	(93)
			547

Company Echo's incremental share of the net carrying value of the assets and liabilities on 1 June 20x8 amounted to approximately $20,000. Acquisition-related transaction costs amounted to approximately $12,000.

Calculate the financial impact of the acquisition of additional interest in Joint Operation Delta on Company Echo and prepare the accounting entries for the transaction.

Analysis

Company Echo applies acquisition accounting under IFRS 3 in respect of the additional interests acquired in accordance with paragraph 21A of IFRS 11. In particular, Company Echo recognizes in both its separate and consolidated financial statements on 1 June 20x8, the incremental increase in its share of the assets and liabilities in Joint Operation Delta in line with the revised contractual agreement. Similarly in this case, Company Echo's additional ownership interest of 10% does not correspond with its incremental share of assets and liabilities in Joint Operation Delta. Its previously held interests in the form of assets and liabilities previously recognized in its separate and consolidated financial statements arising from the first acquisition are not remeasured with this additional acquisition of interests.

Calculate Company Echo's incremental share of the fair value of the assets and liabilities in Joint Operation Delta on 1 June 20x8

Assets/liabilities	Co Echo's incremental share (%)	Fair value at 1 Jun 20x8 ($'000)	Co Echo's incremental share ($'000)
Property, plant and equipment	2%	350	7
Intangible assets (excluding goodwill)	–	85	–
Accounts receivable	15%	300	45
Inventory	5%	275	14
Retirement benefit obligations	15%	(120)	(18)
Accounts payable	5%	(250)	(13)
Contingent liabilities	–	(93)	–
			35
Deferred tax liability (Note 1)			(3)
			32

Note 1: Deferred tax is derived as follows:

$'000

Incremental share of fair value of identifiable assets and liabilities	35
Less: Incremental share of carrying value of assets and liabilities	(20)
Excess of fair value over carrying value	15
Tax rate	20%
Deferred tax liability	3

Calculate the goodwill on acquisition of additional interest in the joint operation

	$'000
Consideration paid	50
Less: Fair value of net identifiable assets	(32)
Goodwill	18

Prepare the accounting entries

In the separate and consolidated financial statements of Company Echo

		$'000	$'000
1 Jun 20x8	Dr Property, plant and equipment	7	
	Dr Goodwill	18	
	Dr Accounts receivable	45	
	Dr Inventory	14	
	Cr Retirement benefit obligation		18
	Cr Accounts payable		13
	Cr Deferred tax liability		3
	Cr Cash at bank		50
	Being acquisition of additional interest in joint operation		
	Dr Transaction costs – profit or loss	12	
	Cr Cash		12
	Payment for transaction costs		

Similarly, the exception described above will apply to acquisition of additional interests in joint operations if the joint operators including the entity that acquires the additional interest are under the common control of the same ultimate controlling parties or parties both before and after the acquisition and that control is not transitory.

Obtaining Control over a Joint Operation whose Activity constitutes Business

A joint operator may acquire additional interests in the joint operation that is a business such that it obtains control over the joint operation. In such a case, it is effectively a business combination achieved in stages. The transaction gives rise

to a significant change in the nature of and economic circumstances of the interest in joint operations such that the joint operator-joint operation relationship is replaced by a parent-subsidiary relationship. The joint operator has given up joint control in exchange for full control over the joint operation. The same treatment will apply when a party to the joint arrangement that has rights to the assets and obligations for the liabilities immediately before the acquisition obtains control over the joint operation (Paragraph 42A of IFRS 11).

Accordingly, such transaction is accounted for as a business combination achieved in stages as discussed in Chapter 7. As set out in paragraph 42A of IFRS 11, the entire previously held interests in the form of the share of assets and liabilities in the joint operation recognized in the joint operator's financial statements immediately before control is obtained is remeasured to fair value[20] at the date where control is obtained. The resulting gain or losses is recognized in profit or loss. Similarly, any amount that was previously recognized in other comprehensive income in respect of the joint operator is recognized on the same basis as would be required if the joint operator had disposed directly of the previously held equity interests. Goodwill is calculated based on the fair value of the remeasured previously held interests and any additional net identifiable asset that is recognized at the date where control is obtained. Refer to Chapter 7 for a more detailed explanation of the accounting treatment and Illustration 7C.3 for the financial reporting impact.

page 616

ILLUSTRATION 7C.3 Obtaining control of a joint operation whose activity constitutes a business

Company X and Y are joint operators in a joint operation set up to mine for oil and gas in Country A. Each company has 50% rights to all the assets, liabilities, revenue generated by and expenses incurred in the joint operation. The joint operation meets the definition of business under IFRS 3.

During the financial year, owing to a change in the laws and legislation in Country A, Company Y had to dispose its interests in the joint operation as it can no longer operate in the country as a joint operator. After evaluating the financial feasibility of the oilfield, Company X decided to purchase 50% ownership interest from Company Y at cash consideration $25 million. With the purchase and sale of ownership interests, the contractual agreement between Company X and Company Y was terminated and Company X obtained control of the joint operation.

The carrying values and fair values of the identifiable assets and liabilities of the joint operation as at the date of acquisition on 1 September 20x8 is set out in the following table. The fair value of the joint operation on 1 September 20x8 as a business is $50 million.

Assets/liabilities	Carrying value of assets/liabilities in joint operation ($'000)	Company X's share of carrying values (50%) ($'000)	Fair value (100%) at date of acquisition ($'000)
Property, plant and equipment	20,000	10,000	35,000
Intangible asset – concession rights	1,500	750	4,000
Inventory	800	400	1,200
Trade and other receivables	400	200	400
Trade and other payables	(250)	(125)	(250)
Provisions (including provision for restoration costs)	(100)	(50)	(230)
Loans payable	(2,000)	(1,000)	(2,000)
Contingent liabilities	–	–	(550)
	20,350	10,175	37,570
Unrecognized goodwill	–	–	12,430
	20,350	10,175	50,000

Assume that Company X has subsidiaries and prepares both separate and consolidated financial statements.

Calculate the financial impact of the step acquisition on Company X and prepare the accounting entries for the transaction. Ignore tax effects.

Analysis

Calculate the financial effect of the remeasurement of the previously-held interests

	$'000	
Fair value of business at the date of acquisition (100%)	50,000	**Note 1**
Co X's share of fair value of business	25,000	
Less: Co X's share of carrying value of assets in joint operation	(10,175)	
Goodwill on acquisition	14,825	

Note 1: The fair value of the business at the date of acquisition includes contingent liabilities and goodwill. This is because in the remeasurement of the previously held interests to fair value at the date where control is obtained, paragraph 41 of IFRS 3 requires the entire previously held interests in the joint operation to be remeasured to fair value.

Calculate the goodwill on acquisition

	$'000
Consideration paid	25,000
Fair value of previously held interests	25,000
	50,000
Less: Fair value of net identifiable assets	(37,570)
Goodwill on acquisition	12,430

Prepare the accounting entries

In the separate financial statements of Company X

		$'000	$'000
1 Sep 20x8	Dr Investment in subsidiary	35,175	
	Dr Trade and other payables	125	
	Dr Provisions	50	
	Dr Loans payable	1,000	
	Cr Property, plant and equipment		10,000
	Cr Intangible asset – concession rights		750
	Cr Inventory		400

	Cr Trade and other receivables		200
	Cr Cash at bank		25,000
	Being derecognition of share of assets and liabilities in joint operation and recognition of investment in subsidiary		

In the consolidated financial statements of Company X

		$'000	$'000
1 Sep 20x8	Dr Property, plant and equipment	35,000	
	Dr Intangible asset – concession rights	4,000	
	Dr Goodwill	12,430	
	Dr Inventory	1,200	
	Dr Trade and other receivables	400	
	Cr Trade and other payables		250
	Cr Provisions		230
	Cr Loans payable		2,000
	Cr Contingent liabilities		550
	Cr Investment in subsidiary		35,175
	Cr Remeasurement gain (profit or loss)		14,825
	Being accounting for step acquisition arising from control obtained over a joint operation whose activity constitutes business		

Note 1: Essentially, the remeasurement gain in the consolidated financial statements comprises of:

	$'000
Fair value uplift of previously held interests on assets and liabilities	8,885
Fair value uplift of previously held interests on unrecognized identifiable assets and liabilities	(275)
Fair value uplift of previously held interests on unrecognized goodwill	6,215
Net remeasurement gain in profit or loss	14,825

Losing Control with Retained Interest in Joint Operation

A parent may sell its interests in its subsidiary (that is a business) to another party such that it loses control. Simultaneously, the selling parent enters into a contractual agreement with that same party to control the business jointly. The other party may also contribute additional assets to the joint arrangement. The contractual agreement is amended to require unanimous consent from both parties in all decisions pertaining to the relevant activities of the business and both parties have rights to the assets and obligations to the liabilities in the joint operation after the transaction. In this scenario, the parent has lost control but obtained joint control over the business post-transaction.

In accounting for this type of transaction, a perceived conflict exists between the requirements of IFRS 10 relating to the accounting for loss of control of a subsidiary and the requirements of IFRS 11 pertaining to the accounting for contribution of assets to a joint operation.

IFRS 10 requires an entity that loses control of a subsidiary but retains an interest in the former subsidiary to account for that retained interest at fair value on the date where control is lost. The re-measurement of that retained interest to fair

value is included in calculating the gain or loss associated with the loss of control of the subsidiary. However, paragraph B34 of IFRS 11 stipulates that an entity that enters into a transaction with a joint operation such as a contribution of assets to a joint operation, in which that entity is a joint operator, is required to recognize gains or losses resulting from the transaction only to the extent of the other parties' interests in the joint operation. The rationale behind this is because such transaction is in essence conducted with the other parties in the joint operation. Accordingly, only the portion of the gains or losses pertaining to the other parties' interests should be recognized in the financial statements of the joint operator. This requirement runs counter to the requirement in IFRS 10 where no restriction is required of the gain or loss to the extent of other parties' interest.

page 619

This conflict is similar to that in IFRS 10 and IAS 28 of which the Board had issued amendments[21] to address the issue in September 2014. Pre-amended IAS 28 similarly requires an investor or joint venturer to restrict the recognition of the gain or loss arising from the contribution or sale of non-monetary assets to the associate or joint venture to the extent of unrelated interests. To address that conflict, the amendments require the entity to recognize full gains or losses with the retained interest remeasured at fair value when the assets contributed to the associate or joint venture meet the definition of business under IFRS 3. If the assets contributed do not qualify as business, partial gains or losses (i.e. restricted to the extent of unrelated interests) are recognized by the investor. The remaining gains or losses are eliminated against the carrying value of the investment in associate or joint venture recognized in the financial statements of the investor or joint venturer. Subsequent to the issuance of these amendments, the Board deliberated on issuing further amendments to address certain unintended consequences[22] arising from applying the amendments. However, the Board subsequently decided to consider all those issues comprehensively as part of a wider research project on the equity method of accounting. At the same time, they also deferred the effective date of the amendments so that entities would not need to change the way in which they apply IAS 28 twice in a short period.

Similarly, the Interpretations Committee also discussed on the accounting treatment for a parent losing control and obtaining joint control in the joint operation. However, the Interpretation Committee decided not to address the issue but instead, recommend to the Board to address this issue concurrently with the issue discussed in the preceding paragraph given the similarities between the two types of transactions.

Pending the Board addressing this conflict on a comprehensive basis, we believe that a similar approach to that of the Amendments to IFRS 10 and IAS 28 should be adopted for transactions in which the parent lost control but obtained joint control in the investee. Specifically, if the joint operation meets the definition of a business under IFRS 3, the entity should recognize the full gain or loss calculated in accordance with B98 to B99 of IFRS 10. The interest retained in this former subsidiary which now takes the form of the entity's interests in the rights to the assets and obligations for liabilities is recognized at fair value in full in the entity's financial statements. This approach is also consistent with the views of a number of the members in the Interpretations Committee as discussed in the March 2016 staff paper on remeasurement of previously held interests—loss of control for the IFRS Interpretations Committee Meeting.

A question arises in the measurement of the retained interest in the joint operation which now takes the form of individual lines of assets and liabilities recognized in the financial statements of the joint operator. Should IFRS 3 be applied in the measurement of the retained interest? Based on the current syntax of IFRS 3 and IFRS 11, it is not clear as to whether that Standard should be applied. In the absence of more specific guidance from the Board, it is our view that entities have an accounting policy choice whether IFRS 3 is applied. Illustration 7C.4 assumes that IFRS 3 is not applied.

ILLUSTRATION 7C.4 Losing control of subsidiary with retained interest in a joint operation

Company Zafron has a wholly-owned subsidiary, Company Sufron whose principal business is in coal exploration, development and production activities in Country Coal. On incorporation of the subsidiary back in 20x4, Company Zafron invested $20 million as initial investment.

page 620

On 1 January 20x9, Company Zafron sold 50% ownership interest in Company Sufron to Company Xena for cash consideration of $30 million. A joint operating agreement was drawn up between the Company Zafron and Company Xena. Specifically, all budgets and work programs going forward will require the unanimous consent of both companies. All the output produced by Company Sufron will be purchased by both companies on a predetermined ratio and both parties will share all costs incurred on the work programmes on an equal basis.

As part of the acquisition of interests in the joint operation, Company Xena is entitled to a share of assets and liabilities in the joint operation as set out in the table below. Also presented in the table are the carrying values and fair values of Company Sufron's assets and liabilities at the date of acquisition.

Assets/liabilities	Co Zafron's share after transaction	Co Xena's share after transaction	Carrying value in Co's Sufron's books ($'000)	Fair value as at date of acquisition ($'000)
Property, plant and equipment	65%	35%	55,000	70,000
Intangible asset (excluding goodwill)	50%	50%	20,000	35,000
Accounts receivable	60%	40%	850	850
Inventory	60%	40%	1,500	2,000
Retirement benefit obligations	75%	25%	(250)	(300)
Accounts payable	55%	45%	(400)	(400)
Provisions (including provision for restoration costs)	50%	50%	(2,000)	(2,500)
			74,700	104,650

Calculate the financial impact of the transaction on Company Zafron and prepare the accounting entries for the transaction. The effects of taxes are ignored for this illustration. Assume that Company Zafron continues to prepare consolidated financial statements after the disposal of Company Sufron.

Analysis

Calculate Company Zafron's share of fair value of assets and liabilities in Company Sufron subsequent to the disposal

Assets/liabilities	Co Zafron's share after transaction	Fair value as at date of acquisition ($'000)	Co Zafron's share of fair value as at date of acquisition ($'000)
Property, plant and equipment	65%	70,000	45,500
Intangible asset (excluding goodwill)	50%	35,000	17,500
Accounts receivable	60%	850	510
Inventory	60%	2,000	1,200
Retirement benefit obligations	75%	(300)	(225)
Accounts payable	55%	(400)	(220)
Provisions (including provision for restoration costs)	50%	(2,500)	(1,250)
		104,650	63,015

Calculate the gain on disposal in the separate and consolidated financial statements

In the separate financial statements of Company Zafron

	$'000
Sales proceeds	30,000
Less: Cost of investment in separate financial statements (Note 1)	(10,000)
Gain on disposal	20,000

Note 1: In the separate financial statements, the carrying value of the investment derecognized is based on the proportionate share of interests disposed. This is derived as 50%*initial cost of $20 million.

In the consolidated financial statements of Company Zafron

	$'000
Sales proceeds	30,000
Add: Fair value of share of assets and liabilities retained (From above)	63,015
Less: Carrying value of net assets derecognized	(74,700)
Gain on disposal	18,135

Prepare the accounting entries

In the separate financial statements of Company Zafron

		$'000	$'000
1 Jan 20x9	Dr Cash at bank	30,000	
	Cr Investment in subsidiary		10,000
	Cr Gain on disposal# (Note 1)		20,000
	Being gain on partial disposal of subsidiary		
	Dr Property, plant and equipment	45,500	
	Dr Intangible asset	17,500	
	Dr Accounts receivable	510	
	Dr Inventory	1,200	
	Dr Gain on disposal# (Note 1)	1,685	
	Cr Retirement benefit obligations		225
	Cr Accounts payable		220
	Cr Provisions		1,250
	Cr Investment in subsidiary		10,000
	Cr Beginning retained earnings (Note 2)		54,700
	Being accounting for share of assets and liabilities in joint operation		

Note 1: Sum of #= Gain on disposal of $18,135.
Note 2: This is derived as:

	$'000
Initial investment	20,000
Carrying value of net assets at 1 Jan 20x9	74,700
Post investment reserves from date of incorporation	54,700

In the consolidated financial statements of Company Zafron

No accounting entry required.

Note 1: No accounting entry is required as Company Zafron has already recognized its share of the fair value assets and liabilities in Company Sufron at 1 January 20x9 in its separate financial statements. These assets and liabilities will be included when the consolidated financial statements are being prepared.

Partial Disposal of Interests in Joint Operations Whilst Maintaining Joint Control

An existing joint operator may partially sell its interests in a joint operation to another party while continuing to maintain joint control. The purchaser may become a joint operator if the contractual agreement is amended to require unanimous consent from all parties including the purchaser on decisions relating to the relevant activities of the joint operation. Conversely, the purchaser may become a party to the joint operation in which it merely participates in the joint operation without having joint control.

IFRS 11 is silent as to how such partial disposal of interests to the rights to assets and obligations to liabilities in the joint operation whilst joint control is maintained should be accounted for and whether a gain or loss on disposal should be recognized in this type of transaction.

Whilst a partial disposal of a subsidiary that does not result in the parent losing control is accounted for as an equity transaction under IFRS 10, consistent with the scenario in which an investor partially disposes its interests in an associate, we do not believe that the principles in IFRS 10 can be extended to that of a partial disposal of interests in a joint operation where the joint operator does not lose joint control. The rationale is because like an associate, the joint operation is not part of the group as defined in IFRS 10. Therefore, the partial disposal of interests in the joint operation is not a transaction with equity owners. Instead, it is a transaction with 3rd parties. Accordingly, the proportionate share of the assets and liabilities disposed should be derecognized with the gain or loss on disposal recognized in the profit or loss. This argument is further augmented by paragraph 48C of IAS 21 which requires the proportionate share of the cumulative amount of exchange differences recognized in other comprehensive income and accumulated in equity to be reclassified to profit or loss in the case of partial disposal of foreign operations. Partial disposal of foreign operations in this case would include partial disposal of interests in a foreign associate or joint arrangement (which includes joint operation). Accordingly, we believe that a gain or loss on disposal should be recognized in the profit or loss when a joint operator makes a partial disposal of its interests in a joint operation.

Specifically, the joint operator should:

1. Derecognize the rights to assets and obligations to liabilities in the joint operation on the basis of the percentage disposed. The assets derecognized would include the share of goodwill and the joint operator's share of unamortized portion of the fair value adjustments at the date of disposal.
2. Recognize the fair value of consideration received from the transaction, event or circumstance that result in the partial disposal of interests in the joint operation.
3. Reclassify to profit or loss[23] or transfer directly to retained earnings[24] if required by other IFRS, the amounts recognized in other comprehensive income in relation to the disposed interests.
4. Recognize any resulting difference as a gain or loss in profit or loss.
5. Account for the remaining interests in the joint operation in accordance with IFRS 11.

page 623

ILLUSTRATION 7C.5 Partial disposal of interests in joint operation whilst maintaining joint control

Companies Class, Gold and Kiss are joint operators in a joint operation. Company Class and Gold holds 33% equity interest each in the joint operation except Company Kiss which holds 34%. On 1 January 20x10, Company Class sold 10% interest in joint operation to Company Symphony for cash consideration of $90,000. In exchange for the cash consideration, Company Symphony is entitled to a share of the assets and liabilities in the joint operation. As a result of the transaction, Company Symphony becomes a party to the joint operation, i.e. it does not have joint control over joint operation with the other companies. Company Class continues to maintain joint control with the rest of the joint operators.

The table below sets out Company Symphony's interests in the assets and liabilities of the joint operation after the acquisition of the interests. Included in the table are also the carrying values of Company Class's share of the respective assets and liabilities of the joint operation as at 1 January 20x10 prior to the transaction.

Assets/liabilities	Co Class's share prior to sale of interests	Co Symphony's share after purchase of interests	Carrying value in Co Class's books at 1 January 20x10 before the transaction ($'000)
Property, plant and equipment	50%	15%	118
Intangible assets	90%	20%	70
Goodwill (Note 1)	100%	-	90
Accounts receivable	55%	18%	60
Inventory	45%	18%	80
Provisions	30%	20%	(14)
Accounts payable	45%	14%	(60)
			344

Note 1: The goodwill arose from the acquisition of the initial interest in the joint operation by Company Class in the past. As this goodwill is recorded in the financial statements of Company Class, Company Class is entitled to 100% of the goodwill recorded.

Calculate the financial impact of the partial disposal of interest in joint operation on Company Class and prepare the accounting entries for the transaction. Ignore tax effects. Assume that Company Class prepares both separate and consolidated financial statements.

Analysis

Calculate the share of assets and liabilities to be derecognized from the separate and consolidated financial statements of Company Class

Assets/liabilities	Co Class's share prior to sale of interests (a)	Co Symphony's share after purchase of interests (b)	Carrying value in Co Class's books at 1 January 20x10 ($'000) (c)	Carrying value of assets/liabilities to be derecognized ($'000) (d)=(b)/(a)*(c)
Property, plant and equipment	50%	15%	118	35
Intangible assets	90%	20%	70	16
Goodwill	100%	-	90	–
Accounts receivable	55%	18%	60	20
Inventory	45%	18%	80	32
Provisions	30%	20%	(14)	(9)
Accounts payable	45%	14%	(60)	(19)
			344	75

Calculate the gain on disposal in the separate and consolidated financial statements

In the separate and consolidated financial statements of Company Class

	$'000
Sales proceeds	90
Less: Carrying value of net assets derecognized	(75)

	Gain on disposal		15

Prepare the accounting entries

In the separate financial statements of Company Class

1 Jan 20x10	Dr Cash at bank	90,000	
	Dr Provisions	9,000	
	Dr Accounts payable	19,000	
	Cr Property, plant and equipment		35,000
	Cr Intangible asset		16,000
	Cr Accounts receivable		20,000
	Cr Inventory		32,000
	Cr Gain on disposal (P/L)		15,000

Being accounting for partial disposal of interests in joint operation

In the consolidated financial statements of Company Class

No accounting entry required.

Note 1: No accounting entry is required as Company Class has derecognized the proportionate share of the assets and liabilities disposed in the joint operation in its separate financial statements. The effects of this entry will be included when the consolidated financial statements are being prepared.

Loss of Joint Control in Joint Operation with Retained Interest in Associate

An existing joint operator may sell its ownership interest partially as well as lose joint control of the joint operation by relinquishing its rights to jointly control the relevant activities of the joint operation with other joint operators. Joint control can also be lost when the contractual agreement between the joint operators is amended in a way such that the entity no longer has joint control. When the contractual agreement that gives rise to joint control is terminated, joint control ceases as well. Subsequent to the sale and losing joint control, the entity reassesses and may conclude that it has significant influence in the former joint operation. For instance, the entity continues to have a seat in the board of directors by virtue of its remaining shareholdings but its consent over decisions on the relevant activities of the joint operation is not required.

Similarly, IFRS 11 does not address how such situations should be accounted for in the financial statements of the selling entity. If the joint operation meets the definition of a business under IFRS 3, given that the joint operator has lost joint control over the joint operation but retained significant influence, we believe that an analogy can be drawn to the scenario where the parent lost control of its subsidiary but retained significant influence.

Although the composition of the group has not change in this case[25] after the transaction (as compared to the scenario where control over a subsidiary ceases), the entity no longer has joint control, i.e. its rights to assets and obligations for liabilities in the joint operation have ceased. Therefore, if the activities of the joint operation constitutes a business under IFRS 3, consistent with the spirit of IFRS 10, it follows accordingly that the carrying values of the entire assets and liabilities associated with the joint operation should be derecognized. A full gain or loss arising from the loss of joint control should be recognized in the profit or loss. The retained interest in the form of an investment in associate should be remeasured at fair value at the date where joint control is lost. This fair value is regarded as the cost on initial recognition of an investment in an associate for which equity accounting in accordance with IAS 28 will apply for purposes of subsequent measurement.

ILLUSTRATION 7C.6 Losing joint control in joint operation with retained interest in associate

Company X and Y are 50:50 joint operators in a joint operation that was established to mine for gold in Country Gold. Although the joint operation is structured through a separate vehicle, the terms of the contractual agreement stipulates specifically that each company has equal rights to the assets, liabilities, revenue generated and expenses incurred in the joint operation.

page 626

As a result of changes in laws and regulations in Country Gold, Company X decided that it was no longer economically viable to continue with the joint operation. Consequently, on 30 June 20x6, it sold 30% of its ownership interest in the joint operation to a third party, Company Z for cash consideration of $170 million.

As part of the transaction, Company Z will take over Company X's entire share of assets and liabilities in the joint operation. The contractual agreement was subsequently amended to exclude Company X from and include Company Z in having joint control with Company Y over the joint operation. Given Company X's expertise in gold mining, it continues to provide inputs on the operational aspect of the joint operation but does not have decision-making rights. Company X assessed that it retains significant influence based on its remaining 20% ownership interest after losing joint control.

As at 30 June 20x6, prior to the transaction, Company X has recognized its share of assets and liabilities in the joint operation in its separate and consolidated financial statements as set out in the table below.

Assets/liabilities	Carrying value at 30 June 20x6 ($'000)
Property, plant and equipment	280,000
Intangible assets – concession rights	25,000
Accounts receivable	800
Inventory	1,200
Other assets	400
Provisions	(750)
Accounts payable	(250)
Loans payable	(30,000)
Tax payable	(1,800)
	274,600

The fair value of Company X's retained interests of 20% amounted to $120 million as at 30 June 20x6.

Calculate the financial impact of the transaction on Company X and prepare the accounting entries for the transaction. Ignore tax effects. Assume that Company X prepares both separate and consolidated financial statements.

Analysis

Calculate the gain on disposal in the separate and consolidated financial statements

In the separate and consolidated financial statements of Company X

	$'000
Sales proceeds	170,000
Add: Fair value of retained interest	120,000
Less: Carrying value of net assets derecognized	(274,600)
Gain on disposal	15,400

page 627

Prepare the accounting entries

In the separate financial statements of Company X

30 Jun 20x6	Dr Investment in associate	120,000	
	Dr Cash at bank	170,000	
	Dr Provisions	750	
	Dr Accounts payable	250	
	Dr Loans payable	30,000	
	Dr Tax payable	1,800	
	Cr Property, plant and equipment		280,000
	Cr Intangible asset – concession rights		25,000
	Cr Accounts receivable		800
	Cr Inventory		1,200
	Cr Other assets		400
	Cr Gain on disposal		15,400
	Being accounting for loss of joint control whilst retaining significant influence		

In the consolidated financial statements of Company X

No accounting entry required.

Note 1: No accounting entry is required as Company X has derecognized its share of the assets and liabilities of joint operation disposed and recognized the fair value of its retained interest in its separate financial statements. The effects of this entry will be included when the consolidated financial statements are being prepared.

Loss of Joint Control in Joint Operation with Retained Interest in Financial Asset

An entity may dispose a significant portion of its interests in the assets and liabilities in a joint operation and lose joint control by relinquishing its rights to jointly control the relevant activities of the joint operation with other joint operators. Following the disposal, it continues to retain some interest in the former joint operation.

Similarly, the entity derecognizes entirely its share of the assets and liabilities in the joint operation as it has lost joint control. If the retained interest in the former joint operation is a financial asset, the retained interest will be remeasured at fair value at the date where joint control is lost. This fair value of the retained interest is regarded as its fair value on initial recognition as a financial asset in accordance with IFRS 9. A gain or loss in respect of the difference between (1) the fair value of any retained interest and any proceeds received from disposing part of the interest in the joint operation and (2) the carrying amount of the assets and liabilities derecognized at the date when joint control was lost is recognized in the profit or loss.

The accounting principles set out above is similar to the scenario described in the preceding section where the joint operator loses joint control but retained significant influence in the former joint operation. Refer to the illustrative example above for the accounting treatment.

CHANGES IN OWNERSHIP INTERESTS FOR JOINT VENTURES

Just like a joint operator, a joint venturer may increase or decrease its ownership interests in its joint venture. As discussed in the preceding sections, a joint venturer accounts for its interests in a joint venture using the equity method in accordance with IAS 28 as it has rights to the net assets of the joint venture. Accordingly, accounting for changes in ownership interests in joint ventures will largely be similar to the accounting for changes in ownership interests in associates as discussed in Chapter 7.

Acquisition of Additional Interests in Joint Venture whilst Maintaining Joint Control

After obtaining joint control in a joint venture, the joint venturer may acquire additional equity interests in that joint venture without gaining control. In other words, the joint venturer continues to have joint control after the acquisition of the additional interests.

As discussed in Chapter 7, IAS 28 does not provide guidance on the accounting for such transactions. However, we believe that the transaction should be accounted for as follows:

1. Fair value of the consideration paid for the additional equity interests is added to the carrying value of the cost of investment in the joint venture.
2. Goodwill and the joint venturer's share of any new fair value adjustments arising from the acquisition of the additional interest is calculated based on the fair value of the consideration given in exchange for the additional interests and the joint venturer's incremental share of the fair value of identifiable assets and liabilities at the date of the additional purchase. The goodwill and unamortized portion of the fair value adjustments for the purchase of additional interests is aggregated with the goodwill and unamortized portion of the fair value adjustments for the existing investment at the end of the financial period.
3. No remeasurement of the previously held interest in the joint venture to fair value is performed. Instead, the joint venturer will continue to apply equity accounting in respect of the previously held interests by recognizing its share of profits and other comprehensive income to the extent of its previously held interests up to the date where the additional interests are acquired. Appropriate adjustments to the joint venturer's share of the profit or loss after acquisition of the additional interest should continue to be made in respect of the joint venturer's share of the fair value adjustments recognized at the date where the joint venture was first acquired (for example additional depreciation based on the fair value of the assets at the date of acquisition of the joint venture).
4. Subsequent to the acquisition of the additional interests, the joint venturer will recognize its share of results and other comprehensive income on the basis of its new shareholding interests in the joint venture. Adjustments pertaining to the amortization of the fair value adjustments related to both the initial acquisition and subsequent purchase of additional interests should continue to be made.

The rationale for the accounting treatment described above is attributable to the fact that there is no change in the nature of the underlying investment after the acquisition of the additional interests. The underlying investment continues to be a joint venture for which the joint venturer has joint control. As control was not obtained with the purchase of the additional interests, the transaction does not constitute a significant economic event that warrants a change in the classification and measurement of the investment as set out in IFRS 3. Hence, no remeasurement of the previously held interests to fair value is carried out and the cost accumulation approach where the fair value of the consideration paid for the additional interest is "accumulated" in the cost of investment in joint venture is most appropriate.

Illustration 7B.3 provides an example on the accounting for an acquisition of additional interest in an associate. This illustration will apply similarly to the acquisition of additional interests in a joint venture.

Obtaining Control of a Business that is a Joint Venture

A joint venturer may obtain control over its existing joint venture through the acquisition of additional interests such that the joint venture becomes a subsidiary. In this case, the joint venturer has effectively acquired a subsidiary in stages.

Paragraph 22(a) of IAS 28 specifically requires the joint venturer to discontinue the use of equity method and account for the investment in accordance with IFRS 3 and IFRS 10. Refer to Chapter 7 for a more detailed discussion of the accounting treatment for business combination achieved in stages.

Partial Disposal of Interests in Joint Ventures Whilst Maintaining Joint Control

A joint venturer may dispose partially its equity interests in a joint venture and continue to have joint control over the joint venture with the other joint venturers.

As discussed in Chapter 7, IAS 28 is silent as to how the cost of the partially disposed investment in joint venture should be measured and whether a gain or loss on disposal should be recognized in such transactions. Whilst IFRS 10 requires a partial disposal of a subsidiary that does not result in the parent losing control to be accounted for as an equity transaction, we believe that this principle cannot be extended to that of a partial disposal of a joint venture where joint control is not lost. The rationale is because the joint venture, as with an associate, is not part of the group as defined in IFRS 10. Therefore, a partial disposal of the joint venture is not a transaction with equity owners. Instead, it is a transaction (i.e. disposal transaction) with third parties. Accordingly, the proportionate cost of investment of the joint venture disposed should be derecognized with the gain or loss on disposal recognized in the profit or loss. This argument is further augmented by paragraph 25 of IAS 28 which provides guidance for transactions in which the joint venturer's ownership interests in a joint venture is reduced but the joint venturer continues to apply the equity method. Specifically, that paragraph requires the joint venturer to reclassify the proportion of the gain or loss (relating to the reduction in the ownership interests) that was previously recognized in other comprehensive income to profit or loss, if that gain or loss would be required to be reclassified to profit or loss on the disposal of the related assets or liabilities. Similarly, paragraph 48C of IAS 21 also requires the proportionate share of the cumulative amount of exchange differences recognized in other comprehensive income and accumulated in equity to be reclassified to profit or loss in the case of partial disposals of foreign operations. Partial disposal of foreign operations in this case would include partial disposal of interests in a foreign associate or joint arrangement (which includes joint venture). Therefore, we believe that a gain or loss on disposal should be recognized in the profit or loss when a joint venturer disposes its investment in the joint venture partially.

Specifically, the joint venturer should:

1. Derecognize the cost of investment in the joint venture on a proportionate basis on the basis of the percentage of shareholdings disposed. The cost of investment derecognized would include the proportionate share of goodwill and the joint venturer's share of unamortized portion of the fair value adjustments at the date of disposal.
2. Recognize the fair value of consideration received from the transaction, event or circumstance that result in the partial disposal.
3. Reclassify on a proportionate basis to profit or loss[26] or transfer directly to retained earnings[27] if page 630 required by other IFRS, the amounts recognized in other comprehensive income in relation to the disposed interests.
4. Recognize any resulting difference as a gain or loss in profit or loss.
5. Account for the remaining interests in the joint venture in accordance with IAS 28.

Illustration 7B.4 provides an example on the accounting for a partial disposal of interest in an associate whilst maintaining significant influence. This illustration will similarly apply to the partial disposal of interests in a joint venture whilst maintaining joint control.

Loss of Joint Control in Joint Venture with Retained Interest in Associate

A joint venturer may lose joint control over its joint venture but retains significant influence in that former joint venture. For instance, the contractual agreement for a joint venture for which the entity is a joint venturer lapses.

In this scenario, paragraph 24 of IAS 28 requires the entity to continue to apply the equity method subsequent to joint control being lost. The retained interest will not be remeasured.

The rationale for this accounting treatment is set out in paragraph 28 of the Basis to Conclusion to IAS 28. Although both the relationship between the investor and investee and the nature of the investment change in this scenario, both type of investments (i.e. investment in associate and investment in joint venture) continues to be accounted for using the equity method. As there is no change in the group boundaries (as associates and joint ventures are not part of the group within the definition of IFRS 10) and no change in the measurement requirement, the Board concluded that losing joint control and retaining significant influence in a former joint venture is not an event that warrants a remeasurement of the retained interest at fair value at the date when joint control is lost.

Paragraph 24 of IAS 28 further stipulates that the same accounting treatment also applies when the reverse happens, i.e. when an investment in an associate becomes an investment in joint venture.

When the loss of joint control is accompanied by an acquisition of additional interests or partial disposal of interests in the previous joint venture, the same accounting principles set out in Chapter 7 in respect of these areas will apply.

The financial reporting effects in this type of transaction will be similar to that set out in Illustration 7B.4 which explains the accounting treatment in a scenario where an investor disposes of its interests in its associate partially whilst maintaining significant influence.

Loss of Joint Control in Joint Venture with Retained Interest in Financial Asset

A joint venturer may dispose a significant portion of its interests in the investment in the joint venture such that it loses joint control but continues to retain some interest in the former joint venture.

As the investment ceases to be a joint venture, paragraph 22 of IAS 28 requires the entity to discontinue the use of the equity method from the date it loses joint control. If the retained interest in the former joint venture is a financial asset, the retained interest is remeasured at fair value at the date where joint control is lost. This fair value of the retained interest is regarded as its fair value on initial recognition as a financial asset in accordance with IFRS 9. The entity will recognize in profit or loss any difference between (1) the fair value of the retained interest and any proceeds received from disposing part of the interest in the joint venture and (2) the carrying value of the investment in the former joint venture at the date the equity method was discontinued.

page 631

Paragraph 22(c) of IAS 28 further requires the entity to account for all amounts previously recognized in other comprehensive income in relation to that investment on the same basis as would have been required if the entity had directly disposed of the related assets or liabilities when the entity discontinues the use of the equity method.

Illustration 7B.5 provides an example on the accounting in the scenario where an investor loses significant influence and retained an interest in the form of a financial asset. This illustration will similarly apply when a joint venturer loses joint control and retained an interest in the form of a financial asset.

page 632

APPENDIX 7D

Deemed Acquisitions and Business Combinations Achieved Without the Transfer of Consideration

Apart from direct acquisitions of equity interests in the acquiree, it is possible for an acquirer to gain control in an acquiree via changes in circumstances. Generally, no consideration is transferred in these transactions. Such form of changes in ownership interests is commonly known as deemed acquisitions in practice. IFRS 3 describes this as a business combination achieved without the transfer of consideration.

Deemed acquisitions may take place due to a number of reasons. Table 7D.1 sets out some scenarios that could give rise to such transactions.

Table 7D.1 Scenarios giving rise to deemed acquisitions

Deemed acquisitions

- Acquiree repurchasing its ordinary shares held by other shareholders in a sufficient quantity for an existing investor (the acquirer) to gain control
- Changes in composition of board of directors such that investor attains control
- Lapsing of minority veto rights that prevented the acquirer from controlling the acquiree previously
- Consensus between the acquirer and acquiree to combine their business solely via contractual agreement. The acquirer in this case neither holds equity interest in the acquiree nor transfers consideration in exchange for control in the acquiree[28].

Paragraph 43 of IFRS 3 requires such transactions to be accounted for using the acquisition method of accounting for business combinations. However, as no consideration is transferred in such cases (i.e. consideration transferred is nil), for the purpose of measuring goodwill or a gain on bargain purchase, paragraph B46 of IFRS 3 requires the acquirer to use the fair value of the acquirer's interest in the acquiree at acquisition date[29] in lieu of the acquisition-date fair value of the consideration.

Notwithstanding the requirement above, it should be noted that in deemed acquisitions transactions for which the acquirer has previously held interests, care should be taken not to double-count the fair value of the acquirer's interest in the acquiree. This is because the fair value of the acquirer's interest in the acquiree would have included the fair value of the acquirer's previously held equity interests in the acquiree at the date of acquisition. Hence, in the measurement of goodwill or gain on bargain purchase in the case of business combination achieved without the transfer of consideration, the fair value of the acquirer's interest in the acquiree is included only once. Therefore, the formula for measuring goodwill will evolve as follows:

page 633

Measurement of goodwill for general business combinations transactions

Goodwill as at acquisition date	=	Consideration transferred	+	Fair value of acquirer's previously held interests at acquisition date	+	Non-controlling interests in the acquiree	–	Fair value of net identifiable assets of acquiree at the date of acquisition

Measurement of goodwill for business combinations achieved without transfer of consideration

Goodwill as at acquisition date	=	Fair value of acquirer's interest in the acquiree at the date of acquisition	+	Non-controlling interests in the acquiree	–	Fair value of net identifiable assets of acquiree at the date of acquisition

ILLUSTRATION 7D.1 Share buyback by associate resulting in deemed acquisition by investor

On 1 January 20x3, Entity X acquired 40,000 shares which represents 40% interest in Entity Y at cash consideration of $500,000. The book and fair values of Entity X's share of net assets of Entity Y as at 1 January 20x3 was $500,000.

On 30 June 20x6, Entity Y embarked on a share buy-back exercise on which 43,000 of the shares held by the other investors were repurchased at cash consideration of $250,000. The shares bought back were cancelled.

As a result of the share repurchase, Entity X has 70% effective interest in Entity Y. Entity X assessed and concluded that it has obtained control over Entity Y.

The summarized financial information of Entity Y from 20x3 to 20x6 is set out below.

Changes in net assets of Entity Y		$
1 Jan 20x4	Net assets at book value	2,000,000
20x4 to 20x5	Profit for the years from 20x4 to 20x5	800,000
31 Dec 20x5	Net assets at book value	2,800,000
30 Jun 20x6	Profit for the period	400,000
	Net assets at book value before share repurchase	3,200,000

	Cash paid for share buy-back	(250,000)
30 Jun 20x6	Net assets at book value after share repurchase	2,950,000
	Net assets at fair value	3,500,000

The fair value of Entity X as a business as at 30 June 20x6 is $3.8 million. Entity X's accounting policy is to measure non-controlling interests at fair value. The fair value of non-controlling interests as at the date of acquisition is $1.14 million.

Calculate the financial impact of the share buy-back on Entity X's shareholding interests and prepare the accounting entries in both the consolidated and separate financial statements of Entity X. Ignore tax effects.

Analysis

Calculate the changes in the effective shareholding interest subsequent to the share buy-back by Entity Y and analyze the impact of the transaction

	Before transaction	After transaction
Shares held by Entity X	40,000	40,000
Shares in issue by Entity Y	100,000	57,000
Percentage shareholdings	40%	70%

Subsequent to the share buy-back by Entity Y, Entity X's effective shareholding interests in Entity X increased from 40% to 70%. This is notwithstanding the fact that Entity X's absolute shareholding interest in Entity Y (40,000 shares) has not change before and after the transaction. Entity X assessed that it has obtained control with 70% as a result of the share buy-back i.e. there was effectively a deemed acquisition. As Entity X had 40% interest previously, this transaction is therefore accounted for as a step acquisition in accordance with IFRS 3. Non-controlling interests at the date of acquisition is 30%.

Calculate the goodwill on acquisition

	$	
Fair value of Entity X's interest in Entity Y	2,660,000	(70%*3,800,000)
Non-controlling interests at fair value	1,140,000	(30%*3,800,000)
	3,800,000	
Less: Fair value of net identifiable assets	(3,500,000)	
Goodwill	300,000	

Calculate the fair value adjustment arising from the transaction

	$
Fair value of net identifiable assets at date of acquisition	3,500,000
Less: Book value of net assets at date of acquisition	(2,950,000)
Excess of fair value over net book value (Note 1)	550,000

Note 1: Excess of fair value of net book value relates to trademark which is amortized over 8 years.

Calculate the remeasurement gain

	$	
Fair value at the date of acquisition	2,660,000	(70%*3,800,000)
Less: Carrying value of previously held interests	(1,280,000)	(Note 1)
Remeasurement gain (charged to profit or loss)	1,380,000	

Note 1: This is derived as:

Investment in Entity Y		$	
1 Jan 20x3	Balance as at beginning	500,000	
31 Dec 20x3 to 30 Jun 20x6	Equity accounting for FY20x3 to 30 Jun 20x6	780,000	(40%*(3,200,000 – 1,250,000*))
30 Jun 20x6	Balance as at end	1,280,000	

* Derived as $500,000/40%

In essence, the remeasurement gain comprises of two components. They are namely (1) the fair value of the 30% interest acquired in the deemed acquisition transaction and (2) the remeasurement gain associated with the previously held interests of 40%. Hence, the remeasurement gain can be analyzed as:

	$	
Fair value of the 30% interest acquired	1,140,000	(30%*3,800,000)
Remeasurement gain on previously held interest of 40%	240,000	((40%*3,800,000) – 1,280,000)
Remeasurement gain (net)	1,380,000	

Prepare accounting entries

In the separate financial statements of Company X

30 Jun 20x6	Dr Investment in subsidiary	500,000	
	Cr Investment in associate		500,000
	Being reclassification of investment in associate to investment in subsidiary with control being obtained		

In the consolidated financial statements of Company X

30 Jun 20x6	Dr Investment in subsidiary	780,000	
	Cr Beginning retained profits		780,000
	Being equity accounting from 20x3 to 30 June 20x6		
	Dr Share capital and reserves	2,950,000	
	Dr Goodwill	300,000	
	Dr Intangible asset – trademark	550,000	

Cr Investment in subsidiary		1,280,000
Cr Remeasurement gain (profit or loss)		1,380,000
Cr Non-controlling interests		1,140,000

Being elimination of share capital and pre-acquisition reserves with control being obtained over Entity Y

Analytical check on the balance in non-controlling interests

Balance in non-controlling interests	$	
NCI's share of carrying value of net assets as at 30 June 20x6	885,000	(30%*2,950,000)
NCI's share of goodwill	90,000	(30%*300,000)
NCI's share of intangible asset	165,000	(30%*550,000)
	1,140,000	

Deemed Disposals

Just like deemed acquisitions, a parent entity may reduce its shareholding interests in its subsidiary by way of changes in circumstances, other than via actual sale of its equity interests. This is known as deemed disposals in practice.

Deemed disposals may take place due to a number of reasons. Set out in Table 7D.2 are some circumstances that could give rise to such transactions.

Table 7D.2 Scenarios resulting in deemed disposals

Deemed disposals
• Parent does not subscribe or subscribes partially to the subsidiary's rights issue • Subsidiary issues ordinary shares upon the exercise of options held by third parties (e.g. employees on exercise of the vested employee share options) • Subsidiary issues additional ordinary shares to third parties other than the investor to raise funding • Changes in composition of the board of directors such that parent loses control • Conversion of convertible loans issued by subsidiary into ordinary shares by third party debt holders

As evident from the table above, the parent entity's absolute shareholding interest in the subsidiary may or may not change after the transaction. However, the relative shareholding interest to other investors will generally reduce after the transaction. In substance, there is a disposal of the parent's interest in the subsidiary. Accordingly, such transactions should be accounted for in accordance with the requirements of IFRS 10.

ILLUSTRATION 7D.2 Issuance of additional shares by subsidiary resulting in parent losing control

On 1 January 20x5, Entity P acquired 80,000 shares (which represents 80%) in Entity S at cash consideration of $1,500,000. The book value and fair value of the net assets of Entity S at the date of acquisition were $800,000 and $1,000,000 respectively.

On 1 January 20x9, Entity S issued 100,000 new shares at $1 each to raise additional funds for expansion. As Entity P did not subscribe to the new shares, its effective shareholding interest in Entity S was diluted after the transaction.

Upon assessment, Entity P concluded that it has lost control in Entity S after the shares issue. Although it has lost control, Entity P has assessed that it retains significant influence post-transaction. The fair value of its retained interest at

the date where it lost control (i.e. 1 January 20x9) is $2 million.

The following table shows the movement of net asset position of Entity S from 1 January 20x5 to 1 January 20x9.

Changes in net assets of Entity S		$
1 Jan 20x5	Net assets at book value	800,000
	Net assets at fair value	1,000,000
20x5 to 20x8	Profit for the years from 20x5 to 20x8	2,500,000
31 Dec 20x8	Net assets at book value	3,300,000
	Net assets at fair value	4,000,000
1 Jan 20x9	Cash received from share issuance	100,000
1 Jan 20x9	Net assets at book value after share issuance	3,400,000

Calculate the financial impact of the share issuance by Entity S on Entity P's shareholding interests and prepare the accounting entries in both the consolidated and separate financial statements of Entity P. Ignore tax effects.

Analysis

Calculate the changes in the effective shareholding interest subsequent to the share issuance by Entity S and analyze the impact of the transaction

	Before transaction	After transaction
Shares held by Entity P	80,000	80,000
Shares in issue by Entity S	100,000	200,000
Percentage shareholdings	80%	40%

Subsequent to the issuance of additional shares by Entity S, Entity P's relative shareholding interest in Entity S decreases from 80% to 40%. In substance, there is effectively a disposal of 40% shareholding interest in Entity S. This is notwithstanding the fact that Entity P's absolute shareholding interest remains at 80,000 shares post transaction. As Entity P has assessed that it has lost control but retained significant influence after the transaction, the transaction is accounted for as a disposal of subsidiary in accordance with paragraph 25 of IFRS 10. The remaining interest retained in the Entity S of 40% is recognized at fair value on the date when control was lost (i.e. 1 January 20x9) and this fair value is regarded as the cost of initial recognition in an associate.

Following from the discussion above, there is, essentially two offsetting financial impact from Entity P's perspective (in the consolidated financial statements of Entity P) as set out in the table below.

	$'000	
Loss on deemed disposal	(3,580)	((80%*3,400,000) + 700,000 + 160,000)
Fair value of retained interest	2,000	
	(1,580)	

First, as Entity P has lost control of Entity S, the assets and liabilities including non-controlling interests are derecognized. The loss on disposal comprises of Entity P's share (80%) of the assets and liabilities of $3.4 million, goodwill on acquisition of $700,000 as well as the carrying value of the fair value adjustment pertaining to intangible asset of $160,000 at the date where control is lost is derecognised as well. Refer to the computation below to derive the balances mentioned.

Second, given that Entity P has retained 40% in Entity S on 1 January 20x9, paragraph 25 of IFRS 10 requires Entity P to recognize the remaining 40% interest at fair value, amounting to $2 million.

The net impact of the above effects is a loss on deemed disposal of $1.58 million. This loss is recognized in the consolidated profit or loss account of Entity P. (Note: NCI's share of 20% of the assets and liabilities of Entity S as well as the balance in NCI are derecognized without any reclassification to the profit or loss account.)

In the separate financial statements of Entity P, there is no accounting impact. No disposal has taken place as Entity P's absolute shareholding interest in Entity S has remained unchanged.

Calculate Entity P's share of goodwill on acquisition

	$	
Consideration paid	1,500,000	
Less: Fair value of net identifiable assets	(800,000)	(80%*1,000,000)
Goodwill (Note 1)	700,000	

Note 1: Goodwill is assumed to be unimpaired on 1 January 20x9.

Calculate Entity P's share of fair value adjustment arising from the transaction

	$	
Fair value of net identifiable assets at date of acquisition	800,000	
Less: Book value of net assets at date of acquisition	(640,000)	(80%*800,000)
Excess of fair value over net book value (Note 1)	160,000	

Note 1: Excess of fair value of net book value relates to intangible asset with indefinite useful life. The intangible asset is not impaired on 1 January 20x9.

Calculate Entity P's share of change in net assets from date of acquisition to date where control is lost

	$	
Change in net assets from date of acquisition to date where control is lost	2,600,000	(2,500,000 + 100,000)
Entity P's share (80%)	2,080,000	(80%*2,600,000)

Prepare accounting entries

In the separate financial statements of Entity P

1 Jan 20x9	Dr Investment in associate	1,500,000	
	Cr Investment in subsidiary		1,500,000
	Being reclassification of investment in subsidiary to investment in associate with lost of control		

In the consolidated financial statements of Entity P

1 Jan 20x9	Dr Investment in associate	500,000	
	Dr Loss on deemed disposal	1,580,000	
	Cr Beginning retained profits (Note 1)		2,080,000

Being accounting for deemed disposal and beginning retained profits

Note 1: The credit entry is required to reinstate Entity P's share of profits generated by Entity S from the date of acquisition to the date where control is lost.

However, if we were to vary the scenario in Illustration 7D.2 above such that Entity S issues 24,000 instead of 100,000 new shares for $100,000. In this case, Entity P's interest is diluted from 80% to 65% after the transaction. Non-controlling interests in Entity S increases from 20% to 35%. It is assumed that non-controlling interests is carried at proportionate share of net identifiable assets.

Entity P assessed and concluded that it continues to have control in Entity S. In this case, as the dilution in shareholding interest does not result in Entity P losing control of Entity S, the transaction will be accounted for as an equity transaction in accordance with paragraph 23 of IFRS 10.

Specifically, Entity P will adjust the carrying amounts of its interests and the non-controlling interests to reflect the effects of the transaction on their relative interests in Entity S. The difference between the amount by which non-controlling interests is adjusted and the cash proceeds received from the share issuance exercise is recognized in equity and attributed to the shareholders of Entity P.

In computing the extent to which the carrying value of non-controlling interests is adjusted, a question invariably arises whether non-controlling interests' share of goodwill is required to be adjusted in equity transactions with non-controlling interests.

Insofar as IFRS 10 is concerned, the IFRS Standard is silent on the accounting treatment. However, it is our view that the proportion of the goodwill balance should be re-allocated between the parent entity and non-controlling interests when their relative interests in the subsidiary changes. This is regardless of whether non-controlling interests are measured at the proportionate share of net identifiable assets or fair value. It should be noted that the carrying value of the goodwill in the consolidated financial statements remains unchanged. The reallocation is manifested in the increase in the carrying value of non-controlling interests by the proportion of goodwill allocated to them.

The reason is because in the absence of doing so, the goodwill balance will not be reflective of the parent entity's ownership interests in the subsidiary after the change in the relative ownership interests. This will consequently have an impact when either an impairment loss on goodwill is recognized or when goodwill is derecognized when control over the subsidiary is lost. If goodwill is not re-allocated, the impairment loss on goodwill recognized or the gain/loss recognized upon the lost of control will not be reflective of the ownership interest applicable to that of the parent and the non-controlling interests. This same treatment is accorded to the carrying value of any fair value adjustments previously arising from the acquisition of the subsidiary at the date where relative ownership interests changes.

Hence, in this case, Entity P will reallocate 15% of the goodwill to non-controlling interests. This is illustrated below.

Calculate the changes in the effective shareholding interest subsequent to the share issuance by Entity S and analyze the impact of the transaction

	Before transaction	After transaction
Shares held by Entity P	80,000	80,000
Shares in issue by Entity S	100,000	124,000
Entity P's relative shareholding	80%	65%
NCI's relative shareholding	20%	35%

Calculate the balance in the non-controlling interests before the share issuance transaction

	$	
NCI's share of carrying value of net assets	660,000	(20%*3,300,000)

NCI's share of fair value adjustment – intangible asset	40,000	(20%*200,000)
	700,000	

Calculate the balance in the non-controlling interests after the share issuance transaction

	$	
NCI's share of carrying value of net assets	1,190,000	(35%*3,400,000)
NCI's original share of fair value adjustment – intangible asset	40,000	(20%*200,000)
	1,230,000	
Re-allocation of goodwill from Entity P	131,250	(15%/80%*700,000)
Re-allocation of fair value adjustment from Entity P	30,000	(15%/80%*160,000)
	1,391,250	

Calculate the change in the non-controlling interests

	$
NCI's balance before the share issuance	700,000
NCI's balance after the share issuance	1,391,250
Change in NCI balance	691,250

Calculate the loss on deemed disposal recognized as equity attributable to shareholders of Entity P

	$	
Share of net assets given up by Entity P	495,000	(15%*3,300,000)
Share of goodwill given up by Entity P	131,250	(From above)
Share of fair value adjustment given up by Entity P	30,000	(From above)
	656,250	
Less: Entity P's share of cash from share issuance	(65,000)	(65%*100,000)
	591,250	Note 1

Note 1: In essence, there are two offsetting financial impact to Entity P's shareholding interest in Entity S arising from the share issuance.

First, with the dilution of relative interests in Entity S from 80% to 65%, Entity P's effective share in the net assets (inclusive of goodwill and the fair value adjustment) of Entity S also reduces by 15% amounting to $656,250 as computed above.

Second, with the increase in the net assets of Entity S from the consideration of $100,000 received from the share issuance, Entity P is entitled to 65% interest of the increase in the net assets i.e. $65,000.

The net impact of the above is a loss of deemed disposal of $591,250 which is adjusted in equity in accordance with paragraph B96 of IFRS 10.

Prepare accounting entries

In the separate financial statements of Entity S

1 Jan 20x9	Dr Cash	100,000	
	Cr Share capital		100,000
	Being cash received from shares issued		

In the separate financial statements of Entity P

No entry required as there is no change in Entity P's absolute shareholding interests.

In the consolidated financial statements of Entity P

1 Jan 20x9	Dr Share capital and reserves	800,000		
	Dr Goodwill	700,000		
	Dr Intangible asset	200,000		
	Cr Investment in subsidiary		1,500,000	
	Cr Non-controlling interests		200,000	
	Being elimination of share capital, reserves and investment in subsidiary			
	Dr Beginning retained profits	500,000		
	Cr Non-controlling interests		500,000	
	Being NCI's share of profits up to 1 Jan 20x9 before share issuance			
	Dr Share capital	100,000		
	Dr Loss on deemed disposal – Equity	591,250		
	Cr Non-controlling interests		691,250	Note 1
	Being adjustment for changes in relative interests			

Note 1: Although goodwill has been re-allocated to the non-controlling interests upon the dilution in interests in Entity S, the carrying value of the goodwill in the consolidated financial statements of $700,000 remains unchanged.

Analytical check for non-controlling interests as at 1 January 20x9

	$	
NCI's share of carrying value of net assets on 1 Jan 20x9	1,190,000	(35%*3,400,000)
NCI's share of goodwill	131,250	(From above)
NCI's share of intangible asset	70,000	(35%*200,000)
	1,391,250	

APPENDIX 7E

Consolidated Cash Flow Statement Involving a Foreign Subsidiary

Scenario:

Co Y is a 90% owned subsidiary of SingCo. The functional currency of Co Y is the USD.

Exchange rates are as follows:	SGD to USD
On date of acquisition of Co Y, 1 Jan 20x3	2.1
Purchase of patents by Co Y, 1 Jan 20x2	2.2
Sale of undervalued inventory, 1 July 20x3	1.95
Purchase of fixed assets by Co Y, 1 Jan 20x4	1.9
Purchase of opening inventory during Dec 20x4	1.8
Purchase of closing inventory during Dec 20x5	1.64
1 Jan 20x5	1.78
Average rate for 20x5	1.68
31 Dec 20x5	1.6
Date of dividend declaration	1.65

Additional information:

- Assume that operating cash flows, sales, purchases, operating expenses, interest, and tax occur evenly throughout the year.
- Foreign Currency Translation Reserve (cumulative translation losses to 31 Dec 20x4) is a loss of $20,000
- Fair value of consideration transferred was $300,000 and fair value of non-controlling interests as at acquisition date was $30,000.
- Undervalued inventory at acquisition date was $20,000. Tax rate was 20%.
- Undervalued inventory was sold on 1 July 20x3
- There was no disposal of fixed assets during 20x5

Required:

(1) Translate the financial statements of Co Y for the year ended 31 Dec 20x5 to SGD.
(2) Prepare the consolidated cash flow statements for the year ended 31 Dec 20x5 in SGD.

The current financial statements of Co Y in USD are shown below:

Income Statement of Co Y for the year ended 31 Dec 20x5		USD	Rate	SGD
Sales		1,000,000		
Cost of Sales				
Opening inventory	160,000			
Purchases	800,000			
	960,000			
Less closing inventory	(240,000)			
Less Cost of Sales		(720,000)		
Gross profit		280,000		
Operating expenses		(50,000)		
Interest expense		(8,000)		
Depreciation		(40,000)		
Amortization of patents		(6,000)		
Profit before tax		176,000		
Tax expense		(100,000)		
Profit after tax		76,000		
Dividends declared		(20,000)		
Profit retained		56,000		
Retained earnings, 1 Jan 20x5		200,000		
Retained earnings, 31 Dec 20x5		256,000		

Statement of Financial Position as at 31 Dec 20x5		USD	Rate	SGD
Fixed assets, net book value		320,000		
Patents		60,000		
Inventory		240,000		
Accounts receivable		100,000		
Cash		10,000		
		730,000		
Accounts payable		374,000		
Share capital		100,000		
Retained earnings				
Pre-acquisition:	120,000			
Post-acquisition:	136,000			
		256,000		
Foreign Currency Translation Reserve (FCTR)				
		730,000		

The comparative statement of financial position of Co Y in USD are shown below:

Statement of Financial Position as at 31 December 20x4	USD
Fixed assets, net book value	240,000
Patents	66,000
Inventory	160,000
Accounts receivable	120,000

Cash	164,000
	750,000
Accounts payable	450,000
Share capital	100,000
Retained earnings	200,000
Foreign Currency Translation Reserve (FCTR)	
	750,000

The cash flow statement of Sing Co for the year ended 31 December 20x5 is as follows:

Statement of Cash Flows for the year ended 31 December 20x5

	Parent SGD
Net profit after tax	320,000
Add back/Less non-cash items	
Depreciation	124,000
Add/(less) decrease/(increase) in working capital	
Change in inventory	230,000
Change in accounts receivable	(45,000)
Change in accounts payble	78,930
Operating cash flows	707,930
Cash flows from (to) financing activities	
Dividends paid	(200,000)
Cash flows from investing activities	
Purchase of fixed assets	(340,000)
Change in cash and cash equivalents during 20x5	167,930
Cash and cash equivalents	
1 Jan 20x5	324,000
31 Dec 20x5	491,930

Solutions:

Income Statement of Co Y for the year ended 31 Dec 20x5		USD	Rate	SGD
Sales		1,000,000	1.68	1,680,000
Cost of Sales				
Opening inventory	160,000			
Purchases	800,000			
	960,000			
Less closing inventory	(240,000)			
Less Cost of Sales		(720,000)	1.68	(1,209,600)
Gross profit		280,000		470,400
Operating expenses		(50,000)	1.68	(84,000)
Interest expense		(8,000)	1.68	(13,440)
Depreciation		(40,000)	1.68	(67,200)
Amortization of patents		(6,000)	1.68	(10,080)
Profit before tax		176,000		295,680
Tax expense		(100,000)	1.68	(168,000)
Profit after tax		76,000		127,680
Dividends declared		(20,000)	1.65	(33,000)
Profit retained		56,000		94,680
Retained earnings, 1 Jan 20x5		200,000	Note 1	344,000
Retained earnings, 31 Dec 20x5		256,000	To BS	438,680

Statement of Financial Position as at 31 Dec 20x5		USD	Rate	SGD
Fixed assets, net book value		320,000	1.60	512,000
Patents		60,000	1.60	96,000
Inventory		240,000	1.60	384,000
Accounts receivable		100,000	1.60	160,000
Cash		10,000	1.60	16,000
		730,000		1,168,000
Accounts payable		374,000	1.60	598,400
Share capital		100,000	2.10	210,000
Retained earnings				
Pre-acquisition:	120,000		2.10	252,000
Post-acquisition:	136,000			186,680
		256,000	From I/S	438,680
Foreign Currency Translation Reserve (FCTR)			Residual	(79,080)
		730,000		1,168,000

Note 1: Reconstruct the statement of financial position as at 31 Dec 20x4 to determine the retained earnings in SGD at the beginning of the year

Statement of Financial Position as at 31 Dec 20x4	USD	Rate	SGD
Net assets	300,000	1.78	534,000
Share capital	100,000	2.10	210,000
Retained earnings	200,000		344,000
Foreign Currency Translation Reserve			(20,000)
	300,000		534,000

Statement of Financial Position as at 31 December 20x4	USD	Rate	SGD
Fixed assets, net book value	240,000	1.78	427,200
Patents	66,000	1.78	117,480
Inventory	160,000	1.78	284,800
Accounts receivable	120,000	1.78	213,600
Cash	164,000	1.78	291,920
	750,000		1,335,000
Accounts payable	450,000	1.78	801,000
Share capital	100,000	2.10	210,000
Retained earnings	200,000		344,000
Foreign Currency Translation Reserve (FCTR)			(20,000)
	750,000		1,335,000

Analysis of movements of assets and liabilities		USD	Rate	SED	
Fixed assets (FA)					
Balance, 1 Jan 20x5		240,000	1.78	427,200	
Purchases (actual rate)	BAL	120,000	1.63	195,600	
Depreciation	PL	(40,000)	1.68	(67,200)	NC
FCTR				(43,600)	NC
Balance, 31 December 20x5		320,000	1.60	512,000	
Patents					
Balance, 1 Jan 20x5		66,000	1.78	117,480	
Amortization of patents	PL	(6,000)	1.68	(10,080)	NC
FCTR				(11,400)	NC
Balance, 31 December 20x5		60,000	1.60	96,000	
Inventory					
Balance, 1 Jan 20x5		160,000	1.78	284,800	
Purchases	PL	800,000	1.68	1,344,000	NC
Cost of sales	PL	(720,000)	1.68	(1,209,600)	NC
FCTR				(35,200)	NC
Balance, 31 Dec 20x5		240,000	1.60	384,000	

Accounts receivable (AR)					
Balance, 1 Jan 20x5		120,000	1.78	213,600	
Sales	PL	1,000,000	1.68	1,680,000	NC
Collections	BAL	(1,020,000)	1.68	(1,713,600)	Cash
FCTR				(20,000)	NC
Balance, 31 Dec 20x5		100,000	1.60	160,000	
Accounts payable (AP)					
Balance, 1 Jan 20x5		450,000	1.78	801,000	
Purchases	PL	800,000	1.68	1,344,000	NC
Cash payments	BAL	(876,000)	1.68	(1,471,680)	Cash
FCTR				(74,920)	NC
Balance, 31 Dec 20x5		374,000	1.60	598,400	
Cash					
Balance, 1 Jan 20x5		164,000	1.78	291,920	
Fixed assets purchases	FA	(120,000)	1.63	(195,600)	
Collections	AR	1,020,000	1.68	1,713,600	
Cash payments	AP	(876,000)	1.68	(1,471,680)	
Operating expenses	PL	(50,000)	1.68	(84,000)	
Interest expense	PL	(8,000)	1.68	(13,440)	
Tax	PL	(100,000)	1.68	(168,000)	
Dividends	EQUITY	(20,000)	1.65	(33,000)	
FCTR				(23,800)	
Balance, 31 Dec 20x5		10,000	1.60	16,000	

Change in FCTR (add all the FCTR from each asset and liability)	(59,080)
FCTR, 1 Jan 20x5	(20,000)
FCTR, 31 Dec 20x5	(79,080)

NC: Non-cash items

Statement of Cash Flows for the year ended 31 December 20x5

		Subsidiary SGD		Parent SGD	Consolidation adjustments	Consolidated Cash Flow
Net profit after tax		127,680		320,000	(29,700)	417,980
Add back/Less non-cash items						
	Depreciation	67,200		124,000		191,200
	Amortization of patents	10,080				10,080
Add/(less) decrease/(increase) in working capital						
	Change in inventory	(134,400)	1,344,000-1,209,600	230,000		95,600
	Change in accounts receivable	33,600	1,680,000-1,713,600	(45,000)		(11,400)
	Change in accounts payable	(127,680)	1,344,000-1,471,680	78,930		(48,750)
Operating cash flows		(23,520)		707,930		654,710
Cash flows from (to) financing activities						
	Dividends paid	(33,000)		(200,000)	33,000	(200,000)
	Dividend paid to NCI				(3,300)	(3,300)
Cash flows from investing activities						
	Purchase of fixed assets	(195,600)		(340,000)		(535,600)
Change in cash and cash equivalents during 20x5		(252,120)		167,930		(84,190)
Cash and cash equivalents						
	1 Jan 20x5	291,920		324,000		615,920
	FCTR on cash	(23,800)				(23,800)
	31 Dec 20x5	16,000		491,930		507,930

PROBLEMS

P7.1 Simultaneous consolidation

Information pertaining to the group structure of A Ltd is as follows:

B Ltd

A Ltd acquired 70% of the share capital of B on 1 January 20x1.

Fair value of consideration transferred	$1,250,000
Share capital of B at date of acquisition	300,000
Retained earnings of B at date of acquisition	700,000

Net assets acquired were at fair values.

Fair value of non-controlling interests of B at date of acquisition	500,000

C Ltd

B Ltd acquired 60% of the issued share capital of C Ltd on 2 January 20x2.

Fair value of consideration transferred	$ 650,000
Share capital of C at date of acquisition	250,000
Retained earnings of C at date of acquisition	500,000

Net assets acquired were at fair values.

Fair value of non-controlling interests of C at date of acquisition 400,000

Extracts of the 20x4 financial statements of the companies are shown below:

	A	B	C
Profit before tax	$12,500,000	$ 750,000	$6,250,000
Tax expense	(2,500,000)	(150,000)	(1,250,000)
Profit after tax	$10,000,000	$ 600,000	$5,000,000
Dividends declared	(5,000,000)	(300,000)	(500,000)
Profit retained for the year	$ 5,000,000	$ 300,000	$4,500,000
Retained earnings, 1 January 20x4	12,500,000	1,500,000	1,000,000
Retained earnings, 31 December 20x4	$17,500,000	$ 1,800,000	$5,500,000

Profit before tax includes dividend income from subsidiaries. The tax rate was 20%.

Required:

1. Prepare the necessary elimination and consolidation journal entries with respect to A's group financial statements for the year ended 31 December 20x4, using the simultaneous method of consolidation.
2. On 1 January 20x5, C acquired 90% of D's share capital. Calculate the total non-controlling interest percentage, both direct and indirect, in D Ltd.

P7.2 Simultaneous consolidation and equity accounting

Extracts of financial statements of P, X, Y and Z for the year ended 31 December 20x5 are as follows:

Income Statement
For the Year Ended 31 December 20x5

	P	X	Y	Z
Net profit before tax	$500,000	$250,000	$160,000	$300,000
Tax	(100,000)	(50,000)	(32,000)	(60,000)
Net profit after tax	$400,000	$200,000	$128,000	$240,000
Dividends declared	(56,000)	(40,000)	(65,000)	(50,000)
Net profit retained	$344,000	$160,000	$ 63,000	$190,000

Statement of Financial Position As at 31 December 20x5				
	P	X	Y	Z
Investment in X, at cost	$ 500,000			
Investment in Y, at cost	600,000			
Investment in Z, at cost			$ 750,000	
Other assets	2,400,000	$2,200,000	500,000	$2,800,000
Total assets	$3,500,000	$2,200,000	$1,250,000	$2,800,000
Share capital	$ 600,000	$ 500,000	$ 350,000	$ 550,000
Other reserves	250,000	350,000		
Retained earnings	800,000	900,000	600,000	750,000
Shareholders' equity	$1,650,000	$1,750,000	$ 950,000	$1,300,000

Additional information:

(a) All intercorporate investments were acquired prior to 1 January 20x5. Y was acquired before Z. There was no change in share capital since the date of acquisition. Ownership interests in investee companies, retained earnings, and fair value of non-controlling interests at the date of acquisition are as follows:

Investee Company	X	Y	Z
Percentage of direct interest held by investor company	40%	60%	80%
Retained earnings of investee company at acquisition date	$400,000	$355,000	$330,000
Fair value of non-controlling interests at acquisition date	Not applicable	$470,000	$220,000

(b) Book values at acquisition date were close to fair values. No goodwill impairment charges were made since acquisition date.

(c) Assume no changes in retained earnings other than profit and dividends.

(d) Assume a tax rate of 20%.

Required:

1. Calculate the amounts of dividend income recorded by each investor company. Assume that dividend income is tax-exempt.
2. Applying the default assumptions relating to "control" and "significance influence," show all relevant adjustment and elimination entries (with narratives and workings) for the year ended 31 December 20x5 that need to be passed to prepare the consolidated financial statements in accordance with the relevant IFRS pronouncements.

P7.3 Simultaneous consolidation

On 2 January 20x1, P Ltd paid $316,000 to acquire 160,000 ordinary shares (issued at $1 per share) in SA Ltd. At that point, SA Ltd's retained earnings were $100,000. SA Ltd had an issued share capital of 200,000 ordinary shares (issued at $1 per share). On 2 January 20x1, SA Ltd acquired a 90% interest in SB Ltd for $204,800. The share capital and retained earnings of SB Ltd were $100,000 and $80,000, respectively at the date of acquisition. The book values of identifiable net assets were close to their fair values. On 31 December 20x1, the retained earnings for SA Ltd and SB Ltd were $130,000 and $100,000, respectively. Tax rate was 20%. Assume that the acquisition costs paid by P Ltd, relative to the equity interests held, are proportionate to the fair value of SA Ltd and SB Ltd.

The cost method is used to account for the investment in subsidiaries. The following information is available for the year ended 31 December 20x2:

Income Statement
For the Year Ended 31 December 20x2

	P	SA	SB
Net profit before tax	$163,680	$65,000	$10,000
Tax	(36,400)	(13,000)	(2,000)
Net profit after tax	$127,280	$52,000	$ 8,000

	P	SA	SB
Dividends declared	$60,000	$29,600	$8,140

Required:

Using the simultaneous method of consolidation, prepare consolidation entries to:

1. Eliminate Investment in SA and SB.
2. Allocate profits to non-controlling interests.
3. Eliminate dividends.

P7.4 Indirect associate

Refer to P7.3. If SA had only "significant influence" over SB, show how the journal entries will differ from P7.3 above.

P7.5 Direct and indirect associate

Refer to P7.3. If SA and SB are only associate companies in the hierarchical structure, show the necessary journal entries that need to be passed by P in its financial statements using the equity method.

P7.6 Comprehensive problem set

The following information relates to the acquisition of S Co, A Co, and B Co.

	Acquired company		
	S Co	A Co	B Co
Acquirer	P Co	P Co	S Co
Date of acquisition	1 January 20x1	1 January 20x2	1 July 20x1
Percentage acquired	90%	40%	60%
Fair value of non-controlling interests	$400,000		$260,000

The following information relates to statements of financial position at date of acquisition.

	S Co Book value	S Co Fair value	A Co Book value	A Co Fair value	B Co Book value	B Co Fair value
Intangible assets	$ 0	$ 250,000				
Fixed assets	1,000,000	1,000,000	$250,000	$290,000	$ 40,000	$ 40,000
Inventory	400,000	450,000	150,000	150,000	60,000	55,000
Accounts receivable	200,000	200,000	120,000	120,000	50,000	50,000
Cash	100,000	100,000	45,000	45,000	10,000	10,000
Total assets	$1,700,000	$2,000,000	$565,000	$605,000	$160,000	$155,000
Contingent liability	$ 0	$ 120,000				
Accounts payable	400,000	400,000	$245,000	$245,000	$100,000	$100,000
Share capital	1,000,000		200,000		50,000	
Retained earnings	300,000		120,000		10,000	
Equity and liabilities	$1,700,000	$2,000,000	$565,000	$605,000	$160,000	$155,000

The following information is available for the year ended 31 December 20x3.

Income Statement
For the Year Ended 31 December 20x3

	P Co	S Co	A Co	B Co
Sales	$8,000,000	$3,500,000	$1,500,000	$200,000
Cost of sales	(5,500,000)	(3,000,000)	(900,000)	(120,000)
Gross profit	$2,500,000	$ 500,000	$ 600,000	$ 80,000
Operating expenses net of other income	(1,000,000)	(300,000)	(400,000)	(50,000)
Net profit before tax	$1,500,000	$ 200,000	$ 200,000	$ 30,000
Tax	(300,000)	(40,000)	(40,000)	(6,000)
Net profit after tax	$1,200,000	$ 160,000	$ 160,000	$ 24,000
Dividends declared	(200,000)	(120,000)	(100,000)	(20,000)
Net profit attributable to shareholders	$1,000,000	$ 40,000	$ 60,000	$ 4,000
Retained earnings, 1 January 20x3	2,500,000	500,000	180,000	25,000
Retained earnings, 31 December 20x3	$3,500,000	$ 540,000	$ 240,000	$ 29,000

Statement of Financial Position As at 31 December 20x3				
	P Co	S Co	A Co	B Co
Investment in S Co	$ 4,200,000			
Investment in A Co	600,000			
Investment in B Co		$ 400,000		
Fixed assets	4,000,000	1,400,000	$400,000	$ 70,000
Inventory	2,000,000	600,000	200,000	75,000
Accounts receivable	1,000,000	500,000	35,000	60,000
Cash	120,000	40,000	20,000	5,000
	$11,920,000	$2,940,000	$655,000	$210,000
Accounts payable	$ 5,420,000	$1,400,000	$215,000	$131,000
Share capital	3,000,000	1,000,000	200,000	50,000
Retained earnings	3,500,000	540,000	240,000	29,000
	$11,920,000	$2,940,000	$655,000	$210,000

Additional information:

(a) Intangible assets of S Co had an estimated useful life of five years from the date of acquisition by P Co. Amortization on a straight line basis was deemed to be the appropriate method of recognizing the decline in value.

(b) Remaining useful life of the fixed assets of A Co at the date of acquisition was five years. Assume nil residual values.

(c) Inventory at the date of acquisition was sold off within the following financial period. page 654

(d) The contingent liability of S Co at the date of acquisition was disposed of as follows: 50% of the contingent liability was settled by S Co in the year following acquisition; the remaining 50% of the contingent liability was deemed as unwarranted in the year of settlement.

(e) The following sales of inventory were made during 20x2 and 20x3:

	20x2	20x3
Sales from S to P	$300,000	
Original cost	240,000	
Gross profit on transfer	$ 60,000	
Percentage unsold to third parties at year-end	60%	10%
Sales from P to S		$600,000
Original cost		450,000
Gross profit on transfer		$150,000
Percentage unsold to third parties at year-end		30%

(f) The following transfers of fixed assets from A to P were made during 20x2:

Date of transfer	1 July 20x2
Original cost	$100,000)
Accumulated depreciation	(20,000)
Net book value at transfer date	$ 80,000

Transfer price	$120,000
Remaining useful life at transfer	4 years
Residual value is nil.	

(g) There was no change in share capital of the acquired companies from acquisition date.
(h) Assume a tax rate of 20%. Recognize tax on fair value adjustments.

Required:

1. Prepare the consolidation and equity accounting entries for the year ended 31 December 20x3.
2. Perform an analytical check on non-controlling interests of S Co as at 31 December 20x3.
3. Perform an analytical check on the investment in A Co as at 31 December 20x3.

P7.7 Comprehensive problem set

The details of acquisition of Y Co, Z Co, and W Co are shown below.

	Y Co	Z Co	W Co
Date of acquisition	1 January 20x3	1 January 20x4	1 July 20x4
Percentage acquired by P Co	90%	30%	
Percentage acquired by Y Co			60%
Shareholders' equity at date of acquisition:			
Share capital	$ 800,000	$200,000	$500,000
Retained earnings	900,000	400,000	340,000
	$1,700,000	$600,000	$840,000
Fair value of non-controlling interests at acquisition date	$ 400,000		$300,000

The fair and book values of net identifiable assets of each company at the date of acquisition are shown below:

	Y Co		Z Co		W Co	
	Book value	Fair value	Book value	Fair value	Book value	Fair value
Inventory	$ 220,000	$ 320,000			$100,000	$ 90,000
Intangible assets				$300,000		
Other net assets	1,480,000	1,480,000	$600,000	600,000	740,000	740,000
Net identifiable assets	$1,700,000	$1,800,000	$600,000	$900,000	$840,000	$830,000

The financial statements of P Co, Y Co, Z Co, and W Co for the year ended 31 December 20x5 are shown below.

Income Statement For the Year Ended 31 December 20x5				
	P Co	Y Co	Z Co	W Co
Profit before tax	$4,200,000	$1,800,000	$ 500,000	$300,000
Tax	(840,000)	(360,000)	(80,000)	(66,000)
Profit after tax	$3,360,000	$1,440,000	$ 420,000	$234,000
Dividends declared	(400,000)	(300,000)	(100,000)	(120,000)
Profit retained	$2,960,000	$1,140,000	$ 320,000	$114,000
Retained earnings, 1 January 20x5	1,200,000	1,200,000	700,000	400,000
Retained earnings, 31 December 20x5	$4,160,000	$2,340,000	$1,020,000	$514,000

page 656

Statement of Financial Position As at 31 December 20x5				
	P Co	Y Co	Z Co	W Co
Fixed assets, net book value	$2,800,000	$2,060,000	$ 700,000	$3,500,000
Investment in Y Co, at cost	2,200,000			
Investment in Z Co, at cost	800,000			
Investment in W Co, at cost		800,000		
Inventory	760,000	500,000	230,000	100,000
Intercompany receivable		100,000		
Accounts receivable	600,000	500,000	300,000	250,000
Cash	45,000	100,000	50,000	30,000
	$7,205,000	$4,060,000	$1,280,000	$3,880,000
Accounts payable	$1,745,000	$ 920,000	$ 60,000	$2,866,000
Intercompany payable	100,000			
Share capital	1,200,000	800,000	200,000	500,000
Retained earnings	4,160,000	2,340,000	1,020,000	514,000
	$7,205,000	$4,060,000	$1,280,000	$3,880,000

Additional information:

(a) Inventory of Y Co at the date of acquisition was disposed of in 20x4 while that of W Co was disposed of in January 20x5.

(b) The intangible asset of Z Co at the date of acquisition had no definite useful life and was tested for impairment on an annual basis. Thus far, no impairment loss arose for the intangible asset.

(c) In November 20x4, Y Co transferred inventory to Z Co at an invoiced price of $200,000. The original cost of the inventory was $140,000.

Percentage resold to third parties during 20x4	20%
Percentage resold to third parties during 20x5	60%
Percentage unsold as at end of 20x5	20%

(d) On 1 January 20x5, Z Co transferred its fixed asset to P Co at a transfer price of $172,000. The original cost of the fixed asset was $180,000 and its accumulated depreciation was $36,000 at the date of transfer. The original useful

life of the asset was five years and the remaining life as at 1 January 20x5 was four years.

(e) Assume at tax rate of 20%. Recognize tax effects on fair value adjustments.

Required:

1. Prepare the consolidation and equity accounting journal entries for P Co and its subsidiaries and associate for the year ended 31 December 20x5.
2. Perform an analytical check on the balances of non-controlling interests as at 31 December 20x5.
3. Perform an analytical check on the balance of the investment in associate account as at 31 December 20x5.
4. Prepare the consolidation worksheet for P Co and its group for the year ended 31 December 20x5.

P7.8 Comprehensive problem set

The financial statements of P Co, Y Co, Z Co, and W Co for the year ended 31 December 20x5 are shown below:

Income Statement
For the Year Ended 31 December 20x5

	P Co	Y Co	Z Co	W Co
Profit before tax	$2,400,000)	$1,200,000	$600,000	$ 480,000
Tax	(480,000)	(240,000)	(150,000)	(96,000)
Profit after tax	$1,920,000	$ 960,000	$450,000	$ 384,000
Retained earnings, 1 January 20x5	1,450,000	900,000	600,000	900,000
Dividends declared	(500,000)	(100,000)	(80,000)	(100,000)
Retained earnings, 31 December 20x5	$2,870,000	$1,760,000	$970,000	$1,184,000

Statement of Financial Position
As at 31 December 20x5

	P Co	Y Co	Z Co	W Co
Fixed assets, net book value	$3,000,000	$2,120,000	$ 750,000	$1,900,000
Investment in Y Co, at cost	2,000,000			
Investment in Z Co, at cost	1,200,000			
Investment in W Co, at cost		1,500,000		
Inventory	960,000	500,000	300,000	500,000
Intercompany receivable		250,000		
Accounts receivable	500,000	450,000	250,000	350,000
Cash	120,000	50,000	90,000	100,000
	$7,780,000	$4,870,000	$1,390,000	$2,850,000
Accounts payable	$2,660,000	$2,110,000	$ 20,000	$ 366,000
Intercompany payable	250,000			
Share capital	2,000,000	1,000,000	400,000	800,000
Retained earnings	2,870,000	1,760,000	970,000	1,184,000
Revaluation reserves				500,000
	$7,780,000	$4,870,000	$1,390,000	$2,850,000

Date of acquisition	1 Jul 20x3	1 Jan 20x4	1 Jan 20x2
Percentage acquired by P Co	90%	30%	
Percentage acquired by Y Co			80%
Shareholders' equity at date of acquisition:			
Share capital	$1,000,000	$400,000	$800,000
Retained earnings	600,000	500,000	500,000
Revaluation reserves			400,000
	$1,600,000	$900,000	$1,700,000

	Y Co	W Co
Fair value of non-controlling interests at 1 July 20x3	$200,000	$375,000
Fair value of non-controlling interests at 1 Jan 20x2		350,000
Retained earnings of W Co at 1 July 20x3 (when P Co acquired Y Co)		750,000
Revaluation reserves of W Co at 1 July 20x3		400,000

Information on net identifiable assets of each company is as follows:

	Y Co		Z Co	
	Book value	Fair value	Book value	Fair value
Inventory	$ 400,000	$ 500,000		
Intangible asset				$300,000
Other net assets	1,200,000	1,200,000	$900,000	900,000
Net identifiable assets	$1,600,000	$1,700,000	$900,000	$1,200,000

The fair value of the net assets of W Co was close to its book value at 1 January 20x2 and 1 July 20x3.

Additional information:

(a) Amortization of fair value adjustments:
   (i) Undervalued inventory at date of acquisition was disposed of in January 20x4.
   (ii) The intangible asset had a useful life of five years from acquisition date.

(b) Z Co transferred its fixed asset to P Co on 1 July 20x5. Details are as follows:

Transfer price invoiced by Z Co to P Co		$500,000
Original cost of the fixed asset	$400,000	
Accumulated depreciation	(250,000)	
Net book value		150,000
Profit on sale recorded by Z Co in 20x5		$350,000
Original useful life		4 years
Remaining life as at 1 July 20x5 (assume nil residual value)		2½ years
Remaining life as at 31 Dec 20x5		2 years

(c) Other transactions were as follows:

Sale of inventory from Y Co to P Co	$150,000
Original cost of inventory	100,000
Gross profit recorded by Y Co	$ 50,000
Date of sale	1 November 20x4
Percentage resold to third parties during 20x4	20%
Percentage resold to third parties during 20x5	70%
Percentage in inventory at end 20x5	10%

(d) Sale of inventory from P Co to Y Co in 20x5:

Transfer price	$200,000
Original cost of inventory	$250,000
Percentage in inventory at end 20x5	40%

The loss on transfer was not indicative of an impairment loss.

(e) Assume a tax rate of 20%. Recognize tax effects on fair value adjustments.

Required:

1. Prepare the consolidation and equity accounting entries for 20x5.
2. Prepare the consolidation worksheet.
3. Perform an analytical check on the year-end balances of non-controlling interests in Y and W and the Investment in Z.

P7.9 Intragroup transfer of assets

P Co had direct and indirect interests in the following companies as at 1 January 20x0:

Companies	Direct investor	Percentage ownership
S Co	P Co	70%
A Co	P Co	40%
B Co	S Co	90%

On 1 January 20x0, a transfer of fixed assets was made as follows:

Transfer of fixed assets

Transfer price		$250,000
Original cost	$240,000	
Accumulated depreciation	(40,000)	
Net book value at date of transfer		(200,000)
Profit on transfer		$ 50,000
Remaining useful life at date of transfer		10 years

Profit figures for the investee companies for the years ended 31 December 20x0 and 20x1 are shown below.

	20x0		20x1	
	Profit before tax	Tax	Profit before tax	Tax
S Co	$800,000	$160,000	$900,000	$180,000
A Co	300,000	60,000	250,000	50,000
B Co	650,000	130,000	500,000	100,000

page 660

Required:

Show the equity accounting and consolidation adjustments as at 31 December 20x0 and 31 December 20x1 under each of the two alternative situations below:

1. Transfer is made from S Co to A Co;
2. Transfer is made from B Co to P Co.

Recognize a full-year depreciation and tax effects at 20%.

P7.10 Business combination achieved in stages

P acquired S in two successive purchases resulting in control on 31 December 20x0. Details of the purchases and other information relating to S are shown below:

Date	Acquisition cost paid for incremental investment	Fair value of S as an entity	Percentage acquired in incremental investment	Cumulative percentage held by P
1 July 20x0	$ 5,000,000	$16,000,000	30%	30%
31 December 20x0	7,000,000	17,500,000	40%	70%
	$12,000,000			
31 December 20x0				
Share capital of S		10,000,000		
Retained earnings of S		5,000,000		
		$15,000,000		
Non-controlling interests of S as at 31 December 20x0		$5,250,000		

Book value of net identifiable assets of S was close to its fair value at each acquisition date. Extracts of S's income statement for the period 1 July 20x0 to 31 December 20x0 are shown below:

Profit before tax	$3,000,000
Tax expense	(600,000)
Profit after tax	$2,400,000
Dividends declared	(400,000)
Profit retained	$2,000,000

Required:

1. Prepare the journal entries in respect of the investment in P's separate financial statements for the year ended 31 December 20x0.
2. Prepare the equity accounting and consolidation entries to be put through in P's consolidated financial statements for the year ended 31 December 20x0.

P7.11 Change in ownership interests

Consider the following alternative scenarios relating to an investor company P Co's investment in another company, known as X Co.

Scenario: P Co increases ownership interest in X Co from 60% to 75%

Details	Ownership interest	Amount
Purchase of initial investment of 60% interest in X Co on 1 January 2010		
Fair value of consideration transferred	60%	$13,000,000
Fair value of non-controlling interests	40%	6,000,000
Purchase of additional investment of 15% interest in X Co on 1 July 2011		
Incremental purchase	15%	4,500,000
Non-controlling interests' balance as at 1 July 2011		7,500,000
Fair value of identifiable net assets of X Co		22,000,000
Book value of identifiable net assets of X Co		19,000,000
Tax rate is 20%		

Required:

Determine the amounts of the following items (if any) arising at the date of the most recent transaction in each scenario:

1. New goodwill
2. Effects that are taken to the Income Statement
   (a) Remeasurement gain or loss
   (b) Profit or loss on sale of investment
3. Equity effects that are not taken to the Income Statement
   (a) Profit or loss arising from changes in ownership
   (b) Changes in non-controlling interests

P7.12 Change in ownership interests

Scenario: P Co decreases ownership interest in X Co from 95% to 70%

Details	Ownership interest	Amount
Purchase of initial investment of 95% interest in X Co on 1 December 2010		
Fair value of consideration transferred	95%	$29,000,000
Fair value of non-controlling interests	5%	1,500,000

P Co's share of 95% of equity of X Co as at 31 December 2011		27,300,000
Divestment of 25% interest in X Co on 31 December 2011		
Sales proceeds	(25%)	12,800,000

Required:

Determine the amounts of the following items (if any) arising at the date of the most recent transaction in each scenario:

1. New goodwill
2. Effects that are taken to the Income Statement
   (a) Remeasurement gain or loss
   (b) Profit or loss on sale of investment
3. Equity effects that are not taken to the Income Statement
   (a) Profit or loss arising from changes in ownership
   (b) Changes in non-controlling interests

P7.13 Change in ownership interests

Scenario: P Co increases ownership interest in X Co from 40% to 60%

Details	Ownership interest	Amount
Purchase of initial investment of 40% interest in X Co on 1 January 2009		
Original cost of previously acquired interests	40%	$5,200,000
P Co's share (40%) of post-acquisition change in equity of X Co for the period from 1 January 2009 to 1 April 2011		900,000
Purchase of additional investment of 20% interest in X Co on 1 April 2011		
Fair value of consideration transferred	20%	4,100,000
Fair value of non-controlling interests	40%	6,500,000
Fair value of previously acquired interests	40%	6,500,000
Fair value of identifiable net assets (*deferred tax has not been considered yet*)		15,000,000
Book value of identifiable net assets		13,000,000
Tax rate		20%

Required:

Determine the amounts of the following items (if any) arising at the date of the most recent transaction in each scenario:

1. New goodwill
2. Effects that are taken to the Income Statement
   (a) Remeasurement gain or loss
   (b) Profit or loss on sale of investment
3. Equity effects that are not taken to the Income Statement
   (a) Profit or loss arising from changes in ownership

(b) Changes in non-controlling interests

P7.14 Change in ownership interests

Scenario: P Co decreases ownership interest in X Co from 80% to 50%

Details	Ownership interest	Amount
Purchase of initial investment of 80% interest in X Co on 1 January 2009		
Fair value of consideration transferred	80%	$16,000,000
Fair value of non-controlling interests	20%	3,600,000
P Co's share (80%) of post-acquisition change in equity of X Co for the period from 1 January 2009 to 1 September 2011		2,300,000
Sale of investment of 30% interest in X Co on 1 September 2011		
Sales proceeds	(30%)	9,000,000
Fair value of retained interests	50%	11,800,000
Original cost of investment	50%	10,000,000

Required:

Determine the amounts of the following items (if any) arising at the date of the most recent transaction in each scenario:

1. New goodwill
2. Effects that are taken to the Income Statement
   (a) Remeasurement gain or loss
   (b) Profit or loss on sale of investment
3. Equity effects that are not taken to the Income Statement
   (a) Profit or loss arising from changes in ownership
   (b) Changes in non-controlling interests

P7.15 and P7.16 Multilevel consolidation and change in ownership interests

The problem set applies to P7.15 and P7.16

The financial statements of P Co and its subsidiaries and associate show below:

Income Statement for year ended 31 December 20x3					
	P Co	X Co	Z Co	Y Co	Consolidated
Profit before tax	3,200,000	2,200,000	856,000	750,000	
Tax	(640,000)	(440,000)	(154,080)	(150,000)	
Profit after tax	2,560,000	1,760,000	701,920	600,000	
Income to NCI					
Dividends declared	(500,000)	(230,000)	(70,000)	(134,000)	
Profit retained	2,060,000	1,530,000	631,920	466,000	
Retained earnings, 1 Jan 20x3	2,560,000	1,245,000	550,000	900,000	
Retained earnings, 31 Dec 20x3	4,620,000	2,775,000	1,181,920	1,366,000	

Statement of Financial Position as at 31 December 20x3					
	P Co	X Co	Z Co	Y Co	Consolidated
Goodwill					
Fixed assets, net book value	2,190,000	1,345,900	880,000	1,345,000	
Investment in X Co	4,375,000				
Investment in Z Co	1,500,000				
Investment in Y Co		900,000			
Inventory	430,000	532,000	250,000	660,000	
Intercompany receivable		100,000			
Other net assets	920,000	1,967,000	1,800,000	1,632,000	
Cash	145,900	50,000	40,000	10,000	
	9,560,900	4,894,900	2,970,000	3,647,000	
Accounts payable	3,240,900	919,900	888,080	1,811,000	
Intercompany payable	100,000				
Share capital	1,600,000	1,200,000	900,000	350,000	
Retained earnings	4,620,000	2,775,000	1,181,920	1,366,000	
Revaluation reserves				120,000	
Capital reserve					
Non-controlling interests					
	9,560,900	4,894,900	2,970,000	3,647,000	

Details relating to information at acquisition date and subsequent changes in ownership are shown below:

	X Co	Z Co	Y Co
Date of acquisition/investment by immediate parent/investor	1 Jan 20x1	1 Jan 20x1	1 July 20x0
Percentage acquired by P Co	80%	30%	
Percentage of retained interests by P Co on 31 Dec 20x2	70%		
Percentage acquired by X Co			70%
Shareholders' equity at date of acquisition/investment			
by immediate parent/investor			
Share capital	1,200,000	900,000	350,000
Retained earnings	700,000	345,000	450,000
Revaluation reserves			90,000
	1,900,000	1,245,000	890,000

	X Co	Y Co
Fair value of direct NCI of X Co and indirect NCI of Y Co at 1 January 20x1	626,300	
Fair value of direct NCI of Y Co at 1 January 20x1		363,500
Fair value of direct NCI of Y Co at 1 July 20x0		421,000
Retained earnings of Y Co at 1 January 20x1 (when X Co was acquired by P Co)		600,000
Revaluation reserves of Y Co at 1 January 20x1 and 31 Dec 20x2		95,000
Proceeds on sale of 10% of X Co at 31 Dec 20x2	1,000,000	
NCI: non-controlling interests		

Fair and book values of net assets of each company at date of acquisition/investment by P Co

	X Co →		Z Co →		Y Co (1 Jan 20x1) →	
	Book value	Fair value	Book value	Fair value	Book value	Fair value
Provision for claim	(160,000)	(220,000)				
Intangible asset				100,000		
Other net assets	2,060,000	2,060,000	1,245,000	1,245,000	1,045,000	1,045,000
Total net assets	1,900,000	1,840,000	1,245,000	1,345,000	1,045,000	1,045,000

Additional information:

1. During 20x2, X Co recognized a litigation settlement of $200,000 in respect of the provision for claims on an award of damages to the plaintiff by the courts. The case is still pending the final outcome following appeal by the plaintiff for a higher award of damages.
2. During August 20x2, Y Co sold excess inventory to P Co as follows:

Transfer price	$120,000
Original cost (Carrying amount)	$90,000
Percentage resold to third parties during 20x2	25%
Percentage resold to third parties during 20x3	65%

3. Intangible assets of Z Co had a remaining lease term of 5 years from date of acquisition.
4. The fair value of net assets of Y Co was close to its book value when acquired by X Co and P Co.
5. On 1 July 20x2, Z Co sold machinery to Y Co as follows:

Transfer price	$180,000
Profit on sale	$40,000
Expected useful life as at 30 June 20x2	8 years
Residual value	Negligible

6. Apply a tax rate of 20% on all appropriate adjustments. Recognize tax effects on fair value adjustments. Companies recognize impairment losses, if any, at the financial year-end.

P7.15 Consolidation and Equity Accounting Entries

Prepare the consolidation and equity accounting entries for the year ended 31 December 20x3.

P7.16 Analytical Checks and End-Results Approach

Prepare the consolidated financial statements for the year ended 31 December 20x3 using the end-result approach through workings (i.e., analytically derive the balances without having to formulate the consolidation and equity accounting entries and worksheets).

P7.17 and P7.18 Multilevel consolidation and change in ownership interests

The problem set applies to P7.17 and P7.18

The financial statements of P Co and its subsidiaries and associate are as follows:

Income Statement for year ended 31 December 20x6					
	P Co	B Co	D Co	C Co	Consolidated
Profit before tax	5,800,000	1,670,000	600,000	420,000	
Tax	(1,160,000)	(334,000)	(120,000)	(81,400)	
Profit after tax	4,640,000	1,336,000	480,000	338,600	
Income to NCI					
Dividends declared	(320,000)	(200,000)	(150,000)	(100,000)	
Profit retained	4,320,000	1,136,000	330,000	238,600	
Retained earnings, 1 Jan 20x6	2,000,000	1,456,000	1,400,000	232,000	
Retained earnings, 31 Dec 20x6	6,320,000	2,592,000	1,730,000	470,600	

Statement of Financial Position as at 31 December 20x6					
	P Co	B Co	D Co	C Co	Consolidated
Goodwill					
Fixed assets, net book value	4,490,000	3,340,000	2,540,000	761,000	
Investment in B Co	2,500,000				
Investment in D Co	1,150,000				
Investment in C Co		800,000			
Inventory	600,000	520,000	448,500	348,900	
Intercompany receivable	1,700,000				
Other net assets	1,125,800	1,880,000	1,973,000	557,200	
Cash	747,000	256,000	189,000	56,000	
	12,312,800	6,796,000	5,150,500	1,723,100	
Accounts payable	2,492,800	1,004,000	2,520,500	676,500	
Bank loans	1,500,000				
Intercompany payable		1,700,000			
Share capital	2,000,000	1,500,000	900,000	420,000	
Retained earnings	6,320,000	2,592,000	1,730,000	470,600	
Revaluation reserves				156,000	
Non-controlling interests					
	12,312,800	6,796,000	5,150,500	1,723,100	

Details relating to information at acquisition date and subsequent changes in ownership are shown below:

	B Co	D Co	C Co
Date of acquisition/investment by immediate parent/investor	1 Aug 20x4	1 Jan 20x5	1 July 20x3
Percentage acquired by P Co	90%	60%	
Percentage of retained interests by P Co		40%	
Percentage acquired by B Co			60%
Shareholders' equity at date of acquisition by immediate parent/investor			
Share capital	1,500,000	900,000	420,000
Retained earnings	650,000	400,000	120,000
Revaluation reserves			60,000
	2,150,000	1,300,000	600,000

	B Co	C Co
Fair value of direct NCI of B and indirect NCI of C at 1 Aug 20x4	250,000	
Fair value of direct NCI of C at 1 Aug 20x4		550,000
Fair value of direct NCI of C at 1 July 20x3		420,000
Retained earnings of C Co at 1 Aug 20x4 (when B Co was acquired by P Co)		180,000
Revaluation reserves of C Co at 1 Aug 20x4		70,000
Fair value of inventory of C Co at 1 July 20x3		120,000
Book value of inventory of C Co at 1 July 20x3		90,000

NCI: Non-controlling interests

Fair and book values of net assets of each company at date of acquisition by P Co

	B Co ----------→		D Co ----------→		C Co (1 Aug 20x4) ---------→	
	Book value	Fair value	Book value	Fair value	Book value	Fair value
Accounts receivable	300,000	220,000				
Inventory					150,000	200,000
Other net assets	1,850,000	1,850,000	1,300,000	1,300,000	520,000	520,000
Total net assets	2,150,000	2,070,000	1,300,000	1,300,000	670,000	720,000

Additional information:

1. D Co was originally a subsidiary of P Co when P Co purchased an initial investment of 60% interest in D Co. On 1 January 20x5, P Co sold 20% interest in D Co, reducing the ownership interest in D Co to 40%.

Date of acquisition of D Co as subsidiary	1 July 20x2

Fair value of consideration transferred at 1 July 20x2	$1,725,000
Fair value of non-controlling interests at 1 July 20x2	$700,000
P Co's share (60%) of post-acquisition change in equity of D Co for the period of 1 July 20x2 to 1 January 20x5	$530,000
Date of loss of control	1 January 20x5
Sales proceeds for 20% interest	$900,000
Fair value of retained interests	$1,800,000

2. The fair value of accounts receivable of B Co was lower than the book value because of an expected loss on a debtor. B Co recognized an incurred loss in 20x6 when the debtor went into bankruptcy. In 20x6, B Co expensed off $280,000 as impairment losses. page 668
3. The undervalued inventory of C Co was disposed as follows:

Sixty percent of the inventory of C Co was sold during 20x5 for	$120,000
Thirty percent was sold during 20x6 for	$60,000

4. P Co provided a long term loan to B Co on 1 September 20x5 to finance the construction of equipment. The use of the loan proceeds met the requirements of IAS 23 *Borrowing Costs*. B Co completed the construction of the equipment on 1 February 20x6 and commenced the depreciation of the equipment over a useful life of 5 years from that date. Residual value is negligible. The following interest income was recognized by P Co:

Interest income from 1 Sep 20x5 to 31 Dec 20x5	$120,000
Interest income from 1 Jan 20x6 to 30 Jan 20x6	$205,000
Interest income from 1 Feb 20x6 to 31 Dec 20x6	$150,000

P Co incurred interest expense on third party loans as follows:

Interest expense from 1 Sep 20x5 to 31 Dec 20x5	$105,000
Interest expense from 1 Jan 20x6 to 30 Jan 20x6	$195,000
Interest expense from 1 Feb 20x6 to 31 Dec 20x6	$126,000

5. There was a sale of inventory from D Co to P Co, with details as follows:

Date of sale	16 July 20x5
Fair value at date of sale	$90,000
Transfer price for sale of inventory	$70,000
Original cost of inventory	$80,000
Percentage resold to third parties during 20x5	25%
Percentage resold to third parties during 20x6	55%
Percentage stock take loss on inventory as at 31 December 20x6	10%

6. Apply a tax rate of 20% on all appropriate adjustments. Recognize tax effects on fair value adjustments. Companies recognize impairment losses, if any, at the financial year-end.

P7.17 Consolidation and Equity Accounting Entries

Prepare the consolidation and equity accounting entries for the year ended 31 December 20x6.

P7.18 Analytical Checks and End-Results Approach

Prepare the consolidated financial statements for the year ended 31 December 20x6 using the end-result approach through workings (i.e., analytically derive the balances without having to formulate the consolidation and equity accounting entries and worksheets).

page 669

P7.19 and P7.20 Multilevel consolidation and changes in ownership interests

The problem set applies to P7.19 and P7.20

The financial statements of P Co and its subsidiaries and associate are shown below:

Income Statement for year ended 31 December 20x6

	P Co	B Co	D Co	C Co	Consolidated
Profit before tax	4,377,000	1,558,000	677,000	489,000	
Tax	(875,400)	(311,600)	(135,400)	(97,800)	
Profit after tax	3,501,600	1,246,400	541,600	391,200	
Income to NCI					
Dividends declared	(200,000)	(130,000)	(50,000)	(80,000)	
Profit retained	3,301,600	1,116,400	491,600	311,200	
Retained earnings, 1 Jan 20x6	2,416,000	850,900	876,600	453,200	
Retained earnings, 31 Dec 20x6	5,717,600	1,967,300	1,368,200	764,400	

Statement of Financial Position as at 31 December 20x6

	P Co	B Co	D Co	C Co	Consolidated
Goodwill					
Intangible assets, net book value					
Fixed assets, net book value	2,844,000	2,063,000	1,459,000	898,000	
Investment in B Co	3,500,000				
Investment in D Co	800,000				
Investment in C Co		820,000			
Inventory	820,000	610,000	510,000	267,000	
Intercompany receivable	1,800,000				
Other net assets	2,384,000	1,382,000	815,450	535,000	
Cash	320,000	700,000	150,000	127,000	
	12,468,000	5,575,000	2,934,450	1,827,000	
Accounts payable	4,050,400	207,700	766,250	382,600	
Bank loans	1,400,000				
Intercompany payable		1,800,000			
Share capital	1,300,000	1,600,000	800,000	550,000	
Retained earnings	5,717,600	1,967,300	1,368,200	764,400	
Revaluation reserves				130,000	
Non-controlling interests					
	12,468,000	5,575,000	2,934,450	1,827,000	

page 670

	B Co	D Co	C Co
Date of acquisition by direct investor	1/8/20x4	1/7/20x2**	1/7/20x3
Date of change from control to significant influence		1/1/20x5*	
Percentage of investment acquired by P Co	90%	60%	
Percentage of retained interests by P Co		40%	
Percentage acquired by B Co			70%
Shareholders' equity as at the above dates			
Share capital	1,600,000	800,000	550,000
Retained earnings	685,000	393,000**	332,000
Revaluation reserves			85,000
	2,285,000	1,193,000	967,000

	B Co	C Co
Fair value of non-controlling interests at 1 Aug 20x4	350,000	320,000
Fair value of non-controlling interests at 1 July 20x3		303,800
Retained earnings of C Co at 1 Aug 20x4 (when B Co was acquired by P Co)		320,000
Revaluation reserves of C Co at 1 Aug 20x4		95,000
Additional revaluation reserves of C Co as at 31 Dec 20x6		35,000

Fair and book values of net assets of B Co and C Co as at 1 Aug 20x4 and D Co as at 1 July 20x2

	B Co----------→		D Co----------→		C Co----------→	
	Book value	Fair value	Book value	Fair value	Book value	Fair value
Provision for loss	–	(200,000)				
Intangible assets	–	150,000				
Inventory					120,000	200,000
Other net assets	2,285,000	2,285,000	1,193,000	1,193,000	845,000	845,000
Total net assets	2,285,000	2,235,000	1,193,000	1,193,000	965,000	1,045,000

** Date of acquisition of D as a subsidiary	1 July 20x2
Fair value of consideration transferred on 1 July 20x2	1,200,000
Fair value of NCI of D Co on 1 July 20x2	700,000
* Date of loss of control of D Co	1 Jan 20x5
Retained earnings of D Co on 1 Jan 20x5	860,000
Sales proceeds for 20% interest	600,000
Fair value of retained interests	1,200,000

Additional information:

1. Intangible assets of B Co had an indefinite useful life. On 31 December 20x6, the annual impairment test revealed a loss of $110,000.

2. Ninety percent of the undervalued inventory of C Co was sold to third parties during 20x5 and the remaining was written down to zero during 20x6.
3. P provided a loan to B to finance the self-construction of a warehouse. The warehouse was ready for use on 1 July 20x6 and had an estimated useful life of 10 years. The following transactions were recorded in P's books:

page 671

Interest income from B	
1 Sept 20x5 to 31 Dec 20x5	190,000
1 Jan 20x6 to 30 June 20x6	285,000
Interest expense paid by P to third party banks	
1 Sept 20x5 to 31 Dec 20x5	160,000
1 Jan 20x6 to 30 June 20x6	240,000

4. Provision for loss of B Co was the present value of the expected loss on a claim by a third party. On 1 December 20x5, court hearings confirmed the actual loss of $100,000, which was recognized by B Co.
5. Apply a tax rate of 20% on all appropriate adjustments. Recognize tax effects on fair value adjustments.

P7.19 Consolidation and Equity Accounting Entries

Prepare the consolidation and equity accounting entries for the year ended 31 December 20x6.

P7.20 Analytical checks and end-results approach

Prepare the consolidated financial statements for the year ended 31 December 20x6 using the end result approach through workings (i.e., analytically derive the balances without having to formulate the consolidation and equity accounting entries and worksheets).

P7.21 and P7.22 Multilevel consolidation and change in ownership interests

The problem set applies to P7.21 and P7.22

The financial statements of P Co and its subsidiaries and associate are as follows:

Income Statement for year ended 31 December 20x3					
	P Co	X Co	Z Co	Y Co	Consolidated
Profit before tax	6,273,000	2,840,000	1,071,000	550,000	
Tax	(1,254,600)	(568,000)	(214,200)	(110,000)	
Profit after tax	5,018,400	2,272,000	856,800	440,000	
Income to NCI					
Dividends declared	(320,000)	(240,000)	(60,000)	(65,000)	
Profit retained	4,698,400	2,032,000	796,800	375,000	
Retained earnings, 1 Jan 20x3	4,530,000	1,280,000	474,000	900,000	
Retained earnings, 31 Dec 20x3	9,228,400	3,312,000	1,270,800	1,275,000	

page 672

Statement of Financial Position as at 31 December 20x3					
	P Co	X Co	Z Co	Y Co	Consolidated
Goodwill					
Fixed assets, net book value	3,630,000	1,987,000	1,510,000	661,000	
Investment in X Co	3,150,000				
Investment in Z Co	1,030,000				
Investment in Y Co		800,000			
Inventory	496,000	678,000	302,000	950,000	
Intercompany receivable		114,000			
Other net assets	2,811,000	1,321,000	1,680,000	822,000	
Cash	365,000	292,000	164,000	99,430	
	11,482,000	5,192,000	3,656,000	2,532,430	
Accounts payable	679,600	680,000	1,585,200	807,430	
Intercompany payable	114,000				
Share capital	1,460,000	1,200,000	800,000	350,000	
Retained earnings	9,228,400	3,312,000	1,270,800	1,275,000	
Revaluation reserves				100,000	
Capital reserve					
Non-controlling interests					
	11,482,000	5,192,000	3,656,000	2,532,430	

	X Co	Z Co	Y Co
Date of acquisition by direct investor	1/1/20x1	1/1/20x1	1/7/20x0
Percentage acquired by P Co on initial investment	90%	30%	
Percentage of retained interests by P Co	70%		
Percentage acquired by X Co			60%
Shareholders' equity at above dates			
Share capital	1,200,000	800,000	350,000
Retained earnings	770,000	349,000	537,000
Revaluation reserves			90,000
	1,970,000	1,149,000	977,000

	X Co	Y Co
Fair value of NCI at 1 January 20x1	320,300	462,000
Fair value of NCI at 1 July 20x0		320,000
Retained earnings of Y Co at 1 January 20x1		
(when X Co was acquired by P Co)		610,000
Revaluation reserves of Y Co at 1 January 20x1		120,000
Revaluation reserves of Y Co decreased on 31 Dec 20x3		(20,000)
Proceeds on sale of 20% of X Co at 31 Dec 20x2	1,150,000	

Fair and book values of net assets of each company at 1 January 20x1

	X Co ----------→		Z Co ----------→		Y Co ---------→	
	Book value	Fair value	Book value	Fair value	Book value	Fair value
Provision for claim		(100,000)				
Intangible asset				400,000		
Other net assets	1,970,000	1,970,000	1,149,000	1,149,000	1,080,000	1,080,000
Total net assets	1,970,000	1,870,000	1,149,000	1,549,000	1,080,000	1,080,000

Additional information:

1. The fair values of identifiable net assets of Y Co were close to their book values on 1 July 20x0 and 1 January 20x1. The estimated useful life of intangible assets of Z Co was 5 years, but the recoverable amount on 31 December 20x3 was assessed to be $100,000.
2. The provision for claims of X Co was settled in full during 20x1 for $170,000.
3. On 1 July 20x3, Z provided new electrical equipment to P Co at the invoiced price of $120,000. The original cost was $200,000 and the net book value was $60,000. The estimated useful life of the equipment was 3 years. Z Co recognized a profit of $$60,000 on the sale. Due to operational changes, the estimated recoverable amount of the equipment as at 31 December 20x3 was $60,000.
4. On 1 October 20x2, Y sold inventory to P Co at the transfer price of $130,000. The original cost and carrying amount of the inventory was $70,000. Percentage resold to third parties were as follows:

20x2	10%
20x3	80%

5. Apply a tax rate of 20% on all appropriate adjustments. Recognize tax effects on fair value adjustments. Companies recognize impairment losses, if any, at the financial year-end.

P7.21 Consolidation and Equity Accounting Entries

Prepare the consolidation and equity accounting entries for the year ended 31 December 20x3.

P7.22 Analytical checks and end-results approach

Prepare the consolidated financial statements for the year ended 31 December 20x3 using the end result approach through workings (i.e., analytically derive the balances without having to formulate the consolidation and equity accounting entries and worksheets).

[1] In this situation, P acquired A as a stand-alone entity. However, if P had purchased A after A acquired B, the fair value of considera-tion transferred must be eliminated against both A's and B's share capital and pre-acquisition retained earnings as at 1 January 20x0. The investment in B must also be eliminated. An example is shown in Illustration 7.2.

[2] In this situation, P acquired A as a stand-alone entity. However, if P had purchased A after A acquired B, the investment must be eliminated against the consolidated equity of A at the date of acquisition. An example is shown in Illustration 7.2.

[3] The check is independent but by no means error-proof. If an element is omitted in both the derivation and the check of non-controlling interests, the check will affirm consistency but not accuracy.

[4] Basis of conclusions, BCZ182, IFRS 10.

[5] IFRS 10 paragraph B98.

[6] An alternative approach is to adjust the unrealized profit from the investor's profit and the investment in associate in a downstream sale.

[7] Such transactions are also known as piecemeal acquisitions of associates or step acquisition of associates in practice.